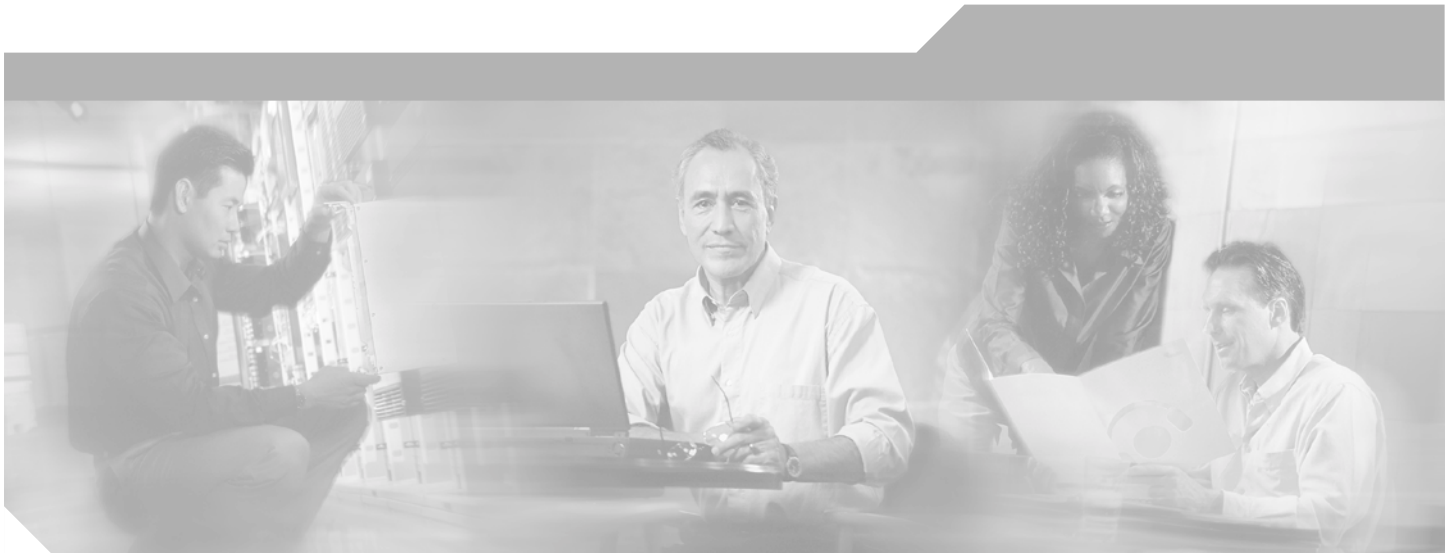


Catalyst 2950 Desktop Switch Command Reference

Cisco IOS Release 12.1(11)EA1
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Preface

Audience

This guide is for the networking professional using the Cisco IOS command-line interface (CLI) to manage the Catalyst 2950 switch, hereafter referred to as *the switch*. Before using this guide, you should have experience working with the Cisco IOS and be familiar with the concepts and terminology of Ethernet and local area networking.

Purpose

The Catalyst 2950 switch is supported by either the standard software image (SI) or the enhanced software image (EI). The enhanced software image provides a richer set of features, including access control lists (ACLs), enhanced quality of service (QoS) features, the Secure Shell (SSH) Protocol, extended-range VLANs, IEEE 802.1W Rapid Spanning Tree Protocol (STP), and the IEEE 802.1S Multiple STP.

The enhanced software image supports these switches:

- Catalyst 2950C-24
- Catalyst 2950G-12-EI
- Catalyst 2950G-24-EI
- Catalyst 2950G-24-EI-DC
- Catalyst 2950G-48-EI
- Catalyst 2950T-24

The standard software image supports these switches:

- Catalyst 2950-12
- Catalyst 2950-24
- Catalyst 2950SX-24

This guide provides the information you need about the CLI commands that have been created or changed for use with the Catalyst 2950 family of switches. For information about the standard IOS Release 12.1 commands, refer to the IOS documentation set available from the Cisco.com home page by selecting **Service and Support** > **Technical Documents**. On the Cisco Product Documentation home page, select **Release 12.1** from the Cisco IOS Software drop-down list.

This guide does not provide procedures for configuring your switch. For detailed configuration procedures, refer to the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

This guide does not describe system messages you might encounter. For more information, refer to the *Catalyst 2950 Desktop Switch System Message Guide* for this release.

Organization

This guide is organized into these chapters:

Chapter 1, "Using the Command-Line Interface," describes how to access the command modes and use the switch CLI to configure software features. It also lists the commands that have the same function but different syntax in software releases earlier than Release 12.1(6)EA2 and in Release 12.1(6)EA2 or later.

Chapter 2, "Cisco IOS Commands," describes in alphabetical order the IOS commands that you use to configure and monitor your switch.

Appendix A, "Debug Commands," describes the **debug** privileged EXEC commands. Debug commands are helpful in diagnosing and resolving internetworking problems.

Conventions

This guide uses these conventions to convey instructions and information:

Command descriptions use these conventions:

- Commands and keywords are in **boldface** text.
- Arguments for which you supply values are in *italic*.
- Square brackets ([ ]) mean optional elements.
- Braces ({ }) group required choices, and vertical bars (|) separate the alternative elements.
- Braces and vertical bars within square brackets ([{ | }]) mean a required choice within an optional element.

Interactive examples use these conventions:

- Terminal sessions and system displays are in `screen` font.
- Information you enter is in **`boldface screen`** font.
- Nonprinting characters, such as passwords or tabs, are in angle brackets (< >).

Notes, cautions, and tips use these conventions and symbols:

Note Means *reader take note*. Notes contain helpful suggestions or references to materials not contained in this manual.

Caution Means *reader be careful*. In this situation, you might do something that could result equipment damage or loss of data.

Timesaver Means the following *will help you solve a problem*. The tips information might not be troubleshooting or even an action, but could be useful information.

Related Publications

These documents provide complete information about the switch and are available from this URL:

http://www.cisco.com/univercd/cc/td/doc/product/lan/cat2950/index.htm

You can order printed copies of documents with a DOC-xxxxxx= number from the Cisco.com sites and from the telephone numbers listed in the "Obtaining Documentation" section on page xiii.

- *Release Notes for the Catalyst 2950 Desktop Switch*, (not orderable but is available on Cisco.com)

Note Switch requirements and procedures for initial configurations and software upgrades tend to change and therefore appear only in the release notes. Before installing, configuring, or upgrading the switch, refer to the release notes on Cisco.com for the latest information.

- *Catalyst 2950 Desktop Switch Software Configuration Guide* (order number DOC-7811380=)
- *Catalyst 2950 Desktop Switch Command Reference* (order number DOC-7811381=)
- *Catalyst 2950 Desktop Switch System Message Guide* (order number DOC-7814233=)
- *Catalyst 2950 Desktop Switch Hardware Installation Guide* (order number DOC-7811157=)
- *Catalyst GigaStack Gigabit Interface Converter Hardware Installation Guide* (order number DOC-786460=)
- Cluster Management Suite (CMS) online help (available only from the switch CMS software)
- *CWDM Passive Optical System Installation Note* (not orderable but is available on Cisco.com)
- *1000BASE-T Gigabit Interface Converter Installation Notes* (not orderable but is available on Cisco.com)

Obtaining Documentation

The following sections explain how to obtain documentation from Cisco Systems.

World Wide Web

You can access the most current Cisco documentation on the World Wide Web at the following URLs:

http://www.cisco.com

Translated documentation is available at the following URL:

http://www.cisco.com/public/countries_languages.shtml

Documentation CD-ROM

Cisco documentation and additional literature are available in a Cisco Documentation CD-ROM package, which is shipped with your product. The Documentation CD-ROM is updated monthly and may be more current than printed documentation. The CD-ROM package is available as a single unit or through an annual subscription.

Ordering Documentation

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- Registered Cisco Direct Customers can order Cisco product documentation from the Networking Products MarketPlace:

  http://www.cisco.com/cgi-bin/order/order_root.pl

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  http://www.cisco.com/go/subscription

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Cisco Systems
Attn: Document Resource Connection
170 West Tasman Drive
San Jose, CA 95134-9883

We appreciate your comments.

Obtaining Technical Assistance

Cisco provides Cisco.com as a starting point for all technical assistance. Customers and partners can obtain documentation, troubleshooting tips, and sample configurations from online tools by using the Cisco Technical Assistance Center (TAC) website. Cisco.com registered users have complete access to the technical support resources on the Cisco TAC website.

Cisco.com

Cisco.com is the foundation of a suite of interactive, networked services that provides immediate, open access to Cisco information, networking solutions, services, programs, and resources at any time, from anywhere in the world.

Cisco.com is a highly integrated Internet application and a powerful, easy-to-use tool that provides a broad range of features and services to help you to

- Streamline business processes and improve productivity
- Resolve technical issues with online support
- Download and test software packages
- Order Cisco learning materials and merchandise
- Register for online skill assessment, training, and certification programs

You can self-register on Cisco.com to obtain customized information and service. To access Cisco.com, go to the following URL:

http://www.cisco.com

Technical Assistance Center

The Cisco TAC is available to all customers who need technical assistance with a Cisco product, technology, or solution. Two types of support are available through the Cisco TAC: the Cisco TAC Web Site and the Cisco TAC Escalation Center.

Inquiries to Cisco TAC are categorized according to the urgency of the issue:

- Priority level 4 (P4)—You need information or assistance concerning Cisco product capabilities, product installation, or basic product configuration.
- Priority level 3 (P3)—Your network performance is degraded. Network functionality is noticeably impaired, but most business operations continue.
- Priority level 2 (P2)—Your production network is severely degraded, affecting significant aspects of business operations. No workaround is available.
- Priority level 1 (P1)—Your production network is down, and a critical impact to business operations will occur if service is not restored quickly. No workaround is available.

Which Cisco TAC resource you choose is based on the priority of the problem and the conditions of service contracts, when applicable.

Cisco TAC Website

The Cisco TAC website allows you to resolve P3 and P4 issues yourself, saving both cost and time. The site provides around-the-clock access to online tools, knowledge bases, and software. To access the Cisco TAC website, go to the following URL:

http://www.cisco.com/tac

All customers, partners, and resellers who have a valid Cisco services contract have complete access to the technical support resources on the Cisco TAC website. The Cisco TAC website requires a Cisco.com login ID and password. If you have a valid service contract but do not have a login ID or password, go to the following URL to register:

http://www.cisco.com/register/

If you cannot resolve your technical issues by using the Cisco TAC website, and you are a Cisco.com registered user, you can open a case online by using the TAC Case Open tool at the following URL:

http://www.cisco.com/tac/caseopen

If you have Internet access, it is recommended that you open P3 and P4 cases through the Cisco TAC website.

Cisco TAC Escalation Center

The Cisco TAC Escalation Center addresses issues that are classified as priority level 1 or priority level 2; these classifications are assigned when severe network degradation significantly impacts business operations. When you contact the TAC Escalation Center with a P1 or P2 problem, a Cisco TAC engineer will automatically open a case.

To obtain a directory of toll-free Cisco TAC telephone numbers for your country, go to the following URL:

http://www.cisco.com/warp/public/687/Directory/DirTAC.shtml

Before calling, please check with your network operations center to determine the level of Cisco support services to which your company is entitled; for example, SMARTnet, SMARTnet Onsite, or Network Supported Accounts (NSA). In addition, please have available your service agreement number and your product serial number.



Using the Command-Line Interface

The Catalyst 2950 switches are supported by Cisco IOS software. This chapter describes how to use the switch command-line interface (CLI) to configure the software features.

For a complete description of the commands that support these features, see Chapter 2, "Cisco IOS Commands." For more information on Cisco IOS Release 12.1, refer to the Cisco IOS Release 12.1 Command Summary.

For task-oriented configuration steps, refer to the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

The switches are preconfigured and begin forwarding packets as soon as they are attached to compatible devices.

By default, all ports belong to virtual LAN (VLAN) 1. Access to the switch itself is also through VLAN 1, which is the default management VLAN. The management VLAN is configurable. You manage the switch by using Telnet, Secure Shell (SSH) Protocol, web-based management, and Simple Network Management Protocol (SNMP) through devices connected to ports assigned to the management VLAN.

Type of Memory

The switch Flash memory stores the Cisco IOS software image, the startup and private configuration files, and helper files.

Platforms

This IOS release runs on a variety of switches. For a complete list, refer to the *Release Notes for the Catalyst 2950 Switch*.

CLI Command Modes

This section describes the CLI command mode structure. Command modes support specific Cisco IOS commands. For example, the **interface** *type_number* command works only when entered in global configuration mode. These are the main command modes:

- User EXEC
- Privileged EXEC
- Global configuration
- Interface configuration
- Config-vlan
- VLAN configuration
- Line configuration

Table 1-1 lists the command modes, how to access each mode, the prompt you see in that mode, and how to exit that mode. The prompts listed assume the default name *Switch*.

Table 1-1 Command Modes Summary

Command Mode	Access Method	Prompt	Exit or Access Next Mode
User EXEC	This is the first level of access. (For the switch) Change terminal settings, perform basic tasks, and list system information.	`Switch>`	Enter the **logout** command. To enter privileged EXEC mode, enter the **enable** command.
Privileged EXEC	From user EXEC mode, enter the **enable** command.	`Switch#`	To exit to user EXEC mode, enter the **disable** command. To enter global configuration mode, enter the **configure** command.
Global configuration	From privileged EXEC mode, enter the **configure** command.	`Switch(config)#`	To exit to privileged EXEC mode, enter the **exit** or **end** command, or press **Ctrl-Z**. To enter interface configuration mode, enter the **interface** command.
Interface configuration	From global configuration mode, specify an interface by entering the **interface** command.	`Switch(config-if)#`	To exit to privileged EXEC mode, enter the **end** command, or press **Ctrl-Z**. To exit to global configuration mode, enter the **exit** command. To enter subinterface configuration mode, specify a subinterface with the **interface** command.
Config-vlan	In global configuration mode, enter the **vlan** *vlan-id* command.	`Switch(config-vlan)#`	To exit to global configuration mode, enter the **exit** command. To return to privileged EXEC mode, enter the **end** command, or press **Ctrl-Z**.

Table 1-1 Command Modes Summary (continued)

Command Mode	Access Method	Prompt	Exit or Access Next Mode
VLAN configuration	From privileged EXEC mode, enter the **vlan database** command.	`Switch(vlan)#`	To exit to privileged EXEC mode, enter the **exit** command.
Line configuration	From global configuration mode, specify a line by entering the **line** command.	`Switch(config-line)#`	To exit to global configuration mode, enter the **exit** command. To return to privileged EXEC mode, enter the **end** command, or press **Ctrl-Z**.

User EXEC Mode

After you access the device, you are automatically in user EXEC command mode. The EXEC commands available at the user level are a subset of those available at the privileged level. In general, use the user EXEC commands to change terminal settings temporarily, to perform basic tests, and to list system information.

The supported commands can vary depending on the version of IOS software in use. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch> ?
```

Privileged EXEC Mode

Because many of the privileged commands configure operating parameters, privileged access should be password-protected to prevent unauthorized use. The privileged command set includes those commands contained in user EXEC mode, as well as the **configure** command through which you access the remaining command modes.

If your system administrator has set a password, you are prompted to enter it before being granted access to privileged EXEC mode. The password does not appear on the screen and is case sensitive.

The privileged EXEC mode prompt is the device name followed by the pound sign (#).

```
Switch#
```

Enter the **enable** command to access privileged EXEC mode:

```
Switch> enable
Switch#
```

The supported commands can vary depending on the version of IOS software in use. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch# ?
```

To return to user EXEC mode, enter the **disable** command.

Global Configuration Mode

Global configuration commands apply to features that affect the device as a whole. Use the **configure** privileged EXEC command to enter global configuration mode. The default is to enter commands from the management console.

When you enter the **configure** command, a message prompts you for the source of the configuration commands:

```
Switch# configure
Configuring from terminal, memory, or network [terminal]?
```

You can specify either the terminal or nonvolatile RAM (NVRAM) as the source of configuration commands.

This example shows you how to access global configuration mode:

```
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
```

The supported commands can vary depending on the version of IOS software in use. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch(config)# ?
```

To exit global configuration command mode and to return to privileged EXEC mode, enter the **end** or **exit** command, or press **Ctrl-Z**.

Interface Configuration Mode

Interface configuration commands modify the operation of the interface. Interface configuration commands always follow a global configuration command, which defines the interface type.

Use the **interface** *type_number.subif* command to access interface configuration mode. The new prompt shows interface configuration mode.

```
Switch(config-if)#
```

The supported commands can vary depending on the version of IOS software in use. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch(config-if)# ?
```

To exit interface configuration mode and to return to global configuration mode, enter the **exit** command. To exit interface configuration mode and to return to privileged EXEC mode, enter the **end** command, or press **Ctrl-Z**.

config-vlan Mode

Use this mode to configure normal-range VLANs (VLAN IDs 1 to 1005) or, when VTP mode is transparent, to configure extended-range VLANs (VLAN IDs 1006 to 4094) when the enhanced software image is installed. When VTP mode is transparent, the VLAN and VTP configuration is saved in the running configuration file, and you can save it to the switch startup configuration file by using the **copy running-config startup-config** privileged EXEC command. The configurations of VLAN IDs 1 to 1005 are saved in the VLAN database if VTP is in transparent or server mode. The extended-range VLAN configurations are not saved in the VLAN database.

Enter the **vlan** *vlan-id* global configuration command to access config-vlan mode:

```
Switch(config)# vlan 2000
Switch(config-vlan)#
```

The supported keywords can vary but are similar to the commands available in VLAN configuration mode. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch(config-vlan)# ?
```

For extended-range VLANs, all characteristics except MTU size must remain at the default setting.

To return to global configuration mode, enter **exit**; to return to privileged EXEC mode, enter **end.** All commands except **shutdown** take effect when you exit config-vlan mode.

VLAN Configuration Mode

You can use the VLAN configuration commands to create or modify VLAN parameters for VLANs 1 to 1005. Enter the **vlan database** privileged EXEC command to access VLAN configuration mode:

```
Switch# vlan database
Switch(vlan)#
```

The supported commands can vary depending on the version of IOS software in use. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch(vlan)# ?
```

To return to privileged EXEC mode, enter the **abort** command to abandon the proposed database. Otherwise, enter **exit** to implement the proposed new VLAN database and to return to privileged EXEC mode.

Line Configuration Mode

Line configuration commands modify the operation of a terminal line. Line configuration commands always follow a line command, which defines a line number. Use these commands to change terminal parameter settings line-by-line or for a range of lines.

Use the **line vty** *line_number* [*ending_line_number*] command to enter line configuration mode. The new prompt indicates line configuration mode.

This example shows how to enter line configuration mode for virtual terminal line 7:

```
Switch(config)# line vty 0 7
```

The supported commands can vary depending on the version of IOS software in use. To view a comprehensive list of commands, enter a question mark (**?**) at the prompt.

```
Switch(config-line)# ?
```

To exit line configuration mode and to return to global configuration mode, use the **exit** command. To exit line configuration mode and to return to privileged EXEC mode, enter the **end** command, or press **Ctrl-Z**.

Command Summary

Table 1-2 lists and describes commands that have the same function but different syntax in software releases earlier than Release 12.1(6)EA2 and in Release 12.1(6)EA2 or later. It lists the commands supported in releases earlier than Release 12.1(6)EA2, the equivalent commands in Release 12.1(6)EA2 or later, and command descriptions.

If you are running Release 12.1(6)EA2 or later, the switch supports the commands in the left column of Table 1-2 only if they are in a saved configuration file. When you save the switch configuration after modifying it, the commands in Table 1-2 are replaced by equivalent commands supported in Release 12.1(6)EA2 or later.

For information about commands listed in the left column of Table 1-2, refer to the *Catalyst 2950 Desktop Switch Command Reference, Cisco IOS Release 12.0(5.2)WC(1)* (April 2001). You can access this document at this URL:

http://www.cisco.com/univercd/cc/td/doc/product/lan/cat2950/2950_wc/index.htm.

Table 1-2 Command Comparison

Command in IOS releases earlier than Release 12.1(6)EA2	Command in Release 12.1(6)EA2 or later	Description
mac-address-table secure	**switchport port-security mac-address**	Adds secure addresses to the MAC address table.
no negotiation auto	**speed nonegotiate**	Disables autonegotiation on 1000BASE-X, -LX, and -ZX Gigabit Interface Converter (GBIC) ports.
port group	**channel-group**	Assigns a port to a Fast EtherChannel or Gigabit EtherChannel port group.
port monitor	**monitor session**	Enables Switch Port Analyzer (SPAN) port monitoring on a port.
port protected	**switchport protected**	Isolates Layer 2 unicast, multicast, and broadcast traffic from other protected ports on the same switch.
port security	**switchport port-security**	Enables port security on a port and restricts the use of the port to a user-defined group of stations.
port security action	**switchport port-security violation**	Specifies the action to take when an address violation occurs on a secure port.
port security max-mac-count	**switchport port-security maximum**	Specifies the maximum number of secure addresses supported by a secure port.
port storm-control	**storm-control**	Enables unicast, multicast, or broadcast storm control on a port, and specifies storm-control parameters on a port.
show mac-address-table secure	**show port-security**	Displays the port security settings for an interface and the secure addresses in the MAC address table.

Table 1-2 Command Comparison (continued)

Command in IOS releases earlier than Release 12.1(6)EA2	Command in Release 12.1(6)EA2 or later	Description
show port group	**show etherchannel**	Displays EtherChannel information for a channel.
show port monitor	**show monitor**	Displays SPAN session information.
show port protected	**show interfaces switchport**	Displays the port protection settings of a port.
show port security	**show port-security**	Displays the port security settings defined for a port.
show port storm-control	**show storm-control**	Displays the packet-storm control information.
spanning-tree rootguard	**spanning-tree guard**	Enables the root guard feature for all VLANs associated with a port.
switchport priority	**mls qos cos**	Defines the default class of service (CoS) value of a port.
switchport priority override	**mls qos cos override**	Assigns the default CoS value to all incoming packets on a port.

Table 1-3 lists and describes the commands that are not supported in Release 12.1(6)EA2 or later. These commands are supported only in software releases earlier than Release 12.1(6)EA2. If you are running Release 12.1(6)EA2 or later, the switch supports the commands listed in Table 1-3 only if they are in a saved configuration file.

Table 1-3 Commands Not Supported in Release 12.1(6)EA2 or Later

Command	Description
clear ip address	Deletes an IP address for a switch without disabling the IP processing.
clear mac-address-table static	Deletes static entries from the MAC address table.
management	Shuts down the current management VLAN interface and enables the new management VLAN interface.
show mac-address-table self	Displays the addresses added by the switch itself to the MAC address table.
spanning-tree protocol	Specifies the Spanning Tree Protocol (STP) to be used for specified spanning-tree instances. In Release 12.1(6)EA2 or later, the switch supports only IEEE Ethernet STP.

For detailed command syntax and descriptions, see Chapter 2, "Cisco IOS Commands." For task-oriented configuration steps, refer to the software configuration guide for this release.



Cisco IOS Commands

aaa authentication dot1x

Use the **aaa authentication dot1x** global configuration command to specify one or more authentication, authorization, and accounting (AAA) methods for use on interfaces running IEEE 802.1X. Use the **no** form of this command to disable authentication.

aaa authentication dot1x {**default**} *method1* [*method2*...]

no aaa authentication dot1x {**default**} *method1* [*method2*...]

Syntax Description

default	Use the listed authentication methods that follow this argument as the default list of methods when a user logs in.
method1 [*method2*...]	At least one of these keywords: • **enable**—Use the enable password for authentication. • **group radius**—Use the list of all Remote Authentication Dial-In User Service (RADIUS) servers for authentication. • **line—**Use the line password for authentication. • **local—**Use the local username database for authentication. • **local-case—**Use the case-sensitive local username database for authentication. • **none—**Use no authentication. The client is automatically authenticated by the switch without using the information supplied by the client.

Defaults

No authentication is performed.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The *method* argument identifies the list of methods that the authentication algorithm tries in the given sequence to validate the password provided by the client. The only method that is truly 802.1X-compliant is the **group radius** method, in which the client data is validated against a RADIUS authentication server. The remaining methods enable AAA to authenticate the client by using locally configured data. For example, the **local** and **local-case** methods use the username and password that are saved in the IOS configuration file. The **enable** and **line** methods use the **enable** and **line** passwords for authentication.

If you specify **group radius**, you must configure the RADIUS server by entering the **radius-server host** global configuration command.

If you are not using a RADIUS server, you can use the **local** or **local-case** methods, which access the local username database to perform authentication. By specifying the **enable** or **line** methods, you can supply the clients with a password to provide access to the switch.

Use the **show running-config** privileged EXEC command to display the configured lists of authentication methods.

Examples

This example shows how to enable AAA and how to create an authentication list for 802.1X. This authentication first tries to contact a RADIUS server. If this action returns an error, the user is allowed access with no authentication.

```
Switch(config)# aaa new model
Switch(config)# aaa authentication dot1x default group radius none
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
aaa new-model	Enables the AAA access control model. For syntax information, refer to **Cisco IOS Security Command Reference for Release 12.1 > Authentication, Authorization, and Accounting > Authentication Commands**.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

access-list (IP extended)

Use the extended version of the **access-list** global configuration command to configure an extended IP access control list (ACL). Use the **no** form of this command to remove an extended IP ACL.

access-list *access-list-number* {**deny** | **permit | remark**} *protocol* {*source source-wildcard* | **host** *source* | **any**} [*operator port*] {*destination destination-wildcard* | **host** *destination* | **any**} [*operator port*] [**dscp** *dscp-value*] [**time-range** *time-range-name*]

no access-list *access-list-number*

This command is available on physical interfaces only if your switch is running the enhanced software image (EI).

Syntax Description

access-list-number	Number of an ACL, from 100 to 199 or from 2000 to 2699.
protocol	Name of an IP protocol. *protocol* can be **ip**, **tcp**, or **udp**.
deny	Deny access if conditions are matched.
permit	Permit access if conditions are matched.
remark	ACL entry comment up to 100 characters.
source source-wildcard | **host** *source* | **any**	Define a source IP address and wildcard. The *source* is the source address of the network or host from which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *source-wildcard* applies wildcard bits to the source. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *source* and *source-wildcard* of *source* 0.0.0.0. • The keyword **any** as an abbreviation for *source* and *source-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a source-wildcard.
destination destination-wildcard | **host** *destination* | **any**	Define a destination IP address and wildcard. The *destination* is the destination address of the network or host to which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *destination-wildcard* applies wildcard bits to the destination. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *destination* and *destination-wildcard* of *destination* 0.0.0.0. • The keyword **any** as an abbreviation for *destination* and *destination-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a destination-wildcard.

operator port	(Optional) Define a source or destination port. The *operator* can be only **eq** (equal). If *operator* is after the source IP address and wildcard, conditions match when the source port matches the defined port. If *operator* is after the destination IP address and wildcard, conditions match when the destination port matches the defined port. The *port* is a decimal number or name of a Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) port. The number can be from 0 to 65535. Use TCP port names only for TCP traffic. Use UDP port names only for UDP traffic.
dscp *dscp-value*	(Optional) Define a Differentiated Services Code Point (DSCP) value to classify traffic. For the *dscp-value*, enter any of the 13 supported DSCP values (0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56), or use the question mark (**?**) to see a list of available values.
time-range *time-range-name*	(Optional) For the **time-range** keyword, enter a meaningful name to identify the time range. For a more detailed explanation of this keyword, refer to the software configuration guide.

Defaults

The default extended ACL is always terminated by an implicit deny statement for all packets.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Plan your access conditions carefully. The ACL is always terminated by an implicit deny statement for all packets.

You can use ACLs to control virtual terminal line access by controlling the transmission of packets on an interface.

Extended ACLs support only the TCP and UDP protocols.

Use the **show ip access-lists** command to display the contents of IP ACLs.

Use the **show access-lists** command to display the contents of all ACLs.



Note For more information about configuring IP ACLs, refer to the “Configuring Network Security with ACLs” chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to configure an extended IP ACL that allows only TCP traffic to the destination IP address 128.88.1.2 with a TCP port number of 25 and how to apply it to an interface:

```
Switch(config)# access-list 102 permit tcp any host 128.88.1.2 eq 25
Switch(config)# interface fastethernet0/8
Switch(config-if)# ip access-group 102 in
```

This is an example of an extended ACL that allows TCP traffic only from two specified networks. The wildcard bits apply to the host portions of the network addresses. Any host with a source address that does not match the ACL statements is denied.

```
access-list 104 permit tcp 192.5.0.0 0.0.255.255 any
access-list 104 permit tcp 128.88.0.0 0.0.255.255 any
```

Note In these examples, all other IP access is implicitly denied.

You can verify your settings by entering the **show ip access-lists** or **show access-lists** privileged EXEC command.

Related Commands

Command	Description
access-list (IP standard)	Configures a standard IP ACL.
ip access-group	Controls access to an interface.
show access-lists	Displays ACLs configured on the switch.
show ip access-lists	Displays IP ACLs configured on the switch.

access-list (IP standard)

Use the standard version of the **access-list** global configuration command to configure a standard IP access control list (ACL). Use the **no** form of this command to remove a standard IP ACL.

access-list *access-list-number* {**deny** | **permit | remark**} {*source source-wildcard* | **host** *source* | **any**}

no access-list *access-list-number*

This command is available on physical interfaces only if your switch is running the enhanced software image (EI).

Syntax Description

access-list-number	Number of an ACL, from 1 to 99 or from 1300 to 1999.
deny	Deny access if conditions are matched.
permit	Permit access if conditions are matched.
remark	ACL entry comment up to 100 characters.
source source-wildcard | **host** *source* | **any**	Define a source IP address and wildcard. The *source* is the source address of the network or host from which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *source-wildcard* applies wildcard bits to the source. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *source* and *source-wildcard* of *source* 0.0.0.0. • The keyword **any** as an abbreviation for *source* and *source-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a source-wildcard.

Defaults

The default standard ACL is always terminated by an implicit deny statement for all packets.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Plan your access conditions carefully. The ACL is always terminated by an implicit deny statement for all packets.

You can use ACLs to control virtual terminal line access by controlling the transmission of packets on an interface.

Use the **show ip access-lists** command to display the contents of IP ACLs.

Use the **show access-lists** command to display the contents of all ACLs.

Note For more information about configuring IP ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to configure a standard IP ACL that allows only traffic from the host network 128.88.1.10 and how to apply it to an interface:

```
Switch(config)# access-list 12 permit host 128.88.1.10
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# ip access-group 12 in
```

This is an example of an standard ACL that allows traffic only from three specified networks. The wildcard bits apply to the host portions of the network addresses. Any host with a source address that does not match the ACL statements is denied.

```
access-list 14 permit 192.5.34.0  0.0.0.255
access-list 14 permit 128.88.0.0  0.0.0.255
access-list 14 permit 36.1.1.0  0.0.0.255
```

Note In these examples, all other IP access is implicitly denied.

You can verify your settings by entering the **show ip access-lists** or **show access-lists** privileged EXEC command.

Related Commands

Command	Description
access-list (IP extended)	Configures an extended IP ACL.
ip access-group	Controls access to an interface.
show access-lists	Displays ACLs configured on the switch.
show ip access-lists	Displays IP ACLs configured on the switch.

boot private-config-file

Use the **boot private-config-file** global configuration command to specify the filename that IOS uses to read and write a nonvolatile copy of the private configuration. Use the **no** form of this command to return to the default setting.

boot private-config-file *filename*

no boot private-config-file

Syntax Description

filename	The name of the private configuration file.

Defaults

The default configuration file is *private-config.text*.

Command Modesb

Global configuration

Command History

Release	Modification
12.1(11)EA1	This command was first introduced.

Usage Guidelines

Filenames are case sensitive.

Examples

This example shows how to specify the name of the private configuration file as *pconfig*:

```
Switch(config)# boot private-config-file pconfig
```

Related Commands

Command	Description
show boot	Displays the settings of the boot environment variables.

channel-group

Use the **channel-group** interface configuration command to assign an Ethernet interface to an EtherChannel group. Use the **no** form of this command to remove an Ethernet interface from an EtherChannel group.

channel-group *channel-group-number* **mode** {**auto** [**non-silent**] | **desirable** [**non-silent**] | **on**}

no channel-group

Syntax Description

channel-group-number	Specify the channel group number. The range is 1 to 6.
mode	Specify the EtherChannel Port Aggregation Protocol (PAgP) mode of the interface.
auto	Enable PAgP only if a PAgP device is detected. Auto mode places an interface into a passive negotiating state, in which the interface responds to PAgP packets it receives but does not initiate PAgP packet negotiation. A channel is formed only with another port group in desirable mode. This setting is the default.
desirable	Unconditionally enable PAgP. Desirable mode places an interface into an active negotiating state, in which the interface initiates negotiations with other interfaces by sending PAgP packets. A channel is formed with another port group in either the desirable or auto mode.
non-silent	(Optional) Used with the **auto** or **desirable** keyword when traffic is expected from the other device.
on	Force the interface to channel without PAgP. With the **on** mode, a usable EtherChannel exists only when an interface group in the **on** mode is connected to another interface group in the **on** mode.

Defaults

No channel groups are assigned.

There is no default mode.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **port group** command.

Usage Guidelines

You must specify the mode when entering this command. If the mode is not entered, an Ethernet interface is not assigned to an EtherChannel group, and an error message appears.

You do not have to create a port-channel interface before assigning a physical interface to a channel group. A port-channel interface is created automatically when the channel group gets its first physical interface.

You do not have to disable the IP address that is assigned to a physical interface that is part of a channel group, but we highly recommend that you do so.

You can create port channels by entering the **interface port-channel** global configuration command or when the channel group gets its first physical interface assignment. The port channels are not created at runtime or dynamically.

Any configuration or attribute changes you make to the port-channel interface are propagated to all interfaces within the same channel group as the port channel (for example, configuration changes are also propagated to the physical interfaces that are not part of the port channel, but are part of the channel group).

With the **on** mode, a usable EtherChannel exists only when a port group in **on** mode is connected to another port group in **on** mode.

If you do not specify **non-silent** with the **auto** or **desirable** mode, silent is assumed. The silent mode is used when the switch is connected to a device that is not PAgP-capable and seldom, if ever, transmits packets. A example of a silent partner is a file server or a packet analyzer that is not generating traffic. In this case, running PAgP on a physical port prevents that port from ever becoming operational; however, it allows PAgP to operate, to attach the interface to a channel group, and to use the interface for transmission. Both ends of the link cannot be set to silent.

Caution You should exercise care when setting the mode to **on** (manual configuration). All ports configured in the **on** mode are bundled in the same group and are forced to have similar characteristics. If the group is misconfigured, packet loss or Spanning Tree Protocol (STP) loops might occur.

Examples

This example shows how to add an interface to the EtherChannel group specified as channel group 1:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# channel-group 1 mode on
```

You can verify your settings by entering the **show etherchannel** or **show running-config** privileged EXEC command.

Related Commands

Command	Description
interface port-channel	Accesses or creates the port channel.
port-channel load-balance	Sets the load distribution method among the ports in the EtherChannel.
show etherchannel	Displays EtherChannel information for a channel.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

class

Use the **class** policy-map configuration command to define a traffic classification for the policy to act on using the class-map name or access group. Use the **no** form of this command to delete an existing class map.

class *class-map-name* [**access-group name** *acl-index-or-name*]

no class *class-map-name*

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

class-map-name	Name of the class map.
access-group name *acl-index-or-name*	(Optional) Number or name of an IP standard or extended access control list (ACL) or name of an extended MAC ACL. For an IP standard ACL, the index range is 1 to 99 and 1300 to 1999; for an IP extended ACL, the index range is 100 to 199 and 2000 to 2699.

Defaults

No policy-map class maps are defined.

Command Modes

Policy-map configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Before you use the **class** command, use the **policy-map** global configuration command to identify the policy map and to enter policy-map configuration mode. After you specify a policy map, you can configure a policy for new classes or modify a policy for any existing classes in that policy map. You attach the policy map to an interface by using the **service-policy** interface configuration command; however, you cannot attach one that uses an ACL classification to the egress direction.

The class name that you specify in the policy map ties the characteristics for that class to the class map and its match criteria as configured by using the **class-map** global configuration command.

The **class** command performs the same function as the **class-map** global configuration command. Use the **class** command when a new classification, which is not shared with any other ports, is needed. Use the **class-map** command when the map is shared among many ports.

Note In a policy map, the class named *class-default* is not supported. The switch does not filter traffic based on the policy map defined by the **class class-default** policy-map configuration command.

After entering the **class** command, you enter policy-map class configuration mode. These configuration commands are available:

- **default**: sets a command to its default.
- **exit**: exits policy-map class configuration mode and returns to policy-map configuration mode.
- **no**: returns a command to its default setting.
- **set**: specifies a Differentiated Services Code Point (DSCP) value to be assigned to the classified traffic. For more information, see the **set** command.
- **police**: defines a policer for the classified traffic. The policer specifies the bandwidth limitations and the action to take when the limits are exceeded. For more information, see the **police** command.

To return to policy-map configuration mode, use the **exit** command. To return to privileged EXEC mode, use the **end** command.



Note For more information about configuring ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to create a policy map named *policy1*. When attached to the ingress port, it matches all the incoming traffic defined in *class1* and polices the traffic at an average rate of 1 Mbps and bursts at 131072 bytes. Traffic exceeding the profile is dropped.

```
Switch(config)# policy-map policy1
Switch(config-pmap)# class class1
Switch(config-pmap-c)# police 1000000 131072 exceed-action drop
Switch(config-pmap-c)# exit
Switch(config-pmap)#
```

You can verify your settings by entering the **show policy-map** privileged EXEC command.

Related Commands

Command	Description
class-map	Creates a class map to be used for matching packets to the class whose name you specify.
match	Defines the match criteria to classify traffic.
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces to specify a service policy.
show policy-map	Displays quality of service (QoS) policy maps.

class-map

Use the **class-map** global configuration command to create a class map to be used for matching packets to the class whose name you specify and to enter class-map configuration mode. Use the **no** form of this command to delete an existing class map and to return to global configuration mode.

class-map *class-map-name* [**match-all**]

no class-map *class-map-name* [**match-all**]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

class-map-name	Name of the class map.
match-all	(Optional) Perform a logical-AND of all matching statements under this class map. All criteria in the class map must be matched.

Defaults

No class maps are defined.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use this command to specify the name of the class for which you want to create or modify class-map match criteria and to enter class-map configuration mode. In this mode, you can enter one **match** command to configure the match criteria for this class.

The **class-map** command and its subcommands are used to define packet classification and marking as part of a globally named service policy applied on a per-interface basis.

In quality of service (QoS) class-map configuration mode, these configuration commands are available:

- **exit**: exits from QoS class-map configuration mode.
- **no**: removes a match statement from a class map.
- **match**: configures classification criteria. For more information, see the **match** class-map configuration command.

Only one match criterion per class map is supported. For example, when defining a class map, only one **match** command can be entered.

Only one access control list (ACL) can be configured in a class map. The ACL can have multiple access control entries (ACEs).

Note The switch does not support any deny conditions in an ACL configured in a class map.

Note For more information about configuring ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to configure the class map named *class1*. *class1* has one match criteria, which is a numbered ACL.

```
Switch(config)# access-list 103 permit tcp any any eq 80
Switch(config)# class-map class1
Switch(config-cmap)# match access-group 103
Switch(config-cmap)# exit
```

You can verify your settings by entering the **show class-map** privileged EXEC command.

Related Commands

Command	Description
class	Defines a traffic classification for the policy to act on by using the class-map name or access group.
match	Defines the match criteria to classify traffic.
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces to specify a service policy.
show class-map	Displays QoS class maps.

clear interface

Use the **clear interface** privileged EXEC command to clear the hardware logic on an interface or a VLAN.

clear interface {*interface-id* | **vlan** *vlan-id*}

Syntax Description

interface-id	ID of the interface.
vlan-id	VLAN ID. Valid VLAN IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to clear the hardware logic on a Gigabit Ethernet interface:

```
Switch# clear interface gigabitethernet0/1
```

This example shows how to clear the hardware logic on a specific VLAN:

```
Switch# clear interface vlan 5
```

You can verify that the interface-reset counter for an interface is incremented by entering the **show interfaces** privileged EXEC command.

clear mac address-table

Use the **clear mac address-table** privileged EXEC command to delete from the MAC address table a specific dynamic address, all dynamic addresses on a particular interface, or all dynamic addresses on a particular VLAN. This command also clears the MAC address notification global counters.

clear mac address-table {**dynamic** [**address** *mac-addr* | **interface** *interface-id* | **vlan** *vlan-id*] | **notification**}

Note Beginning with Release 12.1(11)EA1, the **clear mac address-table** command replaces the **clear mac-address-table** command (with the hyphen). The **clear mac-address-table** command (with the hyphen) will become obsolete in a future release.

Syntax Description

dynamic	Delete all dynamic MAC addresses.
dynamic address *mac-addr*	(Optional) Delete the specified dynamic MAC address.
dynamic interface *interface-id*	(Optional) Delete all dynamic MAC addresses on the specified physical port or port channel.
dynamic vlan *vlan-id*	(Optional) Delete all dynamic MAC addresses for the specified VLAN. Valid IDs are from 1 to 4096 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
notification	Clear the notifications in the history table and reset the counters.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **notification** keyword was added.
12.1(11)EA1	The **clear mac-address-table** command was replaced by the **clear mac address-table** command.

Examples

This example shows how to remove a specific dynamic address from the MAC address table:

```
Switch# clear mac address-table dynamic address 0008.0070.0007
```

You can verify that the information was deleted by entering the **show mac address-table** privileged EXEC command.

Related Commands

Command	Description
mac address-table notification	Enables the MAC address notification feature.
show mac address-table	Displays the MAC address table static and dynamic entries.
show mac address-table notification	Displays the MAC address notification settings for all interfaces or the specified interface.
snmp trap mac-notification	Enables the Simple Network Management Protocol (SNMP) MAC address notification trap on a specific interface.

clear pagp

Use the **clear pagp** privileged EXEC command to clear Port Aggregation Protocol (PAgP) channel-group information.

clear pagp {*channel-group-number* [**counters**] | **counters**}

Syntax Description

channel-group-number	Channel group number. The range is 1 to 6.
counters	Clear traffic counters.

Defaults

This command has no default setting.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to clear channel-group information for a specific group:

```
Switch# clear pagp 4
```

This example shows how to clear channel-group traffic counters:

```
Switch# clear pagp counters
```

You can verify that the information was deleted by entering the **show pagp** privileged EXEC command.

Related Commands

Command	Description
show pagp	Displays PAgP channel-group information.

clear port-security dynamic

Use the **clear port-security dynamic** privileged EXEC command to delete from the MAC address table a specific dynamic secure address or all the dynamic secure addresses on an interface.

clear port-security dynamic [**address** *mac-addr* | **interface** *interface-id*]

Syntax Description

address *mac-addr*	(Optional) Delete the specified dynamic secure MAC address.
interface *interface-id*	(Optional) Delete all the dynamic secure MAC addresses on the specified physical port or port channel.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(11)EA1	This command was first introduced.

Examples

This example shows how to remove a specific dynamic secure address from the MAC address table:

```
Switch# clear port-security dynamic address 0008.0070.0007
```

This example shows how to remove all the dynamic secure addresses learned on a specific interface:

```
Switch# clear port-security dynamic interface gigabitethernet0/1
```

This example shows how to remove all the dynamic secure addresses from the address table:

```
Switch# clear port-security dynamic
```

You can verify that the information was deleted by entering the **show port-security** privileged EXEC command.

Related Commands

Command	Description
switchport port-security	Enables port security on an interface.
switchport port-security mac-address *mac-address*	Configures secure MAC addresses.
switchport port-security maximum *value*	Configures a maximum number of secure MAC addresses on a secure interface.
show port-security	Displays the port security settings defined for an interface or for the switch.

clear spanning-tree detected-protocols

Use the **clear spanning-tree detected-protocols** privileged EXEC command to restart the protocol migration process (force the renegotiation with neighboring switches) on all interfaces or on the specified interface.

clear spanning-tree detected-protocols [**interface** *interface-id*]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

interface *interface-id*	(Optional) Restart the protocol migration process on the specified interface. Valid interfaces include physical ports, VLANs, and port channels. The valid VLAN range is 1 to 4094; do not enter leading zeros. The valid port-channel range is 1 to 6.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

A switch running both the Multiple Spanning Tree Protocol (MSTP) and the Rapid Spanning Tree Protocol (RSTP) supports a built-in protocol migration mechanism that enables it to interoperate with legacy 802.1D switches. If an MSTP and RSTP switch receives a legacy 802.1D configuration bridge protocol data unit (BPDU) with the protocol version set to 0, it sends only 802.1D BPDUs on that port. A multiple spanning-tree (MST) switch can also detect that a port is at the boundary of a region when it receives a legacy BPDU, an MST BPDU (version 3) associated with a different region, or an RST BPDU (version 2).

However, the switch does not automatically revert to the MSTP mode if it no longer receives 802.1D BPDUs because it cannot determine whether the legacy switch has been removed from the link unless the legacy switch is the designated switch. Use the **clear spanning-tree detected-protocols** command in this situation.

Examples

This example shows how to restart the protocol migration process on Fast Ethernet interface 0/1:

```
Switch# clear spanning-tree detected-protocols interface fastethernet0/1
```

clear vmps statistics

Use the **clear vmps statistics** privileged EXEC command to clear the statistics maintained by the VLAN Query Protocol (VQP) client.

clear vmps statistics

Syntax Description

This command has no arguments or keywords.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to clear VLAN Membership Policy Server (VMPS) statistics:

```
Switch# clear vmps statistics
```

You can verify that the information was deleted by entering the **show vmps statistics** privileged EXEC command.

Related Commands

Command	Description
show vmps statistics	Displays the VQP version, reconfirmation interval, retry count, VMPS IP addresses, and the current and primary servers.

clear vtp counters

Use the **clear vtp counters** privileged EXEC command to clear the VLAN Trunking Protocol (VTP) and pruning counters.

clear vtp counters

Syntax Description

This command has no arguments or keywords.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Examples

This example shows how to clear the VTP counters:

```
Switch# clear vtp counters
```

You can verify that the information was deleted by entering the **show vtp counters** privileged EXEC command.

Related Commands

Command	Description
show vtp counters	Displays general information about the VTP management domain, status, and counters.

cluster commander-address

You do not need to enter this command. The command switch automatically provides its MAC address to member switches when these switches join the cluster. The member switch adds this information and other cluster information to its running configuration file. Enter the **no** form of this global configuration command from the member switch console port to remove it from a cluster only during debugging or recovery procedures.

cluster commander-address *mac-address* [**member** *number* **name** *name*]

no cluster commander-address

Syntax Description

mac-address	MAC address of the cluster command switch.
member *number*	(Optional) Number of a configured member switch. The range is from 0 to 15.
name *name*	(Optional) Name of the configured cluster up to 31 characters.

Defaults

The switch is not a member of any cluster.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

A cluster member can have only one command switch.

The member switch retains the identity of the command switch during a system reload by using the *mac-address* parameter.

You can enter the **no** form on a member switch to remove it from the cluster during debugging or recovery procedures. You would normally use this command from the member switch console port only when the member has lost communication with the command switch. With normal switch configuration, we recommend that you remove member switches only by entering the **no cluster member** *n* global configuration command on the command switch.

When a standby command-switch becomes active (becomes the command switch), it removes the cluster commander-address line from its configuration.

Examples

This is an example of text from the running configuration of a cluster member:

```
Switch(config)# show running-config

<output truncated>

cluster commander-address 00e0.9bc0.a500 member 4 name my_cluster

<output truncated>
```

This example shows how to remove a member from the cluster by using the cluster member console:

```
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# no cluster commander-address
```

You can verify your settings by entering the **show cluster** privileged EXEC command.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

cluster discovery hop-count

Use the **cluster discovery hop-count** global configuration command on the command switch to set the hop-count limit for extended discovery of candidate switches. Use the **no** form of this command to set the hop count to the default value.

cluster discovery hop-count *number*

no cluster discovery hop-count

Syntax Description

number	Number of hops from the cluster edge that the command switch limits the discovery of candidates. The range is from 1 to 7.

Defaults

The hop count is set to 3.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Enter this command only on the command switch. This command does not operate on member switches.

If the hop count is set to 1, it disables extended discovery. The command switch discovers only candidates that are one hop from the edge of the cluster. The edge of the cluster is the point between the last discovered member switch and the first discovered candidate switch.

Examples

This example shows how to set the hop count limit to 4. This command is entered on the command switch.

```
Switch(config)# cluster discovery hop-count 4
```

You can verify your settings by entering the **show cluster** privileged EXEC command on the command switch.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.
show cluster candidates	Displays a list of candidate switches.

cluster enable

Use the **cluster enable** global configuration command on a command-capable switch to enable it as the cluster command switch, assign a cluster name, and optionally assign a member number to it. Use the **no** form of this command to remove all members and make the command switch a candidate switch.

cluster enable *name* [*command-switch-member-number*]

no cluster enable

Syntax Description

name	Name of the cluster up to 31 characters. Valid characters include only alphanumerics, dashes, and underscores.
command-switch-member-number	(Optional) Assign a member number to the command switch of the cluster. The range is from 0 to 15.

Defaults

The switch is not a command switch.

No cluster name is defined.

The member number is 0 when this is the command switch.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

This command runs on any command-capable switch that is not part of any cluster. This command fails if a device is already configured as a member of the cluster.

You must name the cluster when you enable the command switch. If the switch is already configured as the command switch, this command changes the cluster name if it is different from the previous name.

Examples

This example shows how to enable the command switch, name the cluster, and set the command switch member number to 4:

```
Switch(config)# cluster enable Engineering-IDF4 4
```

You can verify your settings by entering the **show cluster** privileged EXEC command on the command switch.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.

cluster holdtime

Use the **cluster holdtime** global configuration command on the command switch to set the duration in seconds before a switch (either the command or member switch) declares the other switch down after not receiving heartbeat messages. Use the **no** form of this command to set the duration to the default value.

cluster holdtime *holdtime-in-secs*

no cluster holdtime

Syntax Description

holdtime-in-secs	Duration in seconds before a switch (either a command or member switch) declares the other switch down. The range is from 1 to 300 seconds.

Defaults

The holdtime is 80 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Use this command with the **cluster timer** global configuration command only on the command switch. The command switch propagates the values to all its cluster members so that the setting is consistent among all switches in the cluster.

The holdtime is typically set as a multiple of the interval timer (**cluster timer**). For example, it takes (holdtime-in-secs divided by interval-in-secs) number of heartbeat messages to be missed in a row to declare a switch down.

Examples

This example shows how to change the interval timer and the duration on the command switch:

```
Switch(config)# cluster timer 3
Switch(config)# cluster holdtime 30
```

You can verify your settings by entering the **show cluster** privileged EXEC command.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.

cluster management-vlan

Use the **cluster management-vlan** global configuration command on the command switch to change the management VLAN for the entire cluster. Use the **no** form of this command to change the management VLAN to VLAN 1.

cluster management-vlan *n*

no cluster management-vlan

Syntax Description

n	VLAN ID of the new management VLAN. Valid VLAN IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.

Defaults

The default management VLAN is VLAN 1.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Enter this command only on the command switch. This command changes the management VLAN of the command switch and member switches. Member switches must have either a trunk connection or connection to the new command-switch management VLAN to maintain communication with the command switch.

This command is not written to the configuration file.

Examples

This example shows how to change the management VLAN to VLAN 5 on the entire cluster:

```
Switch(config)# cluster management-vlan 5
```

You can verify your settings by entering the **show interfaces vlan** *vlan-id* privileged EXEC command.

Related Commands

Command	Description
show interfaces	Displays the administrative and operational status of a switching (nonrouting) port.

cluster member

Use the **cluster member** global configuration command on the command switch to add members to a cluster. Use the **no** form of this command to remove members from the cluster.

cluster member [*n*] **mac-address** *H.H.H* [**password** *enable-password*] [**vlan** *vlan-id*]

no cluster member *n*

Syntax Description

n	(Optional) The number that identifies a cluster member. The range is from 0 to 15.
mac-address *H.H.H*	MAC address of the member switch in hexadecimal format.
password *enable-password*	(Optional) Enable password of the candidate switch. The password is not required if there is no password on the candidate switch.
vlan *vlan-id*	(Optional) VLAN ID through which the candidate is added to the cluster by the command switch. The range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.

Defaults

A newly enabled command switch has no associated cluster members.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Enter this command only on the command switch to add a member to or remove a member from the cluster. If you enter this command on a switch other than the command switch, the switch rejects the command and displays an error message.

You must enter a member number to remove a switch from the cluster. However, you do not need to enter a member number to add a switch to the cluster. The command switch selects the next available member number and assigns it to the switch that is joining the cluster.

You must enter the enable password of the candidate switch for authentication when it joins the cluster. The password is not saved in the running or startup configuration. After a candidate switch becomes a member of the cluster, its password becomes the same as the command-switch password.

If a switch does not have a configured host name, the command switch appends a member number to the command-switch host name and assigns it to the member switch.

If you do not specify a VLAN ID, the command switch automatically chooses a VLAN and adds the candidate to the cluster.

Examples

This example shows how to add a switch as member 2 with MAC address 00E0.1E00.2222 and the password *key* to a cluster. The command switch adds the candidate to the cluster through VLAN 3.

```
Switch(config)# cluster member 2 mac-address 00E0.1E00.2222 password key vlan 3
```

This example shows how to add a switch with MAC address 00E0.1E00.3333 to the cluster. This switch does not have a password. The command switch selects the next available member number and assigns it to the switch joining the cluster:

```
Switch(config)# cluster member mac-address 00E0.1E00.3333
```

You can verify your settings by entering the **show cluster members** privileged EXEC command on the command switch.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.
show cluster candidates	Displays a list of candidate switches.
show cluster members	Displays information about the cluster members.

cluster run

Use the **cluster run** global configuration command to enable clustering on a switch. Use the **no** form of this command to disable clustering on a switch.

cluster run

no cluster run

Defaults

Clustering is enabled on all switches.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

When you enter the **no cluster run** command on a command switch, the command switch is disabled. Clustering is disabled, and the switch cannot become a candidate switch.

When you enter the **no cluster run** command on a member switch, it is removed from the cluster. Clustering is disabled, and the switch cannot become a candidate switch.

When you enter the **no cluster run** command on a switch that is not part of a cluster, clustering is disabled on this switch. This switch cannot then become a candidate switch.

Examples

This example shows how to disable clustering on the command switch:

```
Switch(config)# no cluster run
```

You can verify that clustering is disabled by entering the **show cluster** privileged EXEC command.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.

cluster standby-group

Use the **cluster standby-group** global configuration command to enable command switch redundancy by binding the Hot Standby Router Protocol (HSRP) standby group to the cluster. Use the **no** form of this command to unbind the cluster from the HSRP standby group.

cluster standby-group *HSRP-group-name*

no cluster standby-group

Syntax Description

HSRP-group-name	Name of the HSRP group that is bound to the cluster. The group name is limited to 32 characters.

Defaults

The cluster is not bound to any HSRP group.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

You must enter this command only on the command switch. If you enter it on a member switch, an error message appears.

The command switch propagates the cluster-HSRP binding information to all members. Each member switch stores the binding information in its nonvolatile RAM (NVRAM).

The HSRP group name must be a valid standby group; otherwise, the command entry produces an error.

Use the same group name on all members of the HSRP standby group that is to be bound to the cluster. Use the same HSRP group name on all cluster-HSRP capable members for the HSRP group that is to be bound. (When not binding a cluster to an HSRP group, you can use different names on the cluster command and the member switches.)

Examples

This example shows how to bind the HSRP group named *my_hsrp* to the cluster. This command is entered on the command switch.

```
Switch(config)# cluster standby-group my_hsrp
```

This example shows the error message when this command is entered on a command switch and the specified HSRP standby group does not exist:

```
Switch(config)# cluster standby-group my_hsrp
%ERROR:Standby (my_hsrp) group does not exist
```

This example shows the error message when this command is entered on a member switch:

```
Switch(config)# cluster standby-group my_hsrp
%ERROR:This command runs on a cluster command switch
```

You can verify your settings by entering the **show cluster** privileged EXEC command.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.
show standby	Displays standby group information.
standby ip	Enables HSRP on the interface.

cluster timer

Use the **cluster timer** global configuration command on the command switch to set the interval in seconds between heartbeat messages. Use the **no** form of this command to set the interval to the default value.

cluster timer *interval-in-secs*

no cluster timer

Syntax Description

interval-in-secs	Interval in seconds between heartbeat messages. The range is from 1 to 300 seconds.

Defaults

The interval is 8 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Use this command with the **cluster holdtime** global configuration command only on the command switch. The command switch propagates the values to all its cluster members so that the setting is consistent among all switches in the cluster.

The holdtime is typically set as a multiple of the heartbeat interval timer (**cluster timer**). For example, it takes (holdtime-in-secs divided by the interval-in-secs) number of heartbeat messages to be missed in a row to declare a switch down.

Examples

This example shows how to change the heartbeat interval timer and the duration on the command switch.

```
Switch(config)# cluster timer 3
Switch(config)# cluster holdtime 30
```

You can verify your settings by entering the **show cluster** privileged EXEC command.

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.

define interface-range

Use the **define interface-range** global configuration command to create an interface-range macro. Use the **no** form of this command to delete the defined macro.

define interface-range *macro-name interface-range*

no define interface-range *macro-name interface-range*

Syntax Description

macro-name	Name of the interface-range macro; up to 32 characters.
interface-range	Interface range; for valid values for interface ranges, see "Usage Guidelines."

Defaults

This command has no default setting.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The macro name is a 32-character maximum character string.

A macro can contain up to five ranges.

All interfaces in a range must be the same type; that is, all Fast Ethernet ports, all Gigabit Ethernet ports, all EtherChannel ports, or all VLANs, but you can combine multiple interface types in a macro.

When entering the *interface-range*, use this format:

- *type* {*first-interface*} - {*last-interface*}
- You must add a space between the first interface number and the hyphen when entering an *interface-range*. For example, **gigabitethernet 0/1 -2** is a valid range; **gigabitethernet 0/1-2** is not a valid range.

Valid values for *type* and *interface*:

- **vlan** *vlan-id*, where *vlan-id* is from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed.
- **port-channel** *port-channel-number*, where *port-channel-number* is from 1 to 6
- **fastethernet** *interface-id*
- **gigabitethernet** *interface-id*

VLAN interfaces must have been configured with the **interface vlan** command (the **show running-config** privileged EXEC command displays the configured VLAN interfaces). VLAN interfaces not displayed by the **show running-config** command cannot be used in *interface-ranges*.

For physical interfaces, the *interface-id* is defined as a slot/number (where slot is always 0 for the switch), and the range can be entered as *type* **0**/*number* - *number* (for example, **gigabitethernet0/1 - 2**). You can also enter multiple ranges.

When you define a range, you must enter a space before and after the hyphen (-):

```
interface range gigabitethernet0/1 - 2
```

When you define multiple ranges, you must enter a space before and after the comma (,):

```
interface range fastethernet0/3 - 7 , gigabitethernet0/1 - 2
```

Examples

This example shows how to create a multiple-interface macro:

```
Switch(config)# define interface-range macro1 fastethernet 0/3 -7 , gigabitethernet 0/2
```

Related Commands

Command	Description
interface range	Executes a command on multiple ports at the same time.
show running-config	Displays the current operating configuration, including defined macros. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

delete

Use the **delete** privileged EXEC command to delete a file or directory on the Flash memory device.

delete [**/force**] [**/recursive**] *filesystem***:/***file-url*

Syntax Description

/force	(Optional) Suppress the prompt that confirms the deletion.
/recursive	(Optional) Delete the named directory and all subdirectories and the files contained in it.
*filesystem***:**	Alias for a Flash file system. Use **flash:** for the system board Flash device.
/file-url	The path (directory) and filename to delete.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **/force** and **/recursive** keywords were added.

Usage Guidelines

If you use the **/force** keyword, you are prompted once at the beginning of the deletion process to confirm the deletion.

If you use the **/recursive** keyword without the **/force** keyword, you are prompted to confirm the deletion of every file.

The prompting behavior depends on the setting of the **file prompt** global configuration command. By default, the switch prompts for confirmation on destructive file operations. For more information about this command, refer to the *Cisco IOS Command Reference for Release 12.1*.

Examples

This example shows how to delete a file from the switch Flash memory:

```
Switch# delete flash:filename
```

You can verify that the directory was removed by entering the **dir** *filesystem*: privileged EXEC command.

Related Commands

Command	Description
copy	Downloads a file from a source, such as a TFTP server, to a destination, such as the Flash memory.
dir *filesystem*:	Displays a list of files on a file system.
rename	Renames a file.

deny (access-list configuration)

Use the **deny** access-list configuration command to configure conditions for a named or numbered IP access control list (ACL). Use the **no** form of this command to remove a deny condition from the IP ACL.

Use these commands with standard IP ACLs:

deny {*source source-wildcard* | **host** *source* | **any**}

no deny {*source source-wildcard* | **host** *source* | **any**}

Use these commands with extended IP ACLs:

deny *protocol* {*source source-wildcard* | **host** *source* | **any**} [*operator port*] {*destination destination-wildcard* | **host** *source* | **any**} [*operator port*] [**dscp** *dscp-value*] [**time-range** *time-range-name*]

no deny *protocol* {*source source-wildcard* | **host** *source* | **any**} [*operator port*] {*destination destination-wildcard* | **host** *source* | **any**} [*operator port*] [**dscp** *dscp-value*] [**time-range** *time-range-name*]

This command is available on physical interfaces only if your switch is running the enhanced software image (EI).

Syntax Description

protocol	Name of an IP protocol. *protocol* can be **ip**, **tcp**, or **udp**.
source source-wildcard | **host** *source* | **any**	Define a source IP address and wildcard. The *source* is the source address of the network or host from which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *source-wildcard* applies wildcard bits to the source. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *source* and *source-wildcard* of *source* 0.0.0.0. • The keyword **any** as an abbreviation for *source* and *source-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a source-wildcard.
destination destination-wildcard | **host** *destination* | **any**	Define a destination IP address and wildcard. The *destination* is the destination address of the network or host to which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *destination-wildcard* applies wildcard bits to the destination. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *destination* and *destination-wildcard* of *destination* 0.0.0.0. • The keyword **any** as an abbreviation for *destination* and *destination-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a destination-wildcard.

operator port	(Optional) Define a source or destination port. The *operator* can be only **eq** (equal). If *operator* is after the source IP address and wildcard, conditions match when the source port matches the defined port. If *operator* is after the destination IP address and wildcard, conditions match when the destination port matches the defined port. The *port* is a decimal number or name of a Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) port. The number can be from 0 to 65535. Use TCP port names only for TCP traffic. Use UDP port names only for UDP traffic.
dscp *dscp-value*	(Optional) Define a Differentiated Services Code Point (DSCP) value to classify traffic. For the *dscp-value*, enter any of the 13 supported DSCP values (0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56), or use the question mark (**?**) to see a list of available values.
time-range *time-range-name*	(Optional) For the **time-range** keyword, enter a meaningful name to identify the time range. For a more detailed explanation of this keyword, refer to the software configuration guide.

Defaults

There are no specific conditions that deny packets in the named or numbered IP ACL.

The default ACL is always terminated by an implicit deny statement for all packets.

Command Modes

Access-list configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use this command after the **ip access-list** global configuration command to specify deny conditions for an IP ACL. You can specify a source IP address, destination IP address, IP protocol, TCP port, or UDP port. Specify the TCP and UDP port numbers only if *protocol* is **tcp** or **udp** and *operator* is **eq**.



Note For more information about configuring IP ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to create an extended IP ACL and to configure deny conditions for it:

```
Switch(config)# ip access-list extended Internetfilter
Switch(config-ext-nacl)# deny tcp host 190.5.88.10 any
Switch(config-ext-nacl)# deny tcp host 192.1.10.10 any
```

This is an example of a standard ACL that sets a deny condition:

```
ip access-list standard Acclist1
 deny 192.5.34.0  0.0.0.255
 deny 128.88.10.0  0.0.0.255
 deny 36.1.1.0  0.0.0.255
```

Note In these examples, all other IP access is implicitly denied.

You can verify your settings by entering the **show ip access-lists** or **show access-lists** privileged EXEC command.

Related Commands

Command	Description
ip access-list	Defines an IP ACL.
permit (access-list configuration)	Sets conditions for an IP ACL.
ip access-group	Controls access to an interface.
show ip access-lists	Displays IP ACLs configured on the switch.
show access-lists	Displays ACLs configured on a switch.

deny (MAC access-list configuration)

Use the **deny** MAC access-list configuration command to prevent Layer 2 traffic from being forwarded if the conditions are matched. Use the **no** form of this command to remove a deny condition from the MAC named access control list (ACL).

{**permit** | **deny**} {**any** | **host** *src-MAC-addr*} {**any** | **host** *dst-MAC-addr*} [**aarp** | **amber** | **appletalk** | **dec-spanning** | **decnet-iv** | **diagnostic** | **dsm** | **etype-6000** | **etype-8042** | **lat** | **lavc-sca** | **mop-console** | **mop-dump** | **msdos** | **mumps** | **netbios** | **vines-echo** |**vines-ip** | **xns-idp**]

no {**permit** | **deny**} {**any** | **host** *src-MAC-addr*} {**any** | **host** *dst-MAC-addr*} [**aarp** | **amber** | **appletalk** | **dec-spanning** | **decnet-iv** | **diagnostic** | **dsm** | **etype-6000** | **etype-8042** | **lat** | **lavc-sca** | **mop-console** | **mop-dump** | **msdos** | **mumps** | **netbios** | **vines-echo** |**vines-ip** | **xns-idp**]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

any	Keyword to deny any source or destination MAC address.
host *src-MAC-addr*	Define a host MAC address. If the source address for a packet matches the defined address, traffic from that address is denied. MAC address-based subnets are not allowed.
host *dst-MAC-addr*	Define a destination MAC address. If the destination address for a packet matches the defined address, traffic to that address is denied. MAC address-based subnets are not allowed.
aarp	Select Ethertype AppleTalk Address Resolution Protocol that maps a data-link address to a network address.
amber	Select EtherType DEC-Amber.
appletalk	Select EtherType AppleTalk/EtherTalk.
dec-spanning	Select EtherType Digital Equipment Corporation (DEC) spanning tree.
decnet-iv	Select EtherType DECnet Phase IV protocol.
diagnostic	Select EtherType DEC-Diagnostic.
dsm	Select EtherType DEC-DSM.
etype-6000	Select EtherType 0x6000.
etype-8042	Select EtherType 0x8042**.**
lat	Select EtherType DEC-LAT.
lavc-sca	Select EtherType DEC-LAVC-SCA.
mop-console	Select EtherType DEC-MOP Remote Console.
mop-dump	Select EtherType DEC-MOP Dump.
msdos	Select EtherType DEC-MSDOS.
mumps	Select EtherType DEC-MUMPS.

netbios	Select EtherType DEC-Network Basic Input/Output System (NETBIOS).
vines-echo	Select EtherType Virtual Integrated Network Service (VINES) Echo from Banyan Systems.
vines-ip	Select EtherType VINES IP.
xns-idp	Select EtherType Xerox Network Systems (XNS) protocol suite (from 0 to 65535), an arbitrary Ethertype in decimal, hexadecimal, or octal.

Defaults

This command has no defaults. However, the default action for a MAC named ACL is to deny.

Command Modes

MAC access-list configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

When an access control entry (ACE) is added to an ACL, an implied **deny-any-any** condition exists at the end of the list. That is, if there are no matches, the packets are denied. However, before the first ACE is added, the list permits all packets.

These options are not allowed:

- Class of service (CoS)
- Ethertype number of a packet with Ethernet II or Subnetwork Access Protocol (SNAP) encapsulation
- Link Service Access Point (LSAP) number of a packet with 802.2 encapsulation

Note For more information about configuring MAC extended ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to define the MAC named extended ACL to deny NETBIOS traffic from any source to MAC address 00c0.00a0.03fa. Traffic matching this list is denied.

```
Switch(config-ext-macl)# deny any host 00c0.00a0.03fa netbios
```

This example shows how to remove the deny condition from the named MAC extended ACL:

```
Switch(config-ext-macl)# no deny any host 00c0.00a0.03fa netbios
```

You can verify your settings by entering the **show access-lists** privileged EXEC command.

Related Commands

Command	Description
mac access-list extended	Creates an ACL based on MAC addresses for non-IP traffic.
permit (MAC access-list configuration)	Permits Layer 2 traffic to be forwarded if conditions are matched.
show access-lists	Displays ACLs configured on a switch.

dot1x default

Use the **dot1x default** global configuration command to reset the global 802.1X parameters to their default values.

dot1x default

Syntax Description

This command has no arguments or keywords.

Defaults

This command has no default setting.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to reset the global 802.1X parameters:

```
Switch(config)# dot1x default
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
dot1x max-req	Sets the maximum number of times that the switch sends an Extensible Authentication Protocol (EAP)-request/identity frame before restarting the authentication process.
dot1x re-authentication	Enables periodic re-authentication of the client.
dot1x timeout quiet-period	Sets the number of seconds that the switch remains in the quiet state following a failed authentication exchange.
dot1x timeout re-authperiod	Sets the number of seconds between re-authentication attempts.
dot1x timeout tx-period	Sets the number of seconds that the switch waits for a response to an EAP-request/identity frame from the client before retransmitting the request.
show dot1x	Displays the 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x max-req

Use the **dot1x max-req** global configuration command to set the maximum number of times that the switch sends an Extensible Authentication Protocol (EAP)-request/identity frame (assuming that no response is received) before restarting the authentication process. Use the **no** form of this command to return to the default setting.

dot1x max-req *count*

no dot1x max-req

Syntax Description

count	Number of times that the switch sends an EAP-request/identify frame before restarting the authentication process. The range is 1 to 10.

Defaults

The default is 2 times.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You should change the default value of this command only to adjust for unusual circumstances such as unreliable links or specific behavioral problems with certain clients and authentication servers.

Examples

This example shows how to set the number of times that the switch sends an EAP-request/identity frame to 5 before restarting the authentication process:

```
Switch(config)# dot1x max-req 5
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
dot1x timeout tx-period	Sets the number of seconds that the switch waits for a response to an EAP-request/identity frame from the client before retransmitting the request.
show dot1x	Displays the 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x multiple-hosts

Use the **dot1x multiple-hosts** interface configuration command to allow multiple hosts (clients) on an 802.1X-authorized port that has the **dot1x port-control** interface configuration command set to **auto**. Use the **no** form of this command to return to the default setting.

dot1x multiple-hosts

no dot1x multiple-hosts

Syntax Description

This command has no arguments or keywords.

Defaults

Multiple hosts are disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command enables you to attach multiple clients to a single 802.1X-enabled port. In this mode, only one of the attached hosts must be successfully authorized for all hosts to be granted network access. If the port becomes unauthorized (re-authentication fails, or an Extensible Authentication Protocol over LAN [EAPOL]-logoff message is received), all attached clients are denied access to the network.

Examples

This example shows how to enable 802.1X on Fast Ethernet interface 0/1 and to allow multiple hosts:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# dot1x port-control auto
Switch(config-if)# dot1x multiple-hosts
```

You can verify your settings by entering the **show dot1x** [**interface** *interface-id*] privileged EXEC command.

Related Commands

Command	Description
dot1x port-control	Enables manual control of the authorization state of the port.
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x port-control

Use the **dot1x port-control** interface configuration command to enable manual control of the authorization state of the port. Use the **no** form of this command to return to the default setting.

dot1x port-control {**auto** | **force-authorized** | **force-unauthorized**}

no dot1x port-control

Syntax Description

auto	Enable 802.1X authentication on the interface and cause the port to transition to the authorized or unauthorized state based on the 802.1X authentication exchange between the switch and the client.
force-authorized	Disable 802.1X authentication on the interface and cause the port to transition to the authorized state without any authentication exchange required. The port transmits and receives normal traffic without 802.1X-based authentication of the client.
force-unauthorized	Deny all access through this interface by forcing the port to transition to the unauthorized state, ignoring all attempts by the client to authenticate. The switch cannot provide authentication services to the client through the interface.

Defaults

The authorization state is **force-authorized**.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The 802.1X protocol is supported on Layer 2 static-access ports.

You can use the **auto** keyword only if the port is not configured as one of these:

- Trunk port—If you try to enable 802.1X on a trunk port, an error message appears, and 802.1X is not enabled. If you try to change the mode of an 802.1X-enabled port to trunk, the port mode is not changed.
- Dynamic port—A port in dynamic mode can negotiate with its neighbor to become a trunk port. If you try to enable 802.1X on a dynamic port, an error appears, and 802.1X is not enabled. If you try to change the mode of an 802.1X-enabled port to dynamic, the port mode is not changed.
- Dynamic-access port—If you try to enable 802.1X on a dynamic-access (VLAN Query Protocol [VQP]) port, an error appears, and 802.1X is not enabled. If you try to change an 802.1X-enabled port to dynamic VLAN assignment, an error appears, and the VLAN configuration is not changed.
- EtherChannel port—Before enabling 802.1X on the port, you must first remove it from the EtherChannel. If you try to enable 802.1X on an EtherChannel or on an active port in an EtherChannel, an error appears, and 802.1X is not enabled. If you enable 802.1X on a not-yet active port of an EtherChannel, the port does not join the EtherChannel.

- Secure port—You cannot configure a secure port as an 802.1X port. If you try to enable 802.1X on a secure port, an error appears, and 802.1X is not enabled. If you try to change an 802.1X-enabled port to a secure port, an error appears, and the security settings are not changed.
- Switched Port Analyzer (SPAN) destination port—You can enable 802.1X on a port that is a SPAN destination port; however, 802.1X is disabled until the port is removed as a SPAN destination. You can enable 802.1X on a SPAN source port.

To globally disable 802.1X on the switch, you must disable it on each port. There is no global configuration command for this task.

Examples

This example shows how to enable 802.1X on Fast Ethernet interface 0/1:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# dot1x port-control auto
```

You can verify your settings by entering the **show dot1x** privileged EXEC command and checking the Status column in the 802.1X Port Summary section of the output. An *enabled* status means the port-control value is set either to **auto** or to **force-unauthorized**.

Related Commands

Command	Description
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x re-authenticate

Use the **dot1x re-authenticate** privileged EXEC command to manually initiate a re-authentication of all 802.1X-enabled ports or the specified 802.1X-enabled port.

dot1x re-authenticate [**interface** *interface-id*]

Syntax Description

interface *interface-id*	(Optional) Slot and port number of the interface to re-authenticate.

Defaults

There is no default setting.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You can use this command to re-authenticate a client without waiting for the configured number of seconds between re-authentication attempts (re-authperiod) and automatic re-authentication.

Examples

This example shows how to manually re-authenticate the device connected to Fast Ethernet interface 0/1:

```
Switch# dot1x re-authenticate interface fastethernet0/1
Starting reauthentication on FastEthernet0/1.
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x re-authentication

Use the **dot1x re-authentication** global configuration command to enable periodic re-authentication of the client. Use the **no** form of this command to return to the default setting.

dot1x re-authentication

no dot1x re-authentication

Syntax Description

This command has no arguments or keywords.

Defaults

Periodic re-authentication is disabled.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You configure the amount of time between periodic re-authentication attempts by using the **dot1x timeout re-authperiod** global configuration command.

Examples

This example shows how to disable periodic re-authentication of the client:

```
Switch(config)# no dot1x re-authentication
```

This example shows how to enable periodic re-authentication and set the number of seconds between re-authentication attempts to 4000 seconds:

```
Switch(config)# dot1x re-authentication
Switch(config)# dot1x timeout re-authperiod 4000
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
dot1x timeout re-authperiod	Sets the number of seconds between re-authentication attempts.
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x timeout quiet-period

Use the **dot1x timeout quiet-period** global configuration command to set the number of seconds that the switch remains in the quiet state following a failed authentication exchange (for example, the client provided an invalid password). Use the **no** form of this command to return to the default setting.

dot1x timeout quiet-period *seconds*

no dot1x timeout quiet-period

Syntax Description

seconds	Time in seconds that the switch remains in the quiet state following a failed authentication exchange with the client. The range is 0 to 65535 seconds.

Defaults

The default time is 60 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

During the quiet period, the switch does not accept or initiate any authentication requests.

You should change the default value of this command only to adjust for unusual circumstances such as unreliable links or specific behavioral problems with certain clients and authentication servers.

If you want to provide a faster response time to the user, enter a smaller number than the default.

Examples

This example shows how to set the quiet time on the switch to 30 seconds:

```
Switch(config)# dot1x timeout quiet-period 30
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x timeout re-authperiod

Use the **dot1x timeout re-authperiod** global configuration command to set the number of seconds between re-authentication attempts. Use the **no** form of this command to return to the default setting.

dot1x timeout re-authperiod *seconds*

no dot1x timeout re-authperiod

Syntax Description

seconds	Number of seconds between re-authentication attempts. The range is 1 to 4294967295.

Defaults

The default is 3600 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **dot1x timeout re-authperiod** global configuration command affects the behavior of the switch only if you have enabled periodic re-authentication by using the **dot1x re-authentication** global configuration command.

You should change the default value of this command only to adjust for unusual circumstances such as unreliable links or specific behavioral problems with certain clients or authentication servers.

Examples

This example shows how to enable periodic re-authentication and set the number of seconds between re-authentication attempts to 4000 seconds:

```
Switch(config)# dot1x re-authentication
Switch(config)# dot1x timeout re-authperiod 4000
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
dot1x re-authentication	Enables periodic re-authentication of the client.
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

dot1x timeout tx-period

Use the **dot1x timeout tx-period** global configuration command to set the number of seconds that the switch waits for a response to an Extensible Authentication Protocol (EAP)-request /identity frame from the client before retransmitting the request. Use the **no** form of this command to return to the default setting.

dot1x timeout tx-period *seconds*

no dot1x timeout tx-period

Syntax Description

seconds	Time in seconds that the switch should wait for a response to an EAP-request/identity frame from the client before retransmitting the request. The range is 1 to 65535 seconds.

Defaults

The default is 30 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You should change the default value of this command only to adjust for unusual circumstances such as unreliable links or specific behavioral problems with certain clients or authentication servers.

Examples

This example shows how to set 60 as the number of seconds that the switch waits for a response to an EAP-request/identity frame from the client before retransmitting the request:

```
Switch(config)# dot1x timeout tx-period 60
```

You can verify your settings by entering the **show dot1x** privileged EXEC command.

Related Commands

Command	Description
dot1x max-req	Sets the maximum number of times that the switch sends an EAP-request/identity frame before restarting the authentication process.
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

duplex

Use the **duplex** interface configuration command to specify the duplex mode of operation for switch ports. Use the **no** form of this command to return the port to its default value.

duplex {**auto** | **full** | **half**}

no duplex

Syntax Description

auto	Port automatically detects whether it should run in full- or half-duplex mode.
full	Port is in full-duplex mode.
half	Port is in half-duplex mode.

Defaults

For Fast Ethernet and 10/100/1000 ports, the default is **auto**.

For 100BASE-FX ports, the default is **full**.

For the default duplex mode of the Gigabit Interface Converter (GBIC)-module ports, refer to the documentation that came with your GBIC module.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Certain ports, such as GBIC module ports, can be configured to be either full duplex or half duplex. The applicability of this command depends on the device to which the switch is attached.

The 100BASE-FX ports on Catalyst 2950C-24 switches do not support the **duplex** command. These ports only operate in full-duplex and 100-Mbps mode.

For Fast Ethernet ports, setting the port to **auto** has the same effect as specifying **half** if the attached device does not autonegotiate the duplex parameter.

If the speed is set to **auto**, the switch negotiates with the device at the other end of the link for the speed setting and then forces the speed setting to the negotiated value. The duplex setting remains as configured on each end of the link, which could result in a duplex setting mismatch.

Note The 10/100/1000 ports can operate only in the full-duplex mode.

If both the speed and duplex are set to specific values, autonegotiation is disabled.

Note For guidelines on setting the switch speed and duplex parameters, refer to the *Catalyst 2950 Desktop Switch Hardware Installation Guide.*

Examples

This example shows how to set a Fast Ethernet port to half duplex:

```
Switch(config)# interface fastethernet0/11
Switch(config-if)# duplex half
```

This example shows how to set a Gigabit Ethernet port to full duplex:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# duplex full
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
speed	Specifies the port speed.

errdisable detect

Use the **errdisable detect** global configuration command to enable error disable detection. Use the **no** form of this command to disable this feature.

errdisable detect cause {**all** | **dtp-flap** | **gbic-invalid** | **link-flap** | **pagp-flap**}

no errdisable detect cause {**all** | **dtp-flap** | **gbic-invalid** | **link-flap** | **pagp-flap**}

Syntax Description

all	Enable detection for all error disable causes.
dtp-flap	Enable detection for the Dynamic Trunking Protocol (DTP)- flap cause.
gbic-invalid	Enable error detection for an invalid GBIC error-disable cause.
link-flap	Enable detection for the link flap cause.
pagp-flap	Enable detection for the Port Aggregation Protocol (PAgP)-flap cause.

Defaults

The default is **all**, enabled for all causes.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **bpduguard**, **rootguard**, and **udld** keywords were removed. The **gbic-invalid** keyword was added.

Usage Guidelines

A cause (**dtp-flap**, **gbic-invalid**, **link-flap**, and **pagp-flap**) is the reason why the error-disabled state occurred. When a cause is detected on an interface, the interface is placed in error-disabled state, an operational state similar to link-down state. If you do not enable errdisable recovery for the cause, the interface stays in the error-disabled state until you enter the **shutdown** and **no shutdown** interface configuration commands. If you enable the recovery for a cause, the interface is brought out of the error-disabled state and allowed to retry the operation again when all the causes have timed out.

You must enter the **shutdown** and then the **no shutdown** commands to manually recover an interface from the error-disabled state.

Examples

This example shows how to enable error disable detection for the link-flap error-disable cause:

```
Switch(config)# errdisable detect cause link-flap
```

You can verify your settings by entering the **show errdisable detect** privileged EXEC command.

Related Commands

Command	Description
errdisable recovery	Configures the recovery mechanism variables.
show errdisable detect	Displays errdisable detection status.
show interfaces trunk	Displays interface status or a list of interfaces in error-disabled state.

errdisable recovery

Use the **errdisable recovery** global configuration command to configure the recover mechanism variables. Use the **no** form of this command to return to the default setting.

errdisable recovery {**cause** {**all** | **bpduguard** | **dtp-flap** | **gbic-invalid** | **link-flap** | **pagp-flap** | **psecure-violation** | **udld**} | {**interval** *interval*}

no errdisable recovery {**cause** {**all** | **bpduguard** | **dtp-flap** | **gbic-invalid** | **link-flap** | **pagp-flap** | **psecure-violation** | **udld**} | {**interval** *interval*}

Syntax Description

cause	Enable error disable to recover from a specific cause.
all	Enable the timer to recover from all error-disable causes.
bpduguard	Enable the timer to recover from the bridge protocol data unit (BPDU)-guard error-disable state.
dtp-flap	Enable the timer to recover from the Dynamic Trunking Protocol (DTP)-flap error-disable state.
gbic-invalid	Enable the timer to recover from an invalid GBIC error disable state.
link-flap	Enable the timer to recover from the link-flap error-disable state.
pagp-flap	Enable the timer to recover from the Port Aggregation Protocol (PAgP)-flap error-disable state.
psecure-violation	Enable the timer to recover from a port security violation disable state.
udld	Enable the timer to recover from the UniDirectional Link Detection (UDLD) error-disable state.
interval *interval*	Specify the time to recover from specified error-disable state. The range is 30 to 86400 seconds. The same interval is applied to all causes. The default interval is 300 seconds. **Note** The errdisable recovery timer initializes at a random differential from the configured interval value. The difference between the actual timeout value and the configured value can be up to 15 percent of the configured interval.

Note Though visible in the command-line help string, the **channel-misconfig** keyword is not supported.

Defaults

Recovery is disabled for all causes.

The default interval is 300 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **gbic-invalid** and **psecure-violation** keywords were added. The **rootguard** keyword was removed.

Usage Guidelines

A cause (**bpduguard**, **dtp-flap**, **gbic-invalid**, **link-flap**, **pagp-flap**, **psecure-violation**, and **udld**) is defined as the reason why the error-disabled state occurred. When a cause is detected on an interface, the interface is placed in error-disabled state, an operational state similar to link-down state. If you do not enable errdisable recovery for the cause, the interface stays in error-disabled state until you enter a **shutdown** and **no shutdown** interface configuration command. If you enable the recovery for a cause, the interface is brought out of the error-disabled state and allowed to retry the operation again when all the causes have timed out.

Otherwise, you must enter the **shutdown** then **no shutdown** commands to manually recover an interface from the error-disabled state.

Examples

This example shows how to enable the recovery timer for the BPDU guard error-disable cause:

```
Switch(config)# errdisable recovery cause bpduguard
```

This example shows how to set the timer to 500 seconds:

```
Switch(config)# errdisable recovery interval 500
```

You can verify your settings by entering the **show errdisable recovery** privileged EXEC command.

Related Commands

Command	Description
show errdisable recovery	Displays errdisable recovery timer information.
show interfaces status	Displays interface status.

flowcontrol

Use the **flowcontrol** interface configuration command to set the receive or send flow-control value for an interface. When flow control **send** is on for a device and it detects any congestion at its end, it notifies the link partner or the remote device of the congestion by sending a pause frame. When flow control **receive** is on for the remote device and it receives a pause frame, it stops sending any data packets. This prevents any loss of data packets during the congestion period.

Use the **receive off** and **send off** keywords to disable flow control.

flowcontrol {**receive** | **send**} {**desired | off** | **on**}

Note This **flowcontrol** command applies only to 10/100/1000 ports and Gigabit Interface Converter (GBIC)-module ports on the switches.

Syntax Description

receive	Sets whether the interface can receive flow-control packets from a remote device.
send	Sets whether the interface can send flow-control packets to a remote device.
desired	When used with **receive**, allows an interface to operate with an attached device that is required to send flow-control packets or with an attached device that is not required to but can send flow-control packets. When used with **send**, the interface sends flow-control packets to a remote device if the remote device supports it.
off	When used with **receive**, turns off an attached device's ability to send flow-control packets to an interface. When used with **send**, turns off the local port's ability to send flow-control packets to a remote device.
on	When used with **receive**, allows an interface to operate with an attached device that is required to send flow-control packets or with an attached device that is not required to but can send flow-control packets. When used with **send**, the interface sends flow-control packets to a remote device if the remote device supports it.

Defaults

The defaults for 10/100/1000 and GBIC-module ports are **flowcontrol receive off** and **flowcontrol send desired**.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **asymmetric** and **symmetric** keywords were replaced with the **receive**, **send**, **off**, **on**, and **desired** keywords.

Usage Guidelines

Use the **flowcontrol** command only on 10/100/1000 and GBIC-module ports.

We strongly recommend that you do not configure IEEE 802.3X flowcontrol when quality of service (QoS) is configured on the switch. Before configuring flowcontrol on an interface, make sure to disable QoS on the switch.

Note that when used with **receive**, the **on** and **desired** keywords have the same result.

When you use the **flowcontrol** command to set a port to control traffic rates during congestion, you are setting flow control on a port to one of these conditions:

- **receive on** and **send on**: Flow control operates in both directions; pause frames can be sent by both the local device and the remote device to show link congestion.
- **receive on** and **send desired**: The port can receive pause frames and is able to send pause frames if the attached device supports them.
- **receive on** and **send off**: The port cannot send pause frames, but can operate with an attached device that is required to or is able to send pause frames; the port is able to receive pause frames.
- **receive off** and **send on**: The port sends pause frames if the remote device supports them, but cannot receive pause frames from the remote device.
- **receive off** and **send desired**: The port cannot receive pause frames, but can send pause frames if the attached device supports them.
- **receive off** and **send off**: Flow control does not operate in either direction. In case of congestion, no indication is given to the link partner, and no pause frames are sent or received by either device.

Table 2-1 shows the flow control resolution achieved on local and remote ports by a combination of settings. The table assumes that for **receive**, using the **desired** keyword has the same results as using the **on** keyword.

Table 2-1 Flow Control Settings and Local and Remote Port Flow Control Resolution

Flow Control Settings		Flow Control Resolution	
Local Device	**Remote Device**	**Local Device**	**Remote Device**
send on/receive on	**send on/receive on**	Sends and receives	Sends and receives
	send on/receive off	Does not send or receive	Does not send or receive
	send desired/receive on	Sends and receives	Sends and receives
	send desired/receive off	Does not send or receive	Does not send or receive
	send off/receive on	Sends and receives	Receives only
	send off/receive off	Does not send or receive	Does not send or receive
send on/receive off	**send on/receive on**	Does not send or receive	Does not send or receive
	send on/receive off	Does not send or receive	Does not send or receive
	send desired/receive on	Sends only	Receives only
	send desired/receive off	Does not send or receive	Does not send or receive
	send off/receive on	Sends only	Receives only
	send off/receive off	Does not send or receive	Does not send or receive

Table 2-1 Flow Control Settings and Local and Remote Port Flow Control Resolution (continued)

Flow Control Settings		Flow Control Resolution	
Local Device	**Remote Device**	**Local Device**	**Remote Device**
send desired/receive on	**send on/receive on**	Sends and receives	Sends and receives
	send on/receive off	Receives only	Sends only
	send desired/receive on	Sends and receives	Sends and receives
	send desired/receive off	Receives only	Sends only
	send off/receive on	Sends and receives	Receives only
	send off/receive off	Does not send or receive	Does not send or receive
send desired/receive off	**send on/receive on**	Does not send or receive	Does not send or receive
	send on/receive off	Does not send or receive	Does not send or receive
	send desired/receive on	Sends only	Receives only
	send desired/receive off	Does not send or receive	Does not send or receive
	send off/receive on	Sends only	Receives only
	send off/receive off	Does not send or receive	Does not send or receive
send off/receive on	**send on/receive on**	Receives only	Sends and receives
	send on/receive off	Receives only	Sends only
	send desired/receive on	Receives only	Sends and receives
	send desired/receive off	Receives only	Sends only
	send off/receive on	Receives only	Receives only
	send off/receive off	Does not send or receive	Does not send or receive
send off/receive off	**send on/receive on**	Does not send or receive	Does not send or receive
	send on/receive off	Does not send or receive	Does not send or receive
	send desired/receive on	Does not send or receive	Does not send or receive
	send desired/receive off	Does not send or receive	Does not send or receive
	send off/receive on	Does not send or receive	Does not send or receive
	send off/receive off	Does not send or receive	Does not send or receive

Examples

This example shows how to configure the local port to not support any level of flow control by the remote port:

```
Switch(config-if)# flowcontrol receive off
Switch(config-if)# flowcontrol send off
```

You can verify your settings by entering the **show interfaces counters** privileged EXEC command.

Related Commands

Command	Description
show interfaces counters	Displays the interface settings on a switch, including input and output flow control.

interface

Use the **interface** global configuration command to configure an interface type, create a switch virtual interface to be used as the management VLAN interface, and to enter interface configuration mode.

interface {*interface-id* | **vlan** *number*}

no interface {*interface-id* | **vlan** *number*}

Syntax Description

interface-id	Specify the interface type (Fast Ethernet or Gigabit Ethernet) and number.
vlan *number*	VLAN number from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed to be used as the management VLAN. Do not enter leading zeroes.

Defaults

The default management VLAN interface is VLAN 1.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

When creating a management VLAN interface, a space between **vlan** and *number* is accepted.

Only one management VLAN interface can be active.

You cannot delete the management VLAN 1 interface.

You can use the **no shutdown** interface configuration command to shut down the active management VLAN interface and to enable a new one.

You can configure the management VLAN interface on static-access and trunk ports.

Examples

This example shows how to enable the switch to configure interface 2:

```
Switch(config)# interface fastethernet0/2
Switch(config-if)#
```

This example shows how to change the management VLAN from the default management VLAN to VLAN 3. This series of commands should only be entered from the console. If these commands are entered through a Telnet session, the **shutdown** command disconnects the session, and there is no way to use IP to access the system.

```
Switch# configure terminal
Switch(config)# interface vlan 3
Switch(config-if)# ip address 172.20.128.176 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
```

You can verify your settings by entering the **show interfaces** and **show interfaces vlan** *vlan-id* privileged EXEC commands.

Related Commands

Command	Description
show interfaces	Displays the administrative and operational status of a switching (nonrouting) port.
shutdown	Disables a port and shuts down the management VLAN.

interface port-channel

Use the **interface port-channel** global configuration command to access or create the port-channel logical interface for Layer 2 interfaces. Use the **no** form of this command to remove the port channel.

interface port-channel *port-channel-number*

no interface port-channel *port-channel-number*

Syntax Description

port-channel-number	Port-channel number. The range is 1 to 6.

Defaults

No port-channel logical interfaces are defined.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Only one port channel in a channel group is allowed.

Follow these guidelines when you use the **interface port-channel** command:

- If you want to use the Cisco Discovery Protocol (CDP), you must configure it only on the physical interface and not on the port-channel interface.
- On the port-channel interface, if you do not assign a static MAC address or if you assign a static MAC address and then later remove it, the switch automatically assigns a MAC address to the interface.

Examples

This example shows how to create a port-channel interface with a port-channel number of 5:

```
Switch(config)# interface port-channel 5
```

You can verify your settings by entering the **show running-config** or **show etherchannel** *channel-group-number* **detail** privileged EXEC command.

Related Commands

Command	Description
channel-group	Assigns an Ethernet interface to an EtherChannel group.
show etherchannel	Displays EtherChannel information for a channel.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**

interface range

Use the **interface range** global configuration command to enter interface range configuration mode and to execute a command on multiple ports at the same time. Use the **no** form of this command to remove an interface range.

interface range {*port-range* | **macro** *name*}

no interface range {*port-range* | **macro** *name*}

Syntax Description

port-range	Port range. For a list of valid values for *port-range*, see the "Usage Guidelines" section.
macro *name*	Specify the name of a macro.

Defaults

This command has no default setting.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

From the interface range configuration mode, all interface parameters that you enter are applied to all interfaces within the range.

For VLANs, you can use the **interface range** command only on existing VLAN interfaces. To display VLAN interfaces, enter the **show running-config** privileged EXEC command. VLANs not displayed cannot be used in the **interface range** command. The commands that you enter under the **interface range** command are applied to all existing VLAN interfaces in the range.

All configuration changes made to an interface range are saved to nonvolatile RAM (NVRAM), but the interface range itself is not saved to NVRAM.

You can enter the interface range in two ways:

- Specifying up to five interface ranges
- Specifying a previously defined interface-range macro

You can define up to five interface ranges with a single command, with each range separated by a comma (**,**).

All interfaces in a range must be the same type; that is, all Fast Ethernet ports, all Gigabit Ethernet ports, all EtherChannel ports, or all VLANs.

These are the valid values for *port-range* type and interface:

- **vlan** *vlan-id*, where *vlan-id* is from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed.
- **port-channel** *port-channel-number*, where *port-channel-number* is from 1 to 6
- **fastethernet** *interface-id*
- **gigabitethernet** *interface-id*

For physical interfaces, the *interface-id* is defined as a slot/number (where slot is always 0 for the switch), and the range is entered as *type* **0**/*number* - *number* (for example, **gigabitethernet0/1 - 2**). You can also enter multiple ranges.

When you define a range, you must enter a space before and after the hyphen (**-**):

```
interface range gigabitethernet0/1 - 2
```

When you define multiple ranges, you must enter a space before and after the comma (**,**):

```
interface range fastethernet0/3 - 7 , gigabitethernet0/1 - 2
```

You cannot specify both a macro and an interface range in the same command.

A single interface can also be specified in *port-range*. (The command is then similar to the **interface** *interface-id* global configuration command.)

Note For more information about configuring interface ranges, refer to the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to use the **interface range** command to enter interface range configuration mode and to enter commands for two ports:

```
Switch(config)# interface range gigabitethernet0/1 - 2
Switch(config-if-range)#
```

This example shows how to use a port-range macro *macro1* for the same function. The advantage is that you can reuse the *macro1* until you delete it.

```
Switch(config)# define interface-range macro gigabitethernet0/1 - 2
Switch(config)# interface range macro macro1
Switch(config-if-range)#
```

Related Commands

Command	Description
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

ip access-group

Use the **ip access-group** interface configuration command to control access to an interface. Use the **no** form of this command to remove an access group from an interface.

ip access-group {*access-list-number* | *name*} **in**

no ip access-group {*access-list-number* | *name*} **in**

This command is available on physical interfaces only if your switch is running the enhanced software image (EI).

Syntax Description

access-list-number	Number of the IP access control list (ACL), from 1 to 199 or from 1300 to 2699.
name	Name of an IP ACL, specified in the **ip access-list** command.

Defaults

No ACL is applied to the interface.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You can apply IP ACLs only to ingress interfaces. If a MAC access group is already defined for an interface, you cannot apply this command to the interface.

The ACLs can be standard or extended.

For standard ACLs, after receiving a packet, the switch checks the packet source address. If the source address matches a defined address in the ACL and the list permits the address, the switch forwards the packet.

For extended ACLs, after receiving the packet, the switch checks the match conditions in the ACL. If the conditions are matched, the switch forwards the packet.

If the specified ACL does not exist, the switch forwards all packets.

IP access groups can be separated on Layer 2 and Layer 3 interfaces.

Note For more information about configuring IP ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to apply a numbered ACL to an interface:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# ip access-group 101 in
```

You can verify your settings by entering the **show access-lists** or **show ip access-lists** privileged EXEC command.

Related Commands

Command	Description
access-list (IP extended)	Defines an extended IP ACL.
access-list (IP standard)	Defines a standard IP ACL.
deny (access-list configuration)	Configures conditions for an IP ACL.
ip access-list	Defines an IP ACL.
permit (access-list configuration)	Configures conditions for an IP ACL.
show access-lists	Displays ACLs configured on the switch.
show ip access-lists	Displays IP ACLs configured on the switch.

ip access-list

Use the **ip access-list** global configuration command to create an IP access control list (ACL) to be used for matching packets to an ACL whose name or number you specify and to enter access-list configuration mode. Use the **no** form of this command to delete an existing IP ACL and return to global configuration mode.

ip access-list {**extended** | **standard**} {*access-list-number* | *name*}

no ip access-list {**extended** | **standard**} {*access-list-number* | *name*}

This command is available on physical interfaces only if your switch is running the enhanced software image (EI).

Syntax Description

access-list-number	Number of an ACL. For standard IP ACLs, the range is from 1 to 99 and 1300 to 1999. For extended IP ACLs, the range from 100 to 199 and from 2000 to 2699.
name	Name of an ACL. **Note** The ACL name must begin with an alphabetic character to prevent ambiguity with numbered ACLs. A name also cannot contain a space or quotation mark.

Defaults

No named or numbered IP ACLs are defined.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use this command to enter access-list configuration mode and to specify the name or number of the IP ACL for which you want to create or modify ACL match criteria. In this mode, you must enter the **permit** and **deny** commands to configure the permit and deny access conditions for this list.

Use the **ip access-list** command and its subcommands to define packet classification and marking as part of a globally-named service policy applied on a per-interface basis or as an IP access group applied on a per-interface basis.

Specifying **standard** or **extended** with the **ip access-list** command determines the prompt that you get when you enter access-list configuration mode.

Note For more information about configuring IP ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to configure a standard ACL named *Internetfilter1*:

```
Switch(config)# ip access-list standard Internetfilter1
Switch(config-std-nacl)# permit 192.5.34.0  0.0.0.255
Switch(config-std-nacl)# permit 192.5.32.0  0.0.0.255
Switch(config-std-nacl)# exit
```

This example shows how to configure an extended ACL named *Internetfilter2*:

```
Switch(config)# ip access-list extended Internetfilter2
Switch(config-ext-nacl)# permit any 128.8.10.0  0.0.0.255 eq 80
Switch(config-ext-nacl)# permit any 128.5.8.0  0.0.0.255 eq 80
Switch(config-ext-nacl)# exit
```

Note In these examples, all other IP access is implicitly denied.

You can verify your settings by entering the **show access-lists** or **show ip access-lists** privileged EXEC command.

Related Commands

Command	Description
deny (access-list configuration)	Configures conditions for an IP ACL.
ip access-group	Controls access to an interface.
permit (access-list configuration)	Configures conditions for an IP ACL.
service-policy	Applies a policy map to the input of an interface.
show access-lists	Displays ACLs configured on the switch.
show ip access-lists	Displays IP ACLs configured on the switch.

ip address

Use the **ip address** interface configuration command to set an IP address for a switch. Use the **no** form of this command to remove an IP address or to disable IP processing.

ip address *ip-address subnet-mask*

no ip address *ip-address subnet-mask*

Syntax Description

ip-address	IP address.
subnet-mask	Mask for the associated IP subnet.

Defaults

No IP address is defined for the switch.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

A switch can have one IP address.

The IP address of the switch can be accessed only by nodes connected to ports that belong to the management VLAN. The default for the management VLAN is VLAN 1, but you can configure a different VLAN as the management VLAN.

If you remove the IP address through a Telnet or Secure Shell (SSH) session, your connection to the switch is lost.

If your switch receives its IP address from a Bootstrap Protocol (BOOTP) or a Dynamic Host Configured Protocol (DHCP) server and you remove the switch IP address by using the **no ip address** command, IP processing is disabled, and the BOOTP or DHCP server cannot reassign the address.

Examples

This example shows how to configure the IP address for the switch on a subnetted network:

```
Switch(config)# interface vlan 1
Switch(config-if)# ip address 172.20.128.2 255.255.255.0
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

ip igmp snooping

Use the **ip igmp snooping** global configuration command to globally enable Internet Group Management Protocol (IGMP) snooping. Use the **no** form of this command to disable IGMP snooping.

ip igmp snooping

no ip igmp snooping

Syntax Description

This command has no arguments or keywords.

Defaults

IGMP snooping is globally enabled.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

When IGMP snooping is globally enabled, it enables IGMP snooping on all the existing VLAN interfaces. When IGMP snooping is globally disabled, it disables IGMP snooping on all the existing VLAN interfaces.

The configuration is saved in nonvolatile RAM (NVRAM).

Examples

This example shows how to globally enable IGMP snooping:

```
Switch(config)# ip igmp snooping
```

This example shows how to globally disable IGMP snooping:

```
Switch(config)# no ip igmp snooping
```

You can verify your settings commands by entering the **show ip igmp snooping** privileged EXEC command.

Related Commands

Command	Description
ip igmp snooping vlan	Enables IGMP snooping on a VLAN interface.
ip igmp snooping vlan immediate-leave	Enables IGMP Immediate-Leave processing.
ip igmp snooping vlan mrouter	Configures a Layer 2 port as a multicast router port.
ip igmp snooping vlan static	Configures a Layer 2 port as a member of a group.
show ip igmp snooping	Displays the IGMP snooping configuration.

ip igmp snooping vlan

Use the **ip igmp snooping vlan** global configuration command to enable Internet Group Management Protocol (IGMP) snooping on a specific VLAN. Use the **no** form of this command to disable IGMP snooping on a VLAN interface.

ip igmp snooping vlan *vlan-id*

no ip igmp snooping vlan *vlan-id*

Syntax Description

vlan-id	VLAN ID. The range is from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.

Defaults

IGMP snooping is enabled when each VLAN is created.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

This command automatically configures the VLAN if it is not already configured. The configuration is saved in nonvolatile RAM (NVRAM).

Examples

This example shows how to enable IGMP snooping on VLAN 2:

```
Switch(config)# ip igmp snooping vlan 2
```

This example shows how to disable IGMP snooping on VLAN 2:

```
Switch(config)# no ip igmp snooping vlan 2
```

You can verify your settings by entering the **show ip igmp snooping vlan** privileged EXEC command.

Related Commands

Command	Description
ip igmp snooping	Globally enables IGMP snooping. IGMP snooping must be globally enabled in order to be enabled on a VLAN.
ip igmp snooping vlan immediate-leave	Enables IGMP Immediate-Leave processing.
ip igmp snooping vlan mrouter	Configures a Layer 2 port as a multicast router port.
ip igmp snooping vlan static	Configures a Layer 2 port as a member of a group.
show ip igmp snooping	Displays the IGMP snooping configuration.

ip igmp snooping vlan immediate-leave

Use the **ip igmp snooping immediate-leave** global configuration command to enable Internet Group Management Protocol (IGMP) Immediate-Leave processing on a VLAN interface. Use the **no** form of this command to disable Immediate-Leave processing on the VLAN interface.

ip igmp snooping vlan *vlan-id* **immediate-leave**

no ip igmp snooping vlan *vlan-id* **immediate-leave**

Syntax Description

vlan-id	VLAN ID value. The range is between 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.

Defaults

IGMP Immediate-Leave processing is disabled.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Use the Immediate-Leave feature only when there is only one IP multicast receiver present on every port in the VLAN. The Immediate-Leave configuration is saved in nonvolatile RAM (NVRAM).

The Immediate-Leave feature is supported only with IGMP version 2 hosts.

Examples

This example shows how to enable IGMP Immediate-Leave processing on VLAN 1:

```
Switch(config)# ip igmp snooping vlan 1 immediate-leave
```

This example shows how to disable IGMP Immediate-Leave processing on VLAN 1:

```
Switch(config)# no ip igmp snooping vlan 1 immediate-leave
```

You can verify your settings by entering the **show ip igmp snooping vlan** privileged EXEC command.

Related Commands

Command	Description
ip igmp snooping	Enables IGMP snooping.
ip igmp snooping vlan mrouter	Configures a Layer 2 port as a multicast router port.
show ip igmp snooping	Displays the IGMP snooping configuration.
ip igmp snooping vlan static	Configures a Layer 2 port as a member of a group.
show mac address-table multicast	Displays the Layer 2 multicast entries for a VLAN.

ip igmp snooping vlan mrouter

Use the **ip igmp snooping vlan mrouter** global configuration command to add a multicast router port and to configure the multicast router learning method. Use the **no** form of this command to remove the configuration.

ip igmp snooping vlan *vlan-id* **mrouter** {**interface** *interface-id* | **learn** {**cgmp** | **pim-dvmrp**}}

no ip igmp snooping vlan *vlan-id* **mrouter** {**interface** *interface-id* | **learn** {**cgmp** | **pim-dvmrp**}}

Syntax Description

vlan *vlan-id*	Specify the VLAN ID. The range is from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
interface *interface-id*	Specify the interface of the member port that is configured to a static router port.
learn	Specify the multicast router learning method.
cgmp	Set the switch to learn multicast router ports by snooping on Cisco Group Management Protocol (CGMP) packets.
pim-dvmrp	Set the switch to learn multicast router ports by snooping on IGMP queries and Protocol-Independent Multicasting-Distance Vector Multicast Routing Protocol (PIM-DVMRP) packets.

Defaults

The default learning method is **pim-dvmrp**.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

The CGMP learning method is useful for controlling traffic in Cisco router environments.

The configured learning method is saved in nonvolatile RAM (NVRAM).

Static connections to multicast routers are supported only on switch ports.

Examples

This example shows how to configure Fast Ethernet interface 0/6 as a multicast router port:

```
Switch(config)# ip igmp snooping vlan 1 mrouter interface fastethernet0/6
```

This example shows how to specify the multicast router learning method as CGMP:

```
Switch(config)# no ip igmp snooping vlan 1 mrouter learn cgmp
```

You can verify your settings by entering the **show ip igmp snooping mrouter** privileged EXEC command.

Related Commands

Command	Description
ip igmp snooping	Globally enables Internet Group Management Protocol (IGMP) snooping.
ip igmp snooping vlan	Enables IGMP snooping on the VLAN interface.
ip igmp snooping vlan immediate-leave	Configures IGMP Immediate-Leave processing.
ip igmp snooping vlan static	Configures a Layer 2 port as a member of a group.
show ip igmp snooping mrouter	Displays the statically and dynamically learned multicast router ports.

ip igmp snooping vlan static

Use the **ip igmp snooping vlan** *vlan-id* **static** global configuration command to add a Layer 2 port as a member of a multicast group. Use the **no** form of this command to remove the configuration.

ip igmp snooping vlan *vlan-id* **static** *mac-address* **interface** *interface-id*

no ip igmp snooping vlan *vlan-id* **static** *mac-address* **interface** *interface-id*

Syntax Description

vlan *vlan-id*	Specify the VLAN ID. The range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
static *mac-address*	Specify the static group MAC address.
interface *interface-id*	Specify the interface configured to a static router port.

Defaults

None configured.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

The command is used to statically configure the IP multicast group member ports.

The static ports and groups are saved in nonvolatile RAM (NVRAM).

Static connections to multicast routers are supported only on switch ports.

Examples

This example shows how to statically configure a host on an interface:

```
Switch(config)# ip igmp snooping vlan 1 static 0100.5e02.0203 interface fastethernet0/6
Configuring port FastEthernet 0/6 on group 0100.5e02.0203
```

You can verify your settings by entering the **show mac address-table multicast** privileged EXEC command.

Related Commands

Command	Description
ip igmp snooping	Enables Internet Group Management Protocol (IGMP) snooping.
ip igmp snooping vlan	Enables IGMP snooping on the VLAN interface.
ip igmp snooping vlan immediate-leave	Configures IGMP Immediate-Leave processing.
ip igmp snooping vlan mrouter	Configures a Layer 2 port as a multicast router port.
show mac address-table multicast	Displays the Layer 2 multicast entries for a VLAN.

mac access-group

Use the **mac access-group** interface configuration command to apply a named extended MAC access control list (ACL) to an interface. Use the **no** form of this command to remove a MAC ACL from an interface.

mac access-group *name* **in**

no mac access-group *name* **in**

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

name	Name of the MAC extended ACL.

Defaults

No MAC ACL is applied to the interface.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You can apply MAC ACLs only to ingress interfaces. If an IP access group is already defined for an interface, you cannot apply this command to the interface.

After receiving the packet, the switch checks the match conditions in the ACL. If the conditions are matched, the switch forwards the packet.

If the specified ACL does not exist, the switch forwards all packets.

Note For more information about configuring MAC extended ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to apply a MAC extended ACL named *macacl2* to an interface:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# mac access-group macacl2 in
```

You can verify your settings by entering the **show mac access-group** privileged EXEC command.

Related Commands

Command	Description
{**deny (MAC access-list configuration)** | **permit (MAC access-list configuration)**}	Configures a MAC ACL.
show access-lists	Displays the ACLs configured on the switch.
show mac access-group	Displays the MAC ACLs configured on the switch.

mac access-list extended

Use the **mac access-list extended** global configuration command to create an access control list (ACL) based on MAC addresses. Using this command changes the mode to extended MAC access-list configuration mode. Use the **no** form of this command to return to the default setting.

mac access-list extended *name*

no mac access-list extended *name*

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

name	Assign a name to the MAC extended ACL.

Defaults

No MAC ACLs are created.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

MAC-named extended ACLs are used with the **mac access-group** interface configuration command and class maps.

Note For more information about configuring MAC extended ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to enter extended MAC access-list configuration mode and to create a MAC extended ACL named *mac1*:

```
Switch(config)# mac access-list extended mac1
Switch(config-ext-macl)#
```

This example shows how to delete the MAC extended ACL named *mac1*:

```
Switch(config)# no mac access-list extended mac1
```

You can verify your settings by entering the **show access-lists** privileged EXEC command.

Related Commands

Command	Description
class-map	Creates a class map to be used for matching packets to the class whose name you specify and to enter class-map configuration mode.
{**deny (MAC access-list configuration)** | **permit (MAC access-list configuration)**}	Configures a MAC ACL.
mac access-group	Applies a MAC ACL to an interface.
show access-lists	Displays the ACLs configured on the switch.

mac address-table aging-time

Use the **mac address-table aging-time** global configuration command to set the length of time that a dynamic entry remains in the MAC address table after the entry is used or updated. Use the **no** form of this command to return to the default setting. The aging time applies to all VLANs.

mac address-table aging-time [**0** | *10–1000000*]

no mac address-table aging-time [**0** | *10–1000000*]

Note Beginning with Release 12.1(11)EA1, the **mac address-table aging-time** command replaces the **mac-address-table aging-time** command (with the hyphen). The **mac-address-table aging-time** command (with the hyphen) will become obsolete in a future release.

Syntax Description

0	This value disables aging. Static address entries are never aged or removed from the table.
10–100000	Aging time in seconds. The range is 10 to 1000000 seconds.

Defaults

The default is 300 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The aging-time values were modified.
12.1(11)EA1	The **mac-address-table aging-time** command was replaced by the **mac address-table aging-time** command.

Usage Guidelines

If hosts do not send continuously, increase the aging time to record the dynamic entries for a longer time. This reduces the possibility of flooding when the hosts send again.

Examples

This example shows how to set the aging time to 200 seconds:

```
Switch(config)# mac address-table aging-time 200
```

This example shows how to disable aging in VLAN 1.

```
Switch(config)# mac address-table aging-time 0
```

This example shows how to set aging time to 450 seconds for all VLANs for which the user did not specify aging time.

```
Switch(config)# mac address-table aging-time 450
```

You can verify your settings by entering the **show mac address-table** privileged EXEC command.

Related Commands

Command	Description
clear mac address-table	Deletes dynamic entries from the MAC address table.
show mac address-table	Displays the MAC address table.
show mac address-table aging-time	Displays the MAC address table aging time for all VLANs or the specified VLAN.

mac address-table notification

Use the **mac address-table notification** global configuration command to enable the MAC notification feature and configure the notification-trap interval or history table. Use the **no** form of this command to disable this feature.

mac address-table notification [**history-size** *size* | **interval** *interval*]

no mac address-table notification [**history-size** *size* | **interval** *interval*]

Note Beginning with Release 12.1(11)EA1, the **mac address-table notification** command replaces the **mac-address-table notification** command (with the hyphen). The **mac-address-table notification** command (with the hyphen) will become obsolete in a future release.

Syntax Description

history-size *size*	(Optional) Configures the maximum number of entries in the MAC notification history table; valid values are 0 to 500.
interval *interval*	(Optional) Configures the notification-trap interval in seconds; valid values are from 0 to 2147483647. The switch sends the notification traps when this amount of time has elapsed.

Defaults

The MAC notification feature is disabled.

The default trap-interval value is 1 second.

The default number of entries in the history table is 1.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(11)EA1	The **mac-address-table notification** command was replaced by the **mac address-table notification** command.

Usage Guidelines

The MAC address notification feature sends Simple Network Management Protocol (SNMP) traps to the network management system (NMS) whenever a MAC address is added or deleted from the forwarding tables. MAC notifications are generated only for dynamic and secure MAC addresses. Events are not generated for self addresses, multicast addresses, or other static addresses.

When you configure the **history-size** option, the existing MAC address history table is deleted, and a new table is created.

You enable the MAC address notification feature by using the **mac address-table notification** command. You must also enable MAC address notification traps on an interface by using the **snmp trap mac-notification** interface configuration command, and configure the switch to send MAC address traps to the NMS by using the **snmp-server enable traps mac-notification** global configuration command.

Examples

This example shows how to enable the MAC notification feature:

```
Switch(config)# mac address-table notification
```

This example shows how to set the notification-trap interval to 60 seconds:

```
Switch(config)# mac address-table notification interval 60
```

This example shows how to set the number of entries in the history table to 32:

```
Switch(config)# mac address-table notification history-size 32
```

You can verify your settings by entering the **show mac address-table notification** privileged EXEC command.

Related Commands

Command	Description
clear mac address-table notification	Clears the MAC address notification global counters.
show mac address-table notification	Displays the MAC address notification settings for all interfaces or the specified interface.
snmp-server enable traps	Sends the SNMP MAC notification traps when the **mac-notification** keyword is appended.
snmp trap mac-notification	Enables the SNMP MAC notification trap on a specific interface.

mac address-table static

Use the **mac address-table static** global configuration command to add static addresses to the MAC address table. Use the **no** form of this command to remove static entries from the MAC address table.

mac address-table static *mac-addr* **vlan** *vlan-id* **interface** *interface-id*

no mac address-table static *mac-addr* **vlan** *vlan-id* **interface** *interface-id*

Note Beginning with Release 12.1(11)EA1, the **mac address-table static** command replaces the **mac-address-table static** command (with the hyphen). The **mac-address-table static** command (with the hyphen) will become obsolete in a future release.

Syntax Description

mac-addr	Destination MAC address (unicast or multicast) to add to the address table. Packets with this destination address received in the specified VLAN are forwarded to the specified interface.
vlan *vlan-id*	Specify the VLAN for which the packet with the specified MAC address is received. Valid VLAN IDs are 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
interface *interface-id*	Interface to which the received packet is forwarded. Valid interfaces include physical ports and port channels.

Defaults

None configured.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **interface** keyword and parameters were changed.
12.1(11)EA1	The **mac-address-table static** command was replaced by the **mac address-table static** command.

Examples

This example shows how to add the static address 0004.5600.67ab to the MAC address table:

```
Switch(config)# mac address-table static 0004.5600.67ab vlan 1 interface fastethernet0/2
```

This example shows how to add the static address c2f3.220a.12f4 to the MAC address table. When a packet is received in VLAN 4 with this MAC address as its destination, the packet is forwarded to the specified interface:

```
Switch(config)# mac address-table static c2f3.220a.12f4 vlan 4 interface
gigabitethernet0/1
```

You can verify your settings by entering the **show mac address-table** privileged EXEC command.

Related Commands

Command	Description
clear mac address-table	Deletes entries from the MAC address table.
mac address-table aging-time	Sets the length of time that a dynamic entry remains in the MAC address table after the entry is used or updated.
show mac address-table	Displays the MAC address table.
show mac address-table static	Displays static MAC address table entries only.

match

Use the **match** class-map configuration command to define the match criteria to classify traffic. Use the **no** form of this command to remove the match criteria.

match {access-group *acl-index* | **access-group name** *acl-name* | **ip dscp** *dscp-list*}

no match {access-group *acl-index* | **access-group name** *acl-name* | **ip dscp**}

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

access-group *acl-index*	Number of an IP standard or extended access control list (ACL). For an IP standard ACL, the ACL index range is 1 to 99 and 1300 to 1999. For an IP extended ACL, the ACL index range is 100 to 199 and 2000 to 2699.
access-group name *acl-name*	Name of an IP standard or extended ACL or name of an extended MAC ACL. **Note** The ACL name must begin with an alphabetic character to prevent ambiguity with numbered ACLs. A name also cannot contain a space or quotation mark.
ip dscp *dscp-list*	List of up to eight IP Differentiated Services Code Point (DSCP) values for each match statement to match against incoming packets. Separate each value with a space. The supported DSCP values are 0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56.

Defaults

No match criteria are defined.

Command Modes

Class-map configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use the **match** command to specify which fields in the incoming packets are examined to classify the packets. Only IP access groups, MAC access groups, and classification based on DSCP values are supported.

Only one **match** command per class map is supported.

Note For more information about configuring ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to classify traffic on an interface by using the access group named *acl2*:

```
Switch(config)# class-map class2
Switch(config-cmap)# match access-group name acl2
Switch(config-cmap)# exit
```

You can verify your settings by entering the **show class-map** privileged EXEC command.

Related Commands

Command	Description
class	Defines a traffic classification for a policy to act on using the class-map name or access group.
class-map	Creates a class map to be used for matching packets to the class whose name you specify.
ip access-group	Controls access to an interface.
mac access-group	Applies a named extended MAC ACL to an interface.
show class-map	Displays quality of service (QoS) class maps.
show policy-map	Displays QoS policy maps.

mls qos cos

Use the **mls qos cos** interface configuration command to define the default class of service (CoS) value of a port or to assign the default CoS to all incoming packets on the port. Use the **no** form of this command to return to the default setting.

mls qos cos {*default-cos* | **override**}

no mls qos cos {*default-cos* | **override**}

Syntax Description

default-cos	Assign a default CoS value to a port. If the port is CoS trusted and packets are untagged, the default CoS value becomes a CoS value used to select one output queue to index into the CoS-to-Differentiated Services Code Point (DSCP) map. The CoS range is 0 to 7.
override	Override the CoS of the incoming packets, and apply the default CoS value on the port to all incoming packets.

Defaults

The default CoS value for a port is 0.

CoS override is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You can use the default value to assign CoS and DSCP values to all packets entering a port if the port has been configured by using the **override** keyword.

Use the **override** keyword when all incoming packets on certain ports deserve higher or lower priority than packets entering from other ports. Even if a port was previously set to trust DSCP or CoS, this command overrides that trust state, and all the incoming CoS values are assigned the default CoS value configured with the **mls qos cos** command. If an incoming packet is tagged, the CoS value of the packet is modified with the default CoS of the port at the ingress port.

Examples

This example shows how to configure the default port CoS to 4:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# mls qos trust cos
Switch(config-if)# mls qos cos 4
```

This example shows how to assign all the packets entering a port to the default port CoS value of 4:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# mls qos cos 4
Switch(config-if)# mls qos cos override
```

You can verify your settings by entering the **show mls qos interface** privileged EXEC command.

Related Commands

Command	Description
mls qos map	Defines the CoS-to-DSCP map or the DSCP-to-CoS map.
mls qos trust	Configures the port trust state.
show mls qos interface	Displays quality of service (QoS) information.

mls qos map

Use the **mls qos map** global configuration command to define the class of service (CoS)-to-Differentiated Services Code Point (DSCP) map or DSCP-to-CoS map. Use the **no** form of this command to return to the default map.

mls qos map {**cos-dscp** *dscp1...dscp8* | **dscp-cos** *dscp-list* **to** *cos*}

no mls qos map {**cos-dscp** | **dscp-cos**}

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

cos-dscp *dscp1...dscp8*	Define the CoS-to-DSCP map. For *dscp1...dscp8*, enter eight DSCP values that correspond to CoS values 0 to 7. Separate each DSCP value with a space. The supported DSCP values are 0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56.
dscp-cos *dscp-list* **to** *cos*	Define the DSCP-to-CoS map. For *dscp-list*, enter up to 13 DSCP values separated by spaces. Then enter the **to** keyword. The supported DSCP values are 0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56. For *cos*, enter the CoS value to which the DSCP values correspond. The CoS range is 0 to 7.

Defaults

Table 2-2 shows the default CoS-to-DSCP map:

Table 2-2 Default CoS-to-DSCP Map

CoS Value	0	1	2	3	4	5	6	7
DSCP Value	0	8	16	24	32	40	48	56

Table 2-3 shows the default DSCP-to-CoS map:

Table 2-3 Default DSCP-to-CoS Map

DSCP Values	0	8, 10	16, 18	24, 26	32, 34	40, 46	48	56
CoS Value	0	1	2	3	4	5	6	7

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

All the maps are globally defined. You apply all maps to all ports.

If you enter the **mls qos trust cos** command, the default CoS-to-DSCP map is applied.

If you enter the **mls qos trust dscp** command, the default DSCP-to-CoS map is applied.

After a default map is applied, you can define the CoS-to-DSCP or DSCP-to-CoS map by entering consecutive **mls qos map** commands.

The supported DSCP values are 0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56. If the **mls qos trust dscp** command is entered and a packet with an untrusted DSCP value is at an ingress port, the packet CoS value is set to 0.

Note The switches do not support the **dscp-mutation**, **dscp-switch-priority**, **ip-prec-dscp**, and **policed-dscp** options.

Examples

This example shows how to define the DSCP-to-CoS map. DSCP values 16, 18, 24, and 26 are mapped to CoS 1. DSCP values 0, 8, and 10 are mapped to CoS 0.

```
Switch# configure terminal
Switch(config)# mls qos map dscp-cos 16 18 24 26 to 1
Switch(config)# mls qos map dscp-cos 0 8 10 to 0
```

This example shows how to define the CoS-to-DSCP map. CoS values 0 to 7 are mapped to DSCP values 8, 8, 8, 8, 24, 32, 56, and 56.

```
Switch# configure terminal
Switch(config)# mls qos map cos-dscp 8 8 8 8 24 32 56 56
```

You can verify your settings by entering the **show mls qos maps** privileged EXEC command.

Related Commands

Command	Description
mls qos cos	Defines the default CoS value of a port or assigns the default CoS to all incoming packets on the port.
mls qos trust	Configures the port trust state.
show mls qos maps	Displays quality of service (QoS) mapping information.

mls qos trust

Use the **mls qos trust** interface configuration command to configure the port trust state. Ingress traffic can be trusted, and classification is performed by examining the class of service (CoS) or the Differentiated Services Code Point (DSCP) value. Use the **no** form of this command to return a port to its untrusted state.

mls qos trust [**cos** [**pass-through dscp**] | **device cisco-phone** | **dscp**]

no mls qos trust [**cos** [**pass-through dscp**] | **device cisco-phone** | **dscp**]

Syntax Description

cos	(Optional) Classify ingress packets with packet CoS values. For untagged packets, the port default CoS value is used.
cos pass-through dscp	(Optional) Configure the interface to classify ingress packets by trusting the CoS value and to send packets without modifying the DSCP value (pass-through mode).
device cisco-phone	(Optional) Classify ingress packets by trusting the value sent from the Cisco IP phone (trusted boundary).
dscp	(Optional) Classify ingress packets with packet DSCP values (most significant 6 bits of the 8-bit service-type field). For non-IP packets, the packet CoS value is set to 0. This keyword is available only if your switch is running the enhanced software image (EI).

Defaults

The port is not trusted.

Pass-through mode is disabled.

Trusted boundary is disabled.

If no keyword is specified and the switch is running the EI, the default is **dscp**.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(11)EA1	The **device cisco-phone** and **pass-through dscp** keywords were added.

Usage Guidelines

Packets entering a quality of service (QoS) domain are classified at the edge of the QoS domain. When the packets are classified at the edge, the switch port within the QoS domain can be configured to one of the trusted states because there is no need to classify the packets at every switch within the domain. Use this command to specify whether the port is trusted and which fields of the packet to use to classify traffic.

When a port is configured with trust DSCP and the incoming packet is a tagged non-IP packet, the CoS value for the packet is set to 0, and the DSCP-to-CoS map is not applied. For an untagged non-IP packet, the default port CoS value is used.

If DSCP is trusted, the DSCP field of the IP packet is not modified. However, it is still possible that the CoS value of the packet is modified (according to the DSCP-to-CoS map).

If CoS is trusted, the CoS of the packet is not modified, but DSCP can be modified (according to the CoS-to-DSCP map) if it is an IP packet.

To return a port to the untrusted state, use the **no mls qos trust** interface configuration command.

The trusted boundary feature prevents security problems if users disconnect their PCs from networked Cisco IP phones and connect them into the switch port to take advantage of trusted CoS settings. You must globally enable the Cisco Discovery Protocol (CDP) on both the switch and on the interface connected to the IP phone. If the phone is not detected, trusted boundary disables the trust setting on the switch port and prevents misuse of a high-priority queue.

If trusted boundary is enabled and the **no mls qos trust** command is entered, the port returns to the untrusted state and cannot be configured to trust if it is connected to a Cisco IP phone.

To disable trusted boundary, use the **no mls qos trust device** interface configuration command.

In software releases earlier than Release 12.1(11)EA1, the switch is in pass-through mode. It uses the CoS value of incoming packets without modifying the DSCP value and sends the packets from one of the four egress queues. You cannot enable or disable pass-through mode if your switch is running a software release earlier than Release 12.1(11)EA1.

In Release 12.1(11)EA1 or later, pass-through mode is disabled by default. The switch assigns a CoS value of 0 to all incoming packets without modifying the packets. It offers best-effort service to each packet regardless of the packet contents or size and sends it from a single egress queue.

You can enable pass-through mode on a switch running Release 12.1(11)EA1 or later by using the **mls qos trust cos pass-through dscp** interface configuration command. To disable pass-through mode, use the **no mls qos trust cos pass-through** interface configuration command.

Note In software releases earlier than Release 12.1(11)EA1, the **mls qos trust** command is available only when the switch is running the EI.

Examples

This example shows how to configure a port to be a DSCP-trusted port:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# mls qos trust dscp
```

Note The **mls qos trust dscp** command is available only when the switch is running the EI.

This example shows how to specify that the Cisco IP phone is a trusted device:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# mls qos trust device cisco-phone
```

This example shows how to configure the interface to trust the CoS of incoming packets and to send them without modifying the DSCP field:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# mls qos trust cos pass-through dscp
```

You can verify your settings by entering the **show mls qos interface** privileged EXEC command.

Related Commands

Command	Description
mls qos cos	Defines the default CoS value of a port or assigns the default CoS to all incoming packets on the port.
mls qos map	Defines the CoS-to-DSCP map or the DSCP-to-CoS map.
show mls qos interface	Displays QoS information.

monitor session

Use the **monitor session** global configuration command to start a new Switched Port Analyzer (SPAN) or Remote SPAN (RSPAN) session. Use the **no** form of this command to remove the SPAN or the RSPAN session or to remove source or destination interfaces from the SPAN or RSPAN session.

monitor session *session_number* {**destination** {**interface** *interface-id* [**encapsulation** {**dot1q**}] | **remote vlan** *vlan-id* **reflector-port** *interface-id*} | **source** {**interface** *interface-id* [, | -] [**both** | **rx** | **tx**] | **remote vlan** *vlan-id*}}

no monitor session *session_number* {**destination** {**interface** *interface-id* [**encapsulation** {**dot1q**}] | **remote vlan** *vlan-id* **reflector-port** *interface-id*} | **source** {**interface** *interface-id* [, | -] [**both** | **rx** | **tx**] | **remote vlan** *vlan-id*}}

no monitor session {*session_number* | **all** | **local** | **remote**}

To use the RSPAN feature, you must have the enhanced software image (EI) installed on your switch.

Syntax Description

session_number	Specify the session number identified with the SPAN or RSPAN session.
destination interface *interface-id*	Specify the destination interface for a local SPAN session. Valid interfaces are physical ports.
encapsulation	(Optional) Specify the encapsulation header for outgoing packets through a destination port. If encapsulation type is not specified, packets are sent in native form. To reconfigure a destination port in native form, enter the command without the **encapsulation** keyword.
dot1q	Specify the encapsulation type as 802.1Q.
destination remote vlan *vlan-id*	Specify the destination remote VLAN for an RSPAN source session.
reflector-port *interface-id*	Specify the reflector port used for a source RSPAN session.
source interface *interface-id*	Specify the SPAN source interface type, slot, and port number. Valid interfaces include physical ports and port channels.
,	**Note** (Optional) Specify a series of interfaces, or separate a range of interfaces from a previous range. Enter a space after the comma.
-	**Note** (Optional) Specify a range of interfaces. Enter a space before and after the hyphen.
both, rx, tx	(Optional) Specify the traffic direction for each source.
source remote vlan *vlan-id*	Specify the source RSPAN VLAN for an RSPAN destination session.
all, **local**, **remote**	Specify **all**, **local**, or **remote** to clear a SPAN or RSPAN session.

Defaults

On a source interface, the default is to monitor both received and transmitted traffic.

If encapsulation type is not specified on a destination port, packets are sent in native form with no encapsulation.

Command Modes Global configuration

Command History

Release	Modification
12.1(6)EA1	This command was first introduced.
12.1(11)EA1	These RSPAN keywords were added: **encapsulation, destination remote vlan reflector-port**, **source remote vlan**, **all**, **local**, **remote**.

Usage Guidelines Traffic that enters or leaves source ports can be monitored by using SPAN or RSPAN. Traffic routed to source ports cannot be monitored.

You can configure (and store in NVRAM) one local SPAN session or multiple RSPAN sessions on a switch. The number of active sessions and combinations are subject to these restrictions:

- SPAN or RSPAN source (**rx**, **tx**, both): one active session limit. (SPAN and RSPAN are mutually exclusive on a source switch).
- RSPAN source sessions have one destination per session with an RSPAN VLAN associated for that session.
- Each RSPAN destination session has one or more destination interfaces for each RSPAN VLAN that it supports.
- RSPAN destination sessions are limited to two, or one if a local SPAN or a source RSPAN session is configured on the same switch.

You can monitor traffic on a single port or on a series or range of ports. You select a series or range of interfaces by using the [**,** | **-**] options.

If you specify a series of interfaces, you must enter a space after the comma. If you specify a range of interfaces, you must enter a space before and after the hyphen (**-**).

EtherChannel ports cannot be configured as SPAN or RSPAN destination or reflector ports. A physical port that is a member of an EtherChannel group can be used as a source or destination port. It cannot participate in the EtherChannel group while it is configured for SPAN or RSPAN.

A port used as a reflector port cannot be a SPAN or RSPAN source or destination port, nor can a port be a reflector port for more than one session at a time.

A port used as a destination port cannot be a SPAN or RSPAN source or reflector port, nor can a port be a destination port for more than one session at a time.

You can enable 802.1X on a port that is a SPAN or RSPAN destination port; however, 802.1X is disabled until the port is removed as a SPAN destination. (If 802.1X is not available on the port, the switch will return an error message.) You can enable 802.1X on a SPAN or RSPAN source port.

Examples This example shows how to create SPAN session 1 to monitor both sent and received traffic on source interface 0/1 on destination interface 0/8:

```
Switch(config)# monitor session 1 source interface fastEthernet0/1 both
Switch(config)# monitor session 1 destination interface fastEthernet0/8
```

This example shows how to delete a destination port from an existing SPAN session:

```
Switch(config)# no monitor session 2 destination fastEthernet0/4
```

This example shows how to configure RSPAN session 1 to monitor multiple source interfaces and to configure the destination RSPAN VLAN and the reflector-port.

```
Switch(config)# monitor session 1 source interface fastEthernet0/10 tx
Switch(config)# monitor session 1 source interface fastEthernet0/2 rx
Switch(config)# monitor session 1 source interface port-channel 102 rx
Switch(config)# monitor session 1 destination remote vlan 901 reflector-port
fastEthernet 0/1
Switch(config)# end
```

You can verify your settings by entering the **show monitor** privileged EXEC command.

Related Commands

Command	Description
remote-span	Configures an RSPAN VLAN in vlan configuration mode.
show monitor	Displays SPAN and RSPAN session information.

mvr

Use the **mvr** global configuration command without keywords to enable the multicast VLAN registration (MVR) feature on the switch. Use the **no** form of this command to disable MVR and its options. Use the command with keywords to set the MVR mode for a switch, to configure the MVR IP multicast address, to set the maximum time to wait for a query reply before removing a port from group membership, and to specify the MVR multicast VLAN. Use the **no** form of this command to return the switch to the default settings.

mvr [**group** *ip-address* [*count*] | **mode** {**compatible** | **dynamic**} | **querytime** *value* | **vlan** *vlan-id*]

no mvr [**group** *ip-address* | **mode** {**compatible** | **dynamic**} | **querytime** *value* | **vlan** *vlan-id*]

Syntax Description

group *ip-address*	(Optional) Statically configure an MVR group IP multicast address on the switch. Use the **no** form of this command to remove a statically configured IP multicast address or contiguous addresses or, when no IP address is entered, to remove all statically configured MVR IP multicast addresses.
count	(Optional) Configure multiple contiguous MVR group addresses. The range is from 1 to 256. The default is 1.
mode	(Optional) Specify the MVR mode of operation. The default is compatible mode.
compatible	Set MVR mode to provide compatibility with Catalyst 2900 XL and 3500 XL switches. This mode does not allow dynamic membership joins on source ports.
dynamic	Set MVR mode to allow dynamic MVR membership on source ports.
querytime *value*	(Optional) Set the maximum time to wait for Internet Group Management Protocol (IGMP) report memberships on a receiver port. This time applies only to receiver-port leave processing. When an IGMP query is sent from a receiver port, the switch waits for the default or configured MVR query time for an IGMP group membership report before removing the port from multicast group membership. The value is the response time in units of tenths of a second. The default is 5 tenths or one-half second. The range is 1 to 100 tenths of a second. Use the **no** form of the command to return to the default setting.
vlan *vlan-id*	(Optional) Specify the VLAN on which MVR multicast data is expected to be received. This is also the VLAN to which all the source ports belong. The default is VLAN 1. Valid VLAN IDs are 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.

Defaults

MVR is disabled.

The default MVR mode is compatible mode.

No IP multicast addresses are configured on the switch.

The default group IP address count is 0.

The default query response time is 5 tenths of or one-half second.

The default multicast VLAN for MVR is VLAN 1.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

A maximum of 256 MVR multicast groups can be configured on a switch.

Use the **mvr group** command to statically configure all the IP multicast addresses that will take part in MVR. Any multicast data sent to a configured multicast address is sent to all the source ports on the switch and to all receiver ports registered to receive data on that IP multicast address.

Note The **mvr group** command prevents adding IP multicast addresses that cause address aliasing between MVR multicast groups or with the reserved IP multicast addresses (in the range 224.0.0.xx). Each IP multicast address translates to a multicast 48-bit MAC address. If the IP address being configured translates (aliases) to the same 48-bit MAC address as a previously configured IP multicast address or the reserved MAC multicast addresses, the command fails.

The **mvr querytime** parameter applies only to receiver ports.

The **mvr group** and **mvr vlan** commands only apply to ports configured as receiver ports.

If the switch MVR is interoperating with Catalyst 2900 XL or Catalyst 3500 XL switches, set the multicast mode to compatible.

When in compatible mode, MVR does not support IGMP dynamic joins on MVR source ports.

Examples

This example shows how to enable MVR:

```
Switch(config)# mvr
```

This example shows how to disable MVR:

```
Switch(config)# no mvr
```

This example shows how to configure 228.1.23.4 as an IP multicast address:

```
Switch(config)# mvr group 228.1.23.4
```

This command fails because of address aliasing:

```
Switch(config)# mvr group 230.1.23.4

Cannot add this IP address - aliases with previously configured IP address 228.1.23.4.
```

This example shows how to configure ten contiguous IP multicast groups with multicast addresses from 228.1.23.1 to 228.1.23.10:

```
Switch(config)# mvr group 228.1.23.1 10
```

This example shows how to delete the previously configured ten IP multicast addresses:

```
Switch(config)# no mvr group 228.1.23.1 10
```

This example shows how to delete all previously configured IP multicast addresses:

```
Switch(config)# no mvr group
```

This example shows how to set the maximum query response time as 1 second (10 tenths):

```
Switch(config)# mvr querytime 10
```

This example shows how to return the maximum query response time to the default setting of one-half second:

```
Switch(config)# no mvr querytime
```

This example shows how to set VLAN 2 as the multicast VLAN:

```
Switch(config)# mvr vlan 2
```

Use the **show mvr members** privileged EXEC command to display the IP multicast group addresses configured on the switch.

You can verify your settings by entering the **show mvr** privileged EXEC command.

Related Commands

Command	Description
mvr immediate	Enables the Immediate-Leave feature on an interface.
mvr type	Configures a port as a receiver or source port.
mvr vlan group	Configures a receiver port as a member of an MVR group.
show mvr	Displays MVR global parameters or port parameters.
show mvr interface	Displays the configured MVR interfaces with their type, status, and Immediate-Leave configuration.
show mvr interface *interface-id* **member**	Displays all MVR groups of which the interface is a member.
show mvr members	Displays all ports that are members of an MVR multicast group; if the group has no members, its status is shown as Inactive.

mvr immediate

Use the **mvr immediate** interface configuration command to enable the Immediate-Leave feature on an interface. Use the **no** form of this command to disable the feature on the interface.

mvr immediate

no mvr immediate

Syntax Description

This command has no keywords or arguments.

Defaults

The Immediate-Leave feature is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The Immediate-Leave feature applies only to receiver ports. When the Immediate-Leave feature is enabled, a receiver port leaves a multicast group more quickly. When the switch receives an Internet Group Management Protocol (IGMP) leave message from a group on a receiver port, it sends an IGMP query on that port and waits for IGMP group membership reports. If no reports are received in a configured time period, the receiver port is removed from multicast group membership. With the Immediate-Leave feature, an IGMP query is not sent from the receiver port on which the IGMP leave was received. As soon as the leave message is received, the receiver port is removed from multicast group membership, thus speeding up leave latency.

The Immediate-Leave feature should only be enabled on receiver ports to which a single receiver device is connected.

Examples

This example shows how to enable the Immediate-Leave feature on a port:

```
Switch(config-if)# mvr immediate
```

This example shows how to disable the Immediate-Leave feature on a port:

```
Switch(config-if)# no mvr immediate
```

You can verify your settings by entering the **show mvr** privileged EXEC command.

Related Commands

Command	Description
mvr	Enables multicast VLAN registration (MVR).
mvr type	Configures a port as a receiver or source port.
mvr vlan group	Configures a receiver port as a member of an MVR group.
show mvr	Displays MVR global parameters or port parameters.

mvr type

Use the **mvr type** interface configuration command to configure a port as a multicast VLAN registration (MVR) receiver or source port. Use the **no** form of this command to return the port to the default settings.

mvr type {**receiver** | **source**}

no mvr type {**receiver** | **source**}

Syntax Description

receiver	Port that receives multicast data and cannot send multicast data to multicast groups.
source	Port that can send and receive multicast data to multicast groups.

Defaults

A port is configured as neither receiver nor source.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Configure a port as a source port if that port should be able to both send and receive multicast data bound for the configured multicast groups. Multicast data is received on all ports configured as source ports.

Configure a port as a receiver port if that port should only be able to receive multicast data and should not be able to send multicast data to the configured multicast groups. None of the receiver ports receives multicast data unless it sends an Internet Group Management Protocol (IGMP) group join message for a multicast group.

A receiver port configured as a static member of a multicast group remains a member until statically removed from membership.

Note All receiver ports must not be trunk ports and must not belong to the MVR source VLAN.

A port that is not taking part in MVR should not be configured as an MVR receiver port or source port. This port is a normal switch port and is able to send and receive multicast data with normal switch behavior.

Examples

This example shows how to configure a port as an MVR receiver port:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# mvr type receiver
```

This example shows how to configure a port as an MVR source port:

```
Switch(config)# interface fastethernet0/3
Switch(config-if)# mvr type source
```

This example shows how to return a port to the default setting:

```
Switch(config)# interface fastethernet0/5
Switch(config-if)# no mvr type receiver
```

You can verify your settings by entering the **show mvr** privileged EXEC command.

Related Commands

Command	Description
mvr	Enables MVR.
mvr immediate	Enables the Immediate-Leave feature on an interface.
mvr vlan group	Configures a receiver port as a member of an MVR group.
show mvr	Displays MVR global parameters or port parameters.

mvr vlan group

Use the **mvr vlan group** interface configuration command to statically configure a receiver port as a member of a multicast VLAN registration (MVR) group in a particular VLAN. Use the **no** form of this command to remove the port from the MVR group.

mvr vlan *vlan-id* **group** *ip-address*

no mvr vlan *vlan-id* **group** *ip-address*

Syntax Description

vlan *vlan-id*	Specify the VLAN ID to which the receiver port belongs. Valid IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
group *ip-address*	Specify the MVR group address for which the interface is statically configured to be a member.

Defaults

A port is configured as neither receiver nor source.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The receiver port belongs to a multicast VLAN.

The group address is configured as a MVR group address.

Examples

This example shows how to configure a static MVR group entry on port 0/1 in VLAN 10:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# mvr vlan 10 group 225.1.1.1
```

This example shows how to remove an entry on port 0/3 in VLAN 10:

```
Switch(config)# interface fastethernet0/3
Switch(config-if)# no mvr 10 group 255.1.1.2
```

You can verify your settings by entering the **show mvr** privileged EXEC command.

Related Commands

Command	Description
mvr	Enables MVR.
mvr immediate	Enables the Immediate-Leave feature on an interface.
mvr type	Configures a port as a receiver or source port.
show mvr	Displays MVR global parameters or port parameters.

pagp learn-method

Use the **pagp learn-method** interface configuration command to set the source-address learning method of incoming packets received from an EtherChannel port. Use the **no** form of this command to return to the default setting.

pagp learn-method aggregation-port

no pagp learn-method

Syntax Description

aggregation-port	Specify address learning on the logical port-channel. The switch transmits packets to the source by using any of the interfaces in the EtherChannel. This setting is the default. With aggregate-port learning, it is not important on which physical port the packet arrives.

Note Though visible in the command-line help strings, the **physical-port** keyword is not supported.

Defaults

The default is **aggregation-port** (logical port channel).

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The switch supports address learning only on aggregate ports even though the **physical-port** keyword is provided in the command-line interface (CLI). The **pagp learn-method** and the **pagp port-priority** interface configuration commands have no affect on the switch hardware.

Note You should not set the learn method to **physical-port** because the switch is an aggregate-learning device.

If the link partner to the switch is a physical learner that has the **channel-group** interface configuration command set to **auto** or **desirable**, the switch automatically uses the load-distribution method based on the source MAC address, regardless of the configured load-distribution method.

If the link partner to the switch is a physical learner that has the **channel-group** interface configuration command set to **on**, set the load-distribution method based on the source MAC address by using the **port-channel load-balance src-mac** global configuration command.

Examples

This example shows how to set the learning method to **aggregation-port** (the default):

```
Switch(config-if)# pagp learn-method aggregation-port
```

You can verify your settings by entering the **show running-config** or **show pagp** *channel-group-number* **internal** privileged EXEC command.

Related Commands

Command	Description
channel-group	Assigns an Ethernet interface to an EtherChannel group.
pagp port-priority	Selects an interface through which all Port Aggregation Protocol (PAgP) traffic through the EtherChannel is sent.
show pagp	Displays PAgP channel-group information.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

pagp port-priority

You do not need to enter this command. It is documented for informational purposes only. Though visible in the command-line help strings, the switch does not support the **pagp port-priority** command.

Use the **pagp port-priority** interface configuration command to select an interface through which all Port Aggregation Protocol (PAgP) traffic through the EtherChannel is sent. Use the **no** form of this command to return to the default value.

pagp port-priority *priority*

no pagp port-priority

Syntax Description

priority	A priority number ranging from 0 to 255.

Defaults

The default value is 128.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **pagp learn-method** and the **pagp port-priority** interface configuration commands have no affect on the switch hardware.

Note You should not change the port priority because the switch does not support this command.

Related Commands

Command	Description
pagp learn-method	Sets the source-address learning method of incoming packets received from an EtherChannel port.
show pagp	Displays PAgP channel-group information.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

permit (access-list configuration)

Use the **permit** access-list configuration command to configure conditions for a named or numbered IP access control list (ACL). Use the **no** form of this command to remove a permit condition from the IP ACL.

Use these commands with standard IP ACLs:

permit {*source source-wildcard* | **host** *source* | **any**}

no permit {*source source-wildcard* | **host** *source* | **any**}

Use these commands with extended IP ACLs:

permit *protocol* {*source source-wildcard* | **host** *source* | **any**} [*operator port*] {*destination destination-wildcard* | **host** *source* | **any**} [*operator port*] [**dscp** *dscp-value*] [**time-range** *time-range-name*]

no permit *protocol* {*source source-wildcard* | **host** *source* | **any**} [*operator port*] {*destination destination-wildcard* | **host** *source* | **any**} [*operator port*] [**dscp** *dscp-value*] [**time-range** *time-range-name*]

This command is available on physical interfaces only if your switch is running the enhanced software image (EI).

Syntax Description

protocol	Name of an IP protocol. *protocol* can be **ip**, **tcp**, or **udp**.
source source-wildcard | **host** *source* | **any**	Define a source IP address and wildcard. The *source* is the source address of the network or host from which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *source-wildcard* applies wildcard bits to the source. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *source* and *source-wildcard* of *source* 0.0.0.0. • The keyword **any** as an abbreviation for *source* and *source-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a source-wildcard.

destination destination-wildcard | **host** *destination* | **any**	Define a destination IP address and wildcard. The *destination* is the destination address of the network or host to which the packet is being sent, specified in one of these ways: • The 32-bit quantity in dotted-decimal format. The *destination-wildcard* applies wildcard bits to the destination. • The keyword **host**, followed by the 32-bit quantity in dotted-decimal format, as an abbreviation for *destination* and *destination-wildcard* of *destination* 0.0.0.0. • The keyword **any** as an abbreviation for *destination* and *destination-wildcard* of 0.0.0.0 255.255.255.255. You do not need to enter a destination-wildcard.
operator port	(Optional) Define a source or destination port. The *operator* can be only **eq** (equal). If *operator* is after the source IP address and wildcard, conditions match when the source port matches the defined port. If *operator* is after the destination IP address and wildcard, conditions match when the destination port matches the defined port. The *port* is a decimal number or name of a Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) port. The number can be from 0 to 65535. Use TCP port names only for TCP traffic. Use UDP port names only for UDP traffic.
dscp *dscp-value*	(Optional) Define a Differentiated Services Code Point (DSCP) value to classify traffic. For the *dscp-value*, enter any of the 13 supported DSCP values (0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56), or use the question mark (**?**) to see a list of available values.
time-range *time-range-name*	(Optional) For the **time-range** keyword, enter a meaningful name to identify the time range. For a more detailed explanation of this keyword, refer to the software configuration guide.

Defaults

There are no specific conditions that permit packets in a named or numbered IP ACL.

The default ACL is always terminated by an implicit deny statement for all packets.

Command Modes

Access-list configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use this command after the **ip access-list** global configuration command to specify permit conditions for a named or numbered IP ACL. You can specify a source IP address, destination IP address, IP protocol, TCP port, or UDP port. Specify the TCP and UDP port numbers only if *protocol* is **tcp** or **udp** and *operator* is **eq**.

Note For more information about configuring IP ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to create an extended IP ACL and configure permit conditions for it:

```
Switch(config)# ip access-list extended Internetfilter2
Switch(config-ext-nacl)# permit host 36.10.10.5 any
Switch(config-ext-nacl)# permit host 192.1.10.8 any
```

This is an example of a standard ACL that sets permit conditions:

```
ip access-list standard Acclist1
 permit 192.5.34.0  0.0.0.255
 permit 128.88.10.0  0.0.0.255
 permit 36.1.1.0  0.0.0.255
```

Note In these examples, all other IP access is implicitly denied.

You can verify your settings by entering the **show ip access-lists** or **show access-lists** privileged EXEC command.

Related Commands

Command	Description
deny (access-list configuration)	Sets deny conditions for an IP ACL.
ip access-group	Controls access to an interface.
ip access-list	Defines an IP ACL.
show access-lists	Displays ACLs configured on a switch.
show ip access-lists	Displays IP ACLs configured on the switch.

permit (MAC access-list configuration)

Use the **permit** MAC access-list configuration command to allow Layer 2 traffic to be forwarded if the conditions are matched. Use the **no** form of this command to remove a permit condition from the named MAC access control list (ACL).

{**permit** | **deny**} {**any** | **host** *src-MAC-addr*} {**any** | **host** *dst-MAC-addr*} [**aarp** | **amber** | **appletalk** | **dec-spanning** | **decnet-iv** | **diagnostic** | **dsm** | **etype-6000** | **etype-8042** | **lat** | **lavc-sca** | **mop-console** | **mop-dump** | **msdos** | **mumps** | **netbios** | **vines-echo** |**vines-ip** | **xns-idp**]

no {**permit** | **deny**} {**any** | **host** *src-MAC-addr*} {**any** | **host** *dst-MAC-addr*} [**aarp** | **amber** | **appletalk** | **dec-spanning** | **decnet-iv** | **diagnostic** | **dsm** | **etype-6000** | **etype-8042** | **lat** | **lavc-sca** | **mop-console** | **mop-dump** | **msdos** | **mumps** | **netbios** | **vines-echo** |**vines-ip** | **xns-idp**]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

any	Keyword to specify to permit any source or destination MAC address.
host *src-MAC-addr*	Define a host MAC address. If the source address for a packet matches the defined address, traffic from that address is permitted. MAC address-based subnets are not allowed.
host *dst-MAC-addr*	Define a destination MAC address. If the destination address for a packet matches the defined address, traffic to that address is permitted. MAC address-based subnets are not allowed.
aarp	Select Ethertype AppleTalk Address Resolution Protocol that maps a data-link address to a network address.
amber	Select EtherType DEC-Amber.
appletalk	Select EtherType AppleTalk/EtherTalk.
dec-spanning	Select EtherType Digital Equipment Corporation (DEC) spanning tree.
decnet-iv	Select EtherType DECnet Phase IV protocol.
diagnostic	Select EtherType DEC-Diagnostic.
dsm	Select EtherType DEC-DSM.
etype-6000	Select EtherType 0x6000.
etype-8042	Select EtherType 0x8042**.**
lat	Select EtherType DEC-LAT.
lavc-sca	Select EtherType DEC-LAVC-SCA.
mop-console	Select EtherType DEC-MOP Remote Console.
mop-dump	Select EtherType DEC-MOP Dump.
msdos	Select EtherType DEC-MSDOS.
mumps	Select EtherType DEC-MUMPS.
netbios	Select EtherType DEC- Network Basic Input/Output System (NETBIOS).
vines-echo	Select EtherType Virtual Integrated Network Service (VINES) Echo from Banyan Systems.

vines-ip	Select EtherType VINES IP.
xns-idp	Select EtherType Xerox Network Systems (XNS) protocol suite (from 0 to 65535), an arbitrary Ethertype in decimal, hexadecimal, or octal.

Defaults

This command has no defaults. However, the default action for a MAC-named ACL is to deny.

Command Modes

MAC access-list configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

When an access control entry (ACE) is added to an ACL, an implied **deny-any-any** condition exists at the end of the list. That is, if there are no matches, the packets are denied. However, before the first ACE is added, the list permits all packets.

These options are not allowed:

- Class of service (CoS)
- Ethertype number of a packet with Ethernet II or Subnetwork Access Protocol (SNAP) encapsulation
- Link Service Access Point (LSAP) number of a packet with 802.2 encapsulation

Note For more information about configuring MAC extended ACLs, refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to define the named MAC extended ACL to deny NETBIOS traffic from any source to MAC address 00c0.00a0.03fa. Traffic matching this list is allowed.

```
Switch(config-ext-macl)# permit any host 00c0.00a0.03fa netbios
```

This example shows how to remove the permit condition from the named MAC extended ACL:

```
Switch(config-ext-macl)# no permit any host 00c0.00a0.03fa netbios
```

You can verify your settings by entering the **show access-lists** privileged EXEC command.

Related Commands

Command	Description
deny (MAC access-list configuration)	Prevents Layer 2 traffic from being forwarded if conditions are matched.
mac access-list extended	Creates an ACL based on MAC addresses.
show access-lists	Displays ACLs configured on a switch.

police

Use the **police** policy-map class configuration command to define a policer for classified traffic. Use the **no** form of this command to remove an existing policer.

police *rate-bps burst-byte* [**exceed-action** {**drop** | **dscp** *dscp-value*}]

no police *rate-bps burst-byte* [**exceed-action** {**drop** | **dscp** *dscp-value*}]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

rate-bps	Specify average traffic rate in bits per second (bps). For 10/100 ports, the range is 1000000 to 100000000, and the granularity is 1 Mbps. For Gigabit-capable Ethernet ports, the range is 8000000 to 1016000000, and the granularity is 8 Mbps.
burst-byte	Specify the normal burst size in bytes. For 10/100 ports, the burst size values are 4096, 8192, 16384, 32768, and 65536. For Gigabit-capable Ethernet ports, the burst size values are 4096, 8192, 16384, 32768, 65536, 131072, 262144, and 524288.
exceed-action drop	(Optional) When the specified rate is exceeded, specify that the switch drops the packet.
exceed-action dscp *dscp-value*	(Optional) When the specified rate is exceeded, specify that the switch changes the Differentiated Services Code Point (DSCP) of the packet to the specified *dscp-value* and then sends the packet.

Defaults

No policers are defined.

Command Modes

Policy-map class configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You can configure up to six policers on ingress Fast Ethernet ports.

You can configure up to 60 policers on ingress Gigabit-capable Ethernet ports.

Policers cannot be configured on egress Fast Ethernet and Gigabit-capable Ethernet ports.

To return to policy-map configuration mode, use the **exit** command. To return to privileged EXEC mode, use the **end** command.

Note For more information about configuring access control lists (ACLs), refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to configure a policer that sets the DSCP value to 46 if traffic does not exceed a 1-Mbps average rate with a burst size of 65536 bytes and drops packets if traffic exceeds these conditions:

```
Switch(config)# policy-map policy1
Switch(config-pmap)# class class1
Switch(config-pmap-c)# set ip dscp 46
Switch(config-pmap-c)# police 1000000 65536 exceed-action drop
Switch(config-pmap-c)# exit
```

You can verify your settings by entering the **show policy-map** privileged EXEC command.

Related Commands

Command	Description
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces, and enters policy-map configuration mode.
show policy-map	Displays quality of service (QoS) policy maps.

policy-map

Use the **policy-map** global configuration command to create or modify a policy map that can be attached to multiple interfaces and to enter policy-map configuration mode. Use the **no** form of this command to delete an existing policy map and return to global configuration mode.

policy-map *policy-map-name*

no policy-map *policy-map-name*

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

policy-map-name	Name of the policy map.

Defaults

No policy maps are defined.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Entering the **policy-map** command enables the policy-map configuration mode. These configuration commands are available:

- **class**: defines the classification match criteria for the specified class map. For more information, see the **class** command.
- **description**: describes the policy map (up to 200 characters).
- **exit**: exits policy-map configuration mode and returns to global configuration mode.
- **no**: removes a previously defined policy map.
- **rename**: renames the policy map.

Note In a policy map, the class named *class-default* is not supported. The switch does not filter traffic based on the policy map defined by the **class class-default** policy-map configuration command.

To return to global configuration mode, use the **exit** command. To return to privileged EXEC mode, use the **end** command.

Before you can configure policies for classes whose match criteria are defined in a class map, use the **policy-map** command to specify the name of the policy map to be created or modified. Entering this command also enables the policy-map configuration mode in which you can configure or modify the class policies for that policy map.

You can configure class policies in a policy map only if the classes have match criteria defined for them. Use the **class-map** and **match** commands to configure the match criteria for a class. Only one **match** command per class map is supported.

Only one policy map per interface per direction is supported. You can apply the same policy map to multiple interfaces but only in the ingress direction.

Note For more information about configuring access control lists (ACLs), refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to create a policy map called *policy1*. When attached to the ingress direction, it matches all the incoming traffic defined in *class1* and polices the traffic at an average rate of 1 Mbps and bursts at 65536 bytes. Traffic exceeding the profile is dropped.

```
Switch(config)# policy-map policy1
Switch(config-pmap)# class class1
Switch(config-pmap-c)# police 1000000 65536 exceed-action drop
Switch(config-pmap-c)# exit
Switch(config-pmap)#
```

This example shows how to delete *policymap2*:

```
Switch(config)# no policy-map policymap2
```

You can verify your settings by entering the **show policy-map** privileged EXEC command.

Related Commands

Command	Description
class	Defines the default CoS value of a port or assigns the default CoS to all incoming packets on the port.
class-map	Creates a class map to be used for matching packets to the class whose name you specify.
police	Defines a policer for classified traffic.
set	Classifies IP traffic by setting a DSCP value in the packet.
show policy-map	Displays quality of service (QoS) policy maps.

port-channel load-balance

Use the **port-channel load-balance** global configuration command to set the load distribution method among the ports in the EtherChannel. Use the **no** form of this command to reset the load distribution to the default.

port-channel load-balance *method*

no port-channel load-balance

Syntax Description

method	Load distribution method. These are the *method* values: • **dst-mac**—Load distribution using the destination MAC address • **src-mac**—Load distribution using the source MAC address

Defaults

The default method is **src-mac**.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If the link partner to the switch is a physical learner that has the **channel-group** interface configuration command set to **auto** or **desirable**, the switch automatically uses the load-distribution method based on the source MAC address regardless of the configured load-distribution method.

If the link partner to the switch is a physical learner that has the **channel-group** interface configuration command set to **on**, set the load-distribution method based on the source MAC address by using the **port-channel load-balance src-mac** global configuration command.

Examples

This example shows how to set the load-distribution method to **dst-mac**:

```
Switch(config)# port-channel load-balance dst-mac
```

You can verify your settings by entering the **show etherchannel** privileged EXEC command.

Related Commands

Command	Description
channel-group	Assigns an Ethernet interface to an EtherChannel group.
show etherchannel	Displays EtherChannel information for a channel.
show running-config	Displays the configuration information running on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

rcommand

Use the **rcommand** user EXEC command to start a Telnet session and to enter commands from the command switch for a member switch. To end the session, enter the **exit** command.

rcommand {*n* | **commander** | **mac-address** *hw-addr*}

Syntax Description

n	Provide the number that identifies a cluster member. The range is from 0 to 15.
commander	Provide access to the command switch from a member switch.
mac-address *hw-addr*	MAC address of the member switch.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

If the switch is the command switch but the member switch *n* does not exist, an error message appears. To obtain the switch number, enter the **show cluster members** privileged EXEC command on the command switch.

You can use this command to access a member switch from the command-switch prompt or to access a command switch from the member-switch prompt.

For Catalyst 2900 XL, 2950, 3500 XL, and 3550 switches, the Telnet session accesses the member-switch command-line interface (CLI) at the same privilege level as on the command switch. For example, if you enter this command at user level on the cluster command switch, the member switch is accessed at user level. If you use this command on the command switch at privileged level, the command accesses the remote device at privileged level. If you use an intermediate enable-level lower than *privileged*, access to the member switch is at user level.

For Catalyst 1900 and 2820 switches running standard edition software, the Telnet session accesses the menu console (the menu-driven interface) if the command switch is at privilege level 15. If the command switch is at privilege level 1, you are prompted for the password before being able to access the menu console. Command switch privilege levels map to the member switches running standard edition software as follows:

- If the command switch privilege level is from 1 to 14, the member switch is accessed at privilege level 1.
- If the command switch privilege level is 15, the member switch is accessed at privilege level 15.

The Catalyst 1900 and 2820 CLI is available only on switches running Enterprise Edition Software.

This command does not work if the vty lines of the command switch have access-class configurations.

You are not prompted for a password because the member switches inherited the password of the command switch when they joined the cluster.

Examples

This example shows how to start a session with member 3. All subsequent commands are directed to member 3 until you enter the **exit** command or close the session.

```
Switch# rcommand 3
Switch-3# show version
Cisco Internet Operating System Software ...
...
Switch-3# exit
Switch#
```

Related Commands

Command	Description
show cluster members	Displays information about the cluster members.

remote-span

Use the **remote-span** VLAN configuration command to add the Remote Switched Port Analyzer (RSPAN) feature to a VLAN. Use the **no** form of this command to remove the RSPAN feature from the VLAN.

remote-span

no remote-span

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

This command has no arguments or keywords.

Defaults

No RSPAN VLANs are defined.

Command Modes

VLAN configuration

Command History

Release	Modification
12.1(11)EA1	This command was first introduced.

Usage Guidelines

When a VLAN is converted from a normal VLAN to an RSPAN VLAN (or the reverse), the VLAN is first deleted and is then recreated with the new configuration. The RSPAN feature is propagated by VLAN Trunking Protocol (VTP) for VLAN-IDs that are lower than 1005.

Before you configure the RSPAN **remote-span** feature, use the **vlan** (global configuration) command to create the VLAN.

Examples

This example shows how to configure an RSPAN VLAN.

```
Switch(config)# vlan 901
Switch(config-vlan)# remote-span
```

This example shows how to remove the RSPAN feature from a VLAN.

```
Switch(config)# vlan 901
Switch(config-vlan)# no remote-span
```

You can verify your settings by entering the **show vlan** user EXEC command.

Related Commands

Command	Description
monitor session	Enables SPAN and RSPAN monitoring on a port and configures a port as a source or destination port.
vlan (global configuration)	Changes to config-vlan mode where you can configure VLANs 1 to 1005. Do not enter leading zeros.

rmon collection stats

Use the **rmon collection stats** interface configuration command to collect Ethernet group statistics. The Ethernet group statistics include utilization statistics about broadcast and multicast packets and error statistics about Cyclic Redundancy Check (CRC) alignment errors and collisions. Use the **no** form of this command to return to the default setting.

rmon collection stats *index* [**owner** *name*]

no rmon collection stats *index* [**owner** *name*]

Syntax Description

index	Remote Network Monitoring (RMON) collection control index. The range is 1 to 65535.
owner *name*	(Optional) Owner of the RMON collection.

Defaults

The RMON statistics collection is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

The RMON statistics collection command is based on hardware counters.

Examples

This example shows how to collect RMON statistics for the owner root on an interface:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# rmon collection stats 2 owner root
```

You can verify your settings by entering the **show rmon statistics** privileged EXEC command.

Related Commands

Command	Description
show rmon statistics	Displays RMON statistics. For more information on this command, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS System Management Commands > RMON Commands**.

service-policy

Use the **service-policy** interface configuration command to apply a policy map defined by the **policy-map** command to the input of a particular interface. Use the **no** form of this command to remove the policy map and interface association.

service-policy input *policy-map-name*

no service-policy input *policy-map-name*

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

policy-map-name	Apply the specified policy map to the input of an interface.

Defaults

No policy maps are attached to the interface.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Only one policy map per ingress interface is supported.

Service policy maps cannot be defined on egress interfaces.

Note For more information about configuring access control lists (ACLs), refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to apply *plcmap1* to an ingress interface:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# service-policy input plcmap1
```

You can verify your settings by entering the **show policy-map** privileged EXEC command.

Related Commands

Command	Description
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces to specify a service policy.
show policy-map	Displays quality of service (QoS) policy maps.

set

Use the **set** policy-map class configuration command to classify IP traffic by setting a Differentiated Services Code Point (DSCP) value. Use the **no** form of this command to remove traffic classification.

set ip dscp *new-dscp*

no set ip dscp *new-dscp*

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

new-dscp	New DSCP value assigned to the classified traffic. The supported DSCP values are 0, 8, 10, 16, 18, 24, 26, 32, 34, 40, 46, 48, and 56.

Defaults

No traffic classification is defined.

Command Modes

Policy-map class configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **set** command can be used in a policy with a **match** command.

The **set** command sets the DSCP value for in-profile packets.

Note This command does not support IP precedence.

To return to policy-map configuration mode, use the **exit** command. To return to privileged EXEC mode, use the **end** command.

Note For more information about configuring access control lists (ACLs), refer to the "Configuring Network Security with ACLs" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide* for this release.

Examples

This example shows how to assign a DSCP value of 10 to all FTP traffic without any policers:

```
Switch(config)# policy-map policy_ftp
Switch(config-pmap)# class ftp_class
Switch(config-pmap-c)# set ip dscp 10
Switch(config-pmap)# exit
```

You can verify your settings by entering the **show policy-map** privileged EXEC command.

Related Commands

Command	Description
police	Defines a policer for classified traffic.
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces to specify a service policy.
show policy-map	Displays quality of service (QoS) policy maps.

show access-lists

Use the **show access-lists** privileged EXEC command to display access control lists (ACLs) configured on the switch.

show access-lists [*name* | *number*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

name	(Optional) Name of the ACL.
number	(Optional) ACL number. The range is from 1 to 2699.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show access-lists** command:

```
Switch# show access-lists
Standard IP access list testingacl
    permit 10.10.10.2
Standard IP access list wizard_1-1-1-2
    permit 1.1.1.2
Extended IP access list 103
    permit tcp any any eq www
Extended IP access list CMP-NAT-ACL
    Dynamic Cluster-HSRP deny   ip any any
    Dynamic Cluster-NAT permit ip any any
      permit ip host 10.123.222.192 any
      permit ip host 10.228.215.0 any
      permit ip host 10.245.137.0 any
      permit ip host 10.245.155.128 any
      permit ip host 10.221.111.64 any
      permit ip host 10.216.25.128 any
      permit ip host 10.186.122.64 any
      permit ip host 10.169.110.128 any
      permit ip host 10.146.106.192 any
```

Related Commands

Command	Description
access-list (IP extended)	Configures an extended IP ACL on the switch.
access-list (IP standard)	Configures a standard IP ACL on the switch.
ip access-list	Configures an IP ACL on the switch.
mac access-list extended	Creates an ACL based on MAC addresses.
show ip access-lists	Displays the IP ACLs configured on a switch.

show boot

Use the **show boot** privileged EXEC command to display the settings of the boot environment variables.

show boot [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(11)EA1	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show boot** command. Table 2-4 describes each field in the output.

```
Switch# show boot
BOOT path-list:      flash:boot
Config file:         flash:config.text
Private Config file: flash:private-config.text
Enable Break:        no
Manual Boot:         yes
HELPER path-list:
NVRAM/Config file
      buffer size:   32768
```

Table 2-4 show boot Field Descriptions

Field	Description
BOOT path-list	Displays a semicolon-separated list of executable files to load and to execute when automatically booting. If the BOOT environment variable is not set, the system attempts to load and execute the first executable image it can find by using a recursive, depth-first search through the Flash file system. In a depth-first search of a directory, each encountered subdirectory is completely searched before continuing the search in the original directory. If the BOOT variable is set but the specified images cannot be loaded, the system attempts to boot the first bootable file that it can find in the Flash file system.
Config file	Displays the filename that IOS uses to read and write a nonvolatile copy of the system configuration.
Private Config file	Displays the filename that IOS uses to read and write a nonvolatile copy of the private configuration.
Enable Break	Displays whether a break during booting is enabled or disabled. If it is set to *yes*, *on*, or *1*, you can interrupt the automatic boot process by pressing the Break key on the console after the Flash file system is initialized.
Manual Boot	Displays whether the switch automatically or manually boots. If it is set to *no* or *0*, the boot loader attempts to automatically boot the system. If it is set to anything else, you must manually boot the switch from the boot loader mode.
Helper path-list	Displays a semicolon-separated list of loadable files to dynamically load during the boot loader initialization. Helper files extend or patch the functionality of the boot loader.
NVRAM/Config file buffer size	Displays the buffer size that IOS uses to hold a copy of the configuration file in memory. The configuration file cannot be larger than the buffer size allocation.

Related Commands

Command	Description
boot private-config-file	Specifies the filename that IOS uses to read and write a nonvolatile copy of the private configuration.

show class-map

Use the **show class-map** user EXEC command to display quality of service (QoS) class maps, which define the match criteria to classify traffic.

show class-map [*class-map-name*] [| {**begin** | **exclude** | **include**} *expression*]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

class-map-name	(Optional) Display the contents of the specified class map.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If you do not specify a *class-map-name*, all class maps appear.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show class-map test** command:

```
Switch> show class-map test
 Class Map match-all test (id 2)
   Match access-group name testingacl
```

This is an example of output from the **show class-map** command:

```
Switch> show class-map
 Class Map match-all wizard_1-1-1-2 (id 3)
   Match access-group name videowizard_1-1-1-2

 Class Map match-all test (id 2)
   Match access-group name testingacl

 Class Map match-any class-default (id 0)
   Match any

 Class Map match-all class1 (id 5)
   Match access-group  103

 Class Map match-all classtest (id 4)
  Description: This is a test.
   Match access-group name testingacl
```

Related Commands

Command	Description
class-map	Creates a class map to be used for matching packets to the class whose name you specify.
match	Defines the match criteria to classify traffic.

show cluster

Use the **show cluster** privileged EXEC command to display the cluster status and a summary of the cluster to which the switch belongs. This command can be entered on command and member switches.

show cluster [| {**begin** | **exclude** | **include**} *expression*]

Syntax Descriptions

| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

On a member switch, this command displays the identity of the command switch, the switch member number, and the state of its connectivity with the command switch.

On a command switch, this command displays the cluster name and the total number of members. It also shows the cluster status and time since the status changed. If redundancy is enabled, it displays the primary and secondary command-switch information.

If you enter this command on a switch that is not a cluster member, the error message `Not a management cluster member` appears.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output when this command is entered on the active command switch:

```
Switch# show cluster
Command switch for cluster "Switch1"
        Total number of members:        7
        Status:                         1 members are unreachable
        Time since last status change:  0 days, 0 hours, 2 minutes
        Redundancy:                     Enabled
                Standby command switch: Member 1
                Standby Group:          Switch1_standby
                Standby Group Number:   110
        Heartbeat interval:             8
        Heartbeat hold-time:            80
        Extended discovery hop count:   3
```

This is an example of output when this command is entered on a member switch:

```
Switch# show cluster
Member switch for cluster "commander"
        Member number:                  3
        Management IP address:          192.192.192.192
        Command switch mac address:     0000.0c07.ac14
        Heartbeat interval:             8
        Heartbeat hold-time:            80
```

This is an example of output when this command is entered on a member switch that is configured as the standby command switch:

```
Switch# show cluster
Member switch for cluster "commander"
        Member number:                  3 (Standby command switch)
        Management IP address:          192.192.192.192
        Command switch mac address:     0000.0c07.ac14
        Heartbeat interval:             8
        Heartbeat hold-time:            80
```

This is an example of output when this command is entered on the command switch that has lost connectivity from member 1:

```
Switch# show cluster
Command switch for cluster "Switch1"
        Total number of members:        7
        Status:                         1 members are unreachable
        Time since last status change:  0 days, 0 hours, 5 minutes
        Redundancy:                     Disabled
        Heartbeat interval:             8
        Heartbeat hold-time:            80
        Extended discovery hop count:   3
```

This is an example of output when this command is entered on a member switch that has lost connectivity with the command switch:

```
Switch# show cluster
Member switch for cluster "commander"
        Member number:                  <UNKNOWN>
        Management IP address:          192.192.192.192
        Command switch mac address:     0000.0c07.ac14
        Heartbeat interval:             8
        Heartbeat hold-time:            80
```

Related Commands

Command	Description
cluster enable	Enables a command-capable switch as the cluster command switch, assigns a cluster name, and optionally assigns a member number to it.
show cluster candidates	Displays a list of candidate switches.
show cluster members	Displays information about the cluster members.

show cluster candidates

Use the **show cluster candidates** privileged EXEC command on the command switch to display a list of candidate switches.

show cluster candidates [**detail** | **mac-address** *H.H.H.*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

detail	(Optional) Display detailed information for all candidates.
mac-address *H.H.H.*	(Optional) Hexadecimal MAC address of the cluster candidate.
| **begin**	(Optional) Display begins with the line that matches the specified *expression*.
| **exclude**	(Optional) Display excludes lines that match the specified *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

You should only enter this command on a command switch.

If the switch is not a command switch, the command displays an empty line at the prompt.

The SN in the output means *switch member number*. If *E* is in the SN column, it means that the switch is discovered through extended discovery. If *E* does not appear in the SN column, it means that the *switch member number* is the upstream neighbor of the candidate switch. The hop count is the number of devices the candidate is from the command switch.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show cluster candidates** command:

```
Switch# show cluster candidates
                                                          |---Upstream---|
MAC Address    Name         Device Type      PortIf  FEC Hops SN PortIf  FEC
00d0.7961.c4c0 c2950-012    WS-C2950-12      Fa0/5        1   0  Fa0/3
00d0.bbf5.e900 ldf-dist-128 WS-C3524-XL      Fa0/7        1   0  Fa0/24
00e0.1e7e.be80 1900_Switch  1900             3        0   1   0  Fa0/11
00e0.1e9f.7a00 c2924XL-24   WS-C2924-XL      Fa0/5        1   0  Fa0/3
00e0.1e9f.8c00 c2912XL-12-2 WS-C2912-XL      Fa0/4        1   0  Fa0/7
00e0.1e9f.8c40 c2912XL-12-1 WS-C2912-XL      Fa0/1        1   0  Fa0/9
0050.2e4a.9fb0 C3508XL-0032 WS-C3508-XL E
0050.354e.7cd0 C2924XL-0034 WS-C2924-XL E
```

This is an example of output from the **show cluster candidates** command that uses the MAC address of a member switch directly connected to the command switch:

```
Switch# show cluster candidates mac-address 00d0.7961.c4c0
Device 'c2950-12' with mac address number 00d0.7961.c4c0
        Device type:            cisco WS-C2950-12
        Upstream MAC address:   00d0.796d.2f00 (Cluster Member 0)
        Local port:             Fa0/3   FEC number:
        Upstream port:          Fa0/13  FEC Number:
        Hops from cluster edge: 1
        Hops from command device: 1
```

This is an example of output from the **show cluster candidates** command that uses the MAC address of a member switch three hops from the cluster edge:

```
Switch# show cluster candidates mac-address 0010.7bb6.1cc0
Device 'c2950-24' with mac address number 0010.7bb6.1cc0
        Device type:            cisco WS-C2950-24
        Upstream MAC address:   0010.7bb6.1cd4
        Local port:             Fa2/1   FEC number:
        Upstream port:          Fa0/24  FEC Number:
        Hops from cluster edge: 3
        Hops from command device: -
```

This is an example of output from the **show cluster candidates detail** command:

```
Switch# show cluster candidates detail
Device 'c2950-12' with mac address number 00d0.7961.c4c0
        Device type:            cisco WS-C2950-12
        Upstream MAC address:   00d0.796d.2f00 (Cluster Member 1)
        Local port:             Fa0/3   FEC number:
        Upstream port:          Fa0/13  FEC Number:
        Hops from cluster edge: 1
        Hops from command device: 2
    Device '1900_Switch' with mac address number 00e0.1e7e.be80
        Device type:            cisco 1900
        Upstream MAC address:   00d0.796d.2f00 (Cluster Member 2)
        Local port:             3       FEC number: 0
        Upstream port:          Fa0/11  FEC Number:
        Hops from cluster edge: 1
        Hops from command device: 2
Device 'c2924-XL' with mac address number 00e0.1e9f.7a00
        Device type:            cisco WS-C2924-XL
        Upstream MAC address:   00d0.796d.2f00 (Cluster Member 3)
        Local port:             Fa0/5   FEC number:
        Upstream port:          Fa0/3   FEC Number:
        Hops from cluster edge: 1
        Hops from command device: 2
```

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.
show cluster members	Displays information about the cluster members.

show cluster members

Use the **show cluster members** privileged EXEC command on the command switch to display information about the cluster members.

show cluster members [*n* | **detail**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

n	(Optional) Number that identifies a cluster member. The range is from 0 to 15.
detail	(Optional) Display detailed information for all cluster members.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

You should only enter this command on a command switch.

If the cluster has no members, this command displays an empty line at the prompt.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show cluster members** command. The SN in the display means *switch number.*

```
Switch# show cluster members
                                                |---Upstream---|
SN MAC Address    Name         PortIf FEC Hops   SN PortIf  FEC  State
0  0002.4b29.2e00 StLouis1                 0                    Up   (Cmdr)
1  0030.946c.d740 tal-switch-1 Fa0/13      1     0  Gi0/1       Up
2  0002.b922.7180 nms-2820     10      0   2     1  Fa0/18      Up
3  0002.4b29.4400 SanJuan2     Gi0/1       2     1  Fa0/11      Up
4  0002.4b28.c480 GenieTest    Gi0/2       2     1  Fa0/9       Up
```

This is an example of output from the **show cluster members 3** command for cluster member 3:

```
Switch# show cluster members 3
Device 'SanJuan2' with member number 3
        Device type:            cisco WS-C3550-12T
        MAC address:            0002.4b29.4400
        Upstream MAC address:   0030.946c.d740 (Cluster member 1)
        Local port:             Gi0/1   FEC number:
        Upstream port:          Fa0/11  FEC Number:
        Hops from command device: 2
```

This is an example of output from the **show cluster members detail** command:

```
Switch# show cluster members detail
Device 'StLouis1' with member number 0 (Command Switch)
        Device type:            cisco WS-C3550-12T
        MAC address:            0002.4b29.2e00
        Upstream MAC address:
        Local port:                     FEC number:
        Upstream port:                  FEC Number:
        Hops from command device: 0
Device 'tal-switch-14' with member number 1
        Device type:            cisco WS-C3548-XL
        MAC address:            0030.946c.d740
        Upstream MAC address:   0002.4b29.2e00 (Cluster member 0)
        Local port:             Fa0/13  FEC number:
        Upstream port:          Gi0/1   FEC Number:
        Hops from command device: 1
Device 'nms-2820' with member number 2
        Device type:            cisco 2820
        MAC address:            0002.b922.7180
        Upstream MAC address:   0030.946c.d740 (Cluster member 1)
        Local port:             10      FEC number: 0
        Upstream port:          Fa0/18  FEC Number:
        Hops from command device: 2
Device 'SanJuan2' with member number 3
        Device type:            cisco WS-C3550-12T
        MAC address:            0002.4b29.4400
        Upstream MAC address:   0030.946c.d740 (Cluster member 1)
        Local port:             Gi0/1   FEC number:
        Upstream port:          Fa0/11  FEC Number:
        Hops from command device: 2
Device 'Test' with member number 4
        Device type:            cisco SeaHorse
        MAC address:            0002.4b28.c480
        Upstream MAC address:   0030.946c.d740 (Cluster member 1)
        Local port:             Gi0/2   FEC number:
        Upstream port:          Fa0/9   FEC Number:
        Hops from command device: 2
Device 'Palpatine' with member number 5
        Device type:            cisco WS-C2924M-XL
        MAC address:            00b0.6404.f8c0
        Upstream MAC address:   0002.4b29.2e00 (Cluster member 0)
        Local port:             Gi2/1   FEC number:
        Upstream port:          Gi0/7   FEC Number:
        Hops from command device: 1
```

Related Commands

Command	Description
show cluster	Displays the cluster status and a summary of the cluster to which the switch belongs.
show cluster candidates	Displays a list of candidate switches.

show dot1x

Use the **show dot1x** privileged EXEC command to display the 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

show dot1x [**interface** *interface-id*] | [**statistics** [**interface** *interface-id*]] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface *interface-id*	(Optional) Display the 802.1X status for the specified port.
statistics [**interface** *interface-id*]	(Optional) Display 802.1X statistics for the switch or the specified interface.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If you do not specify an interface, global parameters and a summary appear. If you specify an interface, details for that interface appear.

If you specify the **statistics** keyword without the **interface** *interface-id* option, statistics appear for all interfaces. If you specify the **statistics** keyword with the **interface** *interface-id* option, statistics appear for the specified interface.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show dot1x** command:

```
Switch# show dot1x

Global 802.1X Parameters
    reauth-enabled               no
    reauth-period              3600
    quiet-period                 60
    tx-period                    30
    supp-timeout                 30
    server-timeout               30
    reauth-max                    2
    max-req                       2
```

```
802.1X Port Summary
    Port Name               Status      Mode                Authorized
    Gi0/1                   disabled    n/a                 n/a
    Gi0/2                   enabled     Auto (negotiate)    no

    802.1X Port Details
    802.1X is disabled on GigabitEthernet0/1

    802.1X is enabled on GigabitEthernet0/2
      Status                Unauthorized
      Port-control          Auto
      Supplicant            0060.b0f8.fbfb
      Multiple Hosts        Disallowed
      Current Identifier    2

      Authenticator State Machine
        State               AUTHENTICATING
        Reauth Count        1

      Backend State Machine
        State               RESPONSE
        Request Count       0
        Identifier (Server) 2

      Reauthentication State Machine
        State               INITIALIZE
```

Note In the previous example, the supp-timeout, server-timeout, and reauth-max values in the Global 802.1X Parameters section are not configurable.When relaying a request from the Remote Authentication Dial-In User Service (RADIUS) authentication server to the client, the supp-timeout is the amount of time the switch waits for a response before it resends the request. When relaying a response from the client to the RADIUS authentication server, the server-timeout is the amount of time the switch waits for a reply before it resends the response. The reauth-max parameter is the maximum number of times that the switch tries to authenticate the client without receiving any response before the switch resets the port and restarts the authentication process.

In the 802.1X Port Summary section of the example, the Status column shows whether the port is enabled for 802.1X (the **dot1x port-control** interface configuration command is set to **auto** or **force-unauthorized**). The Mode column shows the operational status of the port; for example, if you configure the **dot1x port-control** interface configuration command to **force-unauthorized**, but the port has not transitioned to that state, the Mode column displays *auto*. If you disable 802.1X, the Mode column displays *n/a*.

The Authorized column shows the authorization state of the port. For information about port states, refer to the “Configuring 802.1X Port-Based Authentication” chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide*.

This is an example of output from the **show dot1x interface gigabitethernet0/2** privileged EXEC command. Table 2-5 describes the fields in the example.

```
Switch# show dot1x interface gigabitethernet0/2

802.1X is enabled on GigabitEthernet0/2
  Status                Authorized
  Port-control          Auto
  Supplicant            0060.b0f8.fbfb
  Multiple Hosts        Disallowed
  Current Identifier    3
```

```
  Authenticator State Machine
    State               AUTHENTICATED
    Reauth Count        0

  Backend State Machine
    State               IDLE
    Request Count       0
    Identifier (Server) 2

  Reauthentication State Machine
    State               INITIALIZE
```

Table 2-5 show dot1x interface Field Description

Field	Description
802.1X is enabled on GigabitEthernet0/2	
Status	Status of the port (authorized or unauthorized). The status of a port appears as authorized if the **dot1x port-control** interface configuration command is set to **auto**, and authentication was successful.
Port-control	Setting of the **dot1x port-control** interface configuration command.
Supplicant	Ethernet MAC address of the client, if one exists. If the switch has not discovered the client, this field displays *Not set*.
Multiple Hosts	Setting of the **dot1x multiple-hosts** interface configuration command (allowed or disallowed).
Current Identifier[1]	Each exchange between the switch and the client includes an identifier, which matches requests with responses. This number is incremented with each exchange and can be reset by the authentication server.

1. This field and the remaining fields in the example show internal state information. For a detailed description of these state machines and their settings, refer to the IEEE 802.1X standard.

This is an example of output from the **show dot1x statistics interface gigiabitethernet0/1** command. Table 2-6 describes the fields in the example.

```
Switch# show dot1x statistics interface gigabitethernet0/1

GigabitEthernet0/1

    Rx: EAPOL     EAPOL     EAPOL     EAPOL     EAP       EAP       EAP
        Start     Logoff    Invalid   Total     Resp/Id   Resp/Oth  LenError
        0         0         0         21        0         0         0

        Last      Last
        EAPOLVer  EAPOLSrc
        1         0002.4b29.2a03

    Tx: EAPOL     EAP       EAP
        Total     Req/Id    Req/Oth
        622       445       0
```

Table 2-6 show dot1x statistics Field Descriptions

Field	Description
RX EAPOL[1] Start	Number of valid EAPOL-start frames that have been received
RX EAPOL Logoff	Number of EAPOL-logoff frames that have been received
RX EAPOL Invalid	Number of EAPOL frames that have been received and have an unrecognized frame type
RX EAPOL Total	Number of valid EAPOL frames of any type that have been received
RX EAP[2] Resp/ID	Number of EAP-response/identity frames that have been received
RX EAP Resp/Oth	Number of valid EAP-response frames (other than response/identity frames) that have been received
RX EAP LenError	Number of EAPOL frames that have been received in which the packet body length field is invalid
Last EAPOLVer	Protocol version number carried in the most recently received EAPOL frame
LAST EAPOLSrc	Source MAC address carried in the most recently received EAPOL frame
TX EAPOL Total	Number of EAPOL frames of any type that have been sent
TX EAP Req/Id	Number of EAP-request/identity frames that have been sent
TX EAP Req/Oth	Number of EAP-request frames (other than request/identity frames) that have been sent

1. EAPOL = Extensible Authentication Protocol over LAN
2. EAP = Extensible Authentication Protocol

Related Commands

Command	Description
dot1x default	Resets the global 802.1X parameters to their default values.

show env

Use the **show env** user EXEC command to display fan information for the switch.

show env {**all** | **fan**} [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

all	Display both fan and temperature environmental status.
fan	Display the switch fan status (only available in privileged EXEC mode).
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show env all** command:

```
Switch> show env all
FAN 1 is OK
```

This is an example of output from the **show env fan** command:

```
Switch# show env fan
FAN 1 is FAULTY
```

show errdisable recovery

Use the **show errdisable recovery** user EXEC command to display the error-disable recovery timer information.

show errdisable recovery [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show errdisable recovery** command:

```
Switch> show errdisable recovery
ErrDisable Reason    Timer Status
-----------------    --------------
udld                 Enabled
bpduguard            Enabled
channel-misconfig    Enabled
pagp-flap            Enabled
dtp-flap             Enabled
link-flap            Enabled
psecure-violation    Enabled
gbic-invalid         Enabled

Timer interval:300 seconds

Interfaces that will be enabled at the next timeout:

Interface    Errdisable reason    Time left(sec)
---------    -----------------    --------------
Gi0/4           link-flap             279
```

Related Commands

Command	Description
errdisable recovery	Configures the recover mechanism variables.
show interfaces trunk	Displays interface status or a list of interfaces in error-disabled state.

show etherchannel

Use the **show etherchannel** user EXEC command to display EtherChannel information for a channel.

show etherchannel [*channel-group-number*] {**brief** | **detail** | **load-balance** | **port** | **port-channel** | **summary**} [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

channel-group-number	(Optional) Number of the channel group. Valid numbers range from 1 to 6.
brief	Display a summary of EtherChannel information.
detail	Display detailed EtherChannel information.
load-balance	Display the load-balance or frame-distribution scheme among ports in the port channel.
port	Display EtherChannel port information.
port-channel	Display port-channel information.
summary	Display a one-line summary per channel-group.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **show port group** command.

Usage Guidelines

If you do not specify a *channel-group*, all channel groups appear.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show etherchannel 1 detail** command:

```
Switch> show etherchannel 1 detail
Group state = L2
Ports: 1   Maxports = 8
Port-channels: 1 Max Port-channels = 1
               Ports in the group:
               -------------------
Port: Fa0/3
------------

Port state    = Down Not-in-Bndl
Channel group = 1           Mode = Automatic-Sl     Gcchange = 0
Port-channel  = null        GC   = 0x00000000    Pseudo port-channel = Po1
Port index    = 0           Load = 0x00

Flags:  S - Device is sending Slow hello.  C - Device is in Consistent state.
        A - Device is in Auto mode.        P - Device learns on physical port.
        d - PAgP is down.
Timers: H - Hello timer is running.        Q - Quit timer is running.
        S - Switching timer is running.    I - Interface timer is running.

Local information:
                              Hello    Partner  PAgP     Learning  Group
Port      Flags State   Timers  Interval Count   Priority   Method  Ifindex
Fa0/3     dA    U1/S1           1s       0        200        Any      0

Age of the port in the current state: 10d:23h:07m:37s
               Port-channels in the group:
               ----------------------

Port-channel: Po1
------------

Age of the Port-channel   = 03d:02h:22m:43s
Logical slot/port   = 1/0           Number of ports = 0
GC                  = 0x00000000      HotStandBy port = null
Port state          = Port-channel Ag-Not-Inuse
```

This is an example of output from the **show etherchannel 1 summary** command:

```
Switch> show etherchannel 1 summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        R - Layer3      S - Layer2
        u - unsuitable for bundling
        U - port-channel in use
        d - default port
Group Port-channel  Ports
-----+------------+----------------------------------------------------------
1     Po1(SU)     Fa0/6(Pd) Fa0/15(P)
```

This is an example of output from the **show etherchannel 1 brief** command:

```
Switch> show etherchannel 1 brief
Group state = L2
Ports: 1   Maxports = 8
Port-channels: 1 Max Port-channels = 1
```

This is an example of output from the **show etherchannel 1 port** command:

```
Switch> show etherchannel 1 port
                Ports in the group:
                -------------------
Port: Fa0/3
------------

Port state    = Down Not-in-Bndl
Channel group = 1           Mode = Automatic-Sl     Gcchange = 0
Port-channel  = null        GC   = 0x00000000    Pseudo port-channel = Po1
Port index    = 0           Load = 0x00

Flags:  S - Device is sending Slow hello.  C - Device is in Consistent state.
        A - Device is in Auto mode.        P - Device learns on physical port.
        d - PAgP is down.
Timers: H - Hello timer is running.        Q - Quit timer is running.
        S - Switching timer is running.    I - Interface timer is running.

Local information:
                                Hello    Partner  PAgP     Learning  Group
Port      Flags State   Timers  Interval Count   Priority   Method  Ifindex
Fa0/3     dA    U1/S1           1s       0        200        Any      0

Age of the port in the current state: 10d:23h:13m:21s
```

Related Commands

Command	Description
channel-group	Assigns an Ethernet interface to an EtherChannel group.
interface port-channel	Accesses or creates the port channel.

show file

Use the **show file** privileged EXEC command to display a list of open file descriptors, file information, and file system information.

show file {**descriptors** | **information** {*device*:}*filename* | **systems**} [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

descriptors	Display a list of open file descriptors.
information	Display file information.
*device***:**	Device containing the file. Valid devices include the switch Flash memory.
filename	Name of file.
systems	Display file system information.
| **begin**	(Optional) Display begins with the line that matches the specified *expression*.
| **exclude**	(Optional) Display excludes lines that match the specified *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **descriptors** and **information** keywords were added.

Usage Guidelines

File descriptors are the internal representations of open files. You can use this command to see if another user has a file open.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show file descriptors** command:

```
Switch# show file descriptors
File Descriptors:
FD  Position  Open  PID  Path
0    187392   0001    2  tftp://temp/hampton/c2950g.a
1    184320   030A    2  flash:c2950-i-m.a
```

Table 2-7 describes the fields in the **show file descriptors** command output.

Table 2-7 show file descriptors Field Descriptions

Field	Description
FD	File descriptor. The file descriptor is a small integer used to specify the file once it has been opened.
Position	Byte offset from the start of the file.
Open	Flags supplied when opening the file.
PID	Process ID of the process that opened the file.
Path	Location of the file.

This is an example of output from the **show file information nvram:startup-config** command:

```
Switch# show file information nvram:startup-config
nvram:startup-config:
  type is ascii text
```

Table 2-8 lists the possible file types for the previous example.

Table 2-8 Possible File Types

Field	Description
ascii text	Configuration file or other text file.
coff	Runnable image in coff format.
ebcdic	Text generated on an IBM mainframe.
image (a.out)	Runnable image in a.out format.
image (elf)	Runnable image in elf format.
lzw compression	Lzw compressed file.
tar	Text archive file used by the CIP.

This is an example of output from the **show file systems** command:

```
Switch# show file systems
File Systems:

     Size(b)     Free(b)      Type  Flags  Prefixes
*    7741440      433152     flash     rw   flash:
     7741440      433152   unknown     rw   zflash:
       32768       25316     nvram     rw   nvram:
           -           -   network     rw   tftp:
           -           -    opaque     rw   null:
           -           -    opaque     rw   system:
           -           -    opaque     ro   xmodem:
           -           -    opaque     ro   ymodem:
           -           -   network     rw   rcp:
           -           -   network     rw   ftp:
```

For this example, Table 2-9 describes the fields in the **show file systems** command output. Table 2-10 lists the file system types. Table 2-11 lists the file system flags.

Table 2-9 show file systems Field Descriptions

Field	Description
Size(b)	Amount of memory in the file system, in bytes.
Free(b)	Amount of free memory in the file system, in bytes.
Type	Type of file system.
Flags	Permissions for file system.
Prefixes	Alias for file system.

Table 2-10 File System Types

Field	Description
disk	The file system is for a rotating medium.
flash	The file system is for a Flash memory device.
network	The file system is a network file system, such as TFTP, rcp, or FTP.
nvram	The file system is for an NVRAM device.
opaque	The file system is a locally generated *pseudo* file system (for example, the *system*) or a download interface, such as brimux.
rom	The file system is for a ROM or EPROM device.
tty	The file system is for a collection of terminal devices.
unknown	The file system is of unknown type.

Table 2-11 File System Flags

Field	Description
ro	The file system is Read Only.
wo	The file system is Write Only
rw	The file system is Read/Write.

show interfaces

Use the **show interfaces** privileged EXEC command to display the administrative and operational status of all interfaces or a specified interface.

show interfaces [*interface-id* | **vlan** *vlan-id*] [**accounting** | **description** | **etherchannel** | **pruning** | **stats** | **status** [**err-disabled**] | **switchport** | **trunk**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface-id	(Optional) Valid interfaces include physical ports (including type, slot, and port number) and port channels. The valid port-channel range is 1 to 6.
vlan *vlan-id*	(Optional) VLAN ID. The valid VLAN range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
accounting	(Optional) Display interface accounting information.
description	(Optional) Display the administrative status and description set for an interface.
etherchannel	(Optional) Display interface EtherChannel information.
pruning	(Optional) Display interface trunk VTP pruning information.
stats	(Optional) Display the input and output packets by switching path for the interface.
status	(Optional) Display the status of the interface.
err-disabled	(Optional) Display interfaces in error-disabled state.
switchport	(Optional) Display the administrative and operational status of a switching (nonrouting) port.
trunk	Display interface trunk information. If you do not specify an interface, information for only active trunking ports appears.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Note Though visible in the command-line help strings, the **crb, fair-queue, irb, mac-accounting, precedence, random-detect, rate-limit,** and **shape** options are not supported.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show interfaces accounting** command:

```
Switch# show interfaces accounting
Vlan1
                Protocol    Pkts In   Chars In   Pkts Out  Chars Out
                      IP      17950    2351279       3205     411175
                     ARP       8626     552064         62       3720
Interface Vlan5 is disabled

FastEthernet0/1
                Protocol    Pkts In   Chars In   Pkts Out  Chars Out
           Spanning Tree    2956958  179218508      34383    2131700
                     CDP      14301    5777240      14307    5722418
                     VTP          0          0       1408     145908
                     DTP      28592    1572560          0          0

<output truncated>
```

This is an example of output from the **show interfaces gigabitethernet0/1** command:

```
Switch# show interfaces gigabitethernet0/1
FastEthernet0/1 is up, line protocol is down
  Hardware is Fast Ethernet, address is 0005.7428.09c1 (bia 0005.7428.09c1)
  MTU 1500 bytes, BW 10000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Auto-duplex, Auto-speed
  input flow-control is off, output flow-control is off
  Last input never, output 4d21h, output hang never
  Last clearing of "show interface" counters never
  Input queue:0/75/0/0 (size/max/drops/flushes); Total output drops:0
  Queueing strategy:fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     1 packets input, 64 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
     1 packets output, 64 bytes, 0 underruns
     0 output errors, 0 collisions, 2 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier, 0 PAUSE output
     0 output buffer failures, 0 output buffers swapped out
```

This is an example of output from the **show interfaces gigabitethernet0/2 description** command when the interface has been described as *Connects to Marketing* by using the **description** interface configuration command.

```
Switch# show interfaces gigabitethernet0/2 description
Interface Status         Protocol Description
G10/2 up             down     Connects to Marketing
```

This is an example of output from the **show interfaces fastethernet0/1 pruning** command when pruning is enabled in the VTP domain:

```
Switch# show interfaces fastethernet0/1 pruning

Port      Vlans pruned for lack of request by neighbor
Fa0/1     4,196
```

```
Port      Vlan traffic requested of neighbor
Fa0/1     1,4
```

This is an example of output from the **show interfaces stats** command:

```
Switch# show interfaces stats
Vlan1
          Switching path    Pkts In   Chars In   Pkts Out  Chars Out
               Processor    3224706  223689126    3277307  280637322
            Route cache           0          0          0          0
                  Total    3224706  223689126    3277307  280637322
Interface Vlan5 is disabled

FastEthernet0/1
          Switching path    Pkts In   Chars In   Pkts Out  Chars Out
               Processor    3286423  231672787     179501   17431060
            Route cache           0          0          0          0
                  Total    3286423  231672787     179501   17431060
```

This is an example of output from the **show interfaces status** command. It displays the status of all interfaces.

```
Switch# show interfaces status

Port    Name               Status       Vlan       Duplex  Speed Type
Fa0/1                      notconnect   1            auto   auto 10/100BaseTX
Fa0/2                      notconnect   1            auto   auto 10/100BaseTX
Fa0/3                      disabled     100          auto   auto 10/100BaseTX
Fa0/4                      connected    trunk      a-full  a-100 10/100BaseTX
Fa0/5                      notconnect   1            auto   auto 10/100BaseTX
Fa0/6                      connected    trunk      a-full  a-100 10/100BaseTX
Fa0/7                      notconnect   1            auto   auto 10/100BaseTX
Fa0/8                      connected    1          a-full  a-100 10/100BaseTX
Fa0/9                      disabled     1            auto   auto 10/100BaseTX
Fa0/10                     notconnect   5            auto    100 10/100BaseTX
Fa0/11                     disabled     1            auto   auto 10/100BaseTX
Fa0/12                     disabled     1            auto   auto 10/100BaseTX
Gi0/1                      disabled     1            auto   auto unknown
Gi0/2                      notconnect   1            auto   auto unknown
Po1                        notconnect   1            auto   auto
```

This is an example of output from the **show interfaces status err-disabled** command. It displays the status of interfaces in error-disabled state.

```
switch#show interfaces fa0/15 status err-disabled

Port    Name               Status       Reason
Fa0/15                     err-disabled psecure-violation
```

This is an example of output from the **show interfaces etherchannel** command when port channels are configured on the switch:

```
Switch# show interfaces etherchannel
----
FastEthernet0/1:
Port state    = Up Mstr In-Bndl
Channel group = 1           Mode = On/FEC     Gcchange = 0
Port-channel  = Po1         GC   = 0x00010001    Pseudo port-channel = Po1
Port index    = 0           Load = 0x00

Age of the port in the current state:00d:00h:06m:54s
----
Port-channel1:
Age of the Port-channel   = 09d:22h:45m:14s
```

```
Logical slot/port   = 1/0           Number of ports = 1
GC                  = 0x00010001      HotStandBy port = null
Port state          = Port-channel Ag-Inuse

Ports in the Port-channel:

Index   Load   Port    EC state
------+------+------+------------
  0     00     Fa0/1    on

Time since last port bundled:   00d:00h:06m:54s    Fa0/1
```

This is an example of output from the **show interfaces switchport** command for a single interface. Table 2-12 describes the fields in the output.

```
Switch# show interfaces gigabitethernet0/1 switchport
Name:Fa0/1
Switchport:Enabled
Administrative Mode:static access
Operational Mode:down
Administrative Trunking Encapsulation:dot1q
Negotiation of Trunking:Off
Access Mode VLAN:1 (default)
Trunking Native Mode VLAN:1 (default)
Trunking VLANs Enabled:ALL
Pruning VLANs Enabled:2-1001

Protected:false

Voice VLAN:dot1p (Inactive)
Appliance trust:5
```

Table 2-12 show interfaces switchport Field Descriptions

Field	Description
Name	Displays the port name.
Switchport	Displays the administrative and operational status of the port. In this output, the port is in switchport mode.
Administrative Mode Operational Mode	Displays the administrative and operational mode.
Administrative Trunking Encapsulation Operation Trunking Encapsulation Negotiation of Trunking	Displays the administrative and operational encapsulation method, and whether trunking negotiation is enabled.
Access Mode VLAN	Displays the VLAN ID to which the port is configured.
Trunking Native Mode VLAN Trunking VLANs Enabled Trunking VLANs Active	Lists the VLAN ID of the trunk that is in native mode. Lists the allowed VLANs on the trunk. Lists the active VLANs on the trunk.
Pruning VLANs Enabled	Lists the VLANs that are pruning-eligible.
Protected	Displays whether or not protected port is enabled (True) or disabled (False) on the interface.

Table 2-12 show interfaces switchport Field Descriptions (continued)

Field	Description
Voice VLAN	Displays the VLAN ID on which voice VLAN is enabled.
Appliance trust	Displays the class of service (CoS) setting of the data packets of the IP phone.

This is an example of output from the **show interfaces trunk** command:

```
Switch# show interfaces trunk

Port      Mode         Encapsulation  Status        Native vlan
Fa0/4     on           802.1q         trunking      1
Fa0/6     on           802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/4     1-4094
Fa0/6     1-4094

Port      Vlans allowed and active in management domain
Fa0/4     1-2,51-52
Fa0/6     1-2,51-52

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/4     1
Fa0/6     1-2,51-52
```

This is an example of output from the **show interfaces fastethernet0/1 trunk** command. It displays trunking information for the interface.

```
Switch# show interfaces fastethernet0/1 trunk

Port      Mode         Encapsulation  Status        Native vlan
Fa0/1     desirable    802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-4094

Port      Vlans allowed and active in management domain
Fa0/1     1,4,196,306

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,306
```

Related Commands

Command	Description
switchport access	Configures a port as a static-access or dynamic-access port.
switchport protected	Isolates Layer 2 unicast, multicast, and broadcast traffic from other protected ports on the same switch.
switchport trunk pruning	Configures the VLAN pruning-eligible list for ports in trunking mode.

show interfaces counters

Use the **show interfaces counters** privileged EXEC command to display various counters for a specific interface or for all interfaces.

show interfaces [*interface-id* | **vlan** *vlan-id*] **counters** [**broadcast** | **errors** | **multicast** | **trunk** | **unicast**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface-id	(Optional) ID of the physical interface, including type and slot and port number.
vlan *vlan-id*	(Optional) VLAN number of the management VLAN. Valid IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
broadcast	(Optional) Display discarded broadcast traffic.
errors	(Optional) Display error counters.
multicast	(Optional) Display discarded multicast traffic.
trunk	(Optional) Display trunk counters.
unicast	(Optional) Display discarded unicast traffic.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If you do not enter any keywords, all counters for all interfaces are included.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show interfaces counters** command. It displays all the counters for the switch.

```
Switch# show interfaces counters
Port            InOctets   InUcastPkts   InMcastPkts   InBcastPkts
Gi0/1           23324617         10376        185709        126020
Gi0/2                  0             0             0             0

Port           OutOctets  OutUcastPkts  OutMcastPkts  OutBcastPkts
Gi0/1            4990607         28079         21122            10
Gi0/2            1621568         25337             0             0
```

This is an example of output from the **show interfaces counters broadcast** command. It displays the dropped broadcast traffic for all interfaces.

```
Switch# show interfaces counters broadcast
Port      BcastSuppDiscards
Gi0/1                     1
Gi0/2                     0
```

This is an example of output from the **show interfaces gigabitethernet0/1 counters broadcast** command. It displays the dropped broadcast traffic for an interface.

```
Switch# show interfaces gigabitethernet0/1 counters broadcast

Port      BcastSuppDiscards
Gi0/1                     0
```

This is an example of output from the **show interfaces counters errors** command. It displays the interface error counters for all interfaces.

```
Switch# show interfaces counters errors

Port        Align-Err    FCS-Err   Xmit-Err    Rcv-Err UnderSize
Gi0/1               0          0          0          0         0
Gi0/2               0          0          0          0         0

Port      Single-Col Multi-Col  Late-Col Excess-Col Carri-Sen     Runts    Giants
Gi0/1         0        0       0        0       0       0       0
Gi0/2         0        0       0        0       0       0       0
```

This is an example of output from the **show interfaces counters multicast** command. It displays the dropped multicast traffic for all interfaces.

```
Switch# show interfaces counters multicast

Port      McastSuppDiscards
Gi0/1                     0
Gi0/2                     0
```

This is an example of output from the **show interfaces counters trunk** command. It displays the trunk counters for all interfaces.

```
Switch# show interfaces counters trunk

Port        TrunkFramesTx  TrunkFramesRx  WrongEncap
Gi0/1                   0              0           0
Gi0/2                   0              0           0
```

This is an example of output from the **show interfaces counters unicast** command. It displays the dropped unicast traffic for all interfaces.

```
Switch# show interfaces counters unicast

Port      UcastSuppDiscards
Gi0/1                  6872
Gi0/2                     0
```

Related Commands

Command	Description
show interfaces	Displays interface characteristics.
storm-control	Configures broadcast, multicast, and unicast storm control for an interface.

show ip access-lists

Use the **show ip access-lists** privileged EXEC command to display IP access control lists (ACLs) configured on the switch.

show ip access-lists [*name* | *number*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

name	(Optional) ACL name.
number	(Optional) ACL number. The range is from 1 to 199 and from 1300 to 2699.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show ip access-lists** command:

```
Switch# show ip access-lists
Standard IP access list testingacl
    permit 10.10.10.2
Standard IP access list wizard_1-1-1-2
    permit 1.1.1.2
Extended IP access list 103
    permit tcp any any eq www
Extended IP access list CMP-NAT-ACL
    Dynamic Cluster-HSRP deny   ip any any
    Dynamic Cluster-NAT permit ip any any
      permit ip host 10.245.155.128 any
      permit ip host 10.245.137.0 any
      permit ip host 10.146.106.192 any
      permit ip host 10.216.25.128 any
      permit ip host 10.228.215.0 any
      permit ip host 10.221.111.64 any
      permit ip host 10.123.222.192 any
      permit ip host 10.169.110.128 any
      permit ip host 10.186.122.64 any
```

This is an example of output from the **show ip access-lists 103** command:

```
Switch# show ip access-lists 103
Extended IP access list 103
    permit tcp any any eq www
```

Related Commands

Command	Description
access-list (IP extended)	Configures an extended IP ACL on the switch.
access-list (IP standard)	Configures a standard IP ACL on the switch.
ip access-list	Configures an IP ACL on the switch.
show access-lists	Displays ACLs configured on a switch.

show ip igmp snooping

Use the **show ip igmp snooping** privileged EXEC command to display the Internet Group Management Protocol (IGMP) snooping configuration of the switch or the VLAN.

show ip igmp snooping [**vlan** *vlan-id*] [| {**begin** | **exclude** | **include**} *expression*]

show ip igmp snooping [**vlan** *vlan-id*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

vlan *vlan-id*	(Optional) Keyword and variable to specify a VLAN; valid values are 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Use this command to display snooping characteristics for the switch or for a specific VLAN.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show ip igmp snooping** command:

```
Switch# show ip igmp snooping

vlan 1
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is enabled on this Vlan
  IGMP snooping mrouter learn mode is pim-dvmrp on this Vlan
vlan 2
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is enabled on this Vlan
  IGMP snooping mrouter learn mode is cgmp on this Vlan
```

```
vlan 3
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is disabled on this Vlan
  IGMP snooping mrouter learn mode is cgmp on this Vlan
vlan 4
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is disabled on this Vlan
  IGMP snooping mrouter learn mode is cgmp on this Vlan
vlan 5
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is disabled on this Vlan
  IGMP snooping mrouter learn mode is pim-dvmrp on this Vlan
vlan 33
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is disabled on this Vlan
  IGMP snooping mrouter learn mode is pim-dvmrp on this Vlan
```

This is an example of output from the **show ip igmp snooping vlan 1** command:

```
Switch# show ip igmp snooping vlan 1

vlan 1
----------
  IGMP snooping is globally enabled
  IGMP snooping is enabled on this Vlan
  IGMP snooping immediate-leave is enabled on this Vlan
  IGMP snooping mrouter learn mode is pim-dvmrp on this Vlan
```

Related Commands

Command	Description
ip igmp snooping	Enables IGMP snooping.
ip igmp snooping vlan *vlan-id*	Enables IGMP snooping on the VLAN interface.
ip igmp snooping vlan immediate-leave	Configures IGMP Immediate-Leave processing.
ip igmp snooping vlan mrouter	Configures a Layer 2 port as a multicast router port.
show mac address-table multicast	Displays the Layer 2 multicast entries for a VLAN.

show ip igmp snooping mrouter

Use the **show ip igmp snooping mrouter** privileged EXEC command to display information on dynamically learned and manually configured multicast router ports.

show ip igmp snooping mrouter [**vlan** *vlan-id*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

vlan *vlan-id*	(Optional) Keyword and variable to specify a VLAN; valid values are 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

You can also use the **show mac address-table multicast** command to display entries in the MAC address table for a VLAN that has Internet Group Management Protocol (IGMP) snooping enabled.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show ip igmp snooping mrouter vlan 1** command:



Note In this example, `Fa0/3` is a dynamically learned router port, and `Fa0/2` is a configured static router port.

```
Switch# show ip igmp snooping mrouter vlan 1

Vlan    ports
----    -----
   1    Fa0/2(static), Fa0/3(dynamic)
```

Related Commands

Command	Description
ip igmp snooping	Enables IGMP snooping.
ip igmp snooping vlan	Enables IGMP snooping on the VLAN interface.
ip igmp snooping vlan immediate-leave	Configures IGMP Immediate-Leave processing.
ip igmp snooping vlan mrouter	Configures a Layer 2 port as a multicast router port.
show mac address-table multicast	Displays the Layer 2 multicast entries for a VLAN.

show mac access-group

Use the **show mac access-group** user EXEC command to display the MAC access control lists (ACLs) configured for an interface or a switch.

show mac access-group [**interface** *interface-id*] [**|** {**begin** | **exclude** | **include**} *expression*]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

interface *interface-id*	(Optional) Display the ACLs configured on a specific interface (only available in privileged EXEC mode).
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use the **show mac access-group** command without keywords to display MAC ACLs for all interfaces.

Use this command with the **interface** keyword to display ACLs for a specific interface.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mac access-group** command:

```
Switch> show mac access-group
Interface FastEthernet0/1:
   Inbound access-list is not set
Interface FastEthernet0/2:
   Inbound access-list is not set
Interface FastEthernet0/3:
   Inbound access-list is not set
Interface FastEthernet0/4:
   Inbound access-list is not set
...
Interface FastEthernet0/47:
   Inbound access-list is not set
Interface FastEthernet0/48:
   Inbound access-list is not set
Interface GigabitEthernet0/1:
   Inbound access-list is not set
```

```
Interface GigabitEthernet0/2:
   Inbound access-list is 101
```

This is an example of output from the **show mac access-group interface gigabitethernet 0/2** command:

```
Switch# show mac access-group interface gigabitethernet 0/2
Interface GigabitEthernet0/2:
   Inbound access-list is 101
```

Related Commands

Command	Description
mac access-group	Applies a MAC ACL to an interface.

show mac address-table

Use the **show mac address-table** user EXEC command to display the MAC address table.

show mac address-table [aging-time | **count** | **dynamic** | **static**] [**address** *hw-addr*] [**interface** *interface-id*] [**vlan** *vlan-id*] [| {**begin** | **exclude** | **include**} *expression*]

Note Beginning with Release 12.1(11)EA1, the **show mac address-table** command replaces the **show mac-address-table** command (with the hyphen). The **show mac-address-table** command (with the hyphen) will become obsolete in a future release.

Syntax Description

aging-time	(Optional) Display aging time for dynamic addresses for all VLANs.
count	(Optional) Display the count for different kinds of MAC addresses (only available in privileged EXEC mode).
dynamic	(Optional) Display only the dynamic addresses.
static	(Optional) Display only the static addresses.
address *hw-addr*	(Optional) Display information for a specific address (only available in privileged EXEC mode).
interface *interface-id*	(Optional) Display addresses for a specific interface.
vlan *vlan-id*	(Optional) Display addresses for a specific VLAN. Valid IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **show mac-address table secure** command was replaced by the **show port-security** command. The **self** keyword is not supported in this release or later.
12.1(11)EA1	The **show mac-address-table** command was replaced by the **show mac address-table** command.

Usage Guidelines

This command displays the MAC address table for the switch. Specific views can be defined by using the optional keywords and values. If more than one optional keyword is used, all of the conditions must be true in order for that entry to appear.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mac address-table** command:

```
Switch> show mac-address-table

Dynamic Addresses Count:                9
Secure Addresses (User-defined) Count: 0
Static Addresses (User-defined) Count: 0
System Self Addresses Count:            41
Total MAC addresses:                    50
Non-static Address Table:
Destination Address  Address Type  VLAN  Destination Port
-------------------  ------------  ----  --------------------
0010.0de0.e289       Dynamic          1  FastEthernet0/1
0010.7b00.1540       Dynamic          2  FastEthernet0/5
0010.7b00.1545       Dynamic          2  FastEthernet0/5
0060.5cf4.0076       Dynamic          1  FastEthernet0/1
0060.5cf4.0077       Dynamic          1  FastEthernet0/1
0060.5cf4.1315       Dynamic          1  FastEthernet0/1
0060.70cb.f301       Dynamic          1  FastEthernet0/1
00e0.1e42.9978       Dynamic          1  FastEthernet0/1
00e0.1e9f.3900       Dynamic          1  FastEthernet0/1
```

This is an example of output from the **show mac address-table static interface fastethernet0/2 vlan 1** command:

```
Switch> show  mac address-table static interface fastethernet0/2 vlan 1
vlan   mac address     type     ports
-----+---------------+--------+---------
   1  abcd.2345.0099  static    Fa0/2
   1  abcd.0070.0070  static    Fa0/2
   1  abcd.2345.0099  static    Fa0/2
   1  abcd.2345.0099  static    Fa0/2
   1  00d0.d333.7f34  static    Fa0/2
   1  abcd.2345.0099  static    Fa0/2
   1  0005.6667.0007  static    Fa0/2
```

This is an example of output from the **show mac address-table count vlan 1** command:

```
Switch# show  mac address-table count vlan 1
MAC Entries for Vlan 1 :
Dynamic Address Count: 1
Static Address (User-defined) Count: 41
Total MAC Addresses In Use:42
Remaining MAC addresses: 8150
```

This is an example of output from the **show mac address-table aging-time** command:

```
Switch> show  mac address-table aging-time
Vlan Aging Time
---- ----------
1    450
2    300
3    600
300  450
301  450
```

This is an example of output from the **show mac address-table aging-time vlan 1** command:

```
Switch> show  mac address-table aging-time vlan 1
Vlan Aging Time
---- ----------
1    450
```

Related Commands

Command	Description
clear mac address-table dynamic	Deletes from the MAC address table a specific dynamic address, all dynamic addresses on a particular interface, or all dynamic addresses on a particular VLAN.

show mac address-table multicast

Use the **show mac address-table multicast** user EXEC command to display the Layer 2 multicast entries for the switch or for the VLAN.

show mac address-table multicast [**vlan** *vlan-id*] [**count**] [**igmp-snooping** | **user**] [| {**begin** | **exclude** | **include**} *expression*]

Note Beginning with Release 12.1(11)EA1, the **show mac address-table multicast** command replaces the **show mac-address-table multicast** command (with the hyphen). The **show mac-address-table multicast** command (with the hyphen) will become obsolete in a future release.

Syntax Description

vlan *vlan-id*	(Optional) Specify a VLAN; valid values are 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros. (This keyword is only available in privileged EXEC mode.)
count	(Optional) Display total number of entries for the specified criteria instead of the actual entries (only available in privileged EXEC mode).
igmp-snooping	(Optional) Display only entries learned through Internet Group Management Protocol (IGMP) snooping (only available in privileged EXEC mode).
user	(Optional) Display only the user-configured multicast entries (only available in privileged EXEC mode).
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Defaults

This command has no default setting.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(11)EA1	The **show mac-address-table multicast** command was replaced by the **show mac address-table multicast** command.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mac address-table multicast vlan 1** command:

```
Switch# show  mac address-table multicast vlan 1

Vlan    Mac Address     Type    Ports
----    -----------     ----    -----
   1    0100.5e00.0128  IGMP    Fa0/11
   1    0100.5e01.1111  USER    Fa0/5, Fa0/6, Fa0/7, Fa0/11
```

This is an example of output from the **show mac address-table multicast count** command:

```
Switch# show  mac address-table multicast count
Multicast Mac Entries for all vlans: 10
```

This is an example of output from the **show mac address-table multicast vlan 1 count** command:

```
Switch# show  mac address-table multicast vlan 1 count
Multicast Mac Entries for vlan 1: 2
```

This is an example of output from the **show mac address-table multicast vlan 1 user** command:

```
Switch# show  mac address-table multicast vlan 1 user
vlan   mac address      type        ports
-----+----------------+-------+--------------------
1     0100.5e02.0203    user     Fa0/1,Fa0/2,Fa0/4
```

This is an example of output from the **show mac address-table multicast vlan 1 igmp-snooping count** command:

```
Switch# show  mac address-table multicast vlan 1 igmp-snooping count
Number of igmp-snooping programmed entries : 1
```

show mac address-table notification

Use the **show mac address-table notification** user EXEC command to display parameters for the MAC notification feature.

show mac address-table notification [**interface** *interface-id*] [**|** {**begin** | **exclude** | **include**} *expression*]

Note Beginning with Release 12.1(11)EA1, the **show mac address-table notification** command replaces the **show mac-address-table notification** command (with the hyphen). The **show mac-address-table notification** command (with the hyphen) will become obsolete in a future release.

Syntax Description

interface *interface-id*	(Optional) Specify an interface.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Defaults

This command has no default setting.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(11)EA1	The **show mac-address-table notification** command was replaced by the **show mac address-table notification** command.

Usage Guidelines

Use the **show mac address-table notification** command without keywords to display parameters for all interfaces.

Use this command with the **interface** keyword to display parameters for a specific interface.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mac address-table notification** command:

```
Switch> show mac address-table notification
MAC Notification Feature is Disabled on the switch
```

Related Commands

Command	Description
clear mac address-table notification	Clears the MAC address notification global counters.
mac address-table notification	Enables the MAC notification feature.
snmp trap mac-notification	Enables MAC-notification traps on a port.

show mls masks

Use the **show mls masks** user EXEC command to display the details of the Access Control Parameters (ACPs) used for quality of service (QoS) and security access control lists (ACLs).

show mls masks [**qos | security**] [| {**begin** | **exclude** | **include**} *expression*]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

qos	(Optional) Display ACPs used for QoS ACLs.
security	(Optional) Display ACPs used for security ACLs.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Note ACPs are called masks in the command-line interface (CLI) commands and output.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use the **show mls mask** command without keywords to display all ACPs configured on the switch.

Use this command with the **qos** keyword to display the ACPs used for QoS ACLs.

Use this command with the **security** keyword to display the ACPs used for security ACLs.

Note You can configure up to four ACPs (QoS and security) on a switch.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mls masks** command:

```
Switch> show mls masks

Mask1
        Type : qos
        Fields : ip-sa(0.0.0.255), ip-da(host), dest-port, ip-dscp
        Policymap: pmap1
            Interfaces: Fa0/9, Gi0/1
        Policymap: pmap2
            Interfaces: Fa0/1, Fa0/5, Fa0/13
Mask2
        Type : security
        Fields : mac-sa (host), ethertype, ip-dscp
        Access-group: 3
            Interfaces: Fa0/2, Fa0/6
        Access-group: macag1
            Interfaces: Fa0/16
```

In this example, *Mask 1* is a QoS ACP consisting an IP source address (with wildcard bits 0.0.0.255), an IP destination address, and Layer 4 destination port fields. This ACP is used by the QoS policy maps *pmap1* and *pmap2*.

Mask 2 is a security ACP consisting of a MAC source address and ethertype fields. This ACP is used by the MAC security access groups *3* and *macag1*.

Related Commands

Command	Description
ip access-group	Applies an IP ACL to an interface.
mac access-group	Applies a named extended MAC ACL to an interface.
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces and enters policy-map configuration mode.

show mls qos interface

Use the **show mls qos interface** user EXEC command to display quality of service (QoS) information at the interface level.

show mls qos interface [*interface-id*] [**policers**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface-id	(Optional) Display QoS information for the specified interface.
policers	(Optional) Display all the policers configured on the interface, their settings, and the number of policers unassigned (available only when the switch is running the enhanced software image (EI)).
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Note Though visible in the command-line help strings, the **vlan** *vlan-id* option is not supported.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Though visible in the command-line help string, the **policers** keyword is available only when your switch is running the EI.

Use the **show mls qos interface** command without keywords to display parameters for all interfaces.

Use the **show mls qos interface** *interface-id* command to display the parameters for a specific interface.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mls qos interface** command when the Cisco IP phone is a trusted device:

```
Switch> show mls qos interface fastethernet0/1
FastEthernet0/1
trust state:trust cos
trust mode:trust cos
COS override:dis
default COS:0
pass-through:none
trust device:cisco-phone
```

This is an example of output from the **show mls qos interface** command when pass-through mode is configured on an interface:

```
Switch> show mls qos interface fastethernet0/2
FastEthernet0/2
trust state:not trusted
trust mode:not trusted
COS override:dis
default COS:0
pass-through:dscp
```

Related Commands

Command	Description
mls qos cos	Defines the default class of service (CoS) value of a port or assigns the default CoS to all incoming packets on the port.
mls qos map	Defines the class of service (CoS)-to-Differentiated Services Code Point (DSCP) map and DSCP-to-CoS map.
mls qos trust	Configures the port trust state. Ingress traffic can be trusted and classification is performed by examining the CoS or DSCP value.

show mls qos maps

Use the **show mls qos maps** user EXEC command to display quality of service (QoS) mapping information. Maps are used to generate an internal Differentiated Services Code Point (DSCP) value, which represents the priority of the traffic.

show mls qos maps [**cos-dscp** | **dscp-cos**] [**|** {**begin** | **exclude** | **include**} *expression*]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

cos-dscp	(Optional) Display class of service (CoS)-to-DSCP map.
dscp-cos	(Optional) Display DSCP-to-CoS map.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use the **show mls qos maps** command without keywords to display all maps.

Use this command with the **cos-dscp** keyword to display the CoS-to-DSCP map.

Use this command with the **dscp-cos** keyword to display the DSCP-to-CoS map.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mls qos maps cos-dscp** command:

```
Switch> show mls qos maps cos-dscp

Cos-dscp map:
        cos:  0  1  2  3  4  5  6  7
     --------------------------------
       dscp:  8  8  8  8 24 32 56 56
```

This is an example of output from the **show mls qos maps dscp-cos** command:

```
Switch> show mls qos maps dscp-cos

Dscp-cos map:
       dscp:  0  8 10 16 18 24 26 32 34 40 46 48 56
     ---------------------------------------------
        cos:  0  1  1  1  2  2  3  3  4  4  5  6  7
```

This is an example of output from the **show mls qos maps** command:

```
Switch> show mls qos maps

Dscp-cos map:
       dscp:  0  8 10 16 18 24 26 32 34 40 46 48 56
     ---------------------------------------------
        cos:  0  1  1  2  2  3  7  4  4  5  5  7  7

   Cos-dscp map:
        cos:  0  1  2  3  4  5  6  7
     --------------------------------
       dscp:  0  8 16 24 32 40 48 56
```

Related Commands

Command	Description
mls qos map	Defines the CoS-to-DSCP map and DSCP-to-CoS map.

show monitor

Use the **show monitor** user EXEC command to display Switched Port Analyzer (SPAN) and Remote SPAN (RSPAN) session information.

show monitor [**session** {*session_number* | **all** | **local** | **remote**}] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

session *session_number*	(Optional) Specify the session number identified with this SPAN or RSPAN session.
all	Specify all sessions.
local	Specify local sessions.
remote	Specify remote sessions.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA1	This command was first introduced.
12.1(11)EA1	The **all**, **local**, and **remote** keywords were added.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* are not displayed, but the lines that contain *Output* are displayed.

Examples

This is an example of output for the **show monitor** privileged EXEC command for RSPAN source session 1:

```
Switch# show monitor session 1
Session 1
---------
Type: Remote Source Session
Source Ports:
    RX Only:     Fa0/3
    TX Only:       None
    Both:          None
Source VLANs:
    RX Only:       None
    TX Only:       None
    Both:          None
Source RSPAN VLAN: None
Destination Ports: None
   Encapsulation: Native
Reflector Port:  Fa0/4
Filter VLANs:      None
Dest RSPAN VLAN: 901
```

Related Commands

Command	Description
monitor session	Enables SPAN and RSPAN monitoring on a port and configures a port as a source or destination port.

show mvr

Use the **show mvr** privileged EXEC command without keywords to display the Multicast VLAN Registration (MVR) global parameter values, including whether or not MVR is enabled, the MVR multicast VLAN, the maximum query response time, the number of multicast groups, and the MVR mode (dynamic or compatible).

show mvr [| {**begin | exclude | include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mvr** command:

```
Switch# show mvr
MVR Running: TRUE
MVR multicast vlan: 1
MVR Max Multicast Groups: 256
MVR Current multicast groups: 256
MVR Global query response time: 5 (tenths of sec)
MVR Mode: compatible
```

In the previous example, the maximum number of multicast groups is 256. The MVR mode is either compatible (for interoperability with Catalyst 2900 XL and Catalyst 3500 XL switches) or dynamic (where operation is consistent with Internet Group Management Protocol [IGMP] snooping operation, and dynamic MVR membership on source ports is supported).

Related Commands

Command	Description
mvr	Enables and configures multicast VLAN registration on the switch.
mvr type	Configures an MVR port as a receiver or a source port.
show mvr interface	Displays the configured MVR interfaces, status of the specified interface, or all multicast groups to which the interface belongs.
show mvr members	Displays all ports that are members of an MVR multicast group.

show mvr interface

Use the **show mvr interface** privileged EXEC command without keywords to display the Multicast VLAN Registration (MVR) receiver and source ports. Use the command with keywords to display MVR parameters for a specific receiver port.

show mvr interface [*interface-id* [**members** [**vlan** *vlan-id*]] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface-id	(Optional) Display MVR type, status, and Immediate-Leave setting for the interface.
members	(Optional) Display all MVR groups to which the specified interface belongs.
vlan *vlan-id*	(Optional) Display the VLAN to which the receiver port belongs.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If the entered port identification is a non-MVR port or a source port, the command returns an error message. For receiver ports, it displays the port type, per port status, and Immediate-Leave setting.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mvr interface** command:

```
Switch# show mvr interface
Port    Type            Status          Immediate Leave
----    ----            -------         ---------------
Gi0/1   SOURCE          ACTIVE/UP       DISABLED
Gi0/2   RECEIVER        ACTIVE/DOWN     DISABLED
```

In the previous example, Status is defined as:

- Active means the port is part of a VLAN.
- Up/Down means that the port is forwarding/nonforwarding.
- Inactive means that the port is not part of any VLAN.

This is an example of output from the **show mvr interface gigabitethernet0/2** command:

```
Switch# show mvr interface gigabitethernet0/2
Type: RECEIVER Status: ACTIVE Immediate Leave: DISABLED
```

This is an example of output from the **show mvr interface fastethernet0/6 member** command:

```
Switch# show mvr interface fastethernet0/6 member
239.255.0.0     DYNAMIC ACTIVE
239.255.0.1     DYNAMIC ACTIVE
239.255.0.2     DYNAMIC ACTIVE
239.255.0.3     DYNAMIC ACTIVE
239.255.0.4     DYNAMIC ACTIVE
239.255.0.5     DYNAMIC ACTIVE
239.255.0.6     DYNAMIC ACTIVE
239.255.0.7     DYNAMIC ACTIVE
239.255.0.8     DYNAMIC ACTIVE
239.255.0.9     DYNAMIC ACTIVE
```

Related Commands

Command	Description
mvr	Enables and configures multicast VLAN registration on the switch.
mvr type	Configures an MVR port as a receiver or a source port.
show mvr	Displays the global MVR configuration on the switch.
show mvr members	Displays all receiver ports that are members of an MVR multicast group.

show mvr members

Use the **show mvr members** privileged EXEC command to display all receiver ports that are members of an IP multicast group.

show mvr members [*ip-address*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

ip-address	(Optional) The IP multicast address. If the address is entered, all receiver ports that are members of the multicast group appear. If no address is entered, all members of all Multicast VLAN Registration (MVR) groups are listed. If a group has no members, the group is listed as *Inactive*.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **show mvr members** command only applies to receiver ports. All source ports are members of all multicast groups.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show mvr members** command:

```
Switch# show mvr members
MVR Group IP    Status          Members
------------    ------          -------
239.255.0.1     ACTIVE          Gi0/1(d), Gi0/2(s)
239.255.0.2     INACTIVE        None
239.255.0.3     INACTIVE        None
239.255.0.4     INACTIVE        None
239.255.0.5     INACTIVE        None
239.255.0.6     INACTIVE        None
239.255.0.7     INACTIVE        None
239.255.0.8     INACTIVE        None
239.255.0.9     INACTIVE        None
239.255.0.10    INACTIVE        None

<output truncated>

239.255.0.255   INACTIVE        None
239.255.1.0     INACTIVE        None
```

This is an example of output from the **show mvr members 239.255.0.2** command. It shows how to view the members of the IP multicast group 239.255.0.2.

```
Switch# show mvr member 239.255.0.2
239.255.0.2     ACTIVE          Gi0/1(d), Gi0/2(d)
```

Related Commands

Command	Description
mvr	Enables and configures multicast VLAN registration on the switch.
mvr type	Configures an MVR port as a receiver or a source port.
show mvr	Displays the global MVR configuration on the switch.
show mvr interface	Displays the configured MVR interfaces, status of the specified interface, or all multicast groups to which the interface belongs.

show pagp

Use the **show pagp** user EXEC command to display Port Aggregation Protocol (PAgP) channel-group information.

show pagp [*channel-group-number*] {**counters** | **internal** | **neighbor**} [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

channel-group-number	(Optional) Number of the channel group. Valid numbers range from 1 to 6.
counters	Display traffic information.
internal	Display internal information.
neighbor	Display neighbor information.
| **begin**	(Optional) Display begins with the line that matches the *expression*.
| **exclude**	(Optional) Display excludes lines that match the *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

You can enter any **show pagp** command to display the active port channel information. To display the nonactive information, enter the **show pagp** command with a group number.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show pagp 1 counters** command:

```
Switch> show pagp 1 counters
          Information        Flush
Port      Sent   Recv     Sent   Recv
---------------------------------------
Channel group: 1
  Gi0/1    45     42       0      0
  Gi0/2    45     41       0      0
```

This is an example of output from the **show pagp 1 internal** command:

```
Switch> show pagp 1 internal
Flags:  S - Device is sending Slow hello.  C - Device is in Consistent state.
        A - Device is in Auto mode.
Timers: H - Hello timer is running.        Q - Quit timer is running.
        S - Switching timer is running.    I - Interface timer is running.

Channel group 1
                              Hello    Partner  PAgP     Learning  Group
Port      Flags State   Timers  Interval Count   Priority   Method  Ifindex
Gi0/1     SC    U6/S7   H       30s      1        128        Any      16
Gi0/2     SC    U6/S7   H       30s      1        128        Any      16
```

This is an example of output from the **show pagp 1 neighbor** command:

```
Switch> show pagp 1 neighbor
   Flags:  S - Device is sending Slow hello.  C - Device is in Consistent state.
           A - Device is in Auto mode.        P - Device learns on physical port.

   Channel group 1 neighbors
             Partner             Partner          Partner         Partner Group
   Port      Name                Device ID        Port       Age  Flags   Cap.
   Gi0/1     device-p2           0002.4b29.4600   Gi0/1        9s SC      10001
   Gi0/2     device-p2           0002.4b29.4600   Gi0/2       24s SC      10001
```

Related Commands

Command	Description
clear pagp	Clears PAgP channel-group information.
pagp learn-method	Sets the source-address learning method of incoming packets received from an EtherChannel port.

show policy-map

Use the **show policy-map** user EXEC command to display quality of service (QoS) policy maps, which define classification criteria for incoming traffic. Policy maps can include policers that specify the bandwidth limitations and the action to take if the limits are exceeded.

show policy-map [*policy-map-name* [**class** *class-name*]] [| {**begin** | **exclude** | **include**} *expression*]

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

policy-map-name	(Optional) Display the specified policy-map name.
class *class-name*	(Optional) Display QoS policy actions for a individual class.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Use the **show policy-map** command without keywords to display all policy maps configured on the switch.

Note In a policy map, the class named *class-default* is not supported. The switch does not filter traffic based on the policy map defined by the **class class-default** policy-map configuration command.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show policy-map** command:

```
Switch> show policy-map
 Policy Map bumbum
    Description: this is a description.

 Policy Map wizard_policy3
  class wizard_1-1-1-2
   set ip dscp 34
```

```
 Policy Map test

 Policy Map policytest
  class  classtest
   set ip dscp 20
   police 10000000 8192 exceed-action drop
```

This is an example of output from the **show policy-map policytest** command:

```
Switch> show policy-map policytest
 Policy Map policytest
  class  classtest
   set ip dscp 20
   police 10000000 8192 exceed-action drop
```

This is an example of output from the **show policy-map policytest class classtest** command:

```
Switch> show policy-map policytest class classtest
   set ip dscp 20
   police 10000000 8192 exceed-action drop
```

Related Commands

Command	Description
policy-map	Creates or modifies a policy map that can be attached to multiple interfaces to specify a service policy.

show port-security

Use the **show port-security** privileged EXEC command to display the port security settings defined for an interface or for the switch.

show port-security [**interface** *interface-id*] [**address**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface *interface-id*	(Optional) Display the port security settings for the specified interface.
address	(Optional) Display all the secure addresses on all ports.
| **begin**	(Optional) Display begins with the line that matches the specified *expression*.
| **exclude**	(Optional) Display excludes lines that match the specified *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **show port security** and **show mac-address-table secure** commands.

Usage Guidelines

If you enter this command without keywords, the output includes the administrative and the operational status of all secure ports on the switch.

If you enter an *interface-id*, the **show port-security** command displays port security settings for the interface.

If you enter the **address** keyword, the **show port-security address** command displays the secure MAC addresses for all interfaces and the aging information for each secure address.

If you enter an *interface-id* and the **address** keyword, the **show port-security interface** *interface-id* **address** command displays all the MAC addresses for the interface with aging information for each secure address. You can also use this command to display all the MAC addresses for an interface even if you have not enabled port security on it.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show port-security** command:

```
Switch# show port-security
Secure Port      MaxSecureAddr  CurrentAddr  SecurityViolation  Security
Action
                    (Count)        (Count)      (Count)
------------------------------------------------------------------------------
---
     Fa0/1           11            11            0            Shutdown
     Fa0/5           15            5             0            Restrict
     Fa0/11          5             4             0            Protect
------------------------------------------------------------------------------
---
Total Addresses in System :21
Max Addresses limit in System :1024
```

This is an example of output from the **show port-security interface fastethernet0/2** command:

```
Switch# show port-security interface fastethernet0/2
Port Security :Enabled
Port status :SecureUp
Violation mode :Shutdown
Maximum MAC Addresses :11
Total MAC Addresses :11
Configured MAC Addresses :3
Aging time :20 mins
Aging type :Inactivity
SecureStatic address aging :Enabled
Security Violation count :0
```

This is an example of output from the **show port-security address** command:

```
Switch# show port-security address

Secure Mac Address Table
------------------------------------------------------------------------
Vlan    Mac Address       Type                Ports   Remaining Age
                                                         (mins)
----    -----------       ----                -----   -------------
   1    0001.0001.0001    SecureDynamic       Fa0/1      15 (I)
   1    0001.0001.0002    SecureDynamic       Fa0/1      15 (I)
   1    0001.0001.1111    SecureConfigured    Fa0/1      16 (I)
   1    0001.0001.1112    SecureConfigured    Fa0/1      -
   1    0001.0001.1113    SecureConfigured    Fa0/1      -
   1    0005.0005.0001    SecureConfigured    Fa0/5      23
   1    0005.0005.0002    SecureConfigured    Fa0/5      23
   1    0005.0005.0003    SecureConfigured    Fa0/5      23
   1    0011.0011.0001    SecureConfigured    Fa0/11     25 (I)
   1    0011.0011.0002    SecureConfigured    Fa0/11     25 (I)
------------------------------------------------------------------------
Total Addresses in System :10
Max Addresses limit in System :1024
```

This is an example of output from the **show port-security interface fastethernet0/5 address** command:

```
Switch# show port-security interface fastethernet0/5 address
Secure Mac Address Table
------------------------------------------------------------------------
Vlan    Mac Address       Type                Ports   Remaining Age
                                                         (mins)
----    -----------       ----                -----   -------------
   1    0005.0005.0001    SecureConfigured    Fa0/5      19 (I)
   1    0005.0005.0002    SecureConfigured    Fa0/5      19 (I)
   1    0005.0005.0003    SecureConfigured    Fa0/5      19 (I)
------------------------------------------------------------------------
Total Addresses:3
```

Related Commands

Command	Description
switchport port-security	Enables port security on a port, restricts the use of the port to a user-defined group of stations, and configures secure MAC addresses.

show rps

Use the **show rps** privileged EXEC command to display the status of the Cisco Redundant Power System (RPS).

show rps [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show rps** command. Table 2-13 describes the possible output.

```
Switch# show rps
GREEN
```

Table 2-13 show rps Output Description

Display	Description
BLACK	The RPS is off or not properly connected.
GREEN	The RPS is connected and ready to provide back-up power, if required.
ALT_GREEN_BLACK	The RPS is connected but is unavailable because it is providing power to another device (redundancy has been allocated to a neighboring device).

Table 2-13 show rps Output Description (continued)

Display	Description
ALT_AMBER_BLACK	The internal power supply in the switch has failed, and the RPS is providing power to the switch (redundancy has been allocated to this device).
AMBER	The RPS is in standby mode, or the RPS has detected a failure. Press the Standby/Active button on the RPS to put the RPS in active mode. If the RPS LED on the switch remains amber, the RPS has detected a failure. If the failure is minor, the RPS might be in any of the previously described modes. If the failure is critical, the RPS will be down. RPS failures include these modes: • The RPS +12V or -48V voltages exceed the specified thresholds. • The RPS has a fan failure. • The RPS detects excessive temperature. • The RPS has a faulty connection to the switch.

show running-config vlan

Use the **show running-config vlan** privileged EXEC command to display all or a range of VLAN-related configurations on the switch.

show running-config vlan [*vlan-ids*] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

vlan-ids	(Optional) Display configuration information for a single VLAN identified by VLAN ID number or a range of VLANs separated by a hyphen. For *vlan-id*, the range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* are not displayed, but the lines that contain *Output* are displayed.

Examples

This is an example of output from the **show running-config vlan** command:

```
Switch# show running-config vlan 900-2005
Building configuration...

Current configuration:
!
vlan 907
!
vlan 920
!
vlan 1025
!
vlan 2000
!
vlan 2001
end
```

Related Commands

Command	Description
show running-config	Displays the running configuration on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
vlan (global configuration)	Enters config-vlan mode for creating and editing VLANs. When VLAN Trunking Protocol (VTP) mode is transparent, you can use this mode to create extended-range VLANs (VLAN IDs greater than 1005).
vlan database	Enters VLAN configuration mode for creating and editing normal-range VLANs.

show spanning-tree

Use the **show spanning-tree** user EXEC command to display spanning-tree state information.

show spanning-tree [**active** [**detail**] | **backbonefast** | **blockedports** | **bridge** | **detail** [**active**] | **inconsistentports** | **interface** *interface-id* | **mst** | **pathcost method** | **root** | **summary** [**totals**] | **uplinkfast** | **vlan** *vlan-id*] [| {**begin** | **exclude** | **include**} *expression*]

show spanning-tree vlan *vlan-id* [**active** [**detail**] | **blockedports** | **bridge** | **detail** [**active**] | **inconsistentports** | **interface** *interface-id* | **root** | **summary**] [| {**begin** | **exclude** | **include**} *expression*

show spanning-tree {**vlan** *vlan-id*} **bridge** [**address** | **detail** | **forward-time** | **hello-time** | **id** | **max-age** | **priority** [**system-id**] | **protocol**] [| {**begin** | **exclude** | **include**} *expression*]

show spanning-tree {**vlan** *vlan-id*} **root** [**address** | **cost** | **detail** | **forward-time** | **hello-time** | **id** | **max-age** | **port** | **priority** [**system-id**] [| {**begin** | **exclude** | **include**} *expression*]

show spanning-tree interface *interface-id* [**active** [**detail**] | **cost** | **detail** [**active**] | **inconsistency** | **portfast** | **priority** | **rootcost** | **state**] [| {**begin** | **exclude** | **include**} *expression*]

show spanning-tree mst [**configuration** | *instance-id*] [**detail** | **interface** *interface-id* [**detail**]] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

active [**detail**]	(Optional) Display spanning-tree information only on active interfaces (only available in privileged EXEC mode).
backbonefast	(Optional) Display spanning-tree BackboneFast status (only available in privileged EXEC mode).
blockedports	(Optional) Display blocked port information (only available in privileged EXEC mode).
bridge [**address** | **detail** | **forward-time** | **hello-time** | **id** | **max-age** | **priority** [**system-id**] | **protocol**]	(Optional) Display status and configuration of this switch (only available in privileged EXEC mode).
detail [**active**]	(Optional) Display a detailed summary of interface information (only available in privileged EXEC mode).
inconsistentports	(Optional) Display inconsistent port information (only available in privileged EXEC mode).
interface *interface-id* [**active** [**detail**] | **cost** | **detail** [**active**] | **inconsistency** | **portfast** | **priority** | **rootcost** | **state**]	(Optional) Display spanning-tree information for the specified interface (all options only available in privileged EXEC mode). Enter each interface separated by a space. Ranges are not supported. Valid interfaces include physical ports, VLANs, and port channels. The valid VLAN range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros. The valid port-channel range is 1 to 6.

mst [**configuration** | *instance-id*] [**detail** | **interface** *interface-id* [**detail**]]	These keywords and options are available only if your switch is running the EI. (Optional) Display the multiple spanning-tree (MST) region configuration and status (all options only available in privileged EXEC mode). Valid interfaces include physical ports, VLANs, and port channels. The valid VLAN range is 1 to 4094; do not enter leading zeros. The valid port-channel range is 1 to 6.
pathcost method	(Optional) Display the default path cost method (only available in privileged EXEC mode).
root [**address** | **cost** | **detail** | **forward-time** | **hello-time** | **id** | **max-age** | **port** | **priority** [**system-id**]]	(Optional) Display root switch status and configuration (all options only available in privileged EXEC mode).
summary [**totals**]	(Optional) Display a summary of port states or the total lines of the spanning-tree state section (only available in privileged EXEC mode).
uplinkfast	(Optional) Display spanning-tree UplinkFast status (only available in privileged EXEC mode).
vlan *vlan-id* [**active** [**detail**] | **backbonefast** | **blockedports** | **bridge** [**address** | **detail** | **forward-time** | **hello-time** | **id** | **max-age** | **priority** [**system-id**] | **protocol**]	(Optional) Display spanning-tree information for the specified VLAN (only available in privileged EXEC mode). The range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
| **begin**	(Optional) Display begins with the line that matches the *expression*.
| **exclude**	(Optional) Display excludes lines that match the *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC; indicated keywords available only in privileged EXEC mode

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **active**, **backbonefast**, **blockedports**, **bridge**, **inconsistentports**, **pathcost method**, **root**, **total**, and **uplinkfast** keywords were added.
12.1(9)EA1	The **mst** keyword and options were added. The **brief** keyword was removed, and the **detail** keyword was added.

Usage Guidelines

If the *vlan-id* variable is omitted, the command applies to the spanning-tree instance for all VLANs.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* are not displayed, but the lines that contain *Output* are displayed.

Examples

This is an example of output from the **show spanning-tree active** command:

```
Switch# show spanning-tree active
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32768
             Address     0001.4297.e000
             Cost        57
             Port        1 (GigabitEthernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0002.4b29.7a00
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface        Port ID                     Designated                Port ID
Name             Prio.Nbr      Cost Sts      Cost Bridge ID            Prio.Nbr
---------------- -------- --------- --- --------- -------------------- --------
Gi0/1            128.1           19 FWD        38 32768 0030.9441.62c1 128.25
Gi0/2            128.2           19 FWD        57 32769 0002.4b29.7a00 128.2
Po1              128.65          19 FWD        57 32769 0002.4b29.7a00 128.65
```

This is an example of output from the **show spanning-tree detail** command:

```
Switch> show spanning-tree detail

VLAN0001 is executing the ieee compatible Spanning Tree protocol
 Bridge Identifier has priority 32768, sysid 1, address 0002.4b29.7a00
 Configured hello time 2, max age 20, forward delay 15
 Current root has priority 32768, address 0001.4297.e000
 Root port is 1 (GigabitEthernet0/1), cost of root path is 57
 Topology change flag not set, detected flag not set
 Number of topology changes 2 last change occurred 2d18h ago
         from Port-channel1
 Times:  hold 1, topology change 35, notification 2
         hello 2, max age 20, forward delay 15
 Timers: hello 0, topology change 0, notification 0, aging 300

Port 1 (GigabitEthernet0/1) of VLAN0001 is forwarding
  Port path cost 19, Port priority 128, Port Identifier 128.1.
  Designated root has priority 32768, address 0001.4297.e000
  Designated bridge has priority 32768, address 0030.9441.62c1
  Designated port id is 128.25, designated path cost 38
  Timers: message age 4, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  BPDU: sent 2, received 120638

<output truncated>
```

This is an example of output from the **show spanning-tree interface fastethernet 0/1** command:

```
Switch> show spanning-tree interface fastethernet0/1

Vlan             Port ID                     Designated                Port ID
Name             Prio.Nbr      Cost Sts      Cost Bridge ID            Prio.Nbr
---------------- -------- --------- --- --------- -------------------- --------
VLAN0001         128.1           19 FWD        38 32768 0030.9441.62c1 128.25
```

This is an example of output from the **show spanning-tree summary** command:

```
Switch> show spanning-tree summary
Root bridge for: none.
Extended system ID is enabled
PortFast BPDU Guard is disabled
EtherChannel misconfiguration guard is enabled
UplinkFast is disabled
BackboneFast is disabled
Default pathcost method used is short

Name                 Blocking Listening Learning Forwarding STP Active
-------------------- -------- --------- -------- ---------- ----------
VLAN1                13       0         0        1          14
VLAN2                1        0         0        1          2
VLAN3                1        0         0        1          2

<output truncated>
```

This is an example of output from the **show spanning-tree mst configuration** command:

```
Switch# show spanning-tree mst configuration
Name      [region1]
Revision  1
Instance  Vlans Mapped
--------  ------------------
0         1-9,21-4094
1         10-20
----------------------------
```

This is an example of output from the **show spanning-tree mst interface fastethernet0/1** command:

```
Switch# show spanning-tree mst interface fastethernet0/1

FastEthernet0/1 of MST00 is root forwarding
Edge port: no             (default)        port guard : none        (default)
Link type: point-to-point (auto)           bpdu filter: disable     (default)
Boundary : boundary       (STP)            bpdu guard : disable     (default)
Bpdus sent 5, received 74

Instance role state cost      prio vlans mapped
-------- ---- ----- --------- ---- --------------------------------------------
0        root FWD   200000    128  1,12,14-4094
```

This is an example of output from the **show spanning-tree mst 0** command:

```
Switch# show spanning-tree mst 0
###### MST00        vlans mapped: 1-9,21-4094
Bridge      address 0002.4b29.7a00  priority  32768 (32768 sysid 0)
Root        address 0001.4297.e000  priority  32768 (32768 sysid 0)
            port    Gi0/1           path cost 200038
IST master *this switch
Operational hello time 2, forward delay 15, max age 20, max hops 20
Configured  hello time 2, forward delay 15, max age 20, max hops 20

Interface            role state cost      prio type
-------------------- ---- ----- --------- ---- --------------------------------
GigabitEthernet0/1   root FWD   200000    128  P2P bound(STP)
GigabitEthernet0/2   desg FWD   200000    128  P2P bound(STP)
Port-channel1        desg FWD   200000    128  P2P bound(STP)
```

Related Commands

Command	Description
spanning-tree backbonefast	Enables the BackboneFast feature.
spanning-tree bpdufilter	Prevents a port from sending or receiving bridge protocol data units (BPDUs).
spanning-tree bpduguard	Puts a port in the error-disabled state when it receives a BPDU.
spanning-tree cost	Sets the path cost for spanning-tree calculations.
spanning-tree extend system-id	Enables the extended system ID feature.
spanning-tree guard	Enables the root guard or the loop guard feature for all the VLANs associated with the selected interface.
spanning-tree link-type	Overrides the default link-type setting for rapid spanning-tree transitions to the forwarding state.
spanning-tree loopguard default	Prevents alternate or root ports from becoming the designated port because of a failure that leads to a unidirectional link.
spanning-tree mst configuration	Enters multiple spanning-tree (MST) configuration mode through which the MST region configuration occurs.
spanning-tree mst cost	Sets the path cost for MST calculations.
spanning-tree mst forward-time	Sets the forward-delay time for all MST instances.
spanning-tree mst hello-time	Sets the interval between hello BPDUs sent by root switch configuration messages.
spanning-tree mst max-age	Sets the interval between messages that the spanning tree receives from the root switch.
spanning-tree mst max-hops	Sets the number of hops in an MST region before the BPDU is discarded and the information held for a port is aged.
spanning-tree mst port-priority	Configures an interface priority.
spanning-tree mst priority	Configures the switch priority for the specified spanning-tree instance.
spanning-tree mst root	Configures the MST root switch priority and timers based on the network diameter.
spanning-tree port-priority	Configures an interface priority.
spanning-tree portfast (global configuration)	Globally enables the BPDU filtering or the BPDU guard feature on Port Fast-enabled ports or enables the Port Fast feature on all nontrunking ports.
spanning-tree portfast (interface configuration)	Enables the Port Fast feature on an interface and all its associated VLANs.
spanning-tree uplinkfast	Accelerates the choice of a new root port when a link or switch fails or when the spanning tree reconfigures itself.
spanning-tree vlan	Configures spanning tree on a per-VLAN basis.

show storm-control

Use the **show storm-control** user EXEC command to display the packet-storm control information. This command also displays the action that the switch takes when the thresholds are reached.

show storm-control [*interface-id*] [{**broadcast** | **history** | **multicast** | **unicast** }] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface-id	(Optional) Port for which information is to be displayed.
broadcast	(Optional) Display broadcast storm information.
history	(Optional) Display storm history on a per-port basis.
multicast	(Optional) Display multicast storm information.
unicast	(Optional) Display unicast storm information.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **show port storm-control** command.

Usage Guidelines

If the variable *interface-id* is omitted, the **show storm-control** command displays storm-control settings for all ports on the switch.

You can display broadcast, multicast, or unicast packet-storm information by using the corresponding keyword. When no option is specified, the default is to display broadcast storm-control information.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show storm-control broadcast** command:

```
Switch> show storm-control broadcast

Interface  Filter State   Trap State     Upper    Lower    Current  Traps Sent
---------  -------------  -------------  -------  -------  -------  ----------
Fa0/1      <inactive>     <inactive>     100.00%  100.00%    0.00%           0
Fa0/2      <inactive>     <inactive>     100.00%  100.00%    0.00%           0
Fa0/3      <inactive>     <inactive>     100.00%  100.00%    0.00%           0
Fa0/4      Forwarding     Below rising    30.00%   20.00% 20.32%           17
. . . .
```

Table 2-14 lists the **show storm-control** field descriptions.

Table 2-14 show storm-control Field Descriptions

Field	Description
Interface	Displays the ID of the interface.
Filter State	Displays the status of the filter: • Blocking—Storm control is enabled, action is filter, and a storm has occurred. • Forwarding—Storm control is enabled, and a storm has not occurred. • Inactive—Storm control is disabled. • Shutdown—Storm control is enabled, the action is to shut down, and a storm has occurred. **Note** If an interface is disabled by a broadcast, multicast, or unicast storm, the filter state for all traffic types is *shutdown*.
Trap State	Displays the status of the SNMP trap: • Above rising—Storm control is enabled, and a storm has occurred. • Below rising—Storm control is enabled, and a storm has not occurred. • Inactive—The trap option is not enabled.
Upper	Displays the rising suppression level as a percentage of total available bandwidth.
Lower	Displays the falling suppression level as a percentage of total available bandwidth.
Current	Displays the bandwidth utilization of a specific traffic type as a percentage of total available bandwidth. This field is valid only when storm control is enabled.
Traps Sent	Displays the number traps sent on an interface for a specific traffic type.

This is an example of output from the **show storm-control fastethernet0/4 history** command, which displays the ten most recent storm events for an interface.

```
Switch> show storm-control fastethernet0/4 history

 Interface Fa0/4 Storm Event History

 Event Type          Event Start Time  Duration (seconds)
 ------------------  ----------------  ------------------
 Unicast             04:58:18          206
 Broadcast           05:01:54          n/a
 Multicast           05:01:54          n/a
 Unicast             05:01:54          108
 Broadcast           05:05:00          n/a
 Multicast           05:05:00          n/a
 Unicast             05:06:00          n/a
 Broadcast           05:09:39          n/a
 Multicast           05:09:39          n/a
 Broadcast           05:11:32          172
```

Note The duration field could be *n/a* when a storm is still present or when a new storm of a different type occurs before the current storm ends.

Related Commands

Command	Description
storm-control	Enables broadcast, multicast, or unicast storm control on a port.

show system mtu

Use the **show system mtu** privileged EXEC command to display the global maximum packet size or maximum transmission unit (MTU) set for the switch.

show system mtu [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show system mtu** command:

```
Switch# show system mtu
System MTU size is 1500 bytes
```

Related Commands

Command	Description
system mtu	Sets the MTU size for the switch.

show udld

Use the **show udld** user EXEC command to display UniDirectional Link Detection (UDLD) status for all ports or the specified port.

show udld [*interface-id*] [**|** {**begin** | **exclude** | **include**} *expression*]

Syntax Description

interface-id	(Optional) ID of the interface and port number. Valid interfaces include physical ports and VLANs. The VLAN range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

If you do not enter an *interface-id*, the administrative and the operational UDLD status for all interfaces appear.

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show udld gigabitethernet0/1** command. In this example, UDLD is enabled on both ends of the link, and UDLD detects that the link is bidirectional. Table 2-15 describes the fields in this example.

```
Switch> show udld gigabitethernet0/1
Interface gi0/1
---
Port enable administrative configuration setting: Follows device default
Port enable operational state: Enabled
Current bidirectional state: Bidirectional
Current operational state: Advertisement - Single Neighbor detected
Message interval: 60
Time out interval: 5
    Entry 1
    Expiration time: 146
    Device ID: 1
    Current neighbor state: Bidirectional
    Device name: 0050e2826000
    Port ID: Gi0/2
```

```
Neighbor echo 1 device: SAD03160954
Neighbor echo 1 port: Gi0/1
Message interval: 5
CDP Device name: 066527791
```

Table 2-15 show udld Field Descriptions

Field	Description
Interface	The interface on the local device configured for UDLD.
Port enable administrative configuration setting	How UDLD is configured on the port. If UDLD is enabled or disabled, the port enable configuration setting is the same as the operational enable state. Otherwise, the enable operational setting depends on the global enable setting.
Port enable operational state	Operational state that shows whether UDLD is actually running on this port.
Current bidirectional state	The bidirectional state of the link. An unknown state appears if the link is down or if it is connected to an UDLD-incapable device. A bidirectional state appears if the link is a normal two-way connection to a UDLD-capable device. All other values mean miswiring.
Current operational state	The phase of the UDLD state machine. For a normal bidirectional link, the state machine is usually in the Advertisement phase.
Message interval	How often advertisement messages are sent from the local device. Measured in seconds.
Time out interval	The time period, in seconds, that UDLD waits for echoes from a neighbor device during the detection window.
Entry 1	Information from the first cache entry, which contains a copy of echo information received from the neighbor.
Expiration time	The amount of time in seconds remaining before this cache entry is aged out.
Device ID	The neighbor device identification.
Current neighbor state	The neighbor's state. If both the local and neighbor devices are running UDLD, the neighbor state and the local state is bidirectional. If the link is down or the neighbor is not UDLD-capable, no cache entries appear.
Device name	The neighbor MAC address.
Port ID	The neighbor port ID enabled for UDLD.
Neighbor echo 1 device	The MAC address of the neighbors' neighbor from which the echo originated.
Neighbor echo 1 port	The port number ID of the neighbor from which the echo originated.
Message interval	The rate, in seconds, at which the neighbor is sending advertisement messages.
CDP[1] device name	CDP name of the device.

1. CDP = Cisco Discovery Protocol

This is an example of output from the **show udld** interface configuration command when the aggressive mode is configured:

```
Switch# show udld gi0/1
Interface Gi0/1
---
Port enable administrative configuration setting:Enabled / in aggressive mode
Port enable operational state:Enabled / in aggressive mode
Current bidirectional state:Unknown
Current operational state:Link down
Message interval:7
Time out interval:5
No neighbor cache information stored
```

Related Commands

Command	Description
udld (global configuration)	Enables UDLD on all ports on the switch.
udld (interface configuration)	Enables UDLD on a port.
udld reset	Resets any interface that was shut down by UDLD.

show version

Use the **show version** user EXEC command to display version information for the hardware and firmware.

show version [**|** {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show version** command:

```
Switch> show version

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(9)EA1
Copyright (c) 1986-2002 by cisco Systems, Inc.
Compiled Wed 27-Feb-02 06:51 by antonino
Image text-base:0x80010000, data-base:0x804E2000

ROM:Bootstrap program is C2950 boot loader

Switch uptime is 1 hour, 54 minutes
System returned to ROM by power-on
System image file is "flash:c2950-i6q4l2-mz.121-0.0.9.EA1.bin"

cisco WS-C2950G-12-EI (RC32300) processor with 20830K bytes of memory.
Last reset from system-reset
Running Enhanced Image
12 FastEthernet/IEEE 802.3 interface(s)
2 Gigabit Ethernet/IEEE 802.3 interface(s)

32K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address:00:05:74:28:09:C0
Configuration register is 0xF
```

show vlan

Use the **show vlan** user EXEC command to display the parameters for all configured VLANs or one VLAN (if the VLAN ID or name is specified) on the switch.

show vlan [**brief** | **id** *vlan-id* | **name** *vlan-name* | **remote-span** | **summary**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

brief	(Optional) Display one line for each VLAN with the VLAN name, status, and its ports.
id *vlan-id*	(Optional) Display information about a single VLAN identified by VLAN ID number. For *vlan-id*, the range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
name *vlan-name*	(Optional) Display information about a single VLAN identified by VLAN name. The VLAN name is an ASCII string from 1 to 32 characters.
remote-span	(Optional) Display information about Remote SPAN (RSPAN) VLANs.
summary	(Optional) Display VLAN summary information. This keyword is available only if your switch is running the EI.
| begin	(Optional) Display begins with the line that matches the *expression*.
| exclude	(Optional) Display excludes lines that match the *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Note Though visible in the command-line help string when the EI is installed, the **internal usage**, **ifindex**, and **private-vlan** keywords are not supported.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(9)EA1	The **summary** keyword was added.
12.1(11)EA1	The **remote-span** keyword was added.

Usage Guidelines

Expressions are case sensitive. For example, if you enter | **exclude output**, the lines that contain *output* are not displayed, but the lines that contain *Output* are displayed.

Examples

This is an example of output from the **show vlan** command. Table 2-16 describes each field in the display.

```
Switch> show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/5, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/11, Fa0/12
                                                Gi0/1, Gi0/2
2    VLAN0002                         active
51   VLAN0051                         active
52   VLAN0052                         active
100  VLAN0100                         suspended Fa0/3
400  VLAN0400                         suspended
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        1002   1003
2    enet  100002     1500  -      -      -        -    -        0      0
51   enet  100051     1500  -      -      -        -    -        0      0
52   enet  100052     1500  -      -      -        -    -        0      0
100  enet  100100     1500  -      -      -        -    -        0      0
400  enet  100400     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        1      1003
1003 tr    101003     1500  1005   3276   -        -    srb      1      1002
1004 fdnet 101004     1500  -      -      1        ieee -        0      0
1005 trnet 101005     1500  -      -      15       ibm  -        0      0
Remote SPAN VLANs
------------------------------------------------------------------------------


Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
```

Table 2-16 show vlan Command Output Fields

Field	Description
VLAN	VLAN number.
Name	Name, if configured, of the VLAN.
Status	Status of the VLAN (active or suspend).
Ports	Ports that belong to the VLAN.
Type	Media type of the VLAN.
SAID	Security association ID value for the VLAN.
MTU	Maximum transmission unit size for the VLAN.
Parent	Parent VLAN, if one exists.
RingNo	Ring number for the VLAN, if applicable.

Table 2-16 show vlan Command Output Fields (continued)

Field	Description
BrdgNo	Bridge number for the VLAN, if applicable.
Stp	Spanning Tree Protocol type used on the VLAN.
BrdgMode	Bridging mode for this VLAN—possible values are source-route bridging (SRB) and source-route transparent (SRT); the default is SRB.
Trans1	Translation bridge 1.
Trans2	Translation bridge 2.
AREHops	Maximum number of hops for All-Routes Explorer frames—possible values are 1 through 13; the default is 7.
STEHops	Maximum number of hops for Spanning-Tree Explorer frames—possible values are 1 through 13; the default is 7.
Backup CRF	Status of whether or not the Token Ring concentrator relay function (TrCRF) is a backup path for traffic.

This is an example of output from the **show vlan brief** command:

```
Switch> show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

This is an example of output from the **show vlan id** command. The specified VLAN is in the extended VLAN range.

```
Switch# show vlan id 2005

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
2005 VLAN2005                         active    Fa0/2

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
2005 enet  102005     1500  -      -      -        -    -        0      0
```

This is an example of output from the **show vlan summary** command:

```
Switch> show vlan summary
Number of existing VLANs           : 7
 Number of existing VTP VLANs      : 7
 Number of existing extended VLANs : 0
```

Related Commands

Command	Description
switchport mode	Configures the VLAN membership mode of a port.
vlan (global configuration)	Enables config-vlan mode where you can configure VLANs 1 to 4094 when the EI is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
vlan (VLAN configuration)	Configures VLAN characteristics in the VLAN database. Only available for normal-range VLANs (VLAN IDs 1 to 1005). Do not enter leading zeros.

show vmps

Use the **show vmps** user EXEC command without keywords to display the VLAN Query Protocol (VQP) version, reconfirmation interval, retry count, VLAN Membership Policy Server (VMPS) IP addresses, and the current and primary servers, or use the **statistics** keyword to display client-side statistics.

show vmps [**statistics**] [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

statistics	(Optional) Display VQP client-side statistics and counters.
| **begin**	(Optional) Display begins with the line that matches the *expression*.
| **exclude**	(Optional) Display excludes lines that match the *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show vmps** command:

```
Switch> show vmps
VQP Client Status:
--------------------
VMPS VQP Version:   1
Reconfirm Interval: 60 min
Server Retry Count: 3
VMPS domain server:

Reconfirmation status
---------------------
VMPS Action:         other
```

This is an example of output from the **show vmps statistics** command. Table 2-17 describes each field in the example.

```
Switch> show vmps statistics
VMPS Client Statistics
----------------------
VQP  Queries:               0
VQP  Responses:             0
VMPS Changes:               0
VQP  Shutdowns:             0
VQP  Denied:                0
VQP  Wrong Domain:          0
```

```
VQP  Wrong Version:         0
VQP  Insufficient Resource: 0
```

Table 2-17 show vmps statistics Field Descriptions

Field	Description
VQP Queries	Number of queries sent by the client to the VMPS.
VQP Responses	Number of responses sent to the client from the VMPS.
VMPS Changes	Number of times that the VMPS changed from one server to another.
VQP Shutdowns	Number of times the VMPS sent a response to shut down the port. The client disables the port and removes all dynamic addresses on this port from the address table. You must administratively re-enable the port to restore connectivity.
VQP Denied	Number of times the VMPS denied the client request for security reasons. When the VMPS response denies an address, no frame is forwarded to or from the workstation with that address. (Broadcast or multicast frames are delivered to the workstation if the port on the switch has been assigned to a VLAN.) The client keeps the denied address in the address table as a blocked address to prevent further queries from being sent to the VMPS for each new packet received from this workstation. The client ages the address if no new packets are received from this workstation on this port within the aging time period.
VQP Wrong Domain	Number of times the management domain in the request does not match the one for the VMPS. Any previous VLAN assignments of the port are not changed. This response means that the server and the client have not been configured with the same VTP management domain.
VQP Wrong Version	Number of times the version field in the query packet contains a value that is higher than the version supported by the VMPS. The previous VLAN assignment of the port is not changed. The switches send only VMPS version 1 requests.
VQP Insufficient Resource	Number of times the VMPS is unable to answer the request because of a resource availability problem. If the retry limit has not yet been reached, the client repeats the request with the same server or with the next alternate server, depending on whether the per-server retry count has been reached.

Related Commands

Command	Description
clear vmps statistics	Clears the statistics maintained by the VQP client.
vmps reconfirm (global configuration)	Sends VQP queries to reconfirm all dynamic VLAN assignments with the VMPS.
vmps retry	Configures the per-server retry count for the VQP client.
vmps server	Configures the primary VMPS and up to three secondary servers.

show vtp

Use the **show vtp** user EXEC command to display general information about the VLAN Trunking Protocol (VTP) management domain, status, and counters.

show vtp {**counters** | **status**} [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

counters	Display the VTP statistics for the switch.
status	Display general information about the VTP management domain status.
| **begin**	(Optional) Display begins with the line that matches the *expression*.
| **exclude**	(Optional) Display excludes lines that match the *expression*.
| **include**	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter | **exclude output**, the lines that contain *output* are not displayed, but the lines that contain *Output* are displayed.

Examples

This is an example of output from the **show vtp counters** command. Table 2-18 describes each field in the display.

```
Switch> show vtp counters

VTP statistics:
Summary advertisements received    : 38
Subset advertisements received     : 0
Request advertisements received    : 0
Summary advertisements transmitted : 13
Subset advertisements transmitted  : 3
Request advertisements transmitted : 0
Number of config revision errors   : 0
Number of config digest errors     : 0
Number of V1 summary errors        : 0

VTP pruning statistics:

Trunk            Join Transmitted Join Received    Summary advts received from
                                                   non-pruning-capable device
---------------- ---------------- ---------------- ---------------------------
Fa0/9               827              824              0
Fa0/10              827              823              0
Fa0/11              827              823              0
```

Table 2-18 show vtp counters Field Descriptions

Field	Description
Summary advertisements received	Number of summary advertisements received by this switch on its trunk ports. Summary advertisements contain the management domain name, the configuration revision number, the update timestamp and identity, the authentication checksum, and the number of subset advertisements to follow.
Subset advertisements received	Number of subset advertisements received by this switch on its trunk ports. Subset advertisements contain all the information for one or more VLANs.
Request advertisements received	Number of advertisement requests received by this switch on its trunk ports. Advertisement requests normally request information on all VLANs. They can also request information on a subset of VLANs.
Summary advertisements transmitted	Number of summary advertisements sent by this switch on its trunk ports. Summary advertisements contain the management domain name, the configuration revision number, the update timestamp and identity, the authentication checksum, and the number of subset advertisements to follow.
Subset advertisements transmitted	Number of subset advertisements sent by this switch on its trunk ports. Subset advertisements contain all the information for one or more VLANs.
Request advertisements transmitted	Number of advertisement requests sent by this switch on its trunk ports. Advertisement requests normally request information on all VLANs. They can also request information on a subset of VLANs.
Number of configuration revision errors	Number of revision errors. Whenever you define a new VLAN, delete an existing one, suspend or resume an existing VLAN, or modify the parameters on an existing VLAN, the configuration revision number of the switch increments. Revision errors increment whenever the switch receives an advertisement whose revision number matches the revision number of the switch, but the MD5 digest values do not match. This error means that the VTP password in the two switches is different or that the switches have different configurations. These errors means that the switch is filtering incoming advertisements, which causes the VTP database to become unsynchronized across the network.

Table 2-18 show vtp counters Field Descriptions (continued)

Field	Description
Number of configuration digest errors	Number of MD5 digest errors. Digest errors increment whenever the MD5 digest in the summary packet and the MD5 digest of the received advertisement calculated by the switch do not match. This error usually means that the VTP password in the two switches is different. To solve this problem, make sure the VTP password on all switches is the same. These errors mean that the switch is filtering incoming advertisements, which causes the VTP database to become unsynchronized across the network.
Number of V1 summary errors	Number of version 1 errors. Version 1 summary errors increment whenever a switch in VTP V2 mode receives a VTP version 1 frame. These errors mean that at least one neighboring switch is either running VTP version 1 or VTP version 2 with V2-mode disabled. To solve this problem, change the configuration of the switches in VTP V2-mode to disabled.
Join Transmitted	Number of VTP pruning messages sent on the trunk.
Join Received	Number of VTP pruning messages received on the trunk.
Summary Advts Received from non-pruning-capable device	Number of VTP summary messages received on the trunk from devices that do not support pruning.

This is an example of output from the **show vtp status** command. Table 2-19 describes each field in the display.

```
Switch> show vtp status

VTP Version                     : 2
Configuration Revision          : 0
Maximum VLANs supported locally : 250
Number of existing VLANs        : 5
VTP Operating Mode              : Server
VTP Domain Name                 :
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xBF 0x86 0x94 0x45 0xFC 0xDF 0xB5 0x70
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 172.20.135.196 on interface Vl1 (lowest numbered VLAN interface found)
```

Table 2-19 show vtp status Field Descriptions

Field	Description
VTP Version	Displays the VTP version operating on the switch. By default, the switch implements version 1 but can be set to version 2.
Configuration Revision	Current configuration revision number on this switch.
Maximum VLANs Supported Locally	Maximum number of VLANs supported locally.
Number of Existing VLANs	Number of existing VLANs.

Table 2-19 show vtp status Field Descriptions (continued)

Field	Description
VTP Operating Mode	Displays the VTP operating mode, which can be server, client, or transparent. Server: a switch in VTP server mode is enabled for VTP and sends advertisements. You can configure VLANs on it. The switch guarantees that it can recover all the VLAN information in the current VTP database from nonvolatile RAM (NVRAM) after reboot. By default, every switch is a VTP server. **Note** The switch automatically changes from VTP server mode to VTP client mode if it detects a failure while writing the configuration to NVRAM and cannot return to server mode until the NVRAM is functioning. Client: a switch in VTP client mode is enabled for VTP, can send advertisements, but does not have enough nonvolatile storage to store VLAN configurations. You cannot configure VLANs on it. When a VTP client starts up, it does not send VTP advertisements until it receives advertisements to initialize its VLAN database. Transparent: a switch in VTP transparent mode is disabled for VTP, does not send or learn from advertisements sent by other devices, and cannot affect VLAN configurations on other devices in the network. The switch receives VTP advertisements and forwards them on all trunk ports except the one on which the advertisement was received.
VTP Domain Name	Name that identifies the administrative domain for the switch.
VTP Pruning Mode	Displays whether pruning is enabled or disabled. Enabling pruning on a VTP server enables pruning for the entire management domain. Pruning restricts flooded traffic to those trunk links that the traffic must use to access the appropriate network devices.
VTP V2 Mode	Displays if VTP version 2 mode is enabled. By default, all VTP version 2 switches operate in version 1 mode. Each VTP switch automatically detects the capabilities of all the other VTP devices. A network of VTP devices should be configured to version 2 only if all VTP switches in the network can operate in version 2 mode.
VTP Traps Generation	Displays whether VTP traps are sent to a network management station.
MD5 Digest	A 16-byte checksum of the VTP configuration.
Configuration Last Modified	Displays the date and time of the last configuration modification. Displays the IP address of the switch that caused the configuration change to the database.

Related Commands

Command	Description
clear vtp counters	Clears the VTP and pruning counters.
vtp (global configuration)	Configures the VTP filename, interface name, domain name, and mode. You can save configuration resulting from this command in the switch configuration file.
vtp (privileged EXEC)	Configures the VTP password, pruning, and version.
vtp (VLAN configuration)	Configures the VTP domain name, password, pruning, and mode.

show wrr-queue bandwidth

Use the **show wrr-queue bandwidth** user EXEC command to display the weighted round-robin (WRR) bandwidth allocation for the four class of service (CoS) priority queues.

show wrr-queue bandwidth [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show wrr-queue bandwidth** command:

```
Switch> show wrr-queue bandwidth

WRR Queue   :   1   2   3   4

Bandwidth   :  10  20  30  40
```

Related Commands

Command	Description
show wrr-queue cos-map	Displays the mapping of the CoS to the priority queues.
wrr-queue bandwidth	Assigns WRR weights to the four CoS priority queues.
wrr-queue cos-map	Assigns CoS values to the CoS priority queues.

show wrr-queue cos-map

Use the **show wrr-queue cos-map** user EXEC command to display the mapping of the class of service (CoS) priority queues.

show wrr-queue cos-map [| {**begin** | **exclude** | **include**} *expression*]

Syntax Description

| begin	(Optional) Display begins with the line that matches the specified *expression*.
| exclude	(Optional) Display excludes lines that match the specified *expression*.
| include	(Optional) Display includes lines that match the specified *expression*.
expression	Expression in the output to use as a reference point.

Command Modes

User EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

Expressions are case sensitive. For example, if you enter **| exclude output**, the lines that contain *output* do not appear, but the lines that contain *Output* appear.

Examples

This is an example of output from the **show wrr-queue cos-map** command:

```
Switch> show wrr-queue cos-map

CoS Value      :  0  1  2  3  4  5  6  7

Priority Queue :  1  1  2  2  3  3  4  4
```

Related Commands

Command	Description
show wrr-queue bandwidth	Displays the WRR bandwidth allocation for the four CoS priority queues.
wrr-queue bandwidth	Assigns weighted round-robin (WRR) weights to the four CoS priority queues.
wrr-queue cos-map	Assigns CoS values to the CoS priority queues.

shutdown

Use the **shutdown** interface configuration command to disable a port and to shut down the management VLAN. Use the **no** form of this command to enable a disabled port or to activate the management VLAN.

shutdown

no shutdown

Syntax Description

This command has no arguments or keywords.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

The **shutdown** command for a port causes it to stop forwarding. You can enable the port with the **no shutdown** command.

The **no shutdown** command has no effect if the port is a static-access port assigned to a VLAN that has been deleted, suspended, or shut down. The port must first be a member of an active VLAN before it can be re-enabled.

Only one management VLAN interface can be active at a time. The remaining VLANs are shut down. In the **show running-config** command, the active management VLAN interface is the one without the **shutdown** command displayed.

Examples

This example shows how to disable fixed Fast Ethernet port 0/8 and how to re-enable it:

```
Switch(config)# interface fastethernet0/8
Switch(config-if)# shutdown

Switch(config-if)# no shutdown
```

You can verify your settings by entering the **show interfaces** privileged EXEC command.

shutdown vlan

Use the **shutdown vlan** global configuration command to shut down (suspend) local traffic on the specified VLAN. Use the **no** form of this command to restart local traffic on the VLAN.

shutdown vlan *vlan-id*

no shutdown vlan *vlan-id*

Syntax Description

vlan-id	ID of the VLAN to be locally shut down. Valid IDs are from 2 to 1001. VLANs defined as default VLANs under the VLAN Trunking Protocol (VTP), as well as extended-range VLANs (greater than 1005) cannot be shut down. The default VLANs are 1 and 1002 to 1005. Do not enter leading zeros.

Defaults

No default is defined.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

The **shutdown vlan** command does not change the VLAN information in the VTP database. It shuts down traffic locally, but the switch still advertises VTP information.

Examples

This example shows how to shutdown traffic on VLAN 2:

```
Switch(config)# shutdown vlan 2
```

You can verify your setting by entering the **show vlan** privileged EXEC command.

Related Commands

Command	Description
shutdown (config-vlan mode)	Shuts down local traffic on the VLAN when in config-VLAN mode (accessed by the **vlan** *vlan-id* global configuration command).
vlan (global configuration)	Enables config-vlan mode.
vlan database	Enters VLAN configuration mode.

snmp-server enable traps

Use the **snmp-server enable traps** global configuration command to enable the switch to send Simple Network Management Protocol (SNMP) notification for various trap types to the network management system (NMS). Use the **no** form of this command to return to the default setting.

snmp-server enable traps [**c2900** | **cluster** | **config** | **entity** | **hsrp** | **mac-notification** | **rtr** | **snmp** | **syslog** | **vlan-membership** | **vtp**]

no snmp-server enable traps [**c2900** | **cluster** | **config** | **entity** | **hsrp** | **mac-notification** | **rtr** | **snmp** | **syslog** | **vlan-membership** | **vtp**]

Note Though visible in the command-line help strings, the **snmp-server enable informs** command is not supported. To enable sending SNMP inform notifications, use the **snmp-server enable traps** command combined with the **snmp-server host** *host-addr* **informs** command.

Syntax Description

c2900	(Optional) Enable SNMP configuration traps.
cluster	(Optional) Enable cluster traps.
config	(Optional) Enable SNMP configuration traps.
entity	(Optional) Enable SNMP entity traps.
hsrp	(Optional) Enable Hot Standby Router Protocol (HSRP) traps.
mac-notification	(Optional) Enable MAC address notification traps.
rtr	(Optional) Enable SNMP Response Time Reporter traps.
snmp	(Optional) Enable SNMP traps.
syslog	(Optional) Enable SNMP syslog traps.
vlan-membership	(Optional) Enable SNMP VLAN membership traps.
vtp	(Optional) Enable VLAN Trunking Protocol (VTP) traps.

Defaults

The sending of SNMP traps is disabled.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **vlan-membership** keyword was added.

Usage Guidelines

Specify the host (NMS) that receives the traps by using the **snmp-server host** global configuration command. If no trap types are specified, all trap types are sent.

Use the **snmp-server enable traps** command to enable sending of traps or informs, when supported.

Note Informs are not supported in SNMPv1.

To enable more than one type of trap, you must issue a separate **snmp-server enable traps** command for each trap type.

Examples

This example shows how to send VTP traps to the NMS:

```
Switch(config)# snmp-server enable traps vtp
```

You can verify your setting by entering the **show vtp status** privileged EXEC or the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the running configuration on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
snmp-server host	Specifies the host that receives SNMP traps.

snmp-server host

Use the **snmp-server host** global configuration command to specify the recipient (host) of a Simple Network Management Protocol (SNMP) notification operation. Use the **no** form of this command to remove the specified host.

snmp-server host *host-addr* [**informs** | **traps**] [**version** {**1** | **2c** | **3** [**auth** | **noauth**]}] *community-string* [**c2900** | **cluster** | **config** | **entity** | **hsrp** | **mac-notification** | **rtr** | **snmp** | **tty** | **udp-port** | **vlan-membership** | **vtp**]

no snmp-server host *host-addr* [**informs** | **traps**] [**version** {**1** | **2c** | **3** [**auth** | **noauth**]}] *community-string* [**c2900 | cluster** | **config** | **entity** | **hsrp** | **mac-notification** | **rtr** | **snmp** | **tty** | **udp-port** | **vlan-membership** | **vtp**]

Syntax Description

host-addr	Name or Internet address of the host (the targeted recipient).
informs | **traps**	(Optional) Send SNMP traps or informs to this host.
version 1 | **2c** | **3**	(Optional) Version of SNMP used to send the traps. These keywords are supported: **1**—SNMPv1. This option is not available with informs. **2c**—SNMPv2C. **3**—SNMPv3. These optional keywords can follow the version 3 keyword: • **auth** (Optional). Enables Message Digest 5 (MD5) and Secure Hash Algorithm (SHA) packet authentication. • **noauth** (Default). The noAuthNoPriv security level. This is the default if the [**auth** | **noauth**] keyword choice is not specified.
community-string	Password-like community string sent with the notification operation. Though you can set this string by using the **snmp-server host** command, we recommend that you define this string by using the **snmp-server community** global configuration command before using the **snmp-server host** command.
c2900	(Optional) Send SNMP switch traps.
cluster	(Optional) Send cluster member status traps.
config	(Optional) Send SNMP configuration traps.
entity	(Optional) Send SNMP entity traps.
hsrp	(Optional) Send Hot Standby Router Protocol (HSRP) traps.
mac-notification	(Optional) Send MAC notification traps.
rtr	(Optional) Send SNMP Response Time Reporter traps.
snmp	(Optional) Send SNMP-type traps.
tty	(Optional) Send Transmission Control Protocol (TCP) connection traps.
udp-port	(Optional) Send notification host's User Datagram Protocol (UDP) port number.
vlan-membership	(Optional) Send SNMP VLAN membership traps.
vtp	(Optional) Send VLAN Trunking Protocol (VTP) traps.

Defaults

This command is disabled. No notifications are sent.

If you enter this command with no keywords, the default is to send all trap types to the host. No informs are sent to this host.

If no **version** keyword is present, the default is version 1.

If version 3 is selected and no authentication keyword is entered, the default is the **noauth** (noAuthNoPriv) security level.

Note If the *community-string* is not defined by using the **snmp-server community** global configuration command before using this command, the default form of the **snmp-server community** command is automatically inserted into the configuration. The password (*community-string*) used for this automatic configuration of the **snmp-server community** will be the same as that specified in the **snmp-server host** command.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **cluster**, **mac-notification**, and **rtr** keywords were added.
12.1(9)EA1	The **vlan-membership** keyword was added.
12.1(11)EA1	The version **3** option was added, with the **auth** and **noauth** keywords.

Usage Guidelines

SNMP notifications can be sent as traps or inform requests. Traps are unreliable because the receiver does not send acknowledgments when it receives traps. The sender cannot determine if the traps were received. However, an SNMP entity that receives an inform request acknowledges the message with an SNMP response PDU. If the sender never receives the response, the inform request can be sent again. Thus, informs are more likely to reach their intended destinations.

However, informs consume more resources in the agent and in the network. Unlike a trap, which is discarded as soon as it is sent, an inform request must be held in memory until a response is received or the request times out. Traps are also sent only once, but an inform might be retried several times. The retries increase traffic and contribute to a higher overhead on the network.

If you do not enter an **snmp-server host** command, no notifications are sent. To configure the switch to send SNMP notifications, you must enter at least one **snmp-server host** command. If you enter the command with no keywords, all trap types are enabled for the host. To enable multiple hosts, you must enter a separate **snmp-server host** command for each host. You can specify multiple notification types in the command for each host.

When multiple **snmp-server host** commands are given for the same host and kind of notification (trap or inform), each succeeding command overwrites the previous command. Only the last **snmp-server host** command is in effect. For example, if you enter an **snmp-server host inform** command for a host and then enter another **snmp-server host inform** command for the same host, the second command replaces the first.

The **snmp-server host** command is used with the **snmp-server enable traps** global configuration command. Use the **snmp-server enable traps** command to specify which SNMP notifications are sent globally. For a host to receive most notifications, at least one **snmp-server enable traps** command and

the **snmp-server host** command for that host must be enabled. Some notification types cannot be controlled with the **snmp-server enable traps** command. For example, some notification types are always enabled. Other notification types are enabled by a different command.

The **no snmp-server host** command with no keywords disables traps, but not informs, to the host. To disable informs, use the **no snmp-server host informs** command.

Examples

This example shows how to configure a unique SNMP community string named *comaccess* for traps and prevent SNMP polling access with this string through access-list 10:

```
Switch(config)# snmp-server community comaccess ro 10
Switch(config)# snmp-server host 172.20.2.160 comaccess
Switch(config)# access-list 10 deny any
```

This example shows how to send the SNMP traps to the host specified by the name *myhost.cisco.com*. The community string is defined as *comaccess*:

```
Switch(config)# snmp-server enable traps
Switch(config)# snmp-server host myhost.cisco.com comaccess snmp
```

This example shows how to enable the switch to send all traps to the host *myhost.cisco.com* using the community string *public*:

```
Switch(config)# snmp-server enable traps
Switch(config)# snmp-server host myhost.cisco.com public
```

Related Commands

Command	Description
show running-config	Displays the running configuration on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
snmp-server enable traps	Enables SNMP notification for various trap types.

snmp trap mac-notification

Use the **snmp trap mac-notification** interface configuration command to enable the MAC notification traps on a port. Use the **no** form of this command to disable the traps and to return the port to default settings.

snmp trap mac-notification [**added** | **removed**]

no snmp trap mac-notification [**added** | **removed**]

Syntax Description

added	(Optional) Enable MAC notification traps when a MAC address is added to a port.
removed	(Optional) Enable MAC notification traps when a MAC address is removed from a port.

Defaults

The Simple Network Management Protocol (SNMP) address-addition and address-removal traps are disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

Even though you enable the notification trap for a specific interface by using the **snmp trap mac-notification** command, the trap is generated only when you enter the **snmp-server enable traps mac-notification** and the **mac address-table notification** global configuration commands.

Examples

This example shows how to enable an address-addition trap on a port:

```
Switch(config-if)# snmp trap mac-notification added
```

This example shows how to enable an address-removal trap on a port:

```
Switch(config-if)# snmp trap mac-notification removed
```

You can verify your settings by entering the **show mac address-table notification** privileged EXEC command.

Related Commands

Command	Description
clear mac address-table notification	Clears the MAC address notification global counters.
mac address-table notification	Enables the MAC notification feature on a switch.

Command	Description
show mac address-table notification	Displays MAC notification parameters.
snmp-server enable traps	Enables SNMP notification for various trap types.

spanning-tree backbonefast

Use the **spanning-tree backbonefast** global configuration command to enable the BackboneFast feature. Use the **no** form of this command to return to the default setting.

spanning-tree backbonefast

no spanning-tree backbonefast

Syntax Description

This command has no arguments or keywords.

Defaults

BackboneFast is disabled.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The BackboneFast feature is supported only when the switch is running per-VLAN spanning-tree (PVST).

BackboneFast is started when a root port or blocked port on a switch receives inferior bridge protocol data units (BPDUs) from its designated bridge. An inferior BPDU identifies one switch as both the root bridge and the designated bridge. When a switch receives an inferior BPDU, it means that a link to which the switch is not directly connected (an *indirect* link) has failed (that is, the designated bridge has lost its connection to the root switch). If there are alternate paths to the root switch, BackboneFast causes the maximum aging time on the ports on which it received the inferior BPDU to expire and allows a blocked port to move immediately to the listening state. BackboneFast then transitions the interface to the forwarding state. For more information, refer to the software configuration guide for this release.

Enable BackboneFast on all supported switches to allow the detection of indirect link failures and to start the spanning-tree reconfiguration sooner.

Examples

This example shows how to enable BackboneFast on the switch:

```
Switch(config)# spanning-tree backbonefast
```

You can verify your setting by entering the **show spanning-tree summary** privileged EXEC command.

Related Commands

Command	Description
show spanning-tree summary	Displays a summary of the spanning-tree port states.

spanning-tree bpdufilter

Use the **spanning-tree bpdufilter** interface configuration command to prevent a port from sending or receiving bridge protocol data units (BPDUs). Use the **no** form of this command to return to the default setting.

spanning-tree bpdufilter {**disable** | **enable**}

no spanning-tree bpdufilter

Syntax Description

disable	Disable BPDU filtering on the specified interface.
enable	Enable BPDU filtering on the specified interface.

Defaults

BPDU filtering is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

You can enable the BPDU filtering feature when the switch is operating in the per-VLAN spanning-tree (PVST) or the multiple spanning-tree (MST) mode. The MST mode is available only if you have the enhanced software image (EI) installed on your switch.

Caution Enabling BPDU filtering on an interface is the same as disabling spanning tree on it and can result in spanning-tree loops.

You can globally enable BPDU filtering on all Port Fast-enabled ports by using the **spanning-tree portfast bpdufilter default** global configuration command.

You can use the **spanning-tree bpdufilter** interface configuration command to override the setting of the **spanning-tree portfast bpdufilter default** global configuration command.

Examples

This example shows how to enable the BPDU filtering feature on a port:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# spanning-tree bpdufilter enable
```

You can verify your setting by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree portfast (global configuration)	Globally enables the BPDU filtering or the BPDU guard feature on Port Fast-enabled ports or enables the Port Fast feature on all nontrunking ports.
spanning-tree portfast (interface configuration)	Enables the Port Fast feature on an interface and all its associated VLANs.

spanning-tree bpduguard

Use the **spanning-tree bpduguard** interface configuration command to put a port in the error-disabled state when it receives a bridge protocol data unit (BPDU). Use the **no** form of this command to return to the default setting.

spanning-tree bpduguard {**disable** | **enable**}

no spanning-tree bpduguard

Syntax Description

disable	Disable BPDU guard on the specified interface.
enable	Enable BPDU guard on the specified interface.

Defaults

BPDU guard is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

The BPDU guard feature provides a secure response to invalid configurations because you must manually put the port back in service. Use the BPDU guard feature in a service-provider network to prevent a port from being included in the spanning-tree topology.

You can enable the BPDU guard feature when the switch is operating in the per-VLAN spanning-tree (PVST) or the multiple spanning-tree (MST) mode. The MST mode is available only if you have the enhanced software image (EI) installed on your switch.

You can globally enable BPDU guard on all Port Fast-enabled ports by using the **spanning-tree portfast bpduguard default** global configuration command.

You can use the **spanning-tree bpduguard** interface configuration command to override the setting of the **spanning-tree portfast bpduguard default** global configuration command.

Examples

This example shows how to enable the BPDU guard feature on a port:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# spanning-tree bpduguard enable
```

You can verify your setting by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree portfast (global configuration)	Globally enables the BPDU filtering or the BPDU guard feature on Port Fast-enabled ports or enables the Port Fast feature on all nontrunking ports.
spanning-tree portfast (interface configuration)	Enables the Port Fast feature on an interface and all its associated VLANs.

spanning-tree cost

Use the **spanning-tree cost** interface configuration command to set the path cost for spanning-tree calculations. If a loop occurs, spanning tree considers the path cost when selecting an interface to place in the forwarding state. Use the **no** form of this command to return to the default setting.

spanning-tree [**vlan** *vlan-id*] **cost** *cost*

no spanning-tree [**vlan** *vlan-id*] **cost**

Syntax Description

vlan *vlan-id*	(Optional) VLAN ID associated with a spanning-tree instance. The range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
cost	Path cost can range from 1 to 200000000, with higher values meaning higher costs.

Defaults

The default path cost is computed from the interface bandwidth setting. These are the IEEE default path cost values:

- 10 Mbps—100
- 100 Mbps—19
- 155 Mbps—14
- 1000 Mbps—4
- 1 Gbps—4
- 10 Gbps—2
- Speeds greater than 10 Gbps—1

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(9)EA1	The range for the *cost* variable increased.

Usage Guidelines

When you configure the cost, higher values represent higher costs.

You can set a cost on a VLAN that does not exist. The setting takes effect when the VLAN exists.

Examples

This example shows how to set a path cost of 250 on an interface (an access port):

```
Switch(config)# interface fastethernet0/4
Switch(config-if)# spanning-tree cost 250
```

This example shows how to set a path cost of 300 for VLAN 10 on an interface that is a trunk port:

```
Switch(config-if)# spanning-tree vlan 10 cost 300
```

You can verify your settings by entering the **show spanning-tree interface** *interface-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree interface *interface-id*	Displays spanning-tree information for the specified interface.
spanning-tree port-priority	Configures an interface priority.
spanning-tree vlan priority	Sets the switch priority for the specified spanning-tree instance.

spanning-tree extend system-id

Use the **spanning-tree extend system-id** global configuration command to enable the extended system ID feature.

spanning-tree extend system-id

Note Though visible in the command-line help strings, the **no** version of this command is not supported. You cannot disable the extended system ID feature.

Syntax Description

This command has no arguments or keywords.

Defaults

The extended system ID is enabled.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

In Release 12.1(9)EA1 and later, the switches support the 802.1T spanning-tree extensions, and some of the bits previously used for the switch priority are now used for the extended system ID (VLAN identifier for the per-VLAN spanning-tree [PVST] or an instance identifier for the multiple spanning tree [MST]). In earlier releases, the switch priority is a 16-bit value.

The spanning tree uses the extended system ID, the switch priority, and the allocated spanning-tree MAC address to make the bridge ID unique for each VLAN or multiple spanning-tree instance. With earlier releases, spanning tree used one MAC address per VLAN to make the bridge ID unique for each VLAN.

Support for the extended system ID affects how you manually configure the root switch, the secondary root switch, and the switch priority of a VLAN. For more information, see the "spanning-tree mst root" and the "spanning-tree vlan" sections.

If your network consists of switches that do not support the extended system ID and switches that do support it, it is unlikely that the switch with the extended system ID support will become the root switch. The extended system ID increases the switch priority value every time the VLAN number is greater than the priority of the connected switches running older software.

Related Commands

Command	Description
show spanning-tree summary	Displays a summary of spanning-tree port states.
spanning-tree mst root	Configures the multiple spanning-tree (MST) root switch priority and timers based on the network diameter.
spanning-tree vlan priority	Sets the switch priority for the specified spanning-tree instance.

spanning-tree guard

Use the **spanning-tree guard** interface configuration command to enable root guard or loop guard on all the VLANs associated with the selected interface. Root guard restricts which interface is allowed to be the spanning-tree root port or the path-to-the root for the switch. Loop guard prevents alternate or root ports from becoming designated ports when a failure creates a unidirectional link. Use the **no** form of this command to return to the default setting.

spanning-tree guard {**loop** | **none** | **root**}

no spanning-tree guard

Syntax Description

loop	Enable loop guard.
none	Disable root guard or loop guard.
root	Enable root guard.

Defaults

Root guard is disabled.

Loop guard is configured according to the **spanning-tree loopguard default** global configuration command (globally disabled).

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **spanning-tree rootguard** command.
12.1(9)EA1	The **loop** keyword was added.

Usage Guidelines

You can enable root guard or loop guard when the switch is operating in the per-VLAN spanning-tree (PVST) or the multiple spanning-tree (MST) mode. However, you cannot enable both features at the same time. The MST mode is available only if you have the enhanced software image (EI) installed on your switch.

When root guard is enabled, if spanning-tree calculations cause a port to be selected as the root port, the port transitions to the root-inconsistent (blocked) state to prevent the customer's switch from becoming the root switch or being in the path to the root. The root port provides the best path from the switch to the root switch.

When the **no spanning-tree guard** or the **no spanning-tree guard none** command is entered, root guard is disabled for all VLANs on the selected interface. If this interface is in the root-inconsistent (blocked) state, it automatically transitions to the listening state.

Do not enable root guard on interfaces that will be used by the UplinkFast feature. With UplinkFast, the backup interfaces (in the blocked state) replace the root port in the case of a failure. However, if root guard is also enabled, all the backup interfaces used by the UplinkFast feature are placed in the root-inconsistent state (blocked) and prevented from reaching the forwarding state. The UplinkFast feature is not available when the switch is operating in MST mode.

Loop guard is most effective when it is configured on the entire switched network. When the switch is operating in PVST mode, loop guard prevents alternate and root ports from becoming designated ports, and spanning tree does not send bridge protocol data units (BPDUs) on root or alternate ports. When the switch is operating in MST mode, BPDUs are not sent on nonboundary ports only if the port is blocked by loop guard in all MST instances. On a boundary port, loop guard blocks the port in all MST instances.

To disable root guard or loop guard, use the **spanning-tree guard none** interface configuration command. You cannot enable both root guard and loop guard at the same time.

You can override the setting of the **spanning-tree loopguard default** global configuration command by using the **spanning-tree guard loop** interface configuration command.

Examples

This example shows how to enable root guard on all the VLANs associated with the specified interface:

```
Switch(config)# interface fastethernet0/3
Switch(config-if)# spanning-tree guard root
```

This example shows how to enable loop guard on all the VLANs associated with the specified interface:

```
Switch(config)# interface fastethernet0/3
Switch(config-if)# spanning-tree guard loop
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree cost	Sets the path cost for spanning-tree calculations.
spanning-tree loopguard default	Prevents alternate or root ports from becoming designated ports because of a failure that leads to a unidirectional link.
spanning-tree mst cost	Configures the path cost for MST calculations.
spanning-tree mst port-priority	Configures an interface priority.
spanning-tree mst root	Configures the MST root switch priority and timers based on the network diameter.
spanning-tree port-priority	Configures an interface priority.
spanning-tree vlan priority	Sets the switch priority for the specified spanning-tree instance.

spanning-tree link-type

Use the **spanning-tree link-type** interface configuration command to override the default link-type setting, which is determined by the duplex mode of the port, and to enable Rapid Spanning-Tree Protocol (RSTP) transitions to the forwarding state. Use the **no** form of this command to return to the default setting.

spanning-tree link-type {**point-to-point** | **shared**}

no spanning-tree link-type

Syntax Description

point-to-point	Specify that the link type of a port is point-to-point.
shared	Specify that the link type of a port is shared.

Defaults

The switch derives the link type of a port from the duplex mode. A full-duplex port is considered a point-to-point link, and a half-duplex port is considered a shared link.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

You can override the default setting of the link type by using the **spanning-tree link-type** command; for example, a half-duplex link can be physically connected point-to-point to a single port on a remote switch running RSTP and be enabled for rapid transitions.

Examples

This example shows how to specify the link type as shared (regardless of the duplex setting) and to prevent RSTP rapid transitions to the forwarding state:

```
Switch(config-if)# spanning-tree link-type shared
```

You can verify your settings by entering the **show spanning-tree mst interface** *interface-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst interface *interface-id*	Displays multiple spanning-tree (MST) information for the specified interface.

spanning-tree loopguard default

Use the **spanning-tree loopguard default** global configuration command to prevent alternate or root ports from becoming designated ports because of a failure that leads to a unidirectional link. Use the **no** form of this command to return to the default setting.

spanning-tree loopguard default

no spanning-tree loopguard default

Syntax Description

This command has no arguments or keywords.

Defaults

Loop guard is disabled.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

You can enable the loop guard feature when the switch is operating in the per-VLAN spanning-tree (PVST) or the multiple spanning-tree (MST) mode. The MST mode is available only if you have the enhanced software image (EI) installed on your switch.

Loop guard is most effective when it is configured on the entire switched network. When the switch is operating in PVST mode, loop guard prevents alternate and root ports from becoming designated ports, and spanning tree does not send bridge protocol data units (BPDUs) on root or alternate ports. When the switch is operating in MST mode, BPDUs are not sent on nonboundary ports only if the port is blocked by loop guard in all MST instances. On a boundary port, loop guard blocks the port in all MST instances.

Loop guard operates only on ports that are considered point-to-point by the spanning tree.

You can override the setting of the **spanning-tree loopguard default** global configuration command by using the **spanning-tree guard loop** interface configuration command.

Examples

This example shows how to globally enable loop guard:

```
Switch(config)# spanning-tree loopguard default
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree guard loop	Enables the loop guard feature on all the VLANs associated with the specified interface.

spanning-tree mode

Use the **spanning-tree mode** global configuration command to enable either per-VLAN spanning-tree (PVST) or multiple spanning tree (MST) on your switch. Use the **no** form of this command to return to the default setting.

spanning-tree mode {**mst** | **pvst**}

no spanning-tree mode

Syntax Description

mst	Enable MST. This keyword is available only if your switch is running the enhanced software image (EI).
pvst	Enable PVST.

Defaults

The default mode is PVST.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

You cannot run both PVST and MST at the same time.

Caution Changing spanning-tree modes can disrupt traffic because all spanning-tree instances are stopped for the previous mode and restarted in the new mode.

Examples

This example shows to enable MST on the switch:

```
Switch(config)# spanning-tree mode mst
```

You can verify your setting by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

spanning-tree mst configuration

Use the **spanning-tree mst configuration** global configuration command to enter multiple spanning-tree (MST) configuration mode through which you configure the MST region. Use the **no** form of this command to return to the default settings.

spanning-tree mst configuration

no spanning-tree mst configuration

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

This command has no arguments or keywords.

Defaults

The default mapping is that all VLANs are mapped to the common and internal spanning tree (CIST) instance (instance 0).

The default name is an empty string.

The revision number is 0.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

Entering the **spanning-tree mst configuration** command enables the MST configuration mode. These configuration commands are available:

- **abort**: exits the MST region configuration mode without applying configuration changes.
- **exit**: exits the MST region configuration mode and applies all configuration changes.
- **instance** *instance-id* **vlan** *vlan-range*: maps VLANs to an MST instance. The range for the *instance-id* is 1 to 15; the range for *vlan-range* is 1 to 4094. Do not enter leading zeros.
- **name** *name*: sets the configuration name. The *name* string has a maximum length of 32 characters and is case sensitive.
- **no**: negates the **instance**, **name**, and **revision** commands or sets them to their defaults.
- **private-vlan**: Though visible in the command-line help strings, this command is not supported.
- **revision** *version*: sets the configuration revision number. The range is 0 to 65535.
- **show** [**current** | **pending**]: displays the current or pending MST region configuration.

When you map VLANs to an MST instance, the mapping is incremental, and the range of VLANs specified is added or removed to the existing ones. To specify a range, use a hyphen; for example, **instance 1 vlan 1-63** maps VLANs 1 to 63 to MST instance 1. To specify a series, use a comma; for example, **instance 1 vlan 10, 20, 30** maps VLANs 10, 20, and 30 to MST instance 1.

All VLANs that are not explicitly mapped to an MST instance are mapped to the common and internal spanning tree (CIST) instance (instance 0) and cannot be unmapped from the CIST by using the **no** form of the command.

For two or more switches to be in the same MST region, they must have the same VLAN mapping, the same configuration revision number, and the same name.

Examples

This example shows how to enter MST configuration mode, map VLAN 10 to 20 to MST instance 1, name the region *region1*, set the configuration revision to 1, display the pending configuration, apply the changes, and return to global configuration mode:

```
Switch# spanning-tree mst configuration
Switch(config-mst)# instance 1 vlan 10-20
Switch(config-mst)# name region1
Switch(config-mst)# revision 1
Switch(config-mst)# show pending
Pending MST configuration
Name      [region1]
Revision  1
Instance  Vlans Mapped
--------  --------------------
0         1-9,21-4094
1         10-20
-------------------------------

Switch(config-mst)# exit
Switch(config)#
```

This example shows how to add VLANs 1 to 100 to the ones already mapped (if any) to instance 2, to move VLANs 40 to 60 that were previously mapped to instance 2 to the CIST instance, to add VLAN 10 to instance 10, and to remove all the VLANs mapped to instance 2 and map them to the CIST instance:

```
Switch(config-mst)# instance 2 vlan 1-100
Switch(config-mst)# no instance 2 vlan 40-60
Switch(config-mst)# instance 10 vlan 10
Switch(config-mst)# no instance 2
```

You can verify your settings by entering the **show pending** MST configuration command.

Related Commands

Command	Description
show spanning-tree mst configuration	Displays the MST region configuration.

spanning-tree mst cost

Use the **spanning-tree mst cost** interface configuration command to set the path cost for multiple spanning-tree (MST) calculations. If a loop occurs, spanning tree considers the path cost when selecting an interface to put in the forwarding state. Use the **no** form of this command to return to the default setting.

spanning-tree mst *instance-id* **cost** *cost*

no spanning-tree mst *instance-id* **cost**

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

instance-id	ID associated with a spanning-tree instance. The range is 0 to 15.
cost	Path cost is 1 to 200000000, with higher values meaning higher costs.

Defaults

The default path cost is computed from the interface bandwidth setting. These are the IEEE default path cost values:

- 1000 Mbps—20000
- 100 Mbps—200000
- 10 Mbps—2000000

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

When you configure the cost, higher values represent higher costs.

Examples

This example shows how to set a path cost of 250 on an interface associated with instance 2:

```
Switch(config)# interface fastethernet0/4
Switch(config-if)# spanning-tree mst 2 cost 250
```

You can verify your settings by entering the **show spanning-tree mst interface** *interface-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst interface *interface-id*	Displays MST information for the specified interface.
spanning-tree mst port-priority	Configures an interface priority.
spanning-tree mst priority	Configures the switch priority for the specified spanning-tree instance.

spanning-tree mst forward-time

Use the **spanning-tree mst forward-time** global configuration command to set the forward-delay time for all multiple spanning-tree (MST) instances. The forwarding time determines how long each of the listening and learning states last before the interface begins forwarding. Use the **no** form of this command to return to the default setting.

spanning-tree mst forward-time *seconds*

no spanning-tree mst forward-time

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

seconds	Length of the listening and learning states. The range is 4 to 30 seconds.

Defaults

The default is 15 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

Changing the **spanning-tree mst forward-time** command affects all spanning-tree instances.

Examples

This example shows how to set the spanning-tree forwarding time to 18 seconds for all MST instances:

```
Switch(config)# spanning-tree mst forward-time 18
```

You can verify your settings by entering the **show spanning-tree mst** privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst	Displays MST information.
spanning-tree mst hello-time	Sets the interval between hello bridge protocol data units (BPDUs) sent by root switch configuration messages.
spanning-tree mst max-age	Sets the interval between messages that the spanning tree receives from the root switch.
spanning-tree mst max-hops	Sets the number of hops in a region before the BPDU is discarded.

spanning-tree mst hello-time

Use the **spanning-tree mst hello-time** global configuration command to set the interval between hello bridge protocol data units (BPDUs) sent by root switch configuration messages. Use the **no** form of this command to return to the default setting.

spanning-tree mst hello-time *seconds*

no spanning-tree mst hello-time

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

seconds	Interval between hello BPDUs sent by root switch configuration messages. The range is 1 to 10 seconds.

Defaults

The default is 2 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

After you set the **spanning-tree mst max-age** *seconds* global configuration command, if a switch does not receive BPDUs from the root switch within the specified interval, the switch recomputes the spanning-tree topology. The **max-age** setting must be greater than the **hello-time** setting.

Changing the **spanning-tree mst hello-time** command affects all spanning-tree instances.

Examples

This example shows how to set the spanning-tree hello time to 3 seconds for all MST instances:

```
Switch(config)# spanning-tree mst hello-time 3
```

You can verify your settings by entering the **show spanning-tree mst** privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst	Displays multiple spanning-tree (MST) information.
spanning-tree mst forward-time	Sets the forward-delay time for all MST instances.
spanning-tree mst max-age	Sets the interval between messages that the spanning tree receives from the root switch.
spanning-tree mst max-hops	Sets the number of hops in a region before the BPDU is discarded.

spanning-tree mst max-age

Use the **spanning-tree mst max-age** global configuration command to set the interval between messages that the spanning tree receives from the root switch. If a switch does not receive a bridge protocol data unit (BPDU) message from the root switch within this interval, it recomputes the spanning-tree topology. Use the **no** form of this command to return to the default setting.

spanning-tree mst max-age *seconds*

no spanning-tree mst max-age

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

seconds	Interval between messages the spanning tree receives from the root switch. If a switch does not receive a BPDU message from the root switch within this interval, it recomputes the spanning-tree topology. The range is 6 to 40 seconds.

Defaults

The default is 20 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

After you set the **spanning-tree mst max-age** *seconds* global configuration command, if a switch does not receive BPDUs from the root switch within the specified interval, the switch recomputes the spanning-tree topology. The **max-age** setting must be greater than the **hello-time** setting.

Changing the **spanning-tree mst max-age** command affects all spanning-tree instances.

Examples

This example shows how to set the spanning-tree max-age to 30 seconds for all MST instances:

```
Switch(config)# spanning-tree mst max-age 30
```

You can verify your settings by entering the **show spanning-tree mst** privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst	Displays multiple spanning-tree (MST) information.
spanning-tree mst forward-time	Sets the forward-delay time for all MST instances.
spanning-tree mst hello-time	Sets the interval between hello BPDUs sent by root switch configuration messages.
spanning-tree mst max-hops	Sets the number of hops in a region before the BPDU is discarded.

spanning-tree mst max-hops

Use the **spanning-tree mst max-hops** global configuration command to set the number of hops in a region before the bridge protocol data unit (BPDU) is discarded and the information held for a port is aged. Use the **no** form of this command to return to the default setting.

spanning-tree mst max-hops *hop-count*

no spanning-tree mst max-hops

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

hop-count	Number of hops in a region before the BPDU is discarded. The range is 1 to 40 hops.

Defaults

The default is 20 hops.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

The root switch of the instance always sends a BPDU (or M-record) with a cost of 0 and the hop count set to the maximum value. When a switch receives this BPDU, it decrements the received remaining hop count by one and propagates the decremented count as the remaining hop count in the generated M-records. A switch discards the BPDU and ages the information held for the port when the count reaches 0.

Changing the **spanning-tree mst max-hops** command affects all spanning-tree instances.

Examples

This example shows how to set the spanning-tree max-hops to 10 for all MST instances:

```
Switch(config)# spanning-tree mst max-hops 10
```

You can verify your settings by entering the **show spanning-tree mst** privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst	Displays multiple spanning-tree (MST) information.
spanning-tree mst forward-time	Sets the forward-delay time for all MST instances.
spanning-tree mst hello-time	Sets the interval between hello BPDUs sent by root switch configuration messages.
spanning-tree mst max-age	Sets the interval between messages that the spanning tree receives from the root switch.

spanning-tree mst port-priority

Use the **spanning-tree mst port-priority** interface configuration command to configure an interface priority. If a loop occurs, the Multiple Spanning Tree Protocol (MSTP) can determine which interface to put in the forwarding state. Use the **no** form of this command to return to the default setting.

spanning-tree mst *instance-id* **port-priority** *priority*

no spanning-tree mst *instance-id* **port-priority**

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

instance-id	ID associated with a spanning-tree instance. The range is 0 to 15.
priority	Number from 0 to 255. The lower the number, the higher the priority.

Defaults

The default is 128.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

You can assign higher priority values (lower numerical values) to interfaces that you want selected first and lower priority values (higher numerical values) that you want selected last. If all interfaces have the same priority value, the MST puts the interface with the lowest interface number in the forwarding state and blocks other interfaces.

Examples

This example shows how to increase the likelihood that the interface associated with spanning-tree instance 20 is placed into the forwarding state if a loop occurs:

```
Switch(config)# interface fastethernet0/2
Switch(config-if)# spanning-tree mst 20 port-priority 0
```

You can verify your settings by entering the **show spanning-tree mst interface** *interface-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst interface *interface-id*	Displays MST information for the specified interface.
spanning-tree mst cost	Sets the path cost for MST calculations.
spanning-tree mst priority	Sets the switch priority for the specified spanning-tree instance.

spanning-tree mst priority

Use the **spanning-tree mst priority** global configuration command to set the switch priority for the specified spanning-tree instance. Use the **no** form of this command to return to the default setting.

spanning-tree mst *instance-id* **priority** *priority*

no spanning-tree mst *instance-id* **priority**

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

instance-id	ID associated with a spanning-tree instance. The range is 0 to 15.
priority	Set the switch priority for the specified spanning-tree instance. This setting affects the likelihood that the switch is selected as the root switch. A lower value increases the probability that the switch is selected as the root switch. The range is 0 to 61440 in increments of 4096. Valid priority values are 0, 4096, 8192, 12288, 16384, 20480, 24576, 28672, 32768, 36864, 40960, 45056, 49152, 53248, 57344, and 61440. All other values are rejected.

Defaults

The default is 32768.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Examples

This example shows how to set the spanning-tree priority to 8192 for multiple spanning-tree (MST) instance 20:

```
Switch(config)# spanning-tree mst 20 priority 8192
```

You can verify your settings by entering the **show spanning-tree mst** *instance-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst *instance-id*	Displays MST information for the specified interface.
spanning-tree mst cost	Sets the path cost for MST calculations.
spanning-tree mst port-priority	Configures an interface priority.

spanning-tree mst root

Use the **spanning-tree mst root** global configuration command to configure the multiple spanning-tree (MST) root switch priority and timers based on the network diameter. Use the **no** form of this command to return to the default setting.

spanning-tree mst *instance-id* **root** {**primary** | **secondary**} [**diameter** *net-diameter* [**hello-time** *seconds*]]

no spanning-tree mst *instance-id* **root**

This command is available only if your switch is running the enhanced software image (EI).

Syntax Description

instance-id	ID associated with a MST instance. The range is 0 to 15.
root primary	Force this switch to be the root switch.
root secondary	Set this switch to be the root switch should the primary root switch fail.
diameter *net-diameter*	Set the maximum number of switches between any two end stations. The range is 2 to 7. This keyword is available only for MST instance 0.
hello-time *seconds*	Set the interval between hello bridge protocol data units (BPDUs) sent by the root switch configuration messages. The range is 1 to 10 seconds. This keyword is available only for MST instance 0.

Defaults

The primary root switch priority is 24576.

The secondary root switch priority is 28672.

The hello time is 2 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

Use the **spanning-tree mst** *instance-id* **root** command used only on backbone switches.

When you enter the **spanning-tree mst** *instance-id* **root** command, the software tries to set a high enough priority to make this switch the root of the spanning-tree instance. Because of the extended system ID support, the switch sets the switch priority for the instance to 24576 if this value will cause this switch to become the root for the specified instance. If any root switch for the specified instance has a switch priority lower than 24576, the switch sets its own priority to 4096 less than the lowest switch priority. (4096 is the value of the least-significant bit of a 4-bit switch priority value.)

When you enter the **spanning-tree mst** *instance-id* **root secondary** command, because of support for the extended system ID, the software changes the switch priority from the default value (32768) to 28672. If the root switch fails, this switch becomes the next root switch (if the other switches in the network use the default switch priority of 32768 and are therefore unlikely to become the root switch).

Examples

This example shows how to configure the switch as the root switch for instance 10 with a network diameter of 4:

```
Switch(config)# spanning-tree mst 10 root primary diameter 4
```

This example shows how to configure the switch as the secondary root switch for instance 10 with a network diameter of 4:

```
Switch(config)# spanning-tree mst 10 root secondary diameter 4
```

You can verify your settings by entering the **show spanning-tree mst** *instance-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree mst *instance-id*	Displays MST information for the specified instance.
spanning-tree mst forward-time	Sets the forward-delay time for all MST instances.
spanning-tree mst hello-time	Sets the interval between hello BPDUs sent by root switch configuration messages.
spanning-tree mst max-age	Sets the interval between messages that the spanning tree receives from the root switch.
spanning-tree mst max-hops	Sets the number of hops in a region before the BPDU is discarded.

spanning-tree port-priority

Use the **spanning-tree port-priority** interface configuration command to configure an interface priority. If a loop occurs, spanning tree can determine which interface to put in the forwarding state. Use the **no** form of this command to return to the default setting.

spanning-tree [**vlan** *vlan-id*] **port-priority** *priority*

no spanning-tree [**vlan** *vlan-id*] **port-priority**

Syntax Description

vlan *vlan-id*	(Optional) VLAN ID associated with a spanning-tree instance. The range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
priority	Number from 0 to 255. The lower the number, the higher the priority.

Defaults

The default is 128.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

If the variable *vlan-id* is omitted, the command applies to the spanning-tree instance associated with VLAN 1.

You can set the priority on a VLAN that has no interfaces assigned to it. The setting takes effect when you assign the interface to the VLAN.

Examples

This example shows how to increase the likelihood that the Fast Ethernet interface 0/2 will be put in the forwarding state if a loop occurs:

```
Switch(config)# interface fastethernet0/2
Switch(config-if)# spanning-tree vlan 20 port-priority 0
```

You can verify your settings by entering the **show spanning-tree interface** *interface-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree interface *interface-id*	Displays spanning-tree information for the specified interface.
spanning-tree cost	Sets the path cost for spanning-tree calculations.
spanning-tree vlan priority	Sets the switch priority for the specified spanning-tree instance.

spanning-tree portfast (global configuration)

Use the **spanning-tree portfast** global configuration command to globally enable bridge protocol data unit (BPDU) filtering on Port Fast-enabled ports, the BPDU guard feature on Port Fast-enabled ports, or the Port Fast feature on all nontrunking ports. The BPDU filtering feature prevents the switch port from sending or receiving BPDUs. The BPDU guard feature puts Port Fast-enabled ports that receive BPDUs in an error-disabled state. Use the **no** form of this command to return to the default setting.

spanning-tree portfast {**bpdufilter default** | **bpduguard default** | **default**}

no spanning-tree portfast {**bpdufilter default** | **bpduguard default** | **default**}

Syntax Description

bpdufilter default	Globally enable BPDU filtering on Port Fast-enabled ports and prevent the switch port connected to end stations from sending or receiving BPDUs.
bpduguard default	Globally enable the BPDU guard feature on Port Fast-enabled ports and place the ports that receive BPDUs in an error-disabled state.
default	Globally enable the Port Fast feature on all nontrunking ports. When the Port Fast feature is enabled, the port changes directly from a blocking state to a forwarding state without making the intermediate spanning-tree state changes.

Defaults

The BPDU filtering, the BPDU guard, and the Port Fast features are disabled on all ports unless they are individually configured.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **bpdufilter default** and **default** keywords were added.

Usage Guidelines

You can enable these features when the switch is operating in the per-VLAN spanning-tree (PVST) or the multiple spanning-tree (MST) mode. The MST mode is available only if you have the enhanced software image (EI) installed on your switch.

Use the **spanning-tree portfast bpdufilter default** global configuration command to globally enable BPDU filtering on ports that are Port Fast-enabled (the ports are in a Port Fast-operational state). The ports still send a few BPDUs at link-up before the switch begins to filter outbound BPDUs. You should globally enable BPDU filtering on a switch so that hosts connected to switch ports do not receive BPDUs. If a BPDU is received on a Port Fast-enabled port, the port loses its Port Fast-operational status and BPDU filtering is disabled.

You can override the **spanning-tree portfast bpdufilter default** global configuration command by using the **spanning-tree bdpufilter** interface configuration command.

Caution Enabling BPDU filtering on an interface is the same as disabling spanning tree on it and can result in spanning-tree loops.

Use the **spanning-tree portfast bpduguard default** global configuration command to globally enable BPDU guard on ports that are in a Port Fast-operational state. In a valid configuration, Port Fast-enabled ports do not receive BPDUs. Receiving a BPDU on a Port Fast-enabled port signals an invalid configuration, such as the connection of an unauthorized device, and the BPDU guard feature puts the port in the error-disabled state. The BPDU guard feature provides a secure response to invalid configurations because you must manually put the port back in service. Use the BPDU guard feature in a service-provider network to prevent an access port from participating in the spanning tree.

You can override the **spanning-tree portfast bpduguard default** global configuration command by using the **spanning-tree bdpuguard** interface configuration command.

Use the **spanning-tree portfast default** global configuration command to globally enable the Port Fast feature on all nontrunking ports. Configure Port Fast only on ports that connect to end stations; otherwise, an accidental topology loop could cause a data packet loop and disrupt switch and network operation. A Port Fast-enabled port moves directly to the spanning-tree forwarding state when linkup occurs without waiting for the standard forward-delay time.

You can override the **spanning-tree portfast default** global configuration command by using the **spanning-tree portfast** interface configuration command. You can use the **no spanning-tree portfast default** global configuration command to disable Port Fast on all ports unless they are individually configured with the **spanning-tree portfast** interface configuration command.

Examples

This example shows how to globally enable the BPDU filtering feature:

```
Switch(config)# spanning-tree portfast bpdufilter default
```

This example shows how to globally enable the BPDU guard feature:

```
Switch(config)# spanning-tree portfast bpduguard default
```

This example shows how to globally enable the Port Fast feature on all nontrunking ports:

```
Switch(config)# spanning-tree portfast default
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree bpdufilter	Prevents a port from sending or receiving BPDUs.
spanning-tree bpduguard	Puts a port in the error-disabled state when it receives a BPDU.
spanning-tree portfast (interface configuration)	Enables the Port Fast feature on an interface in all its associated VLANs.

spanning-tree portfast (interface configuration)

Use the **spanning-tree portfast** interface configuration command to enable the Port Fast feature on an interface in all its associated VLANs. When the Port Fast feature is enabled, the interface changes directly from a blocking state to a forwarding state without making the intermediate spanning-tree state changes. Use the **no** form of this command to return to the default setting.

spanning-tree portfast [**disable** | **trunk**]

no spanning-tree portfast

Syntax Description

disable	(Optional) Disable the Port Fast feature on the specified interface.
trunk	(Optional) Enable the Port Fast feature on a trunking interface.

Defaults

The Port Fast feature is disabled on all interfaces; however, it is automatically enabled on dynamic-access ports.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(9)EA1	The **disable** and **trunk** keywords were added.

Usage Guidelines

Use this feature only on interfaces that connect to end stations; otherwise, an accidental topology loop could cause a data packet loop and disrupt switch and network operation.

You can enable this feature when the switch is operating in the per-VLAN spanning-tree (PVST) or the multiple spanning-tree (MST) mode.

This feature affects all VLANs on the interface.

A port with the Port Fast feature enabled is moved directly to the spanning-tree forwarding state without waiting the standard forward-time delay.

You can use the **spanning-tree portfast default** global configuration command to globally enable the Port Fast feature on all nontrunking interfaces. However, the **spanning-tree portfast** interface configuration command can override the global setting.

If you configure the **spanning-tree portfast default** global configuration command, you can enable Port Fast on a port that is not a trunk port by using the **no spanning-tree portfast** interface configuration command.

The **no spanning-tree portfast** interface configuration command is the same as the **spanning-tree portfast disable** interface configuration command.

Examples

This example shows how to enable the Port Fast feature on an interface:

```
Switch(config)# interface fastethernet0/2
Switch(config-if)# spanning-tree portfast
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree bpdufilter	Prevents a port from sending or receiving bridge protocol data units (BPDUs).
spanning-tree bpduguard	Puts a port in the error-disabled state when it receives a BPDU.
spanning-tree portfast (global configuration)	Globally enables the BPDU filtering or the BPDU guard feature on Port Fast-enabled ports or enables the Port Fast feature on all nontrunking ports.

spanning-tree stack-port

Use the **spanning-tree stack-port** interface configuration command to enable cross-stack UplinkFast (CSUF) on an interface and to accelerate the choice of a new root port when a link or switch fails or when spanning tree reconfigures itself. Use the **no** form of this command to return to the default setting.

spanning-tree stack-port

no spanning-tree stack-port

Syntax Description

This command has no arguments or keywords.

Defaults

CSUF is disabled on all interfaces.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command is effective only if you enable the UplinkFast feature by using the **spanning-tree uplinkfast** global configuration command.

Use this command only on access switches.

The CSUF feature is supported only when the switch is running per-VLAN spanning-tree (PVST).

You can enable CSUF only on one stack-port Gigabit Interface Converter (GBIC) interface. The stack port connects to the GigaStack GBIC multidrop backbone. If you try to enable CSUF on a Fast Ethernet or a copper-based Gigabit Ethernet port, you receive an error message.

If CSUF is already enabled on an interface and you try to enable it on another interface, you receive an error message. You must disable CSUF on the first interface before enabling it on a new interface.

Examples

This example shows how to enable CSUF on the GBIC interface:

```
Switch(config)# interface gigabitethernet0/1
Switch(config-if)# spanning-tree stack-port
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
spanning-tree uplinkfast	Accelerates the choice of a new root port when a link or switch fails or when spanning tree reconfigures itself.

spanning-tree uplinkfast

Use the **spanning-tree uplinkfast** global configuration command to accelerate the choice of a new root port when a link or switch fails or when the spanning tree reconfigures itself. Use the **no** form of this command to return to the default setting.

spanning-tree uplinkfast [**max-update-rate** *pkts-per-second*]

no spanning-tree uplinkfast [**max-update-rate**]

Syntax Description

max-update-rate *pkts-per-second*	(Optional) The number of packets per second at which update packets are sent. The range is 0 to 65535.

Defaults

UplinkFast is disabled.

The update rate is 150 packets per second.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The *pkts-per-second* range was changed to 0 to 65535.

Usage Guidelines

Use this command only on access switches.

The UplinkFast feature is supported only when the switch is running per-VLAN spanning-tree (PVST).

When you enable UplinkFast, it is enabled for the entire switch and cannot be enabled for individual VLANs.

When UplinkFast is enabled, the switch priority of all VLANs is set to 49152. If you change the path cost to a value less than 3000 and you enable UplinkFast or UplinkFast is already enabled, the path cost of all interfaces and VLAN trunks is increased by 3000 (if you change the path cost to 3000 or above, the path cost is not altered). The changes to the switch priority and the path cost reduces the chance that the switch will become the root switch.

When UplinkFast is disabled, the switch priorities of all VLANs and path costs of all interfaces are set to default values if you did not modify them from their defaults.

When spanning tree detects that the root port has failed, UplinkFast immediately switches over to an alternate root port, changing the new root port directly to FORWARDING state. During this time, a topology change notification is sent.

Do not enable the root guard on interfaces that will be used by the UplinkFast feature. With UplinkFast, the backup interfaces (in the blocked state) replace the root port in the case of a failure. However, if root guard is also enabled, all the backup interfaces used by the UplinkFast feature are placed in the root-inconsistent state (blocked) and prevented from reaching the forwarding state.

If you set the max-update-rate to 0, station-learning frames are not generated, so the spanning-tree topology converges more slowly after a loss of connectivity.

Examples

This example shows how to enable UplinkFast:

```
Switch(config)# spanning-tree uplinkfast
```

You can verify your setting by entering the **show spanning-tree summary** privileged EXEC command.

Related Commands

Command	Description
show spanning-tree summary	Displays a summary of the spanning-tree port states.
spanning-tree stack-port	Enables cross-stack UplinkFast (CSUF) on an interface and accelerates the choice of a new root port when a link or switch fails or when spanning tree reconfigures itself.
spanning-tree vlan root primary	Forces this switch to be the root switch.

spanning-tree vlan

Use the **spanning-tree vlan** global configuration command to configure spanning tree on a per-VLAN basis. Use the **no** form of this command to return to the default setting.

spanning-tree vlan *vlan-id* {**forward-time** *seconds* | **hello-time** *seconds* | **max-age** *seconds* | **priority** *priority* | {**root** {**primary** | **secondary**} [**diameter** *net-diameter* [**hello-time** *seconds*]]}}

no spanning-tree vlan *vlan-id* [**forward-time** | **hello-time** | **max-age** | **priority** | **root**]

Syntax Description

vlan-id	VLAN ID associated with a spanning-tree instance. The range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
forward-time *seconds*	Set the forward-delay time for the specified spanning-tree instance. The forwarding time determines how long each of the listening and learning states last before the interface begins forwarding. The range is 4 to 30 seconds.
hello-time *seconds*	Set the interval between hello bridge protocol data units (BPDUs) sent by the root switch configuration messages. The range is 1 to 10 seconds.
max-age *seconds*	Set the interval between messages the spanning tree receives from the root switch. If a switch does not receive a BPDU message from the root switch within this interval, it recomputes the spanning-tree topology. The range is 6 to 40 seconds.
priority *priority*	Set the switch priority for the specified spanning-tree instance. This setting affects the likelihood that the switch is selected as the root switch. A lower value increases the probability that the switch is selected as the root switch. The range is 0 to 61440 in increments of 4096. Valid priority values are 4096, 8192, 12288, 16384, 20480, 24576, 28672, 32768, 36864, 40960, 45056, 49152, 53248, 57344, and 61440. All other values are rejected.
root primary	Force this switch to be the root switch.
root secondary	Set this switch to be the root switch should the primary root switch fail.
diameter *net-diameter*	Set the maximum number of switches between any two end stations. The range is 2 to 7.

Defaults

Spanning tree is enabled on all VLANs.

The forward-delay time is 15 seconds.

The hello time is 2 seconds.

The max-age is 20 seconds.

The primary root switch priority is 24576.

The secondary root switch priority is 28672.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(9)EA1	The **priority** *priority* range changed from 1 to 65535 to 1 to 61440 (in increments of 4096).

Usage Guidelines

Disabling the STP causes the VLAN to stop participating in the spanning-tree topology. Interfaces that are administratively down remain down. Received BPDUs are forwarded like other multicast frames. The VLAN does not detect and prevent loops when STP is disabled.

You can disable the STP on a VLAN that is not currently active and verify the change by using the **show running-config** or the **show spanning-tree vlan** *vlan-id* privileged EXEC command. The setting takes effect when the VLAN is activated.

When disabling or re-enabling the STP, you must use a single command lint to specify each VLAN that you want to disable or enable.

When a VLAN is disabled and then enabled, all assigned VLANs continue to be its members. However, all spanning-tree bridge parameters are returned to their previous settings (the last setting before the VLAN was disabled).

You can enable spanning-tree options on a VLAN that has no interfaces assigned to it. The setting takes effect when you assign interfaces to it.

When setting the **max-age** *seconds*, if a switch does not receive BPDUs from the root switch within the specified interval, it recomputes the spanning-tree topology. The **max-age** setting must be greater than the **hello-time** setting.

The **spanning-tree vlan** *vlan-id* **root** command should be used only on backbone switches.

When you enter the **spanning-tree vlan** *vlan-id* **root** command, the software checks the switch priority of the current root switch for each VLAN. Because of the extended system ID support, the switch sets the switch priority for the specified VLAN to 24576 if this value will cause this switch to become the root for the specified VLAN. If any root switch for the specified VLAN has a switch priority lower than 24576, the switch sets its own priority for the specified VLAN to 4096 less than the lowest switch priority. (4096 is the value of the least-significant bit of a 4-bit switch priority value.)

When you enter the **spanning-tree vlan** *vlan-id* **root secondary** command, because of support for the extended system ID, the software changes the switch priority from the default value (32768) to 28672. If the root switch should fail, this switch becomes the next root switch (if the other switches in the network use the default switch priority of 32768, and therefore, are unlikely to become the root switch).

Examples

This example shows how to disable the STP on VLAN 5:

```
Switch(config)# no spanning-tree vlan 5
```

You can verify your setting by entering the **show spanning-tree** privileged EXEC command. In this instance, VLAN 5 does not appear in the list.

This example shows how to set the spanning-tree forwarding time to 18 seconds for VLAN 20:

```
Switch(config)# spanning-tree vlan 20 forward-time 18
```

This example shows how to set the spanning-tree hello-delay time to 3 seconds for VLAN 20:

```
Switch(config)# spanning-tree vlan 20 hello-time 3
```

This example shows how to set spanning-tree max-age to 30 seconds for VLAN 20:

```
Switch(config)# spanning-tree vlan 20 max-age 30
```

This example shows how to reset the **max-age** parameter to the default value for spanning-tree instance 100:

```
Switch(config)# no spanning-tree vlan 100 max-age
```

This example shows how to set the spanning-tree priority to 8192 for VLAN 20:

```
Switch(config)# spanning-tree vlan 20 priority 8192
```

This example shows how to configure the switch as the root switch for VLAN 10 with a network diameter of 4:

```
Switch(config)# spanning-tree vlan 10 root primary diameter 4
```

This example shows how to configure the switch as the secondary root switch for VLAN 10 with a network diameter of 4:

```
Switch(config)# spanning-tree vlan 10 root secondary diameter 4
```

You can verify your settings by entering the **show spanning-tree vlan** *vlan-id* privileged EXEC command.

Related Commands

Command	Description
show spanning-tree vlan	Displays spanning-tree information.
spanning-tree cost	Sets the path cost for spanning-tree calculations.
spanning-tree guard	Enables the root guard or the loop guard feature for all the VLANs associated with the selected interface.
spanning-tree port-priority	Sets an interface priority.
spanning-tree portfast (global configuration)	Globally enables the BPDU filtering or the BPDU guard feature on Port Fast-enabled ports or enables the Port Fast feature on all nontrunking ports.
spanning-tree portfast (interface configuration)	Enables the Port Fast feature on an interface in all its associated VLANs.
spanning-tree uplinkfast	Enables the UplinkFast feature, which accelerates the choice of a new root port.

speed

Use the **speed** interface configuration command to specify the speed of a port. Use the **no** form of this command to return the port to its default value.

speed {**10** | **100** | **1000** | **auto**| **nonegotiate**}

no speed

Note You cannot configure speed or duplex mode on Gigabit Interface Converter (GBIC) ports, but for certain types of GBICs, you can configure speed to not negotiate (**nonegotiate**) if they are connected to a device that does not support autonegotiation.

Syntax Description

10	Port runs at 10 Mbps.
100	Port runs at 100 Mbps.
1000	Port runs at 1000 Mbps (only valid for Gigabit Ethernet ports).
auto	Port automatically detects whether it should run at 10 or 100 Mbps on Fast Ethernet ports or at 10, 100, or 1000 Mbps on 10/100/1000 ports.
nonegotiate	Autonegotiation is disabled and the port runs at 1000 Mbps. This option is valid and visible only on 1000BASE-X, -LX, and -ZX GBIC ports. Gigastack GBICs and 1000BASE-T GBICs do not suport disabling of autonegotiation.

Defaults

For Fast Ethernet and 10/100/1000 ports, the default is **auto**.

For 100BASE-FX ports, the default is 100 Mbps.

For Gigabit Interface Converter (GBIC)-module ports, the default is 1000 Mbps.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(11)EA1	The **nonegotiate** keyword was added.

Usage Guidelines

Fast Ethernet ports, except for 100BASE-FX ports, can be configured at 10 or 100 Mbps. The 10/100/1000 ports can be configured at 10, 100, or 1000 Mbps and operate only in the full-duplex mode.

You cannot configure the speed on GBIC interfaces, but you can configure the speed to not negotiate **(nonegotiate)** for the 1000BASE-SX, -LX, or -ZX GBICs, if they are connected to devices that do not support autonegotiation. GBIC-module ports support only 1000 Mbps. The speed values of 10 Mbps and 100 Mbps are not supported.

If the speed is set to **auto**, the switch negotiates with the device at the other end of the link for the speed setting and then forces the speed setting to the negotiated value. The duplex setting remains as configured on each end of the link, which could result in a duplex setting mismatch. If both the speed and duplex are set to specific values, autonegotiation is disabled.

Note The 100BASE-FX ports on Catalyst 2950C-24 switches do not support the **speed** command. These ports operate only in 100-Mbps and full-duplex mode.

Note For guidelines on setting the switch speed and duplex parameters, refer to "Configuring the Switch Ports" chapter in the *Catalyst 2950 Desktop Switch Software Configuration Guide.*

Examples

This example shows how to set FastEthernet port 1 to 100 Mbps:

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# speed 100
```

You can verify your settings by entering the **show running-config** privileged EXEC command.

Related Commands

Command	Description
duplex	Specifies the duplex mode of operation for switch ports.
show running-config	Displays the current operating configuration. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.

storm-control

Use the **storm-control** interface configuration command to enable broadcast, multicast, or unicast storm control on a port and to specify the action taken when a storm occurs on a port. Use the **no** form of this command to disable storm control for broadcast, multicast, or unicast traffic and disable the specified storm-control action.

storm-control {{{**broadcast** | **multicast** | **unicast**} **level** *level* [*lower level*]} | **action** {**shutdown** | **trap**}}

no storm-control {{**broadcast** | **multicast** | **unicast**} **level**} | **action**}

Syntax Description

{**broadcast** | **multicast** | **unicast**}	Determines the type of packet-storm suppression. • **broadcast**—Enable broadcast storm control on the port. • **multicast**—Enable multicast storm control on the port. • **unicast**—Enable unicast storm control on the port.
level *level* [*lower level*]	Defines the rising and falling suppression levels. • *level*—Rising suppression level as a percent of total bandwidth, up to two decimal places; valid values are from 0 to 100 percent. Block the flooding of storm packets when the value specified for *level* is reached. • *lower level*—(Optional) Falling suppression level as a percent of total bandwidth, up to two decimal places; valid values are from 0 to 100. This value must be less than the rising supression value.
action	Action taken when a storm occurs on a port. The default action is to filter traffic and not send an Simple Network Management Protocol (SNMP) trap.
shutdown	Disables the port during a storm.
trap	Sends an SNMP trap when a storm occurs.

Defaults

Broadcast, multicast, and unicast storm control are disabled.

The default action is to filter traffic and to not send an SNMP trap.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced **port storm-control** command.

Usage Guidelines

Use the **storm-control** command to enable or disable broadcast, multicast, or unicast storm control on a port. After a port is disabled during a storm, use the **no shutdown** interface configuration command to enable the port.

The suppression levels are entered as a percentage of total bandwidth. A suppression value of 100 percent means that no limit is placed on the specified traffic type. This feature is enabled only when the rising suppression level is less than 100 percent. If no other storm-control configuration is specified, the default action is to filter the traffic causing the storm and to send no SNMP trap.

When a storm occurs and the action is to filter traffic, if the falling suppression level is not specified, the switch blocks all traffic until the traffic rate drops below the rising suppression level. If the falling suppression level is specified, the switch blocks traffic until the traffic rate drops below this level.

When a multicast or unicast storm occurs and the action is to filter traffic, the switch blocks all traffic (broadcast, multicast, and unicast traffic) and sends only Spanning Tree Protocol (STP) packets.

When a broadcast storm occurs and the action is to filter traffic, the switch blocks only broadcast traffic.

The **trap** and **shutdown** options are independent of each other.

Examples

This example shows how to enable broadcast storm control on a port with a 75.67 percent rising suppression level:

```
Switch(config-if)# storm-control broadcast level 75.67
```

This example shows how to enable multicast storm control on a port with a 87 percent rising suppression level and a 65 percent falling suppression level:

```
Switch(config-if)# storm-control multicast level 87 65
```

This example shows how to enable the **shutdown** action on a port:

```
Switch(config-if)# storm-control action shutdown
```

This example shows how to enable the **trap** action on a port:

```
Switch(config-if)# storm-control action trap
```

This example shows how to disable the **shutdown** action on a port:

```
Switch(config-if)# no storm-control action shutdown
```

You can verify your settings by entering the **show storm-control** privileged EXEC command.

Related Commands

Command	Description
show storm-control	Displays the packet-storm control information.

switchport access

Use the **switchport access** interface configuration command to configure a port as a static-access or dynamic-access port. If the mode is set to access, the port operates as a member of the configured VLAN. If set to dynamic, the port starts discovery of its VLAN assignment based on the incoming packets it receives. Use the **no** form of this command to reset the access mode to the default VLAN for the switch.

switchport access vlan {*vlan-id* | **dynamic**}

no switchport access

Syntax Description

access vlan *vlan-id*	Configure the interface as a static-access port; valid values are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed. Do not enter leading zeros.
access vlan dynamic	Specify that the access mode VLAN is dependent on the VLAN Membership Policy Server (VMPS) protocol. The port is assigned to a VLAN based on the source MAC address of a host (or hosts) connected to the port. The switch sends every new MAC address received to the VMPS server to obtain the VLAN name to which the dynamic-access port should be assigned. If the port already has a VLAN assigned and the source has already been approved by the VMPS, the switch forwards the packet to the VLAN.

Defaults

All ports are in static-access mode in VLAN 1 if the port is not connected to a device running Dynamic Trunking Protocol (DTP). The default access VLAN for an access port is VLAN 1.

All ports are dynamic trunk ports.

A dynamic-access port is initially a member of no VLAN and receives its assignment based on the packet it receives.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **dynamic** keyword was added.

Usage Guidelines

The **no switchport access** command resets the access mode VLAN to the appropriate default VLAN for the device.

The port must be in access mode before the **switchport access vlan** command can take effect. For more information, see the **switchport mode** command.

An access port can be assigned to only one VLAN.

The VMPS server (such as a Catalyst 6000 series switch) must be configured before a port is configured as dynamic.

These restrictions apply to dynamic-access ports:

- The software implements the VLAN Query Protocol (VQP) client, which can query a VMPS such as a Catalyst 6000 series switch. The Catalyst 3550 switches are not VMPS servers. The VMPS server must be configured before a port is configured as dynamic.
- Use dynamic-access ports only to connect end stations. Connecting them to switches or routers that use bridging protocols can cause a loss of connectivity.
- Configure the network so that Spanning Tree Protocol (STP) does not put the dynamic-access port in an STP blocking state. The Port Fast feature is automatically enabled on dynamic-access ports.
- Dynamic-access ports can only be in one VLAN and do not use VLAN tagging.
- Dynamic-access ports cannot be configured as:
  - Members of an EtherChannel port group (dynamic-access ports cannot be grouped with any other port, including other dynamic ports).
  - Source or destination ports in a static address entry.
  - Monitor ports.

Examples

This example shows how to assign a port already in access mode to VLAN 2 (instead of the default VLAN 1):

```
Switch(config-if)# switchport access vlan 2
```

You can verify your settings by entering the **show interfaces** *interface-id* **switchport** privileged EXEC command and examining information in the Administrative Mode and Operational Mode rows.

Related Commands

Command	Description
show interfaces switchport	Displays the administrative and operational status of a switching (nonrouting) port, including port blocking and port protection settings.
switchport mode	Configures the VLAN membership mode of a port.

switchport mode

Use the **switchport mode** interface configuration command to configure the VLAN membership mode of a port. Use the **no** form of this command to reset the mode to the appropriate default for the device.

switchport mode {**access** | **dynamic** {**auto** | **desirable**} | **trunk**}

no switchport mode

Syntax Description

access	Set the port to access mode (either static-access or dynamic-access depending on the setting of the **switchport access vlan** inteface configuration command). The port is set to access unconditionally and operates as a nontrunking, single VLAN interface that transmits and receives nonencapsulated (non-tagged) frames. An access port can be assigned to only one VLAN.
dynamic auto	Set the interface trunking mode dynamic parameter to **auto** to specify that the interface convert the link to a trunk link.
dynamic desirable	Set the interface trunking mode dynamic parameter to **desirable** to specify that the interface actively attempt to convert the link to a trunk link.
trunk	Set the port to trunk unconditionally. The port is a trunking VLAN Layer 2 interface. The port transmits and receives encapsulated (tagged) frames that identify the VLAN of origination. A trunk is a point-to-point link between two switches or between a switch and a router.

Defaults

The default mode is **dynamic desirable**.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **dynamic auto** and **dynamic desirable** keywords were added.

Usage Guidelines

Configuration by using the **access** or **trunk** keywords takes affect only when you configure the port in the appropriate mode by using the **switchport mode** command. The static-access and trunk configurations are saved, but only one configuration is active at a time.

If you enter **access** mode, the interface changes into permanent nontrunking mode and negotiates to convert the link into a nontrunk link even if the neighboring interface does not agree to the change.

If you enter **dynamic auto** mode, the interface converts the link to a trunk link if the neighboring interface is set to **trunk** or **desirable** mode.

If you enter **dynamic desirable** mode, the interface becomes a trunk interface if the neighboring interface is set to **trunk**, **desirable**, or **auto** mode.

If you enter **trunk** mode, the interface changes into permanent trunking mode and negotiates to convert the link into a trunk link even if the interface connecting to it does not agree to the change.

The **no switchport mode** form resets the mode to **dynamic desirable**.

Trunk ports cannot coexist on the same switch.

To autonegotiate trunking, the interfaces must be in the same VTP domain. Trunk negotiation is managed by the Dynamic Trunking Protocol (DTP), which is a point-to-point protocol. However, some internetworking devices might forward DTP frames improperly, which could cause misconfigurations. To avoid this, you should configure interfaces connected to devices that do not support DTP to not forward DTP frames, which turns off DTP.

- If you do not intend to trunk across those links, use the **switchport mode access** interface configuration command to disable trunking.
- To enable trunking to a device that does not support DTP, use the **switchport mode trunk** and **switchport nonegotiate** interface configuration commands to cause the interface to become a trunk but to not generate DTP frames.

Examples

This example shows how to configure a port for access mode:

```
Switch(config-if)# switchport mode access
```

This example shows how set the interface to dynamic desirable mode:

```
Switch(config-if)# switchport mode dynamic desirable
```

This example shows how to configure a port for trunk mode:

```
Switch(config-if)# switchport mode trunk
```

You can verify your settings by entering the **show interfaces** *interface-id* **switchport** privileged EXEC command and examining information in the Administrative Mode and Operational Mode rows.

Related Commands

Command	Description
show interfaces switchport	Displays the administrative and operational status of a switching (nonrouting) port, including port blocking and port protection settings.
switchport access	Configures a port as a static-access port.

switchport nonegotiate

Use the **switchport nonegotiate** interface configuration command to specify that Dynamic Trunking Protocol (DTP) negotiation packets are not sent on the Layer 2 interface. The switch does not engage in DTP negotiation on this interface. Use the **no** form of this command to return to the default setting.

switchport nonegotiate

no switchport nonegotiate

Syntax Description

This command has no arguments or keywords.

Defaults

The default is to use DTP negotiation to determine trunking status.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **no** form of the **switchport nonegotiate** command removes **nonegotiate** status.

This command is valid only when the interface switchport mode is access or trunk (configured by using the **switchport mode access** or the **switchport mode trunk** interface configuration command). This command returns an error if you attempt to execute it in **dynamic** (**auto** or **desirable**) mode.

Internetworking devices that do not support DTP might forward DTP frames improperly and cause misconfigurations. To avoid this, you should turn off DTP by using the **switchport no negotiate** command to configure the interfaces connected to devices that do not support DTP to not forward DTP frames.

When you enter the **switchport nonegotiate** command, DTP negotiation packets are not sent on the interface. The device does or does not trunk according to the **mode** parameter given: **access** or **trunk**.

- If you do not intend to trunk across those links, use the **switchport mode access** interface configuration command to disable trunking.
- To enable trunking ona device that does not support DTP, use the **switchport mode trunk** and **switchport nonegotiate** interface configuration commands to cause the interface to become a trunk but to not generate DTP frames.



Note On GigaStack GBICs, dynamic trunking is supported only when only port of a GigaStack GBIC is being used. If trunking is required on a GigaStack GBIC where both ports are in use, use the **switchport mode trunk** and **switchport nonegotiate** interface configuration commands on both GBIC interfaces to cause the interfaces to become trunks.

Examples

This example shows how to cause an interface to refrain from negotiating trunking mode and to act as a trunk or access port (depending on the **mode** set):

```
Switch(config-if)# switchport nonegotiate
```

You can verify your setting by entering the **show interfaces** *interface-id* **switchport** privileged EXEC command.

Related Commands

Command	Description
show interfaces switchport	Displays the administrative and operational status of a switching (nonrouting) port, including port blocking and port protection settings.
switchport mode	Configures the VLAN membership mode of a port.

switchport port-security

Use the **switchport port-security** interface configuration command without keywords to enable port security on an interface. Use the keywords to configure secure MAC addresses, a maximum number of secure MAC addresses, or the violation mode. Use the **no** form of this command to disable port security or to set the parameters to their default states.

switchport port-security [**mac-address** *mac-address*] | [**mac-address sticky** [*mac-address*]] | [**maximum** *value*] | [**violation** {**protect** | **restrict** | **shutdown**}]

no switchport port-security [**mac-address** *mac-address*] | [**mac-address sticky** [*mac-address*]] | [**maximum** *value*] | [**violation** {**protect** | **restrict** | **shutdown**}]

Syntax Description

mac-address *mac-address*	(Optional) Specify a secure MAC address for the port by entering a 48-bit MAC address. You can add additional secure MAC addresses up to the maximum value configured.
mac-address sticky [*mac-address*]	(Optional) Enable the interface for *sticky learning* by entering only the **mac-address sticky** keywords. When sticky learning is enabled, the interface adds all secure MAC addresses that are dynamically learned to the running configuration and converts these addresses to sticky secure MAC addresses. Specify a sticky secure MAC address by entering the **mac-address sticky** *mac-address* keywords.
maximum *value*	(Optional) Set the maximum number of secure MAC addresses for the interface. The range is from 1 to 132. The default is 1.
violation	(Optional) Set the security violation mode or the action to be taken if port security is violated. The default is **shutdown**.
protect	Set the security violation protect mode. When port secure MAC addresses reach the limit allowed on the port, packets with unknown source addresses are dropped until you remove a sufficient number of secure MAC addresses to drop below the maximum value.
restrict	Set the security restrict mode. In this mode, a port security violation causes a trap notification to be sent to the network management station.
shutdown	Set the security violation shutdown mode. In this mode, a port security violation causes the interface to immediately become error-disabled, and an SNMP trap notification is sent. When a secure port is in the error-disabled state, you can bring it out of this state by entering the **errdisable recovery cause psecure-violation** global configuration command, or you can manually re-enable it by entering the **shutdown** and **no shut down** interface configuration commands.

Defaults

Port security is disabled.

When port security is enabled, if no keywords are entered, the default maximum number of secure MAC addresses is 1.

Sticky learning is disabled.

The default violation mode is **shutdown**.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **port security** and **mac-address-table secure** commands.
12.1(11)EA1	The **mac-address sticky** [*mac-address*] option was added.

Usage Guidelines

After you set the maximum number of secure MAC addresses allowed on a port, you can add secure addresses to the address table by manually configuring all of them, by allowing the port to dynamically configure all of them, or by configuring a number of MAC addresses and allowing the rest to be dynamically configured.

You can delete dynamic secure MAC addresses from the address table by entering the **clear port-security dynamic** privileged EXEC command.

You can enable sticky learning on an interface by using the **switchport port-security mac-address sticky** interface configuration command. When you enter this command, the interface converts all the dynamic secure MAC addresses, including those that were dynamically learned before sticky learning was enabled, to sticky secure MAC addresses. It adds all the sticky secure MAC addresses to the running configuration.

If you disable sticky learning by using the **no switchport port-security mac-address sticky** interface configuration command, or the running configuration is removed, the sticky secure MAC addresses remain part of the running configuration but are removed from the address table. The addresses that were removed can be dynamically reconfigured and added to the address table as dynamic addresses.

You can configure sticky secure MAC addresses by using the **switchport port-security mac-address sticky** *mac-address* interface configuration command. These addresses are added to the running configuration. If port security is disabled, the sticky secure MAC addresses remain in the running configuration.

If you disable sticky learning, when you enter the **switchport port-security mac-address sticky** *mac-address* interface configuration command, an error message appears, and the sticky secure MAC address is not added to the running configuration.

You can delete sticky secure MAC addresses from the address table by disabling sticky learning, which converts the sticky secure MAC addresses to dynamic secure addresses. After disabling sticky learning, delete the dynamic secure addresses.

If you save the sticky secure MAC addresses in the configuration file, when the switch restarts or the interface shuts down, the interface does not need to relearn these addresses. If you do not save the sticky secure addresses, they are lost.

Note If sticky learning is not enabled, when the switch restarts or the port shuts down, all the addresses that were dynamically learned are removed.

A security violation occurs when the maximum number of secure MAC addresses have been added to the address table and a station whose MAC address is not in the address table attempts to access the interface, or when a station whose MAC address is configured as a secure MAC address on another secure port attempts to access the interface.

If you enable port security on a voice VLAN port and if there is a PC connected to the IP phone, you should set the maximum allowed secure addresses on the port to more than 1.

You cannot configure static secure MAC addresses in the voice VLAN.

If you specify **restrict** or **shutdown**, use the **snmp-server host** global configuration command to configure the Simple Network Management Protocol (SNMP) trap host to receive traps.

You can enable port security on a interface only if the port is not configured as one of these:

- Trunk ports—If you try to enable port security on a trunk port, an error message appears, and port security is not enabled. If you try to change the mode of a secure port to trunk, the port mode is not changed.
- Dynamic port—A port in dynamic mode can negotiate with its neighbor to become a trunk port. If you try to enable port security on a dynamic port, an error message appears, and port security is not enabled. If you try to change the mode of a secure port to dynamic, the port mode is not changed.
- Dynamic-access port—If you try to enable port security on a dynamic-access (VLAN Query Protocol [VQP]) port, an error message appears, and port security is not enabled. If you try to change a secure port to dynamic VLAN assignment, an error message appears, and the VLAN configuration is not changed.
- EtherChannel port—Before enabling port security on the port, you must first remove it from the EtherChannel. If you try to enable port security on an EtherChannel or on an active port in an EtherChannel, an error message appears, and port security is not enabled. If you enable port security on a not-yet active port of an EtherChannel, the port does not join the EtherChannel.
- 802.1X port—You cannot configure an 802.1X port as a secure port. If you try to enable port security on an 802.1X port, an error message appears, and port security is not enabled. If you try to change a secure port to an 802.1X port, an error message appears, and the 802.1X settings are not changed.
- Switched Port Analyzer (SPAN) destination port—You can enable port security on a port that is a SPAN destination port; however, port security is disabled until the port is removed as a SPAN destination. You can enable port security on a SPAN source port.

When a secure port is in the error-disabled state, you can bring it out of this state by entering the **errdisable recovery cause psecure-violation** global configuration command, or you can manually re-enable it by entering the **shutdown** and **no shut down** interface configuration commands.

Examples

This example shows how to enable port security:

```
Switch(config-if)# switchport port-security
```

This example shows how to set the action that the port takes when an address violation occurs:

```
Switch(config-if)# switchport port-security violation shutdown
```

This example shows how to set the maximum number of addresses that a port can learn to 20.

```
Switch(config-if)# switchport port-security maximum 20
```

This example shows how to enable sticky learning and to enter two sticky secure MAC addresses:

```
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security mac-address sticky 0000.0000.4141
Switch(config-if)# switchport port-security mac-address sticky 0000.0000.000f
```

You can verify your settings by entering the **show port-security** privileged EXEC command.

Related Commands

Command	Description
clear port-security dynamic	Deletes from the MAC address table a specific dynamic secure address or all the dynamic secure addresses on an interface.
show port-security	Displays the port security settings defined for the port.

switchport port-security aging

Use the **switchport port-security aging** interface configuration command to set the aging time and type for secure address entries or to change the aging behavior for statically configured secure addresses on a particular port. Use the **no** form of this command to disable port security aging or to set the parameters to their default states.

switchport port-security aging {**static** | **time** *time* | **type** {**absolute | inactivity**}}

no switchport port-security aging {**static | time | type**}

Syntax Description

static	Enable aging for statically configured secure addresses on this port.
time *time*	Specify the aging time for this port. The valid range is from 0 to 1440 minutes. If the time is 0, aging is disabled for this port.
type absolute	Set the aging type as absolute aging. All the secure addresses on this port age out exactly after the time *(*minutes) specified and are removed from the secure address list.
type inactivity	Set the aging type as inactivity aging. The secure addresses on this port age out only if there is no data traffic from the secure source address for the specified time period.

Defaults

The port security aging feature is disabled. The default time is 0 minutes.

The default aging type is absolute.

The default static aging behavior is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

To enable secure address aging for a particular port, set the port aging time to a value other than 0.

To allow limited-time access to particular secure addresses, set the aging type as **absolute**. When the aging time lapses, the secure addresses are deleted.

To allow continuous access to a limited number of secure addresses, set the aging type as **inactivity**. This removes the secure address when it becomes inactive, and other addresses can become secure.

To allow unlimited access to a secure address, configure it as a secure address, and disable aging for the statically configured secure address by using the **no switchport port-security aging static** interface configuration command.

Examples

This example sets the aging time as 2 hours for absolute aging for all the secure addresses on Fast Ethernet interface 0/1.

```
Switch(config)# interface fastethernet0/1
Switch(config-if)# switchport port-security aging time 120
```

This example sets the aging time as 2 minutes for inactivity aging type for configured secure addresses on Fast Ethernet interface 0/2.

```
Switch(config)# interface fastethernet0/2
Switch(config-if)# switchport port-security aging time 2
Switch(config-if)# switchport port-security aging type inactivity
Switch(config-if)# switchport port-security aging static
```

This example shows how to disable aging for configured secure addresses.

```
Switch(config-if)# no switchport port-security aging static
```

Related Commands

Command	Description
show port-security	Displays the port security settings defined for the port.
switchport port-security	Enables port security on a port, restricts the use of the port to a user-defined group of stations, and configures secure MAC addresses.

switchport priority extend

Use the **switchport priority extend** interface configuration command to set a port priority for the incoming untagged frames or the priority of frames received by the IP phone connected to the specified port. Use the **no** form of this command to return to the default setting.

switchport priority extend {**cos** *value* | **none** | **trust**}

no switchport priority extend

Syntax Description

cos *value*	Set the IP phone port to override the priority received from PC or the attached device. The class of service (CoS) value is a number from 0 to 7. Seven is the highest priority. The default is 0.
none	The IP phone is not instructed what to do with the priority.
trust	Set the IP phone port to trust the priority received from PC or the attached device.

Defaults

The port priority is not set, and the default value for untagged frames received on the port is 0.

The IP phone connected to the port is not instructed (**none**) what to do with the priority.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Examples

This example shows how to configure the IP phone connected to the specified port to trust the received 802.1P priority:

```
Switch(config-if)# switchport priority extend trust
```

You can verify your settings by entering the **show interfaces** *interface-id* **switchport** privileged EXEC command.

Related Commands

Command	Description
show interfaces	Displays the administrative and operational status of a switching (nonrouting) port.
switchport voice vlan	Configures the voice VLAN on the port.

switchport protected

Use the **switchport protected** interface configuration command to isolate unicast, multicast, and broadcast traffic at Layer 2 from other protected ports on the same switch. Use the **no** form of this command to return to the default setting.

switchport protected

no switchport protected

Syntax Description

This command has no keywords or arguments.

Defaults

No protected port is defined. All ports are nonprotected.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced. It replaced the **port protected** command.

Usage Guidelines

The switchport protection feature is local to the switch; communication between protected ports on the same switch is possible only through a Layer 3 device. To prevent communication between protected ports on different switches, you must configure the protected ports for unique VLANs on each switch and configure a trunk link between the switches. A protected port is different from a secure port.

A protected port does not forward any unicast, multicast, or broadcast traffic to any other protected port. A protected port continues to forward unicast, multicast, and broadcast traffic to unprotected ports and vice versa.

Port monitoring does not work if both the monitor and monitored ports are protected ports.

Examples

This example shows how to enable a protected port on Fast Ethernet interface 0/3:

```
Switch(config)# interface fastethernet0/3
Switch(config-if)# switchport protected
```

You can verify your settings by entering the **show interfaces switchport** privileged EXEC command.

Related Commands

Command	Description
show interfaces switchport	Displays the administrative and operational staus of a switching port.

switchport trunk

Use the **switchport trunk** interface configuration command to set the trunk characteristics when the interface is in trunking mode. Use the **no** form of this command to reset all of the trunking characteristics to the defaults. Use the **no** form with keywords to reset those characteristics to the defaults.

switchport trunk {{**allowed vlan** *vlan-list*} | {**native vlan** *vlan-id*} | {**pruning vlan** *vlan-list*}}

no switchport trunk {{**allowed vlan** *vlan-list*} | {**native vlan** *vlan-id*} | {**pruning vlan** *vlan-list*}}

Syntax Description

allowed vlan *vlan-list*	Set the list of allowed VLANs that can receive and send traffic on this interface in tagged format when in trunking mode. See the following *vlan-list* format. The **none** keyword is not valid. The default is **all**.
native vlan *vlan-id*	Set the native VLAN for sending and receiving untagged traffic when the interface is in 802.1Q trunking mode. Valid IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
pruning vlan *vlan-list*	Set the list of VLANs that are enabled for VTP pruning when in trunking mode. The **all** keyword is not valid.

The *vlan-list* format is **all** | **none** | [**add** | **remove** | **except**] *vlan-atom* where:

- **all** specifies all VLANs from 1 to 4094 when the EI is installed and 1 to 1005 when the SI is installed. Do not enter leading zeros. This keyword is not allowed on commands that do not permit all VLANs in the list to be set at the same time.
- **none** means an empty list. This keyword is not allowed on commands that require certain VLANs to be set or at least one VLAN to be set.
- **add** adds the defined list of VLANs to those currently set instead of replacing the list. Valid IDs are from 2 to 1001; extended-range VLAN IDs are valid in some cases.

  Note You can add extended-range VLANs to the allowed VLAN list, but not to the pruning-eligible VLAN list.

  Separate nonconsecutive VLAN IDs with a comma; do not enter a space after the comma. Use a hyphen to designate a range of IDs; do not enter a space before or after the hyphen. Do not enter leading zeros.
- **remove** removes the defined list of VLANs from those currently set instead of replacing the list. Valid IDs are from 2 to 1001; extended-range VLAN IDs are valid in some cases.

  Note You cannot remove VLAN 1 or VLANs 1002 to 1005 from the list. You can remove extended-range VLANs (VLAN IDs greater than 1005) from the allowed VLAN list, but you cannot remove them from the pruning-eligible list.

  Separate nonconsecutive VLAN IDs with a comma; do not enter a space after the comma. Use a hyphen to designate a range of IDs; do not enter a space before or after the hyphen. Do not enter leading zeros.

- **except** lists the VLANs that should be calculated by inverting the defined list of VLANs. (VLANs are added except the ones specified.) Valid IDs are from 2 to 1001. Separate nonconsecutive VLAN IDs with a comma; do not enter a space after the comma. Use a hyphen to designate a range of IDs; do not enter a space before or after the hyphen. Do not enter leading zeros.
- *vlan-atom* is either a single VLAN number from 1 to 4094 when the EI is installed and 1 to 1005 when the SI is installed, a list of nonconsecutive VLANs, or a continuous range of VLANs described by two VLAN numbers, the lower one first, separated by a hyphen.

  For a list of nonconsecutive VLAN IDs, separate the VLAN IDs with a comma. Do not enter a space after the comma. Do not enter leading zeroes.

  For a continuous range of VLAN IDs, use a hyphen to designate the range. Do not enter a space before or after the hyphen. Do not enter leading zeroes.

  These are examples showing how to specify one or more VLANs:

  – Single VLAN—101
  – List of nonconsecutive VLANs—10,12,14,16,18
  – Continuous range of VLANs—10-15
  – List of VLAN continuous ranges—10-15,20-24
  – List of nonconsecutive VLANs and VLAN continuous ranges—8,11,20-24,44

Defaults

VLAN 1 is the default native VLAN ID on the port.

The default for all VLAN lists is to include all VLANs.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

A trunk port cannot be a secure port or a monitor port. However, a static-access port can monitor a VLAN on a trunk port. The VLAN monitored is the one associated with the static-access port.

Allowed VLAN:

- The **no** form of the **allowed vlan** command resets the list to the default list, which allows all VLANs.
- You cannot remove VLAN 1 or VLANs 1002 to 1005 from the allowed VLAN list.

Native VLANs:

- All untagged traffic received on an 802.1Q trunk port is forwarded with the native VLAN configured for the port.
- If a packet has a VLAN ID that is the same as the sending port native VLAN ID, the packet is sent without a tag; otherwise, the switch sends the packet with a tag.
- The **no** form of the **native vlan** command resets the native mode VLAN to the appropriate default VLAN for the device.

Trunk Pruning:

- The pruning-eligible list applies only to trunk ports.
- Each trunk port has its own eligibility list.
- If you do not want a VLAN to be pruned, remove it from the pruning-eligible list. VLANs that are pruning-ineligible receive flooded traffic.
- VLAN 1, VLANs 1002 to 1005, and extended-range VLANs (VLANs 1006 to 4094) cannot be pruned.

Note The switch does not support Inter-Switch Link (ISL) trunking.

Examples

This example shows how to configure VLAN 3 as the default port to send all untagged traffic:

```
Switch(config-if)# switchport trunk native vlan 3
```

This example shows how to add VLANs 1, 2, 5, and 6 to the allowed list:

```
Switch(config-if)# switchport trunk allowed vlan add 1,2,5,6
```

This example shows how to remove VLANs 3 and 10 to 15 from the pruning-eligible list:

```
Switch(config-if)# switchport trunk pruning vlan remove 3,10-15
```

You can verify your settings by entering the **show interfaces** *interface-id* **switchport** privileged EXEC command.

Related Commands

Command	Description
show interfaces switchport	Displays the administrative and operational status of a switching (nonrouting) port.
switchport mode	Configures the VLAN membership mode of a port.

switchport voice vlan

Use the **switchport voice vlan** interface configuration command to configure voice VLAN on the port. Use the **no** form of this command to return to the default setting.

switchport voice vlan {*vlan-id* **| dot1p | none | untagged**}

no switchport voice vlan

Syntax Description

vlan-id	VLAN used for voice traffic. Valid IDs are from 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1001 when the standard software image (SI) is installed. Do not enter leading zeros.
dot1p	The telephone uses priority tagging and uses VLAN 0 (the native VLAN). By default, the Cisco IP phone forwards the voice traffic with an 802.1P priority of 5.
none	The telephone is not instructed through the CLI about the voice VLAN. The telephone uses the configuration from the telephone key pad.
untagged	The telephone does not tag frames and uses VLAN 4095. The default for the telephone is untagged.

Defaults

The switch default is not to automatically configure the telephone (**none**).

The telephone default is not to tag frames.

Command Modes

Interface configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

You should configure voice VLAN on access ports.

If you enable port security on a voice VLAN port and if there is a PC connected to the IP phone, you should set the maximum allowed secure addresses on the port to more than 1.

You cannot configure static secure MAC addresses in the voice VLAN.

The Port Fast feature is automatically enabled when voice VLAN is configured. When you disable voice VLAN, the Port Fast feature is not automatically disabled.

Examples

This example shows how to configure VLAN 2 as the voice VLAN:

```
Switch(config-if)# switchport voice vlan 2
```

You can verify your settings by entering the **show interfaces** *interface-id* **switchport** privileged EXEC command.

Related Commands

Command	Description
show interfaces *interface-id* **switchport**	Displays the administrative and operational status of a switching (nonrouting) port.
switchport priority extend	Determines how the device connected to the specified port handles priority traffic received on its incoming port.

system mtu

Use the **system mtu** global configuration command to set the maximum packet size or maximum transmission unit (MTU) size for the switch. Use the **no** form of this command to restore the global MTU value to its original default value.

system mtu *bytes*

no system mtu

Syntax Description

bytes	Packet size in bytes. For valid values, see the "Usage Guidelines" section.

Defaults

The default MTU size is 1500 bytes.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This table lists the valid system MTU values for the switches:

Switch	MTU size
Catalyst 2950G-12-EI	1500 to 1530 bytes
Catalyst 2950G-24-EI	
Catalyst 2950G-24-EI-DC	
Catalyst 2950G-48-EI	
Other Catalyst 2950 switches	1500 bytes

The size of frames that can be received by the switch CPU is limited to 1500 bytes, no matter what value was entered with the **system mtu** command. Although frames that are forwarded or routed typically are not received by the CPU, in some cases packets are sent to the CPU, such as traffic sent to control traffic, Simple Network Management Protocol (SNMP), Telnet, or routing protocols.

If you enter a value that is outside of the range for the switch, the value is not accepted.



Note You cannot set the MTU on a per-interface basis.

Examples

This example shows how to set the maximum packet size to 1528 bytes:

```
Switch(config)# system mtu 1528
Switch(config)# exit
```

This example shows the response when you try to set a switch to an out-of-range number:

```
Switch(config)# system mtu 2000
                           ^
% Invalid input detected at '^' marker.
```

You can verify your settings by entering the **show system mtu** privileged EXEC command.

Related Commands

Command	Description
show system mtu	Displays the maximum packet size set for the switch.

udld (global configuration)

Use the **udld** global configuration command to enable aggressive or normal mode in the UniDirectional Link Detection (UDLD) and to set the configurable message timer. Use the **no** form of this command to disable aggressive or normal mode UDLD on all fiber-optic ports.

udld {**aggressive** | **enable** | **message time** *message-timer-interval*}

no udld {**aggressive** | **enable** | **message time**}

Syntax Description

aggressive	Enable UDLD in aggressive mode on all fiber-optic interfaces.
enable	Enable UDLD in normal mode on all fiber-optic interfaces.
message time *message-timer-interval*	Configure the period of time between UDLD probe messages on ports that are in the advertisement phase and are determined to be bidirectional. The range is from 7 to 90 seconds.

Defaults

UDLD is disabled on all fiber-optic interfaces.

The message timer is set at 60 seconds.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **udld enable** global configuration command was changed to **udld** (global configuration).

Usage Guidelines

Use the **udld** global configuration command to enable UDLD only on fiber-optic ports.

In normal mode, if UDLD is in the advertisement or in the detection phase and all the neighbor cache entries are aged out, UDLD restarts the link-up sequence to resynchronize with any potentially out-of-sync neighbors.

If you enable aggressive mode, when all the neighbors of a port have aged out either in the advertisement or in the detection phase, UDLD restarts the link-up sequence to resynchronize with any potentially out-of-sync neighbor. UDLD shuts down the port if, after the fast train of messages, the link state is still undetermined. Use aggressive mode on point-to-point links where no failure between two neighbors is allowed. In this situation, UDLD probe packets can be considered as a heart beat whose presence guarantees the health of the link. Conversely, the loss of the heart beat means that the link must be shut down if it is not possible to re-establish a bidirectional link.

If you change the message time between probe packets, you are making a trade-off between the detection speed and the CPU load. By decreasing the time, you can make the detection-response faster but increase the load on the CPU.

This command affects fiber-optic interfaces only. Use the **udld** interface configuration command to enable UDLD on other interface types.

Examples

This example shows how to enable UDLD on all fiber-optic interfaces:

```
Switch(config)# udld enable
```

You can verify your settings by entering the **show udld** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the running configuration on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
show udld	Displays the UDLD status for all ports or the specified port.
udld (interface configuration)	Enables UDLD on an individual interface or prevents a fiber-optic interface from being enabled by the **udld** global configuration command.
udld reset	Resets any interface shut down by UDLD and permits traffic to again pass through.

udld (interface configuration)

Use the **udld** interface configuration command to enable UniDirectional Link Detection (UDLD) on an individual interface or to prevent a fiber-optic interface from being enabled by the **udld** global configuration command. Use the **no** form of this command to return to the **udld** global configuration command setting or disable UDLD if entered on a nonfiber-optic port.

udld {**aggressive** | **enable** | **disable**}

no udld {**aggressive** | **enable** | **disable**}

Syntax Description

aggressive	Enable UDLD in aggressive mode on the specified interface.
disable	Disable UDLD on the specified interface. This keyword applies only to fiber-optic interfaces.
enable	Enable UDLD in normal mode on the specified interface.

Defaults

On fiber-optic interfaces, UDLD is not enabled, in aggressive mode, or disabled. For this reason, fiber-optic interfaces enable UDLD according to the state of the **udld enable** or **udld aggressive** global configuration command.

On nonfiber-optic interfaces, UDLD is disabled.

Command Modes

Interface configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **aggressive** keyword was added.

Usage Guidelines

UDLD is supported on fiber- and copper-based Ethernet ports.

A UDLD-capable port cannot detect a unidirectional link if it is connected to a UDLD-incapable port of another switch.

This setting overrides the global UDLD configuration on the switch.

In normal mode, if UDLD is in the advertisement or in the detection phase and all the neighbor cache entries are aged out, UDLD restarts the link-up sequence to resynchronize with any potentially out-of-sync neighbors.

If you enable aggressive mode, when all the neighbors of a port have aged out either in the advertisement or in the detection phase, UDLD restarts the link-up sequence to resynchronize with any potentially out-of-sync neighbor. UDLD shuts down the port if, after the fast train of messages, the link state is still undetermined. Use aggressive mode on point-to-point links where no failure between two neighbors is allowed. In this situation, UDLD probe packets can be considered as a heart beat whose presence guarantees the health of the link. Conversely, the loss of the heart beat means that the link must be shut down if it is not possible to re-establish a bidirectional link.

Use the **no udld enable** command on fiber-optic ports to return control of UDLD to the **udld enable** global configuration command or to disable UDLD on nonfiber-optic ports.

Use the **udld aggressive** command on fiber-optic ports to override the settings of the **udld enable** or **udld aggressive** global configuration command. Use the **no** form on fiber-optic ports to remove this setting and to return control of UDLD enabling to the **udld** global configuration command or to disable UDLD on nonfiber-optic ports.

The **disable** keyword is supported on fiber-optic ports only. Use the **no** form of this command to remove this setting and to return control of UDLD to the **udld** global configuration command.

If the switch software detects a GBIC module change and the interface changes from fiber optic to nonfiber optic or from nonfiber optic to fiber optic, all configurations are maintained.

Examples

This example shows how to enable UDLD on an interface:

```
Switch(config)# interface fastethernet0/2
Switch(config-if)# udld enable
```

This example shows how to disable UDLD on a fiber-optic interface despite the setting of the **udld** global configuration command:

```
Switch(config)# interface gigabitethernet0/2
Switch(config-if)# udld disable
```

You can verify your settings by entering the **show running-config** or **show udld** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the running configuration on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
show udld	Displays UDLD status for all ports or the specified port.
udld (global configuration)	Enables UDLD on all fiber-optic ports on the switch.
udld reset	Resets all interfaces shut down by UDLD and permits traffic to again pass through.

udld reset

Use the **udld reset** privileged EXEC command to reset all interfaces shut down by UniDirectional Link Detection (UDLD) and to permit traffic to again pass through. Other features, such as spanning tree, Port Aggregation Protocol (PAgP), and Dynamic Trunking Protocol (DTP), still have their normal effects, if enabled.

udld reset

Syntax Description

This command has no keywords or arguments.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

If the interface configuration is still enabled for UDLD, these ports begin to run UDLD again and might shut down for the same reason if the problem has not been corrected.

Examples

This example shows how to reset all interfaces that have been shut down by UDLD:

```
Switch# udld reset
1 ports shutdown by UDLD were reset.
```

You can verify your settings by entering the **show udld** privileged EXEC command.

Related Commands

Command	Description
show running-config	Displays the running configuration on the switch. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference for Release 12.1 > Cisco IOS File Management Commands > Configuration File Commands**.
show udld	Displays UDLD status for all ports or the specified port.
udld (interface configuration)	Enables UDLD on an individual interface or prevents a fiber-optic interface from being enabled by the **udld** global configuration command.
udld (global configuration)	Enables UDLD on all fiber-optic ports on the switch.

vlan (global configuration)

Use the **vlan** global configuration command to add a VLAN and enter the config-vlan mode. Use the **no** form of this command to delete the VLAN. Configuration information for normal-range VLANs (VLAN IDs 1 to 1005) is always saved in the VLAN database. When VLAN Trunking Protocol (VTP) mode is transparent, you can create extended-range VLANs (VLAN IDs greater than 1005), and the VTP mode and domain name and the VLAN configuration are saved in the switch running configuration file. You can save configurations in the switch startup configuration file by entering the **copy running-config startup-config** privileged EXEC command.

vlan *vlan-id*

no vlan *vlan-id*

Syntax Description

vlan-id	ID of the VLAN to be added and configured. For *vlan-id*, the range is 1 to 4094 when the enhanced software image (EI) is installed and 1 to 1005 when the standard software image (SI) is installed; do not enter leading zeros. You can enter a single VLAN ID, a series of VLAN IDs separated by commas, or a range of VLAN IDs separated by hyphens.

Defaults

This command has no default settings.

Command Modes

Global configuration

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.
12.1(11)EA1	The **remote-span** configuration command was added.

Usage Guidelines

You must use the **vlan** *vlan-id* global configuration command to add extended-range VLANs (VLAN IDs 1006 to 4094). Before configuring VLANs in the extended range, you must use the **vtp transparent** global configuration or VLAN configuration command to put the switch in VTP transparent mode. Extended-range VLANs are not learned by VTP and are not added to the VLAN database, but when VTP mode is transparent, VTP mode and domain name and all VLAN configurations are saved in the running configuration, and you can save them in the switch startup configuration file.

When you save the VLAN and VTP configurations in the startup configuration file and reboot the switch, the configuration is determined in these ways:

- If both the VLAN database and the configuration file show the VTP mode as transparent and the VTP domain names match, the VLAN database is ignored. The VTP and VLAN configurations in the startup configuration file are used. The VLAN database revision number remains unchanged in the VLAN database.
- If the VTP mode is server, or if the startup VTP mode or domain names do not match the VLAN database, the VTP mode and the VLAN configuration for the first 1005 VLANs use the VLAN database information.

- If the image on the switch or the configuration file is earlier than IOS release 12.1(9)EA1, the switch reboots with information in the VLAN database.

If you try to create an extended-range VLAN when the switch is not in VTP transparent mode, the VLAN is rejected, and you receive an error message.

If you enter an invalid VLAN ID, you receive an error message and do not enter config-vlan mode.

Entering the **vlan** command with a VLAN ID enables config-vlan mode. When you enter the VLAN ID of an existing VLAN, you do not create a new VLAN, but you can modify VLAN parameters for that VLAN. The specified VLANs are added or modified when you exit the config-vlan mode. Only the **shutdown** command (for VLANs 1 to 1005) takes effect immediately.

These configuration commands are available in config-vlan mode. The **no** form of each command returns the characteristic to its default state.

Note Although all commands are visible, the only config-vlan command supported on extended-range VLANs is **mtu** *mtu-size*. For extended-range VLANs, all other characteristics must remain at the default state.

- **are** *are-number*: defines the maximum number of all-routes explorer (ARE) hops for this VLAN. This keyword applies only to TrCRF VLANs. Valid values are from 0 to 13. The default is 7. If no value is entered, 0 is assumed to be the maximum.
- **backupcrf**: specifies the backup CRF mode. This keyword applies only to TrCRF VLANs.
  - **enable** backup CRF mode for this VLAN.
  - **disable** backup CRF mode for this VLAN (the default).
- **bridge** {*bridge-number*| **type**}: specifies the logical distributed source-routing bridge, the bridge that interconnects all logical rings having this VLAN as a parent VLAN in FDDI-NET, Token Ring-NET, and TrBRF VLANs. Valid bridge numbers are from 0 to 15. The default bridge number is 0 (no source-routing bridge) for FDDI-NET, TrBRF, and Token Ring-NET VLANs. The **type** keyword applies only to TrCRF VLANs and is one of these:
  - **srb** (source-route bridging)
  - **srt** (source-route transparent) bridging VLAN
- **exit:** applies changes, increments the VLAN database revision number (VLANs 1 to 1005 only), and exits config-vlan mode.
- **media**: defines the VLAN media type. See Table 2-20 for valid commands and syntax for different media types.

Note The switch supports only Ethernet ports. You configure only FDDI and Token Ring media-specific characteristics for VLAN Trunking Protocol (VTP) global advertisements to other switches. These VLANs are locally suspended.

  - **ethernet** is Ethernet media type (the default).
  - **fddi** is FDDI media type.
  - **fd-net** is FDDI network entity title (NET) media type.
  - **tokenring** is Token Ring media type if the VTP v2 mode is disabled, or TrCRF if the VTP version 2 (v) mode is enabled.
  - **tr-net** is Token Ring network entity title (NET) media type if the VTP v2 mode is disabled or TrBRF media type if the VTP v2 mode is enabled.

- **mtu** *mtu-size*: specifies the maximum transmission unit (MTU) (packet size in bytes). Valid values are from 1500 to 18190. The default is 1500 bytes.
- **name** *vlan-name*: names the VLAN with an ASCII string from 1 to 32 characters that must be unique within the administrative domain. The default is *VLANxxxx* where *xxxx* represents four numeric digits (including leading zeros) equal to the VLAN ID number.
- **no:** negates a command or returns it to the default setting.
- **parent** *parent-vlan-id*: specifies the parent VLAN of an existing FDDI, Token Ring, or TrCRF VLAN. This parameter identifies the TrBRF to which a TrCRF belongs and is required when defining a TrCRF. Valid values are from 0 to 1005. The default parent VLAN ID is 0 (no parent VLAN) for FDDI and Token Ring VLANs. For both Token Ring and TrCRF VLANs, the parent VLAN ID must already exist in the database and be associated with a Token Ring-NET or TrBRF VLAN.
- **remote-span**: adds the Remote SPAN (RSPAN) trait to the VLAN. When the RSPAN trait is added to an existing VLAN, the VLAN is first removed and then recreated with the RSPAN trait. Any access ports are deactivated until the RSPAN trait is removed. The new RSPAN VLAN is propagated via VTP for VLAN-IDs less than 1005. This command is available only if your switch is running the EI.
- **ring** *ring-number*: defines the logical ring for an FDDI, Token Ring, or TrCRF VLAN. Valid values are from 1 to 4095. The default for Token Ring VLANs is 0. For FDDI VLANs, there is no default.
- **said** *said-value*: specifies the security association identifier (SAID) as documented in IEEE 802.10. The value is an integer from 1 to 4294967294 that must be unique within the administrative domain. The default value is 100000 plus the VLAN ID number.
- **shutdown:** shuts down VLAN switching on the VLAN. This command takes effect immediately. Other commands take effect when you exit config-vlan mode.
- **state**: specifies the VLAN state:
  - **active** means the VLAN is operational (the default).
  - **suspend** means the VLAN is suspended. Suspended VLANs do not pass packets.
- **ste** *ste-number*: defines the maximum number of spanning-tree explorer (STE) hops. This keyword applies only to TrCRF VLANs. Valid values are from 0 to 13. The default is 7.
- **stp type**: defines the spanning-tree type for FDDI-NET, Token Ring-NET, or TrBRF VLANs. For FDDI-NET VLANs, the default STP type is **ieee**. For Token Ring-NET VLANs, the default STP type is **ibm**. For FDDI and Token Ring VLANs, the default is no type specified.
  - **ieee** for IEEE Ethernet STP running source-route transparent (SRT) bridging.
  - **ibm** for IBM STP running source-route bridging (SRB).
  - **auto** for STP running a combination of source-route transparent bridging (IEEE) and source-route bridging (IBM).
- **tb-vlan1** *tb-vlan1-id* and **tb-vlan2** *tb-vlan2-id*: specifies the first and second VLAN to which this VLAN is translationally bridged. Translational VLANs translate FDDI or Token Ring to Ethernet, for example. Valid values are from 0 to 1005. If no value is specified, 0 (no transitional bridging) is assumed.

Table 2-20 Valid Commands and Syntax for Different Media Types

Media Type	Valid Syntax
Ethernet	**name** *vlan-name*, **media ethernet, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **remote-span**, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id*
FDDI	**name** *vlan-name*, **media fddi, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **ring** *ring-number*, **parent** *parent-vlan-id*, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id*
FDDI-NET	**name** *vlan-name*, **media fd-net, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **bridge** *bridge-number*, **stp type** {**ieee** | **ibm** | **auto**}, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id* If VTP v2 mode is disabled, do not set the **stp type** to **auto**.
Token Ring	VTP v1 mode is enabled. **name** *vlan-name*, **media tokenring, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **ring** *ring-number*, **parent** *parent-vlan-id*, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id*
Token Ring concentrator relay function (TrCRF)	VTP v2 mode is enabled. **name** *vlan-name*, **media tokenring, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **ring** *ring-number*, **parent** *parent-vlan-id*, **bridge type** {**srb** | **srt**}, **are** *are-number*, **ste** *ste-number*, **backupcrf** {**enable** | **disable**}, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id*
Token Ring-NET	VTP v1 mode is enabled. **name** *vlan-name*, **media tr-net, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **bridge** *bridge-number*, **stp type** {**ieee** | **ibm**}, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id*
Token Ring bridge relay function (TrBRF)	VTP v2 mode is enabled. **name** *vlan-name*, **media tr-net, state** {**suspend** | **active**}, **said** *said-value*, **mtu** *mtu-size*, **bridge** *bridge-number*, **stp type** {**ieee** | **ibm** | **auto**}, **tb-vlan1** *tb-vlan1-id*, **tb-vlan2** *tb-vlan2-id*

Table 2-21 describes the rules for configuring VLANs.

Table 2-21 VLAN Configuration Rules

Configuration	Rule
VTP v2 mode is enabled, and you are configuring a TrCRF VLAN media type.	Specify a parent VLAN ID of a TrBRF that already exists in the database. Specify a ring number. Do not leave this field blank. Specify unique ring numbers when TrCRF VLANs have the same parent VLAN ID. Only one backup concentrator relay function (CRF) can be enabled.
VTP v2 mode is enabled, and you are configuring VLANs other than TrCRF media type.	Do not specify a backup CRF.

Table 2-21 VLAN Configuration Rules (continued)

Configuration	Rule
VTP v2 mode is enabled, and you are configuring a TrBRF VLAN media type.	Specify a bridge number. Do not leave this field blank.
VTP v1 mode is enabled.	No VLAN can have an STP type set to auto. This rule applies to Ethernet, FDDI, FDDI-NET, Token Ring, and Token Ring-NET VLANs.
Add a VLAN that requires translational bridging (values are not set to zero).	The translational bridging VLAN IDs that are used must already exist in the database. The translational bridging VLAN IDs that a configuration points to must also contain a pointer to the original VLAN in one of the translational bridging parameters (for example, Ethernet points to FDDI, and FDDI points to Ethernet). The translational bridging VLAN IDs that a configuration points to must be different media types than the original VLAN (for example, Ethernet can point to Token Ring). If both translational bridging VLAN IDs are configured, these VLANs must be different media types (for example, Ethernet can point to FDDI and Token Ring).

Examples

This example shows how to add an Ethernet VLAN with default media characteristics. The default includes a *vlan-name* of *VLANxxx*, where *xxxx* represents four numeric digits (including leading zeros) equal to the VLAN ID number. The default **media** option is **ethernet**; the **state** option is **active**. The default *said-value* variable is 100000 plus the VLAN ID; the *mtu-size* variable is 1500; the **stp-type** option is **ieee**. When you enter the **exit** config-vlan configuration command, the VLAN is added if it did not already exist; otherwise, this command does nothing.

This example shows how to create a new VLAN with all default characteristics and enter config-vlan mode:

```
Switch(config)# vlan 200
Switch(config-vlan)# exit
Switch(config)#
```

This example shows how to create a new extended-range VLAN (when the EI is installed) with all the default characteristics, to enter config-vlan mode, and to save the new VLAN in the switch startup configuration file:

```
Switch(config)# vtp mode transparent
Switch(config)# vlan 2000
Switch(config-vlan)# end
Switch# copy running-config startup config
```

You can verify your setting by entering the **show vlan** privileged EXEC command.

Related Commands

Command	Description
show running-config vlan	Displays all or a range of VLAN-related configurations on the switch.
show vlan	Displays the parameters for all configured VLANs or one VLAN (if the VLAN ID or name is specified) in the administrative domain.
vlan (VLAN configuration)	Configures normal-range VLANs in the VLAN database.

vlan (VLAN configuration)

Use the **vlan** VLAN configuration command to configure VLAN characteristics for a normal-range VLAN (VLAN IDs 1 to 1005) in the VLAN database. You access VLAN configuration mode by entering the **vlan database** privileged EXEC command. Use the **no** form of this command without additional parameters to delete a VLAN. Use the **no** form with parameters to change its configured characteristics.

vlan *vlan-id* [**are** *are-number*] [**backupcrf** {**enable** | **disable**}] [**bridge** *bridge-number* | **type** {**srb** | **srt**}] [**media** {**ethernet** | **fddi** | **fdi-net** | **tokenring** | **tr-net**}] [**mtu** *mtu-size*] [**name** *vlan-name*] [**parent** *parent-vlan-id*] [**ring** *ring-number*] [**said** *said-value*] [**state** {**suspend** | **active**}] [**ste** *ste-number*] [**stp type** {**ieee** | **ibm** | **auto**}] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]

no vlan *vlan-id* [**are** *are-number*] [**backupcrf** {**enable** | **disable**}] [**bridge** *bridge-number* | **type** {**srb** | **srt**}] [**media** {**ethernet** | **fddi** | **fdi-net** | **tokenring** | **tr-net**}] [**mtu** *mtu-size*] [**name** *vlan-name*] [**parent** *parent-vlan-id*] [**ring** *ring-number*] [**said** *said-value*] [**state** {**suspend** | **active**}] [**ste** *ste-number*] [**stp type** {**ieee** | **ibm** | **auto**}] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]

Extended-range VLANs (with VLAN IDs from 1006 to 4094) cannot be added or modified by using these commands. To add extended-range VLANs, use the **vlan (global configuration)** command to enter config-vlan mode.

Note The switch supports only Ethernet ports. You configure only FDDI and Token Ring media-specific characteristics for VLAN Trunking Protocol (VTP) global advertisements to other switches. These VLANs are locally suspended.

Syntax Description

vlan-id	ID of the configured VLAN. Valid IDs are from 1 to 1005 and must be unique within the administrative domain. Do not enter leading zeros.
are *are-number*	(Optional) Specify the maximum number of all-routes explorer (ARE) hops for this VLAN. This keyword applies only to TrCRF VLANs. Valid values are from 0 to 13. If no value is entered, 0 is assumed to be the maximum.
backupcrf {**enable** | **disable**}	(Optional) Specify the backup CRF mode. This keyword applies only to TrCRF VLANs. • **enable** backup CRF mode for this VLAN. • **disable** backup CRF mode for this VLAN.
bridge *bridge-number*| **type** {**srb** | **srt**}	(Optional) Specify the logical distributed source-routing bridge, the bridge that interconnects all logical rings having this VLAN as a parent VLAN in FDDI-NET, Token Ring-NET, and TrBRF VLANs. Valid bridge numbers are from 0 to 15. The **type** keyword applies only to TrCRF VLANs and is one of these: • **srb** (source-route bridging) • **srt** (source-route transparent) bridging VLAN

media {**ethernet** | **fddi** | **fd-net** | **tokenring** | **tr-net**}	(Optional) Specify the VLAN media type. Table 2-22 lists the valid syntax for each media type. • **ethernet** is Ethernet media type (the default). • **fddi** is FDDI media type. • **fd-net** is FDDI network entity title (NET) media type. • **tokenring** is Token Ring media type if the VTP v2 mode is disabled, or TrCRF if the VTP v2 mode is enabled. • **tr-net** is Token Ring network entity title (NET) media type if the VTP v2 mode is disabled or TrBRF media type if the VTP v2 mode is enabled.
mtu *mtu-size*	(Optional) Specify the maximum transmission unit (MTU) (packet size in bytes). Valid values are from 1500 to 18190.
name *vlan-name*	(Optional) Specify the VLAN name, an ASCII string from 1 to 32 characters that must be unique within the administrative domain.
parent *parent-vlan-id*	(Optional) Specify the parent VLAN of an existing FDDI, Token Ring, or TrCRF VLAN. This parameter identifies the TrBRF to which a TrCRF belongs and is required when defining a TrCRF. Valid values are from 0 to 1005.
ring *ring-number*	(Optional) Specify the logical ring for an FDDI, Token Ring, or TrCRF VLAN. Valid values are from 1 to 4095.
said *said-value*	(Optional) Enter the security association identifier (SAID) as documented in IEEE 802.10. The value is an integer from 1 to 4294967294 that must be unique within the administrative domain.
state {**suspend** | **active**}	(Optional) Specify the VLAN state: • If **active**, the VLAN is operational. • If **suspend**, the VLAN is suspended. Suspended VLANs do not pass packets.
ste *ste-number*	(Optional) Specify the maximum number of spanning-tree explorer (STE) hops. This keyword applies only to TrCRF VLANs. Valid values are from 0 to 13.
stp type {**ieee** | **ibm** | **auto**}	(Optional) Specify the spanning-tree type for FDDI-NET, Token Ring-NET, or TrBRF VLAN. • **ieee** for IEEE Ethernet STP running source-route transparent (SRT) bridging. • **ibm** for IBM STP running source-route bridging (SRB). • **auto** for STP running a combination of source-route transparent bridging (IEEE) and source-route bridging (IBM).
tb-vlan1 *tb-vlan1-id* and **tb-vlan2** *tb-vlan2-id*	(Optional) Specify the first and second VLAN to which this VLAN is translationally bridged. Translational VLANs translate FDDI or Token Ring to Ethernet, for example. Valid values are from 0 to 1005. Zero is assumed if no value is specified.

Table 2-22 shows the valid syntax options for different media types.

Table 2-22 Valid Syntax for Different Media Types

Media Type	Valid Syntax
Ethernet	**vlan** *vlan-id* [**name** *vlan-name*] **media ethernet** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]
FDDI	**vlan** *vlan-id* [**name** *vlan-name*] **media fddi** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**ring** *ring-number*] [**parent** *parent-vlan-id*] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]
FDDI-NET	**vlan** *vlan-id* [**name** *vlan-name*] **media fd-net** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**bridge** *bridge-number*] [**stp type** {**ieee** | **ibm** | **auto**}] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*] If VTP v2 mode is disabled, do not set the **stp type** to **auto**.
Token Ring	VTP v1 mode is enabled. **vlan** *vlan-id* [**name** *vlan-name*] **media tokenring** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**ring** *ring-number*] [**parent** *parent-vlan-id*] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]
Token Ring concentrator relay function (TrCRF)	VTP v2 mode is enabled. **vlan** *vlan-id* [**name** *vlan-name*] **media tokenring** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**ring** *ring-number*] [**parent** *parent-vlan-id*] [**bridge type** {**srb** | **srt**}] [**are** *are-number*] [**ste** *ste-number*] [**backupcrf** {**enable** | **disable**}] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]
Token Ring-NET	VTP v1 mode is enabled. **vlan** *vlan-id* [**name** *vlan-name*] **media tr-net** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**bridge** *bridge-number*] [**stp type** {**ieee** | **ibm**}] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]
Token Ring bridge relay function (TrBRF)	VTP v2 mode is enabled. **vlan** *vlan-id* [**name** *vlan-name*] **media tr-net** [**state** {**suspend** | **active**}] [**said** *said-value*] [**mtu** *mtu-size*] [**bridge** *bridge-number*] [**stp type** {**ieee** | **ibm** | **auto**}] [**tb-vlan1** *tb-vlan1-id*] [**tb-vlan2** *tb-vlan2-id*]

Table 2-23 describes the rules for configuring VLANs.

Table 2-23 VLAN Configuration Rules

Configuration	Rule
VTP v2 mode is enabled, and you are configuring a TrCRF VLAN media type.	Specify a parent VLAN ID of a TrBRF that already exists in the database. Specify a ring number. Do not leave this field blank. Specify unique ring numbers when TrCRF VLANs have the same parent VLAN ID. Only one backup concentrator relay function (CRF) can be enabled.
VTP v2 mode is enabled, and you are configuring VLANs other than TrCRF media type.	Do not specify a backup CRF.

Table 2-23 VLAN Configuration Rules (continued)

Configuration	Rule
VTP v2 mode is enabled, and you are configuring a TrBRF VLAN media type.	Specify a bridge number. Do not leave this field blank.
VTP v1 mode is enabled.	No VLAN can have an STP type set to auto. This rule applies to Ethernet, FDDI, FDDI-NET, Token Ring, and Token Ring-NET VLANs.
Add a VLAN that requires translational bridging (values are not set to zero).	The translational bridging VLAN IDs that are used must already exist in the database. The translational bridging VLAN IDs that a configuration points to must also contain a pointer to the original VLAN in one of the translational bridging parameters (for example, Ethernet points to FDDI, and FDDI points to Ethernet). The translational bridging VLAN IDs that a configuration points to must be different media types than the original VLAN (for example, Ethernet can point to Token Ring). If both translational bridging VLAN IDs are configured, these VLANs must be different media types (for example, Ethernet can point to FDDI and Token Ring).

Defaults

The ARE value is 7.

Backup CRF is disabled.

The bridge number is 0 (no source-routing bridge) for FDDI-NET, TrBRF, and Token Ring-NET VLANs.

The **media** type is **ethernet**.

The default *mtu size* is 1500 bytes.

The *vlan-name* variable is *VLANxxxx*, where *xxxx* represents four numeric digits (including leading zeros) equal to the VLAN ID number.

The parent VLAN ID is 0 (no parent VLAN) for FDDI and Token Ring VLANs. For TrCRF VLANs, you must specify a parent VLAN ID. For both Token Ring and TrCRF VLANs, the parent VLAN ID must already exist in the database and be associated with a Token Ring-NET or TrBRF VLAN.

The *ring number* for Token Ring VLANs is 0. For FDDI VLANs, there is no default.

The *said value* is 100000 plus the VLAN ID.

The state is **active**.

The STE value is 7.

The STP type is **ieee** for FDDI-NET and **ibm** for Token Ring-NET VLANs. For FDDI and Token Ring VLANs, the default is no type specified.

The *tb-vlan1-id* and *tb-vlan2-id* variables are zero (no translational bridging).

Command Modes VLAN configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines You can only use this command mode for configuring normal-range VLANs, that is, VLAN IDs 1 to 1005.

Note To configure extended-range VLANs (VLAN IDs 1006 to 4094), use the **vlan** global configuration command.

VLAN configuration is always saved in the VLAN database. If VTP mode is transparent, it is also saved in the switch running configuration file, along with the VTP mode and domain name. You can then save it in the switch startup configuration file by using the **copy running-config startup-config** privileged EXEC command.

When you save VLAN and VTP configuration in the startup configuration file and reboot the switch, the configuration is determined in these ways:

- If both the VLAN database and the configuration file show the VTP mode as transparent and the VTP domain names match, the VLAN database is ignored. The VTP and VLAN configurations in the startup configuration file are used. The VLAN database revision number remains unchanged in the VLAN database.
- If the VTP mode is server, or if the startup VTP mode or domain names do not match the VLAN database, the VTP mode and the VLAN configuration for the first 1005 VLANs use VLAN database information.
- If the image on the switch or the configuration file is earlier than IOS release 12.1(9)EA1, the switch reboots with information in the VLAN database.

The following are the results of using the **no vlan** commands:

- When the **no vlan** *vlan-id* form is used, the VLAN is deleted. Deleting VLANs automatically resets to zero any other parent VLANs and translational bridging parameters that refer to the deleted VLAN.
- When the **no vlan** *vlan-id* **bridge** form is used, the VLAN source-routing bridge number returns to the default (0). The **vlan** *vlan-id* **bridge** command is used only for FDDI-NET and Token Ring-NET VLANs and is ignored in other VLAN types.
- When the **no vlan** *vlan-id* **media** form is used, the media type returns to the default (**ethernet**). Changing the VLAN media type (including the **no** form) resets the VLAN MTU to the default MTU for the type (unless the **mtu** keyword is also present in the command). It also resets the VLAN parent and translational bridging VLAN to the default (unless the **parent**, **tb-vlan1**, or **tb-vlan2** are also present in the command).
- When the **no vlan** *vlan-id* **mtu** form is used, the VLAN MTU returns to the default for the applicable VLAN media type. You can also modify the MTU using the **media** keyword.
- When the **no vlan** *vlan-id* **name** *vlan-name* form is used, the VLAN name returns to the default name (*VLANxxxx*, where *xxxx* represent four numeric digits [including leading zeros] equal to the VLAN ID number).

- When the **no vlan** *vlan-id* **parent** form is used, the parent VLAN returns to the default (0). The parent VLAN resets to the default if the parent VLAN is deleted or if the **media** keyword changes the VLAN type or the VLAN type of the parent VLAN.
- When the **no vlan** *vlan-id* **ring** form is used, the VLAN logical ring number returns to the default (0).
- When the **no vlan** *vlan-id* **said** form is used, the VLAN SAID returns to the default (100,000 plus the VLAN ID).
- When the **no vlan** *vlan-id* **state** form is used, the VLAN state returns to the default (**active)**.
- When the **no vlan** *vlan-id* **stp type** form is used, the VLAN spanning-tree type returns to the default (**ieee**).
- When the **no vlan** *vlan-id* **tb-vlan1** or **no vlan** *vlan-id* **tb-vlan2** form is used, the VLAN translational bridge VLAN (or VLANs, if applicable) returns to the default (0). Translational bridge VLANs must be a different VLAN type than the affected VLAN, and if two are specified, the two must be different VLAN types from each other. A translational bridge VLAN resets to the default if the translational bridge VLAN is deleted, if the **media** keyword changes the VLAN type, or if the **media** keyword changes the VLAN type of the corresponding translation bridge VLAN.

Examples

This example shows how to add an Ethernet VLAN with default media characteristics. The default includes a *vlan-name* of *VLANxxx*, where *xxxx* represents four numeric digits (including leading zeros) equal to the VLAN ID number. The default **media** option is **ethernet**; the **state** option is **active**. The default *said-value* variable is 100000 plus the VLAN ID; the *mtu-size* variable is 1500; the **stp-type** option is **ieee**. When you enter the **exit** or **apply** vlan configuration command, the VLAN is added if it did not already exist; otherwise, this command does nothing.

```
Switch(vlan)# vlan 2
VLAN 2 added:
    Name: VLAN0002
Switch(vlan)# exit
APPLY completed.
Exiting....
```

This example shows how to modify an existing VLAN by changing its name and MTU size:

```
Switch(vlan)# no vlan name engineering mtu 1200
```

You can verify your settings by entering the **show vlan** privileged EXEC command.

Related Commands

Command	Description
show vlan	Displays the parameters for all configured VLANs or one VLAN (if the VLAN ID or name is specified) in the administrative domain.
vlan (global configuration)	Enters config-vlan mode for configuring normal-range and extended-range VLANs.

vlan database

Use the **vlan database** privileged EXEC command to enter VLAN configuration mode. From this mode, you can add, delete, and modify VLAN configurations for normal-range VLANs and globally propagate these changes by using the VLAN Trunking Protocol (VTP). Configuration information is saved in the VLAN database.

vlan database

Note VLAN configuration mode is only valid for VLAN IDs 1 to 1005.

Syntax Description

This command has no arguments or keywords.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

You can use the VLAN database configuration commands to configure VLANs 1 to 1005. To configure extended-range VLANs (VLAN IDs 1006 to 4094) (when the enhanced software image (EI) is installed), use the **vlan (global configuration)** command to enter config-vlan mode. You can also configure VLAN IDs 1 to 1005 by using the **vlan** global configuration command.

To return to the privileged EXEC mode from the VLAN configuration mode, enter the **exit** command.

Note This command mode is different from other modes because it is session-oriented. When you add, delete, or modify VLAN parameters, the changes are not applied until you exit the session by entering the **apply** or **exit** command. When the changes are applied, the VTP configuration version is incremented. You can also *not* apply the changes to the VTP database by entering **abort**.

Once you are in VLAN configuration mode, you can access the VLAN database and make changes by using these commands:

- **vlan**: accesses subcommands to add, delete, or modify values associated with a single VLAN. For more information, see the **vlan (VLAN configuration)** command.
- **vtp**: accesses subcommands to perform VTP administrative functions. For more information, see the **vtp (VLAN configuration)** command.

When you have modified VLAN or VTP parameters, you can use these editing buffer manipulation commands:

- **abort**: exits the mode without applying the changes. The VLAN configuration that was running before you entered VLAN configuration mode continues to be used.
- **apply**: applies current changes to the VLAN database, increments the database configuration revision number, propagates it throughout the administrative domain, and remains in VLAN configuration mode.

  Note You cannot use this command when the switch is in VTP client mode.

- **exit**: applies all configuration changes to the VLAN database, increments the database configuration number, propagates it throughout the administrative domain, and returns to privileged EXEC mode.
- **no**: negates a command or set its defaults; valid values are **vlan** and **vtp**.
- **reset**: abandons proposed changes to the VLAN database, resets the proposed database to the implemented VLAN database on the switch, and remains in VLAN configuration mode.
- **show**: displays VLAN database information.
- **show changes** [*vlan-id*]: displays the differences between the VLAN database on the switch and the proposed VLAN database for all normal-range VLAN IDs (1 to 1005) or the specified VLAN ID (1 to 1005).
- **show current** [*vlan-id*]: displays the VLAN database on the switch or on a selected VLAN (1 to 1005).
- **show proposed** [*vlan-id*]: displays the proposed VLAN database or a selected VLAN (1 to 1005) from the proposed database. The proposed VLAN database is not the running configuration until you use the **exit** or **apply** VLAN configuration command.

You can verify that VLAN database changes have been made or aborted by using the **show vlan** privileged EXEC command. This output is different from the **show** VLAN database configuration command output.

Examples

This example shows how to enter the VLAN configuration mode from the privileged EXEC mode and to display VLAN database information:

```
Switch# vlan database
Switch(vlan)# show
Name: default
    Media Type: Ethernet
    VLAN 802.10 Id: 100001
    State: Operational
    MTU: 1500
    Translational Bridged VLAN: 1002
    Translational Bridged VLAN: 1003

Name: VLAN0002
    Media Type: Ethernet
    VLAN 802.10 Id: 100002
    State: Operational
    MTU: 1500

Name: fddi-default
    Media Type: FDDI
    VLAN 802.10 Id: 101002
    State: Operational
```

```
    MTU: 1500
    Bridge Type: SRB
    Ring Number: 0
    Translational Bridged VLAN: 1
    Translational Bridged VLAN: 1003

<output truncated>
```

This is an example of output from the **show changes** command:

```
Switch(vlan)# show changes

DELETED:
Name: VLAN0004
    Media Type: Ethernet
    VLAN 802.10 Id: 100004
    State: Operational
    MTU: 1500

DELETED:
Name: VLAN0006
    Media Type: Ethernet
    VLAN 802.10 Id: 100006
    State: Operational
    MTU: 1500

MODIFIED:
Current State: Operational
    Modified State: Suspended
```

This example shows how to display the differences between VLAN 7 in the current database and the proposed database.

```
Switch(vlan)# show changes 7

MODIFIED:
Current State: Operational
    Modified State: Suspended
```

This is an example of output from the **show current 20** command. It displays only VLAN 20 of the current database.

```
Switch(vlan)# show current 20
Name: VLAN0020
    Media Type: Ethernet
    VLAN 802.10 Id: 100020
    State: Operational
    MTU: 1500
```

Related Commands

Command	Description
show vlan	Displays the parameters for all configured VLANs in the administrative domain.
shutdown vlan	Shuts down (suspends) local traffic on the specified VLAN.
vlan (global configuration)	Enters config-vlan mode for configuring normal-range and extended-range VLANs.

vmps reconfirm (global configuration)

Use the **vmps reconfirm** global configuration command to change the reconfirmation interval for the VLAN Query Protocol (VQP) client.

vmps reconfirm *interval*

Syntax Description

interval	Reconfirmation interval for VQP client queries to the VLAN Membership Policy Server (VMPS) to reconfirm dynamic VLAN assignments. The interval range is from 1 to 120 minutes.

Defaults

The default reconfirmation interval is 60 minutes.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to set the VQP client to reconfirm dynamic VLAN entries every 20 minutes:

```
Switch(config)# vmps reconfirm 20
```

You can verify your settings by entering the **show vmps** privileged EXEC command and examining information in the Reconfirm Interval row.

Related Commands

Command	Description
show vmps	Displays VQP and VMPS information.
vmps reconfirm (privileged EXEC)	Sends VQP queries to reconfirm all dynamic VLAN assignments with the VMPS.

vmps reconfirm (privileged EXEC)

Use the **vmps reconfirm** privileged EXEC command to immediately send VLAN Query Protocol (VQP) queries to reconfirm all dynamic VLAN assignments with the VLAN Membership Policy Server (VMPS).

vmps reconfirm

Syntax Description

This command has no arguments or keywords.

Defaults

No default is defined.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to send VQP queries to the VMPS:

```
Switch# vmps reconfirm
```

You can verify your settings by entering the **show vmps** privileged EXEC command and examining the VMPS Action row of the Reconfirmation Status section. The **show vmps** command shows the result of the last time the assignments were reconfirmed either as a result of the reconfirmation timer expired or because the **vmps reconfirm** command was entered.

Related Commands

Command	Description
show vmps	Displays VQP and VMPS information.
vmps reconfirm (global configuration)	Changes the reconfirmation interval for the VLAN Query Protocol (VQP) client.

vmps retry

Use the **vmps retry** global configuration command to configure the per-server retry count for the VLAN Query Protocol (VQP) client.

vmps retry *count*

Syntax Description

count	Number of attempts to contact the VLAN Membership Policy Server (VMPS) by the client before querying the next server in the list. The retry range is from 1 to 10.

Defaults

The default retry count is 3.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Examples

This example shows how to set the retry count to 7:

```
Switch(config)# vmps retry 7
```

You can verify your settings by entering the **show vmps** privileged EXEC command and examining information in the Server Retry Count row.

Related Commands

Command	Description
show vmps	Displays VQP and VMPS information.

vmps server

Use the **vmps server** global configuration command to configure the primary VLAN Membership Policy Server (VMPS) and up to three secondary servers. Use the **no** form of this command to remove a VMPS server.

vmps server *ipaddress* [**primary**]

no vmps server [*ipaddress*]

Syntax Description

ipaddress	IP address or host name of the primary or secondary VMPS servers. If you specify a host name, the Domain Name System (DNS) server must be configured.
primary	(Optional) Determines whether primary or secondary VMPS servers are being configured.

Defaults

No primary or secondary VMPS servers are defined.

Command Modes

Global configuration

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The first server entered is automatically selected as the primary server whether or not the **primary** keyword is entered. The first server address can be overridden by using **primary** in a subsequent command.

If a member switch in a cluster configuration does not have an IP address, the cluster does not use the VMPS server configured for that member switch. Instead, the cluster uses the VMPS server on the command switch, and the command switch proxies the VMPS requests. The VMPS server treats the cluster as a single switch and uses the IP address of the command switch to respond to requests.

When using the **no** form without specifying the *ipaddress*, all configured servers are deleted. If you delete all servers when dynamic-access ports are present, the switch cannot forward packets from new sources on these ports because it cannot query the VMPS.

Examples

This example shows how to configure the server with IP address 191.10.49.20 as the primary VMPS server. The servers with IP addresses 191.10.49.21 and 191.10.49.22 are configured as secondary servers.

```
Switch(config)# vmps server 191.10.49.20 primary
Switch(config)# vmps server 191.10.49.21
Switch(config)# vmps server 191.10.49.22
```

This example shows how to delete the server with IP address 191.10.49.21:

```
Switch(config)# no vmps server 191.10.49.21
```

You can verify your settings by entering the **show vmps** privileged EXEC command and examining information in the VMPS Domain Server row.

Related Commands

Command	Description
show vmps	Displays VQP and VMPS information.

vtp (global configuration)

Use the **vtp** global configuration command to set or modify the VLAN Trunking Protocol (VTP) configuration characteristics. Use the **no** form of this command to remove the settings or to return to the default settings.

vtp {**domain** *domain-name* | **file** *filename* | **interface** *name* | **mode** {**client** | **server** | **transparent**} | **password** *password* | **pruning** | **version** *number*}

no vtp {**file** | **interface** | **mode** | **password** | **pruning** | **version** }

Syntax Description

domain *domain-name*	Specify the VTP domain name, an ASCII string from 1 to 32 characters that identifies the VTP administrative domain for the switch. The domain name is case sensitive.
file *filename*	Specify the IOS file system file where the VTP VLAN configuration is stored.
interface *name*	Specify the name of the interface providing the VTP ID updated for this device.
mode	Specify the VTP device mode as client, server, or transparent.
client	Place the switch in VTP client mode. A switch in VTP client mode is enabled for VTP, and can send advertisements, but does not have enough nonvolatile storage to store VLAN configurations. You cannot configure VLANs on the switch. When a VTP client starts up, it does not send VTP advertisements until it receives advertisements to initialize its VLAN database.
server	Place the switch in VTP server mode. A switch in VTP server mode is enabled for VTP and sends advertisements. You can configure VLANs on the switch. The switch can recover all the VLAN information in the current VTP database from nonvolatile storage after reboot.
transparent	Place the switch in VTP transparent mode. A switch in VTP transparent mode is disabled for VTP, does not send advertisements or learn from advertisements sent by other devices, and cannot affect VLAN configurations on other devices in the network. The switch receives VTP advertisements and forwards them on all trunk ports except the one on which the advertisement was received. When VTP mode is transparent, the mode and domain name are saved in the switch running configuration file, and you can save them in the switch startup configuration file by entering the **copy running-config startup config** privileged EXEC command.
password *password*	Set the administrative domain password for the generation of the 16-byte secret value used in MD5 digest calculation to be sent in VTP advertisements and to validate received VTP advertisements. The password can be an ASCII string from 1 to 32 characters. The password is case sensitive.
pruning	Enable VTP pruning on the switch.
version *number*	Set VTP version to version 1 or version 2.

Defaults

The default filename is *flash:vlan.dat*.

The default mode is server mode.

No domain name or password is defined.

No password is configured.

Pruning is disabled.

The default version is version 1.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.
12.1(6)EA2	The **if-id** keyword was added.
12.1(9)EA1	The **domain** and **mode** keywords were added. The **if-id** keyword was replaced by the **interface** keyword.
12.1(11)EA1	The **password**, **pruning**, and **version** keywords were added.

Usage Guidelines

When you save VTP mode and domain name and VLAN configurations in the switch startup configuration file and reboot the switch, the VTP and VLAN configurations are determined by these conditions:

- If both the VLAN database and the configuration file show the VTP mode as transparent and the VTP domain names match, the VLAN database is ignored. The VTP and VLAN configurations in the startup configuration file are used. The VLAN database revision number remains unchanged in the VLAN database.
- If the startup VTP mode is server mode, or the startup VTP mode or domain names do not match the VLAN database, VTP mode and VLAN configuration for the first 1005 VLANs are determined by VLAN database information, and VLANs greater than 1005 are configured from the switch configuration file.
- If the image on the switch or the configuration file is earlier than IOS Release 12.1(9)EA1, the switch reboots using the information in the VLAN database.

The **vtp file** *filename* cannot be used to load a new database; it renames only the file in which the existing database is stored.

Follow these guidelines when configuring a VTP domain name:

- The switch is in the no-management-domain state until you configure a domain name. While in the no-management-domain state, the switch does not send any VTP advertisements even if changes occur to the local VLAN configuration. The switch leaves the no-management-domain state after it receives the first VTP summary packet on any port that is trunking or after you configure a domain name by using the **vtp domain** command. If the switch receives its domain from a summary packet, it resets its configuration revision number to 0. After the switch leaves the no-management-domain state, it can not be configured to re-enter it until you clear the nonvolatile RAM (NVRAM) and reload the software.
- Domain names are case-sensitive.

- After you configure a domain name, it cannot be removed. You can only reassign it to a different domain.

Follow these guidelines when setting VTP mode:

- The **no vtp mode** command returns the switch to VTP server mode.
- The **vtp mode server** command is the same as **no vtp mode** except that it does not return an error if the switch is not in client or transparent mode.
- If the receiving switch is in client mode, the client switch changes its configuration to duplicate the configuration of the server. If you have switches in client mode, be sure to make all VTP or VLAN configuration changes on a switch in server mode. If the receiving switch is in server mode or transparent mode, the switch configuration is not changed.
- Switches in transparent mode do not participate in VTP. If you make VTP or VLAN configuration changes on a switch in transparent mode, the changes are not propagated to other switches in the network.
- If you change the VTP or VLAN configuration on a switch that is in server mode, that change is propagated to all the switches in the same VTP domain.
- The **vtp mode transparent** command disables VTP from the domain but does not remove the domain from the switch.
- The VTP mode must be transparent for you to add extended-range VLANs or for VTP and VLAN information to be saved in the running configuration file.
- If extended-range VLANs are configured on the switch and you attempt to set the VTP mode to server or client, you receive an error message, and the configuration is not allowed.
- VTP can be set to either server or client mode only when dynamic VLAN creation is disabled.

Follow these guidelines when setting a VTP password:

- Passwords are case sensitive. Passwords should match on all switches in the same domain.
- When you use the **no vtp password** form of the command, the switch returns to the no-password state.

Follow these guidelines when setting VTP pruning:

- VTP pruning removes information about each pruning-eligible VLAN from VTP updates if there are no stations belonging to that VLAN.
- If you enable pruning on the VTP server, it is enabled for the entire management domain for VLAN IDs 1 to 1005.
- Only VLANs in the pruning-eligible list can be pruned.
- Pruning is supported with VTP version 1 and version 2.

Follow these guidelines when setting the VTP version:

- Toggling the version 2 (v2) mode state modifies parameters of certain default VLANs.
- Each VTP switch automatically detects the capabilities of all the other VTP devices. To use version 2, all VTP switches in the network must support version 2; otherwise, you must configure them to operate in VTP version 1 mode.
- If all switches in a domain are VTP version 2-capable, you need only to configure version 2 on one switch; the version number is then propagated to the other version-2 capable switches in the VTP domain.
- If you are using VTP in a Token Ring environment, VTP version 2 must be enabled.

- If you are configuring a Token Ring bridge relay function (TrBRF) or Token Ring concentrator relay function (TrCRF) VLAN media type, you must use version 2.
- If you are configuring a Token Ring or Token Ring-NET VLAN media type, you must use version 1.

You cannot save password, pruning, and version configurations in the switch configuration file.

Examples

This example shows how to rename the filename for VTP configuration storage to *vtpfilename*:

```
Switch(config)# vtp file vtpfilename
```

This example shows how to clear the device storage filename:

```
Switch(config)# no vtp file vtpconfig
Clearing device storage filename.
```

This example shows how to specify the name of the interface providing the VTP updater ID for this device:

```
Switch(config)# vtp interface fastethernet
```

This example shows how to set the administrative domain for the switch:

```
Switch(config)# vtp domain OurDomainName
```

This example shows how to place the switch in VTP transparent mode:

```
Switch(config)# vtp mode transparent
```

This example shows how to configure the VTP domain password:

```
Switch(config)# vtp password ThisIsOurDomain'sPassword
```

This example shows how to enable pruning in the VLAN database:

```
Switch(config)# vtp pruning
Pruning switched ON
```

This example shows how to enable version 2 mode in the VLAN database:

```
Switch(config)# vtp version 2
```

You can verify your settings by entering the **show vtp status** privileged EXEC command.

Related Commands

Command	Description
show vtp status	Displays the VTP statistics for the switch and general information about the VTP management domain status.
vtp (VLAN configuration)	Configures most VTP characteristics.

vtp (privileged EXEC)

Use the **vtp** privileged EXEC command to configure the VLAN Trunking Protocol (VTP) password, pruning, and version. Use the **no** form of this command to return to the default settings.

vtp {**password** *password* | **pruning** | **version** *number*}

no vtp {**password** | **pruning** | **version**}

Note Beginning with release 12.1(11)EA1, these keywords are available in the **vtp** global configuration command. This command will become obsolete in a future release.

Syntax Description

password *password*	Set the administrative domain password for the generation of the 16-byte secret value used in MD5 digest calculation to be sent in VTP advertisements and to validate received VTP advertisements. The password can be an ASCII string from 1 to 32 characters. The password is case sensitive.
pruning	Enable VTP pruning on the switch.
version *number*	Set VTP version to version 1 or version 2.

Defaults

No password is configured.

Pruning is disabled.

The default version is version 1.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

Passwords are case sensitive. Passwords should match on all switches in the same domain.

When you use the **no vtp password** form of the command, the switch returns to the no-password state.

VTP pruning removes information about each pruning-eligible VLAN from VTP updates if there are no stations belonging to that VLAN.

If you enable pruning on the VTP server, it is enabled for the entire management domain for VLAN IDs 1 to 1005.

Only VLANs in the pruning-eligible list can be pruned.

Pruning is supported with VTP version 1 and version 2.

Toggling the version 2 (v2) mode state modifies parameters of certain default VLANs.

Each VTP switch automatically detects the capabilities of all the other VTP devices. To use version 2, all VTP switches in the network must support version 2; otherwise, you must configure them to operate in VTP version 1 mode.

If all switches in a domain are VTP version 2-capable, you need only to configure version 2 on one switch; the version number is then propagated to the other version-2 capable switches in the VTP domain.

If you are using VTP in a Token Ring environment, VTP version 2 must be enabled.

If you are configuring a Token Ring bridge relay function (TrBRF) or Token Ring concentrator relay function (TrCRF) VLAN media type, you must use version 2.

If you are configuring a Token Ring or Token Ring-NET VLAN media type, you must use version 1.

You cannot save password, pruning, and version configuration in the switch configuration file.

Examples

This example shows how to configure the VTP domain password:

```
Switch# vtp password ThisIsOurDomain'sPassword
```

This example shows how to enable pruning in the VLAN database:

```
Switch# vtp pruning
Pruning switched ON
```

This example shows how to enable version 2 mode in the VLAN database:

```
Switch# vtp version 2
```

You can verify your setting by entering the **show vtp status** privileged EXEC command.

Related Commands

Command	Description
show vtp status	Displays the VTP statistics for the switch and general information about the VTP management domain status.
switchport trunk pruning	Configures the VLAN pruning-eligible list for ports in trunking mode.
vtp (global configuration)	Configures the VTP filename, interface, domain-name, and mode, which can be saved in the switch configuration file.
vtp (VLAN configuration)	Configures all VTP characteristics but cannot be saved to the switch configuration file.

vtp (VLAN configuration)

Use the **vtp** VLAN configuration command to configure VLAN Trunking Protocol (VTP) characteristics. You access VLAN configuration mode by entering the **vlan database** privileged EXEC command. Use the **no** form of this command to return to the default settings, disable the characteristic, or remove the password.

vtp {**domain** *domain-name* | **password** *password* | **pruning** | **v2-mode** | {**server** | **client** | **transparent**}}

no vtp {**client** | **password** | **pruning** | **transparent** | **v2-mode**}

Note VTP configuration in VLAN configuration mode is saved in the VLAN database when applied.

Syntax Description

domain *domain-name*	Set the VTP domain name by entering an ASCII string from 1 to 32 characters that identifies the VTP administrative domain for the switch. The domain name is case sensitive.
password *password*	Set the administrative domain password for the generation of the 16-byte secret value used in MD5 digest calculation to be sent in VTP advertisements and to validate received VTP advertisements. The password can be an ASCII string from 1 to 32 characters. The password is case sensitive.
pruning	Enable pruning in the VTP administrative domain. VTP pruning causes information about each pruning-eligible VLAN to be removed from VTP updates if there are no stations belonging to that VLAN.
v2-mode	Enable VLAN Trunking Protocol (VTP) version 2 in the administrative domains.
client	Place the switch in VTP client mode. A switch in VTP client mode is enabled for VTP, can send advertisements, but does not have enough nonvolatile storage to store VLAN configurations. You cannot configure VLANs on it. When a VTP client starts up, it does not send VTP advertisements until it receives advertisements to initialize its VLAN database.
server	Place the switch in VTP server mode. A switch in VTP server mode is enabled for VTP and sends advertisements. You can configure VLANs on it. The switch can recover all the VLAN information in the current VTP database from nonvolatile storage after reboot.
transparent	Place the switch in VTP transparent mode. A switch in VTP transparent mode is disabled for VTP, does not send advertisements or learn from advertisements sent by other devices, and cannot affect VLAN configurations on other devices in the network. The switch receives VTP advertisements and forwards them on all trunk ports except the one on which the advertisement was received.

Defaults

The default mode is server mode.

No domain name is defined.

No password is configured.

Pruning is disabled.

VTP version 2 (v2 mode) is disabled.

Command Modes

VLAN configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

If VTP mode is transparent, the mode and domain name are saved in the switch running configuration file, and you can save the configuration in the switch startup configuration file by using the **copy running-config startup-config** privileged EXEC command.

Follow these guidelines when setting VTP mode:

- The **no vtp client** and **no vtp transparent** forms of the command return the switch to VTP server mode.
- The **vtp server** command is the same as **no vtp client** or **no vtp transparent** except that it does not return an error if the switch is not in client or transparent mode.
- If the receiving switch is in client mode, the client switch changes its configuration to duplicate the configuration of the server. If you have switches in client mode, make sure to make all VTP or VLAN configuration changes on a switch in server mode. If the receiving switch is in server mode or transparent mode, the switch configuration is not changed.
- Switches in transparent mode do not participate in VTP. If you make VTP or VLAN configuration changes on a switch in transparent mode, the changes are not propagated to other switches in the network.
- If you make a change to the VTP or VLAN configuration on a switch in server mode, that change is propagated to all the switches in the same VTP domain.
- The **vtp transparent** command disables VTP from the domain but does not remove the domain from the switch.
- The VTP mode must be transparent for you to add extended-range VLANs or for the VTP and the VLAN configurations to be saved in the running configuration file.
- If extended-range VLANs are configured on the switch and you attempt to set the VTP mode to server or client, you receive an error message and the configuration is not allowed.
- VTP can be set to either server or client mode only when dynamic VLAN creation is disabled.

Follow these guidelines when configuring a VTP domain name:

- The switch is in the no-management-domain state until you configure a domain name. While in the no-management-domain state, the switch does not send any VTP advertisements even if changes occur to the local VLAN configuration. The switch leaves the no-management-domain state after receiving the first VTP summary packet on any port that is currently trunking or after configuring a domain name using the **vtp domain** command. If the switch receives its domain from a summary packet, it resets its configuration revision number to zero. After the switch leaves the no-management-domain state, it can never be configured to reenter it until you clear the nonvolatile RAM (NVRAM) and reload the software.

- Domain names are case sensitive.
- After you configure a domain name, it cannot be removed. You can reassign it only to a different domain.

Follow these guidelines when configuring a VTP password:

- Passwords are case sensitive. Passwords should match on all switches in the same domain.
- When the **no vtp password** form of the command is used, the switch returns to the no-password state.

Follow these guidelines when enabling VTP pruning:

- If you enable pruning on the VTP server, it is enabled for the entire management domain.
- Only VLANs included in the pruning-eligible list can be pruned.
- Pruning is supported with VTP version 1 and version 2.

Follow these guidelines when enabling VTP version 2 (v2-mode):

- Toggling the version (v2-mode) state modifies certain parameters of certain default VLANs.
- Each VTP switch automatically detects the capabilities of all the other VTP devices. To use VTP version 2, all VTP switches in the network must support version 2; otherwise, you must configure them to operate in VTP version 1 (**no vtp v2-mode**).
- If all switches in a domain are VTP version 2-capable, you need only to enable VTP version 2 on one switch; the version number is then propagated to the other version-2 capable switches in the VTP domain.
- If you are using VTP in a Token Ring environment or configuring a Token Ring bridge relay function (TrBRF) or Token Ring concentrator relay function (TrCRF) VLAN media type, VTP version 2 (**v2-mode**) must be enabled.
- If you are configuring a Token Ring or Token Ring-NET VLAN media type, you must use VTP version 1.

Examples

This example shows how to place the switch in VTP transparent mode:

```
Switch(vlan)# vtp transparent
Setting device to VTP TRANSPARENT mode.
```

This example shows how to set the administrative domain for the switch:

```
Switch(vlan)# vtp domain OurDomainName
Changing VTP domain name from cisco to OurDomainName
```

This example shows how to configure the VTP domain password:

```
Switch(vlan)# vtp password private
Setting device VLAN database password to private.
```

This example shows how to enable pruning in the proposed new VLAN database:

```
Switch(vlan)# vtp pruning
Pruning switched ON
```

This example shows how to enable V2 mode in the proposed new VLAN database:

```
Switch(vlan)# vtp v2-mode
V2 mode enabled.
```

You can verify your settings by entering the **show vtp status** privileged EXEC command.

Related Commands

Command	Description
show vtp status	Displays the VTP statistics for the switch and general information about the VTP management domain status.
switchport trunk pruning	Configures the VLAN pruning-eligible list for ports in trunking mode.
vtp (global configuration)	Configures the VTP filename, interface, domain-name, and mode.

wrr-queue bandwidth

Use the **wrr-queue bandwidth** global configuration command to assign weighted round-robin (WRR) weights to the four class of service (CoS) priority queues. Use the **no** form of this command to disable the WRR scheduler and enable the strict priority scheduler.

wrr-queue bandwidth *weight1...weight4*

no wrr-queue bandwidth

Syntax Description

weight1...weight4	The ratio of weight1, weight2, weight3, and weight4 determines the weights of the WRR scheduler. Ranges are 1 to 255.

Defaults

WRR is disabled. The strict priority is the default scheduler.

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

WRR allows bandwidth sharing at the egress port. This command defines the bandwidths for egress WRR through scheduling weights.

Examples

This example shows how to assign WRR weights of 10, 20, 30, and 40 to the CoS priority queues 1, 2, 3, and 4:

```
Switch(config)# wrr-queue bandwidth 10 20 30 40
```

This example shows how to disable the WRR scheduler and enable the strict priority scheduler:

```
Switch(config)# no wrr-queue bandwidth
```

You can verify your settings by entering the **show wrr-queue bandwidth** privileged EXEC command.

Related Commands

Command	Description
wrr-queue cos-map	Assigns CoS values to the CoS priority queues.
show wrr-queue bandwidth	Displays the WRR bandwidth allocation for the four CoS priority queues.
show wrr-queue cos-map	Displays the mapping of the CoS to the CoS priority queues.

wrr-queue cos-map

Use the **wrr-queue cos-map** global configuration command to assign class of service (CoS) values to the CoS priority queues. Use the **no** form of this command to set the CoS map to default setting.

wrr-queue cos-map *quid cos1...cosn*

no wrr-queue cos-map

Syntax Description

quid	The queue id of the CoS priority queue. Ranges are 1 to 4 where 1 is the lowest CoS priority queue.
cos1...cosn	The CoS values that are mapped to the queue id.

Defaults

These are the default CoS values:

CoS Value	CoS Priority Queues
0, 1	1
2, 3	2
4, 5	3
6, 7	4

Command Modes

Global configuration

Command History

Release	Modification
12.0(5.2)WC(1)	This command was first introduced.

Usage Guidelines

CoS assigned at the ingress port is used to select a CoS priority at the egress port.

Examples

This example shows how to map CoS values 0, 1, and 2 to CoS priority queue 1, value 3 to CoS priority queue 2, values 4 and 5 to CoS priority 3, and values 6 and 7 to CoS priority queue 4:

```
Switch(config)# wrr-queue cos-map 1 0 1 2
Switch(config)# wrr-queue cos-map 2 3
Switch(config)# wrr-queue cos-map 3 4 5
Switch(config)# wrr-queue cos-map 4 6 7
```

This example shows how to map CoS values 0, 1, 2, and 3 to CoS priority queue 2:

```
Switch(config)# wrr-queue cos-map 2 0 1 2 3
```

After entering the **wrr-queue cos-map 2 0 1 2 3** command, if all other priority queues use their default setting, this is the new mapping:

CoS Value	CoS Priority Queue
Not applied	1
0, 1, 2, 3	2
4, 5	3
6, 7	4

In the previous example, CoS priority queue 1 is no longer used because no CoS value is assigned to the queue.

You can set the CoS values to the default values by entering the **no wrr-queue cos-map** global configuration command.

You can verify your settings by entering the **show wrr-queue cos-map** privileged EXEC command.

Related Commands

Command	Description
wrr-queue bandwidth	Assigns weighted round-robin (WRR) weights to the four CoS priority queues.
show wrr-queue bandwidth	Displays the WRR bandwidth allocation for the four CoS priority queues.
show wrr-queue cos-map	Displays the mapping of the CoS to the priority queues.



Debug Commands

This appendix describes the Catalyst 2950-specific **debug** privileged EXEC commands. These commands are helpful in diagnosing and resolving internetworking problems and should be used only with the guidance of Cisco technical support staff.

Caution Because debugging output is assigned high priority in the CPU process, it can render the system unusable. For this reason, use the **debug** commands only to troubleshoot specific problems or during troubleshooting sessions with Cisco technical support staff. It is best to use the **debug** commands during periods of lower network traffic and fewer users. Debugging during these periods decreases the likelihood that increased **debug** command processing overhead will affect system use.

debug dot1x

Use the **debug dot1x** privileged EXEC command to enable debugging of the 802.1X feature. Use the **no** form of this command to disable debugging output.

debug dot1x {**all** | **authsm** | **backend** | **besm** | **core** | **reauthsm**}

no debug dot1x {**all** | **authsm** | **backend** | **besm** | **core** | **reauthsm**}

Syntax Description

all	Enable debugging of all conditions.
authsm	Enable debugging of the authenticator state machine, which is responsible for controlling access to the network through 802.1X-enabled ports.
backend	Enable debugging of the interaction between the 802.1X process and the switch (Remote Authentication Dial-In User Service [RADIUS] client).
besm	Enable debugging of the backend state machine, which is responsible for relaying authentication request between the client and the authentication server.
core	Enable debugging of the 802.1X process, which includes 802.1X initialization, configuration, and the interaction with the port manager module.
reauthsm	Enable debugging of the re-authentication state machine, which manages periodic re-authentication of the client.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **undebug dot1x** command is the same as the **no debug dot1x** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show dot1x	Displays 802.1X statistics, administrative status, and operational status for the switch or for the specified interface.

debug etherchannel

Use the **debug etherchannel** privileged EXEC command for EtherChannel/Port Aggregation Protocol (PAgP) shim debugging. This shim is the software module that is the interface between the PAgP software module and the port manager software module. Use the **no** form of this command to disable debugging output.

debug etherchannel [**all** | **detail** | **error** | **event** | **idb | linecard**]

no debug etherchannel [**all** | **detail** | **error** | **event** | **idb | linecard**]

Syntax Description

all	(Optional) Display all EtherChannel debug messages.
detail	(Optional) Display detailed EtherChannel debug messages.
error	(Optional) Display EtherChannel error debug messages.
event	(Optional) Debug major EtherChannel event messages.
idb	(Optional) Debug PAgP interface descriptor block messages.
linecard	(Optional) Keyword to debug Switch-Module Configuration Protocol messages to the line card.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If you do not specify a keyword, all debug messages appear.

The **undebug etherchannel** command is the same as the **no debug etherchannel** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show etherchannel	Displays EtherChannel information for the channel.

debug pagp

Use the **debug pagp** privileged EXEC command to debug Port Aggregation Protocol (PAgP) activity. Use the **no** form of this command to disable debugging output.

debug pagp [**all** | **event** | **fsm** | **misc** | **packet**]

no debug pagp [**all** | **event** | **fsm** | **misc** | **packet**]

Syntax Description

all	(Optional) Enable all PAgP debugging.
event	(Optional) Enable debugging of PAgP events.
fsm	(Optional) Enable debugging of the PAgP finite state machine.
misc	(Optional) Enable miscellaneous PAgP debugging.
packet	(Optional) Enable PAgP packet debugging.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug pagp** command is the same as **no debug pagp** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show pagp	Displays PAgP channel-group information.

debug pm

Use the **debug pm** privileged EXEC command to debug port manager (PM) activity. The port manager is a state machine that controls all the logical and physical interfaces. All features, such as VLANs and UniDirectional Link Detection (UDLD), work with the port manager to provide switch functions. Use the **no** form of this command to disable debugging output.

debug pm {**all** | **assert** | **card** | **cookies** | **etherchnl** | **messages** | **port** | **registry** | **sm** | **span** | **split** | **vlan** | **vp**}

no debug pm {**all** | **assert** | **card** | **cookies** | **etherchnl** | **messages** | **port** | **registry** | **sm** | **span** | **split** | **vlan** | **vp**}

Syntax Description

all	Display all PM debugging messages.
assert	Debug assert messages.
card	Debug line-card related events.
cookies	Enable internal PM cookie validation.
etherchnl	Debug EtherChannel-related events.
messages	Debug PM messages.
port	Debug port-related events.
registry	Debug PM registry invocations.
sm	Debug state-machine related events.
span	Debug spanning-tree related events.
split	Debug split-processor.
vlan	Debug VLAN-related events.
vp	Debug virtual-port related events.

Note Though visible in the command-line help strings, the **scp** and **pvlan** keywords are not supported.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **undebug pm** command is the same as the **no debug pm** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.

debug spanning-tree

Use the **debug spanning-tree** privileged EXEC command to debug spanning-tree activities. Use the **no** form of this command to disable debugging output.

debug spanning-tree {**all** | **backbonefast** | **bpdu** | **bpdu-opt** | **config** | **csuf** | **etherchannel** | **events** | **exceptions** | **general** | **mstp** | **pvst+** | **root** | **snmp** | **switch** | **uplinkfast**}

no debug spanning-tree {**all** | **backbonefast** | **bpdu** | **bpdu-opt** | **config** | **csuf** | **etherchannel** | **events** | **exceptions** | **general** | **mstp** | **pvst+** | **root** | **snmp** | **switch** | **uplinkfast**}

Syntax Description

all	Display all spanning-tree debugging messages.
backbonefast	Debug Backbonefast events.
bpdu	Debug spanning-tree bridge protocol data units (BPDUs).
bpdu-opt	Debug optimized BPDU handling.
config	Debug spanning-tree configuration changes.
csuf	Debug cross-stack UplinkFast activity.
etherchannel	Debug EtherChannel support.
events	Debug spanning-tree topology events.
exceptions	Debug spanning-tree exceptions.
general	Debug general spanning-tree activity.
mstp	Debug Multiple Spanning Tree Protocol events.
pvst+	Debug Per-VLAN Spanning Tree Plus (PVST+) events.
root	Debug spanning-tree root events.
snmp	Debug spanning-tree Simple Network Management Protocol (SNMP) handling.
switch	Debug switch shim commands. This shim is the software module that is the interface between the generic Spanning Tree Protocol (STP) code and the platform-specific code of various switch platforms.
uplinkfast	Debug UplinkFast events.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **mstp** and **csuf** keywords were added.

Usage Guidelines

The **undebug spanning-tree** command is the same as the **no debug spanning-tree** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug spanning-tree backbonefast

Use the **debug spanning-tree backbonefast** privileged EXEC command to enable debugging of spanning-tree BackboneFast events. Use the **no** form of this command to disable debugging output.

debug spanning-tree backbonefast [**detail** | **exceptions**]

no debug spanning-tree backbonefast [**detail** | **exceptions**]

Syntax Description

detail	(Optional) Display detailed BackboneFast debugging messages.
exceptions	(Optional) Enable debugging of spanning-tree BackboneFast exceptions.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug spanning-tree backbonefast** command is the same as the **no debug spanning-tree backbonefast** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug spanning-tree bpdu

Use the **debug spanning-tree bpdu** privileged EXEC command to enable debugging of received and transmitted spanning-tree bridge protocol data units (BPDUs). Use the **no** form of this command to disable debugging output.

debug spanning-tree bpdu [**receive | transmit**]

no debug spanning-tree bpdu [**receive | transmit**]

Syntax Description

receive	(Optional) Enable receive BPDU debugging.
transmit	(Optional) Enable transmit BPDU debugging.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug spanning-tree bpdu** command is the same as the **no debug spanning-tree bpdu** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug spanning-tree bpdu-opt

Use the **debug spanning-tree bpdu-opt** privileged EXEC command to enable debugging of optimized spanning-tree bridge protocol data units (BPDUs) handling. Use the **no** form of this command to disable debugging output.

debug spanning-tree bpdu-opt [**detail** | **packet**]

no debug spanning-tree bpdu-opt [**detail** | **packet**]

Syntax Description

detail	(Optional) Debug detailed optimized BPDU handling.
packet	(Optional) Debug packet-level optimized BPDU handling.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug spanning-tree bpdu-opt** command is the same as the **no debug spanning-tree bpdu-opt** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug spanning-tree mstp

Use the **debug spanning-tree mstp** privileged EXEC command to enable debugging of the Multiple Spanning Tree Protocol (MSTP) software. Use the **no** form of this command to disable debugging output.

debug spanning-tree mstp {**all** | **boundary** | **bpdu-rx** | **bpdu-tx** | **errors** | **flush** | **init** | **migration** | **pm** | **proposals** | **region** | **roles** | **sanity_check** | **sync** | **tc** | **timers**}

no debug spanning-tree mstp {**all** | **boundary** | **bpdu-rx** | **bpdu-tx** | **errors** | **flush** | **init** | **migration** | **pm** | **proposals** | **region** | **roles** | **sanity_check** | **sync** | **tc** | **timers**}

Syntax Description

all	Enable all the debugging messages.
boundary	Debug flag changes at these boundaries: • An MST region and a single spanning-tree region running RSTP • An MST region and a single spanning-tree region running 802.1d • An MST region and another MST region with a different configuration
bpdu-rx	Debug the received MST bridge protocol data units (BPDUs)
bpdu-tx	Debug the transmitted MST BPDUs.
errors	Debug MSTP errors.
flush	Debug the port flushing mechanism.
init	Debug the initialization of the MSTP data structures.
migration	Debug the protocol migration state machine.
pm	Debug MSTP port manager events.
proposals	Debug handshake messages between the designated and root switch.
region	Debug the region synchronization between the switch processor (SP) and the route processor (RP).
roles	Debug MSTP roles.
sanity_check	Debug the received BPDU sanity check messages.
sync	Debug the port synchronization events.
tc	Debug topology change notification events.
timers	Debug the MSTP timers for start, stop, and expire events.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(9)EA1	This command was first introduced.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug spanning-tree mstp** command is the same as the **no debug spanning-tree mstp** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug spanning-tree switch

Use the **debug spanning-tree switch** privileged EXEC command to enable debugging of the software interface between the Spanning Tree Protocol (STP) software module and the port manager software module. Use the **no** form of this command to disable debugging output.

debug spanning-tree switch {**all** | **errors** | **general** | **helper** | **pm** | **rx** {**decode** | **errors** | **interrupt** | **process**} | **state** | **tx** [**decode**]}

no debug spanning-tree switch {**all** | **errors** | **general** | **helper** | **pm** | **rx** {**decode** | **errors** | **interrupt** | **process**} | **state** | **tx** [**decode**]}

Syntax Description

all	Enable all the debugging messages.
errors	Enable debugging of error messages for the interface between the spanning-tree software module and the port manager software module.
general	Enable debugging of general events.
helper	Enable debugging of the spanning-tree helper task, which handles bulk spanning-tree updates.
pm	Enable debugging of port manager events.
rx	Display received bridge protocol data unit (BPDU) handling debugging messages. The keywords have these meanings: **decode—**Enable debugging of received packets. **errors—**Enable debugging of receive errors. **interrupt—**Enable debugging of interrupt service requests (ISRs). **process—**Enable debugging of process receive BPDUs. **state—**Enable debugging of spanning-tree port state changes.
tx [**decode**]	Display transmitted BPDU handling debugging messages.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **helper** keyword was added.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug spanning-tree switch** command is the same as the **no debug spanning-tree switch** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug spanning-tree uplinkfast

Use the **debug spanning-tree uplinkfast** privileged EXEC command to enable debugging of spanning-tree UplinkFast events. Use the **no** form of this command to disable debugging output.

debug spanning-tree uplinkfast [**exceptions**]

no debug spanning-tree uplinkfast [**exceptions**]

Syntax Description

exceptions	(Optional) Enable debugging of spanning-tree UplinkFast exceptions.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

This command can be entered only from the switch console.

The **undebug spanning-tree uplinkfast** command is the same as the **no debug spanning-tree uplinkfast** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, select **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show spanning-tree	Displays spanning-tree state information.

debug sw-vlan

Use the **debug sw-vlan** privileged EXEC command to debug VLAN manager activities. Use the **no** form of this command to disable debugging output.

debug sw-vlan {**badpmcookies** | **cfg-vlan** {**bootup** | **cli**} | **events** | **ifs** | **management** | **notification** | **packets** | **registries** | **vtp**}

no debug sw-vlan {**badpmcookies** | **cfg-vlan** {**bootup** | **cli**} | **events** | **ifs** | **management** | **notification** | **packets** | **registries** | **vtp**}

Syntax Description

badpmcookies	Display VLAN manager incidents of bad port manager cookies.
cfg-vlan bootup	Debug config-vlan messages generated when the switch is booting up.
cfg-vlan cli	Debug messages generated when the CLI is in config-vlan mode.
events	Debug VLAN manager events.
ifs	Debug VLAN manager IOS file system (IFS) error tests.
management	Debug VLAN manager management of internal VLANs.
notification	Debug VLAN manager notifications.
packets	Debug packet handling and encapsulation processes.
registries	Debug VLAN manager registries.
vtp	Debug the VLAN Trunking Protocol (VTP).

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.
12.1(9)EA1	The **cfg-vlan** keyword was added.

Usage Guidelines

The **undebug sw-vlan** command is the same as the **no debug sw-vlan** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show vlan	Displays the parameters for all configured VLANs or one VLAN (if the VLAN name or ID is specified) in the administrative domain.
show vtp	Displays general information about VTP management domain, status, and counters.

debug sw-vlan ifs

Use the **debug sw-vlan ifs** privileged EXEC command to enable VLAN manager IOS file system (IFS) error tests. Use the **no** form of this command to disable debugging output.

debug sw-vlan ifs {**open** {**read** | **write**} | **read** {**1** | **2** | **3** | **4**} | **write**}

no debug sw-vlan ifs {**open** {**read** | **write**} | **read** {**1** | **2** | **3** | **4**} | **write**}

Syntax Description

open	Enable VLAN manager IFS debugging of errors in an IFS file open operation.
read	Enable debugging of errors that occurred when opening the IFS VLAN configuration file in order to read it.
write	Enable debugging of errors that occurred when opening the IFS VLAN configuration file in order to write to it.
read	Enable debugging of errors that occurred when performing an IFS file read operation.
{**1** | **2** | **3** | **4**}	Specify the file read operation.
write	Enable debugging of errors that occurred when performing an IFS file write operation.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

When determining the file read operation, Operation **1** reads the file header, which contains the header verification word and the file version number. Operation **2** reads the main body of the file, which contains most of the domain and VLAN information. Operation **3** reads type length version (TLV) descriptor structures. Operation **4** reads TLV data.

The **undebug sw-vlan ifs** command is the same as the **no debug sw-vlan ifs** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show vlan	Displays the parameters for all configured VLANs or one VLAN (if the VLAN name or ID is specified) in the administrative domain.

debug sw-vlan notification

Use the **debug sw-vlan notification** privileged EXEC command to enable debugging messages that trace the activation and deactivation of Inter-Link Switch (ISL) VLAN IDs. Use the **no** form of this command to disable debugging output.

debug sw-vlan notification {**accfwdchange** | **allowedvlancfgchange** | **fwdchange** | **linkchange** | **modechange** | **pruningcfgchange** | **statechange**}

no debug sw-vlan notification {**accfwdchange** | **allowedvlancfgchange** | **fwdchange** | **linkchange** | **modechange** | **pruningcfgchange** | **statechange**}

Syntax Description

accfwdchange	Enable VLAN manager notification of aggregated access interface Spanning Tree Protocol (STP) forward changes.
allowedvlancfgchange	Enable VLAN manager notification of changes to the allowed VLAN configuration.
fwdchange	Enable VLAN manager notification of STP forwarding changes.
linkchange	Enable VLAN manager notification of interface link-state changes.
modechange	Enable VLAN manager notification of interface mode changes.
pruningcfgchange	Enable VLAN manager notification of changes to the pruning configuration.
statechange	Enable VLAN manager notification of interface state changes.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

The **undebug sw-vlan notification** command is the same as the **no debug sw-vlan notification** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show vlan	Displays the parameters for all configured VLANs or one VLAN (if the VLAN name or ID is specified) in the administrative domain.

debug sw-vlan vtp

Use the **debug sw-vlan vtp** privileged EXEC command to enable debugging messages to be generated by the VLAN Trunking Protocol (VTP) code. Use the **no** form of this command to disable debugging output.

debug sw-vlan vtp {**events** | **packets** | **pruning** [**packets** | **xmit**] | **xmit**}

no debug sw-vlan vtp {**events** | **packets** | **pruning** [**packets** | **xmit**] | **xmit**}

Syntax Description

events	Display general-purpose logic flow and detailed VTP debugging messages generated by the VTP_LOG_RUNTIME macro in the VTP code.
packets	Display the contents of all incoming VTP packets that have been passed into the VTP code from the IOS VTP platform-dependent layer, except for pruning packets.
pruning	Enable debugging message to be generated by the pruning segment of the VTP code.
packets	(Optional) Display the contents of all incoming VTP pruning packets that have been passed into the VTP code from the IOS VTP platform-dependent layer.
xmit	(Optional) Display the contents of all outgoing VTP packets that the VTP code requests the IOS VTP platform-dependent layer to send.
xmit	Display the contents of all outgoing VTP packets that the VTP code requests the IOS VTP platform-dependent layer to send, except for pruning packets.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

If no further parameters are entered after the **pruning** keyword, VTP pruning debugging messages appear. They are generated by the VTP_PRUNING_LOG_NOTICE, VTP_PRUNING_LOG_INFO, VTP_PRUNING_LOG_DEBUG, VTP_PRUNING_LOG_ALERT, and VTP_PRUNING_LOG_WARNING macros in the VTP pruning code.

The **undebug sw-vlan vtp** command is the same as the **no debug sw-vlan vtp** command.

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show vtp	Displays general information about VTP management domain, status, and counters.

debug udld

Use the **debug udld** privileged EXEC command to display the UniDirectional Link Detection (UDLD) debug messages. Use the **no** form of this command to disable UDLD debugging.

debug udld {**events** | **packets** | **registries**}

no debug udld {**events** | **packets** | **registries**}

Syntax Description

events	Enable debugging messages for UDLD process events as they occur.
packets	Enable debugging messages for the UDLD process as it receives packets from the packet queue and tries to transmit them at the request of the UDLD protocol code.
registries	Enable debugging messages for the UDLD process as it processes registry calls from the UDLD process-dependent module and other feature modules.

Defaults

Debugging is disabled.

Command Modes

Privileged EXEC

Command History

Release	Modification
12.1(6)EA2	This command was first introduced.

Usage Guidelines

For **debug udld events**, these debugging messages appear:

- General UDLD program logic flow
- State machine state changes
- Program actions for the set and clear ErrDisable state
- Neighbor cache additions and deletions
- Processing of configuration commands
- Processing of link-up and link-down indications

For **debug udld packets**, these debugging messages appear:

- General packet processing program flow on receipt of an incoming packet
- Indications of the contents of the various pieces of packets received (such as type length versions [TLVs]) as they are examined by the packet reception code
- Packet transmission attempts and the outcome

For **debug udld registries**, these categories of debugging messages appear:

- Sub-block creation
- Fiber-port status changes
- State change indications from the port manager software
- MAC address registry calls

Related Commands

Command	Description
show debugging	Displays information about the types of debugging that are enabled. For syntax information, refer to **Cisco IOS Configuration Fundamentals Command Reference For IOS Release 12.1 > Cisco IOS System Management Commands > Troubleshooting Commands**.
show udld	Displays UDLD administrative and operational status for all ports or the specified port.



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